

Decomposition-aware global optimization: certified coordinate grids, conditional recourse, and structural limits

Abstract

Small treewidth alone does not make nonconvex mixed-integer optimization tractable: box-constrained nonconvex quadratic programming is strongly NP-hard at treewidth two. We show that two further quantities make tree decompositions useful for certified global optimization: upper bounds L_i on the curvature of the objective F along each coordinate, and quadratic growth. Subtracting $L_i w^2/8$ at each grid node whose longest adjacent grid interval has length w turns exact dynamic programming over a product grid into a valid lower bound. Min-marginals then filter the grids, and the stage grids form a certificate, checked by recomputation, that needs no growth assumption. Under growth in the L_i -weighted norm, graded grids keep $O(\sqrt{\bar{\kappa}} \log(n+2))$ nodes per coordinate at every accuracy, where $\bar{\kappa}$ is a scale-invariant ratio of curvature to growth. For mixed-integer box quadratic programs with bag size p and input length I , this certifies accuracy 2^{-q} in $f(p, \bar{\kappa})(I+q+1)^5$ bit operations; under growth at a unique minimizer x^* , with the Euclidean ratio $\kappa \geq \bar{\kappa}$, computing an exact minimizer takes $f_1(p, \kappa)I^{O(1)}$ bit operations. The growth constant is never needed. Growth with ratio κ forces the Hessian block of the interior continuous coordinates to have smallest eigenvalue at least $2 \max_i L_i/\kappa$, and every other local minimizer x' to satisfy $F(x') - \min F \geq \max_i L_i \|x' - x^*\|^2/\kappa$. Unless $P = NP$, neither parameter can be dropped and the dependence on κ cannot be polylogarithmic; under the randomized exponential-time hypothesis, the exponent of κ in a bound $f(p)\kappa^{a(p)}I^{O(1)}$ cannot be $o(p)$. We extend the method to exact partial minimization, totally unimodular coupling constraints and several minimizers, and test an exact-arithmetic implementation.

Keywords: global optimization; mixed-integer quadratic programming; tree decomposition; dynamic programming; quadratic growth; certificates; parameterized complexity

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1 Introduction

Optimization models in process systems engineering, energy networks and optimal control are often built from local pieces. A flowsheet links unit models along streams, a multiperiod model links consecutive time stages, and a network model links quantities at adjacent nodes. The objective is then a sum of terms that each depend on a few variables, and the graph recording which variables share a term often has small treewidth. For variables with finitely many values, small treewidth is exactly what dynamic programming needs: nonserial dynamic programming, bucket elimination and junction-tree message passing find a global optimum in time exponential only in the width of a tree decomposition [20, 50, 114].

Mixed-integer nonlinear models have continuous variables and nonconvex terms, and there the situation is less clear. Spatial branch-and-bound solvers relax individual nonconvex terms by convex underestimators and branch on variable domains [16, 17, 139, 171]; they do not use a tree decomposition to organize the search. Discretizing the continuous variables and running a dynamic

program gives neither a valid bound nor a running time that scales well with the accuracy: a uniform grid of mesh h on a unit interval has $\Theta(1/h)$ points, so tables grow like h^{-p} in the bag size p . Sparse polynomial optimization exploits the same structure through linear and semidefinite hierarchies [24, 121, 184]; the known bounds on their size are polynomial in the inverse accuracy [24, 117].

Small treewidth alone does not make continuous nonconvex problems tractable. Del Pia and Khajavirad [53, Theorems 1–3] recently showed that box-constrained nonconvex quadratic programming (box QP) is solvable in strongly polynomial time on forests, but strongly NP-hard at treewidth two even with small integer coefficients, and that quartic minimization is strongly NP-hard on paths. This paper asks which additional quantitative information makes tree decompositions useful for *certified* global optimization of mixed-integer problems, and what fails when it is absent.

Results. We consider $\min\{F(x) : x \in X\}$, with optimal value F^* , where $F = \sum_a f_a(x_{S_a})$ is a sum of factors, X is a box in which some coordinates are integer, and a tree decomposition of the scopes S_a with maximum bag size p is given. Two quantities govern the results: upper bounds L_i on the curvature of F along each coordinate (for a quadratic, $L_i = \max\{H_{ii}, 0\}$), and quadratic growth of F away from a minimizer x^* . *Point growth* $F(x) - F^* \geq g\|x - x^*\|^2$ on X gives the condition number $\kappa = \max\{1, L/g\}$ with $L = \max_i L_i$. *Weighted growth* $F(x) - F^* \geq \gamma \sum_i L_i (x_i - x_i^*)^2$ on X gives $\bar{\kappa} = \max\{1, 1/\gamma\}$. Point growth implies weighted growth with $\bar{\kappa} \leq \kappa$, and $\bar{\kappa}$ does not change when continuous coordinates are rescaled (Definition 3.2).

Choose finite grids in the coordinates and subtract, at each node v of coordinate i , the unary correction $L_i w^2/8$, where w is the length of the longest grid interval ending at v . The minimum of the corrected objective over the grid equals the best vertex lower bound of Bajaj and Hasan [12], taken with per-coordinate curvature bounds, over all cells of the product grid (Proposition 4.2), so it is a lower bound on F^* . Because the corrections are unary, dynamic programming on the given decomposition computes this minimum and all coordinate min-marginals without enumerating cells. Grid intervals whose endpoint min-marginals exceed the value of a known feasible point are removed, and the grids of the successive stages form a *path certificate* (Theorem 4.7).

Growth enters only the running time. Under weighted growth, grids whose spacing grows linearly away from the current corrected minimizer (*graded grids*), combined with this filtering, keep $O(\theta^{-1} \log(n+2))$ nodes per coordinate at every stage, independently of the accuracy, when the grading θ satisfies $8\bar{\kappa}\theta^2 \leq 1$ (Lemma 5.5); filtered uniform grids can need order $\sqrt{n\kappa}$ nodes per coordinate (Corollary 5.11). The algorithm CT (Algorithm 5.6) tries $\theta = 2^{-\mu}$ for $\mu = 2, 3, \dots$ with a cap on the grid size, so the growth constant need not be known. Below, I denotes the input length and $q \geq 0$ an integer.

Theorem 1.1 (Main results). *Let F be an explicit rational quadratic on a mixed box X , given as a sum of factors with a tree decomposition of maximum bag size p , and let $L_i = \max\{0, \partial_{ii}F\}$.*

- (a) *On every instance, $CT(2^{-q})$ returns a feasible rational point $\hat{x}$ and a path certificate showing $\beta \leq F^* \leq F(\hat{x}) \leq \beta + 2^{-q}$. The certificate is checked by recomputing one dynamic program per stage in exact arithmetic, and its validity uses no convexity, uniqueness or growth (Theorems 4.7 and 5.7).*
- (b) *If F has weighted growth with condition number $\bar{\kappa}$, every grid on which $CT(2^{-q})$ builds tables has $O(\sqrt{\bar{\kappa}} \log(n+2))$ nodes per coordinate, independently of q , and $CT(2^{-q})$ uses at most $f(p, \bar{\kappa})(I+q+1)^5$ bit operations, where $f(p, t) = c_0(c_1 p \sqrt{t})^p t(1 + \log_2 t)^2$ for absolute constants c_0, c_1 . The certificate has size, and is checked in time, within the same bound. Neither the growth constant nor $\bar{\kappa}$ is an input (Lemma 5.5 and Theorem 5.7).*

- (c) *EX* (Algorithm 6.13), which runs $CT(2^{-q})$ for $q = 1, 2, 4, \dots$ and snaps each incumbent to a rational stationary point of a nearby face of the box until an exact test accepts it, returns a global minimizer and F^* on every instance, with a certificate whose validity uses neither growth nor uniqueness. If F has point growth with condition number κ , *EX* uses at most $f_1(p, \kappa)(I + 1)^{C_1}$ bit operations, for a computable function f_1 and an absolute constant C_1 (Theorem 6.14).

In parameterized terms, part (b) is a fixed-parameter bound in the parameters p and $\lceil \bar{\kappa} \rceil$, which *CT* neither receives nor computes; here $\bar{\kappa}$ is the smallest weighted condition number of the instance, which exists (Section 3 and Remark 5.8). To the best of our knowledge, no earlier result gives a running time of the form $f(p, \kappa) \text{poly}(I + q)$ for certified approximation, or $f(p, \kappa) \text{poly}(I)$ for exact solution, of nonconvex or mixed-integer box quadratic programs (Section 2). Under point growth, the approximation result extends, with κ in place of $\bar{\kappa}$, to factors that are polynomials of fixed degree (Corollary 6.22), but continuous polynomial objectives can have irrational minimizers (Example 6.23). Exactness itself needs no uniqueness. Every rational mixed box QP has *set growth*: $F(x) - F^* \geq g_S \text{dist}(x, \mathcal{S})^2$ on X for some $g_S > 0$, where $\mathcal{S}$ is the set of minimizers; we put $\kappa_S = \max\{1, L/g_S\}$. A certified gap of 2^{-k} with $k = \text{poly}(I) + \max\{0, \log_2(1/g_S)\}$ lets the snapping step return an exact minimizer (Theorem 6.9 and Remark 6.10); point growth in part (c) only bounds the time needed to reach this accuracy. Table 1 lists these results together with the extensions described below.

Scope of the growth hypothesis. For a box QP, minimality alone confines nonconvexity: at every minimizer, the Hessian block $H_{J_0 J_0}$ of the set J_0 of continuous coordinates strictly inside their bounds is positive semidefinite, so every direction of negative curvature involves active bounds or integer coordinates. Growth adds quantitative conditions (Lemma 3.3). Locally, point growth forces $H_{J_0 J_0} \succeq 2gI$, so $\kappa \geq 2L/\lambda_{\min}(H_{J_0 J_0})$ if $J_0 \neq \emptyset$, and weighted growth forces $H_{J_0 J_0} \succeq 2\gamma \text{diag}(L_i)_{i \in J_0}$. Globally, every other local minimizer x' satisfies $F(x') - F^* \geq g\|x' - x^*\|^2$, so $\kappa \geq L\|x' - x^*\|^2/(F(x') - F^*)$. Within these conditions an instance can have an indefinite Hessian, $\kappa \leq \frac{21}{4}$ and exponentially many strict local minima (Example 5.12). The condition number can be exponential in I , already for $n = 2$ (Section 3). We do not claim that κ or $\bar{\kappa}$ is moderate for the application models mentioned above; they enter only the running-time bounds, never the validity of the output.

Lower bounds. Parts (a)–(c) of Theorem 1.2 assume $P \neq NP$, the exponential-time hypothesis (ETH) that, for some $c > 0$, no algorithm decides 3-SAT on n' variables in time $2^{cn'}$ [101], or its randomized version (rETH), which makes the same claim for algorithms with error probability at most $1/3$ [56]; part (d) is unconditional.

Theorem 1.2 (Lower bounds).

- (a) *Unless $P = NP$, no algorithm computes, for every rational continuous box QP with a supplied tree decomposition of maximum bag size three and a unique minimizer, F^* within 2^{-q} in time polynomial in $I + q + \log \kappa$. The hard instances have $\log \kappa = O(I)$ (Proposition 10.1 and Corollary 10.2).*
- (b) *Computing F^* within $\frac{1}{2}$ is NP-hard for box QPs with $\bar{\kappa} \leq \kappa \leq 2$, continuous or all-binary, by a reduction from maximum independent set; under ETH it cannot be done in time $2^{o(p)} I^{O(1)}$ (Proposition 10.5).*

Table 1: Main results (lower bounds: Theorem 1.2), with a tree decomposition unless stated otherwise. Bounds are in bit operations, except where they count table entries or maximum flows; “exact” marks the bound for computing an exact minimizer. Certificates are valid without the growth hypotheses, which enter only the running times. With $\kappa(L', g') = \max\{1, L'/g'\}$, $\kappa_V = \kappa(L^V, g)$ in the row of Theorem 7.2 and $\kappa_V = \kappa(L^+, g)$, with the reduced curvatures L^+ , in the row of Theorem 7.9; in these two rows the exact bounds with κ_V assume a quadratic F with point growth. The function f' is computable, κ_K is the condition number of the value function on the core (Proposition 7.19), and the degrees of the polynomials and C_1 are absolute constants, except in the row of Theorem 7.2, where the degrees depend on the time bounds of the oracles. In the TU row, $K = \max\{K_Z, s/\eta + 1, r(5 + 2\sqrt{n_c \kappa_c})\}$, n_c is the number of continuous coordinates and $K_Z \leq I$ bounds the number of values of a discrete coordinate.

Result	Setting and hypothesis	Parameters	Bound
Thm. 5.7	mixed box QP; weighted growth	$p, \bar{\kappa}$	$f(p, \bar{\kappa})(I + q + 1)^5$
Thm. 6.14	as above; point growth	p, κ	exact $f_1(p, \kappa)(I + 1)^{C_1}$
Thm. 7.2	value function, exact polynomial-time oracles; point growth of V	p, κ_V	$f(p, \kappa_V) \text{poly}(\kappa_V, I + q)$; exact $f(p, \kappa_V) \text{poly}(\kappa_V, I)$
Thm. 7.9	private convex QP blocks, certified responses; point growth of V	p, κ_V	as above; exact $2^{O(p)} \text{poly}(I)$ if $L^+ = 0$
Thms. 7.15, 7.18, D.2	residual concave in each coordinate, balanced; core of k continuous coordinates, no decomposition; core growth	k, κ_K	one maximum flow per query; exact $8^k(1 + \sqrt{k\kappa_K})^k \text{poly}(I)$
Thm. 7.24	balanced box QP, no decomposition; point growth	n, κ	$(n\kappa(I + q + 1))^{O(1)}$; exact $(n\kappa(I + 1))^{O(1)}$
Thm. 8.9	TU coupling, aligned data, exactly feasible output; set growth, at most r optimal values per coordinate	$p, \kappa_c, s/\eta, r$	$O(I + q)$ levels of K^p table entries
Thm. 9.4	finite optimal set; set growth	p, r, κ_S	$K_S^p \text{poly}(I + q)$, $K_S = O(r\sqrt{n\kappa_S})$; exact $K_S^p \text{poly}(I + \log \kappa_S)$
Thm. 9.7, Cor. 9.8	$H_{ii} \leq 0$ for all i , mixed box; no growth; output: a description of the optimal set	p	$2^{O(p)} \text{poly}(I)$
Thm. 9.14	continuous box QP, diagonal Lagrangian certificate; set growth; outputs also an exact description of the optimal set	p, κ_S	exact $f'(p, \kappa_S) \text{poly}(I)$

(c) Under $r\text{ETH}$ there are no computable f and computable nondecreasing unbounded $\psi \geq 1$ such that some algorithm outputs, for every integer box QP with a supplied tree decomposition of maximum bag size p and a unique minimizer, a feasible point within $\frac{1}{2}$ of F^* in time $f(p)(\kappa I)^{p/\psi(p)}$ (Proposition 10.6).

(d) For $p \geq 1$ and $\kappa \geq 8p$, every deterministic algorithm that accesses F only through its values and returns a point with gap below Lp/κ for all functions on $[0, 1]^p$ with upper coordinate curvature L , a unique minimizer and point growth L/κ makes at least $(\kappa/(8p))^{p/2} - 1$ evaluations on one of them (Proposition 10.3).

Thus, unless $P = NP$, neither parameter can be dropped and the dependence on κ cannot be polylogarithmic; under ETH the dependence on p is exponential even for $\kappa \leq 2$; and under $r\text{ETH}$ the exponent of κ in a bound $f(p)\kappa^{a(p)} \text{poly}(I)$ cannot be $o(p)$, already for integer box quadratics, whereas Theorem 1.1(b) has exponent $p/2 + O(1)$. Because $\bar{\kappa} \leq \kappa$, the same holds for $\bar{\kappa}$. In the

value-oracle model of part (d), CT attains the optimal exponent $p/2$ of κ (Remark 10.4).

Section 10 proves further limits. The expanding-box chain of length m (Example 5.13) has bag size three and $\kappa \leq 80$, and CT certifies accuracy 2^{-q} on it in time polynomial in m and q , but the exact Bellman message for its last state has at least 2^{m-1} quadratic pieces (Proposition 10.7). Growth towards a continuum of minimizers does not bound the grids produced by filtering, and under growth towards an unknown optimal set, a deterministic algorithm that queries values and derivatives needs at least of order $\varepsilon^{-n/2}$ queries for accuracy ε , already when $\kappa_S = 1$. A path of local affine equalities makes the problem NP-hard at bag size three and $\kappa = 1$, and matching separator moments of any finite order gives no local error bound in terms of negative curvature. As far as we know, the expanding-box chain, the unique-minimizer hardness with explicit growth (a refinement of [53, Remark 2]), the rETH bound (a quadratic encoding of multicolored clique with a unique minimizer) and the moment obstruction are new; the other statements adapt standard constructions, among them the running-sum encoding of Subset Sum of Bienstock and Muñoz [24, Appendix A] (see also [38, Example 1.1]).

Conditional recourse. A stiff convex term inflates the condition number even when the subproblem it defines is easy. Exact minimization over some variables for fixed values of the others (*recourse*) preserves upper coordinate curvatures and growth, so the certificate and the growth analysis apply to the value function whenever it is a sum of factors that can be evaluated exactly in polynomial time (Theorem 7.2). Piecewise-affine optimal responses of private convex quadratic blocks certify a reduction of the coordinate curvatures of the value function that is the largest possible for the block (Theorem 7.9 and Proposition 7.8), but recourse does not reduce the parameter to the ratio of negative curvature to growth in general (Proposition 7.11). For a quadratic whose recourse problem is concave in each coordinate and submodular after sign changes, each recourse value is one minimum cut with a flow certificate (Theorems 7.15 and 7.18), and for balanced box quadratics the grid problem itself is a minimum cut, so no tree decomposition is needed (Theorem 7.24). Parametric quadratic programming, cut representations of submodular quadratic functions and threshold encodings of ordered labels are classical; what is new is the curvature and growth accounting that keeps them inside the certificate.

Coupling constraints. The certificate rests on rounding each coordinate independently, which breaks under coupling constraints. Totally unimodular (TU) coupling constraints with mesh-aligned data admit exactly feasible correlated rounding, which restores the certificate and, for quadratics, exact output. Under set growth with finitely many optimal values per coordinate, the algorithm is polynomial for fixed bag size p when the condition number κ_c (built from a curvature bound for the continuous variables and the set-growth constant), the box widths measured in mesh units (s/η) and the number r of optimal values per coordinate are polynomially bounded (Theorem 8.9). Because its meshes are uniform, it is not fixed-parameter tractable in (p, κ_c) (Section 8.7), and its pseudopolynomial first grid cannot be removed unless $P = NP$ (Proposition 10.10).

Several minimizers. One graded grid can need exponential time already with two isolated minimizers (Proposition 9.1). If the optimal set is finite, unions of uniform cells keep $O(r\sqrt{n\kappa_S})$ nodes per coordinate, where r bounds the number of optimal values of a coordinate (Theorem 9.4). For quadratics with nonpositive Hessian diagonal on a mixed box, a dynamic program over the box endpoints describes the whole optimal set as a constraint problem on the same tree decomposition. From it, uniqueness of the minimizer, the dimension of the optimal set and, if it is finite, the number of minimizers are computed in $2^{O(p)} \text{poly}(I)$ bit operations (Theorem 9.7 and Corollary 9.8).

This extends Rosenberg’s face description of the optimal set of a multilinear function [161] and zero-residual descriptions of all optimal labelings of discrete problems [183, 187] to strictly concave coordinates and mixed boxes, in factored form. For continuous box quadratics with a classical diagonal Lagrangian certificate of optimality [106, 125], which is the same at every minimizer, an exact minimizer and an exact description of the optimal set are found in $f'(p, \kappa_S)$ $\text{poly}(I)$ bit operations, for a computable function f' , without knowledge of g_S (Theorem 9.14).

Organization. Section 2 reviews related work, and Section 3 fixes the model, the curvature bounds and the growth conditions. Sections 4–6 prove Theorem 1.1, Sections 7–9 treat the extensions of Table 1, Section 10 proves Theorem 1.2 and the further limits, Section 11 reports experiments, and Section 12 lists open problems. The appendices follow the order of the sections and contain the longer proofs and secondary results.

2 Related work

Tractability of sparse nonconvex quadratic programs. For binary quadratic and pseudo-Boolean optimization, bounded treewidth gives polynomial algorithms by nonserial dynamic programming [20, 42, 50] and exact lift-and-project or linear formulations of size exponential only in the width [25, 122, 181]; sparse moment and sum-of-squares hierarchies exploit the same structure for polynomial problems [121, 184]. For continuous box QP, the tractable classes of unbounded treewidth of Del Pia and Khajavirad [53] and of Khajavirad [110] take variables with nonpositive diagonal to be binary (Remark 7.10 compares Khajavirad’s elimination of the continuous components with our value factors). The exact piecewise-quadratic messages of Del Pia and Khajavirad’s forest algorithm, also used for total-variation problems on trees [116, 120], are compared with our grids in Section 10.4. Box QP is also polynomial for fixed Hessian rank [93]. For convex quadratics with indicator variables, Bhatena et al. give dynamic programs on trees [22] and on graphs of bounded treewidth, the latter linear in n for fixed width, margin, volume growth and condition numbers [21]; this is the closest treewidth-and-conditioning result we know (see also [23]). For integer QP, bounded treewidth together with bounded domains gives fixed-parameter tractability [65]; parameterized by the number of variables, the problem is fixed-parameter tractable with bounded coefficients [126] and $W[1]$ -hard without them [91]; see [74] for integer linear programming. In fixed dimension, integer QP in the plane [54] and, according to a recent preprint, mixed-integer QP (MIQP) [6] are solved exactly in polynomial time. Approximation schemes polynomial in $1/\varepsilon$ exist in fixed dimension [49, 92], for sparse polynomial problems with constraints [24, Theorem 4], for QP with a fixed number of negative eigenvalues in exact real arithmetic [178], and for MIQP with fixed rank and a fixed number of integer variables [51] or, in the bit model, a fixed number of negative eigenvalues and of integer variables [52]. Several of these works explain why, in their settings, the dependence on $1/\varepsilon$ cannot be polylogarithmic unless $P = NP$ [24, 52, 178], from the NP-hardness of Subset Sum or of QP with one negative eigenvalue [151]. Running times that depend on a condition number of the instance are standard in numerical computation [33].

Grid and vertex bounds, filtering and certificates. Lower bounds from function values on a grid are classical for Lipschitz functions [147, 157, 167]; with second-order information the error is quadratic in the mesh [30, 48]. Separable quadratic perturbations that convexify a function (α BB) have maximum separation $\frac{1}{4} \sum_i \alpha_i (u_i - \ell_i)^2$ [2, 4, 134]; perturbations of the opposite sign make it edge-concave, with a vertex-polyhedral envelope [89, 136, 138, 170], and Bajaj and Hasan evaluate such an underestimator, built from an upper bound on the diagonal Hessian entries, at

the 2^n vertices of a box [12]. Piecewise-linear and sawtooth relaxations of squares apply the same second-order error termwise inside a mixed-integer program (MIP) [13, 14, 59].

Removing a region whose relaxation value exceeds an incumbent is optimality-based bound tightening and branch-and-reduce [16, 79, 158, 163], cost-based domain filtering in constraint programming [70] and dead-end elimination in protein design [57, 72]; thresholded min-marginals, computed by two passes of message passing [3, 47, 113, 183], prune states in structured prediction cascades [186]. Exact rational MIP [41, 66] and independently checkable branch-and-bound certificates [37] guard MIP results against floating-point error, and our path certificate follows the latter, with a tree that is a path; interval methods and safe bounds control rounding errors instead [86, 87, 149]. Rational sum-of-squares certificates give exact lower bounds for polynomial objectives [107, 155], but these bounds are values of relaxations, whereas a path certificate brackets F^* within any prescribed 2^{-q} .

Accuracy-independent work under growth. With second-order convergent bounds, the number of branch-and-bound boxes near a nondegenerate minimizer is independent of the tolerance (the cluster problem), but local, asymptotic and possibly exponential in the dimension and in the conditioning of the Hessian [61, 108, 148, 185]. Lemmas 5.4 and 5.5 are a global, nonasymptotic form of this phenomenon in which the exponential dependence falls on the bag size, and $8\bar{\kappa}\theta^2 \leq 1$ plays the role of the bound on the second-order prefactor. Optimistic optimization attains exponential rates when the growth order at the optimum matches the assumed smoothness [141]; for Lipschitz functions with small near-optimal sets, branch-and-bound needs $O(\log(1/\varepsilon))$ function values [154], and Bachoc, Cesari and Gerchinovitz characterize, instance by instance, the number of evaluations needed to find and certify an ε -optimal point [11]. Quadratic growth and error bounds give linear convergence of local methods [60, 109, 130, 145], and restarts remove the need to know the growth constant [162], as our capped trials do.

Proximity and scaling give algorithms logarithmic in the domain size for linear and separable convex integer programs [40, 67, 80, 96, 100]. The closest classical analogue of our stages is the scaling algorithm of Hochbaum and Shanthikumar. For a separable convex objective over $\{x : Ax \geq b\}$, each iteration minimizes a piecewise-linear interpolation of the objective on a grid of $O(n\Delta_A)$ points per variable, where Δ_A bounds the absolute values of the subdeterminants of A . The grid lies in a box around the previous solution, and a proximity theorem allows the box to be halved. For continuous variables, $\lceil \log_2(s/(2\varepsilon)) \rceil$ such grids, with s the initial box width, give a point within ε of an optimal solution in the maximum norm, guaranteed but not certified [96, Section 1.3 and Theorems 1.1, 3.8 and 4.4]. For totally unimodular A and integral data, the integer problem is solved by one linear program over the interpolation at all integer points [137], or by one such linear program per scale [96, Theorem 4.3]; Lemma 8.3 uses the same integrality. Our objectives are nonseparable and nonconvex. The retained region comes from min-marginal filtering, which needs no assumption and yields a certificate; growth only bounds its size. The grid problems are solved by dynamic programming over a tree decomposition.

Adaptive discretization. Coarse-to-fine dynamic programming refines superstates with optimistic costs along the current optimal path, so that each coarse problem is a lower bound [68, 160]. Interval methods give discrete lower bounds for continuous Markov random fields [153] and, for optimal power flow on trees, a number of bound evaluations polynomial in $1/\varepsilon$ [62, Theorem 2]. Decision diagrams give outer approximations of integer nonlinear programs [45] and, on partitions of continuous domains with spatial branch-and-bound, a global method for bounded mixed-integer nonlinear programs [46]. Continuous distributed constraint optimization [94, 175], variable-resolution

control [142], corridor methods [90], two-level dynamic programming [124], adaptive multivariate partitioning with bound tightening [144] and re-centered discretizations of fixed size [83] also refine discretizations adaptively. None of these methods certifies a given accuracy with a number of states independent of that accuracy.

Exact rational output. A rational QP that attains its minimum has a global minimizer of polynomial encoding length [55, 177]; the stationary-polytope argument of Section 6 follows Vavasis, and Remark 6.11 discusses NP membership. Rounding to an exact rational is classical [82, Section 5.1], [119], and in linear programming the optimal face is identified once the gap is below a threshold set by the encoding length [135, 188]; Lemma 6.8 does this for box QP, with set growth turning a small gap into proximity to the optimal set. Uniqueness of a QP minimizer implies quadratic growth [39], and growth towards the optimal set of a QP over a polytope follows from the error bound of Luo and Sturm [129, Theorem 3.3]. For polynomials of degree three or more, irrational minimizers and encoding-size phenomena limit exact output [26].

Value functions and multiparametric programming. Eliminating private variables through their optimal response is standard in Benders decomposition [19, 78] and in condensing for model predictive control [7, 27, 105]. Multiparametric quadratic programming gives piecewise-affine minimizers and piecewise-quadratic values on polyhedral critical regions [18, 172], also with nonunique minimizers [152, 169]; value functions of parameters that enter the objective linearly over a fixed feasible set are concave [69, 133], and infima preserve semiconcavity [34]. We use these facts for a global upper coordinate curvature bound of the reduced objective. Low negative inertia with convex recourse is treated by spatial branching in [128, 178].

Cuts and submodularity. Binary quadratic functions with nonpositive pair coefficients are minimized by one minimum cut [29, 103, 115, 156], also after sign switching when the signed interaction graph is balanced [84, 88, 150]. Variables with nonpositive diagonal can be fixed to endpoints [53, 110, 161], so continuous quadratics that are submodular and concave in each coordinate reduce to such cuts [32], and pairwise problems submodular with respect to label orders reduce to cuts on threshold encodings [95, 102, 164]. For general continuous submodular quadratics, Burer, Natarajan and Willemsen show that a semidefinite relaxation is exact up to dimension three and give a four-variable gap; whether the problem is polynomial when the dimension is part of the input is open [32, footnote 2] (for each fixed dimension it is, by enumerating faces [177]). A sign-restricted case is exact [112], and discretization methods [8–10] are polynomial in $1/\varepsilon$; for balanced box quadratics under point growth, Theorem 7.24 is polynomial in I , κ and $\log(1/\varepsilon)$, which does not settle the open question because κ can be exponential in I . Partial minimization preserves submodularity [173, 174], which underlies the mixed continuous–discrete methods of [31, 85], with certificates from greedy extreme bases [43, 63, 104].

Constraints, optimal sets and limits. Integrality of totally unimodular fibers is the Hoffman–Kruskal theorem [97, 165], and marginal-preserving rounding under hard constraints is classical [35, 73]; Remark 8.11 compares Section 8 with the linear programs of Bienstock and Muñoz [24]. Section 9.2 relates our description of the optimal set to Rosenberg’s face criterion and to tree reparameterizations [161, 183, 187]. Diagonal Lagrangian certificates for box QP are given in [15, 106, 125], and exactness of relaxations of box QP is characterized in [159]; sparse second-order cone and semidefinite formulations are exact under graph conditions [58, 111, 168], and chordal sparsity guarantees positive semidefinite completions [81, 176]. Marginals that agree on separators

glue into a joint measure on tree-structured families [180, 182], but locally consistent moments need not [166, 182], and complete parametric descriptions can have exponentially many pieces [77, 131, 143].

3 Problem class, decompositions and conditioning

3.1 Model

We study

$$\min_{x \in X} F(x) = \sum_{a \in \mathcal{A}} f_a(x_{S_a}), \quad X = \prod_{i=1}^n X_i, \quad (3.1)$$

where $X_i = [\ell_i, u_i]$ for $i \in I_C$ (continuous coordinates) and $X_i = [\ell_i, u_i] \cap \mathbb{Z}$ for $i \in I_Z$ (integer coordinates). Integer endpoints are rounded inward, an empty domain makes the problem infeasible, and a coordinate with $\ell_i = u_i$ is substituted out; so we assume $\ell_i < u_i$ for all i , with $\ell_i, u_i \in \mathbb{Z}$ for $i \in I_Z$. We write $s_i = u_i - \ell_i$, $s = \max_i s_i$, and $\bar{X} = \prod_i [\ell_i, u_i]$ for the *continuous hull* of X . Each factor f_a depends only on the coordinates in its scope $S_a \subseteq [n] = \{1, \dots, n\}$ and is defined and continuous on the projection of $\bar{X}$, including fractional values of integer coordinates. We write $F^* = \min_X F$, $\mathcal{S}$ for the set of global minimizers, and $\mathcal{S}_i = \{x_i : x \in \mathcal{S}\}$ for its projection on coordinate i .

Tree decompositions. A tree decomposition of the scopes is a tree T with node set $\mathcal{T}$, $N = |\mathcal{T}|$, and bags $B_t \subseteq [n]$ such that every scope S_a is contained in some bag and, for each coordinate i , the tree nodes whose bags contain i form a nonempty connected subtree. Each factor is assigned to one bag containing its scope and each coordinate to one *home bag* containing it. We write $p = \max_t |B_t|$ for the maximum bag size, so the width of the decomposition is $p - 1$. The decomposition is part of the input; its validity is checked directly. A decomposition of width at most $2 \text{tw} + 1$, where tw is the treewidth of the interaction graph, can be computed in time $2^{O(\text{tw})}n$ [118], and good heuristic decompositions are standard [28]. No bound on the number of bags containing a coordinate is assumed.

Encoding. For complexity statements the data are rational: box endpoints, and factors given as explicit lists of monomials with rational coefficients. The input length I is the binary length of these data, including the decomposition and the assignment of factors to bags. Running times are counted in bit operations, and a target accuracy is written 2^{-q} with $q \in \mathbb{Z}_{\geq 0}$.

3.2 Coordinate curvature

The certificate of Section 4 needs only one-dimensional upper curvature information.

Definition 3.1 (Coordinate curvature). Numbers $L_1, \dots, L_n \geq 0$ are *upper coordinate curvatures* of F if, for every i and every $x \in \bar{X}$, the function $t \mapsto F(x + te_i) - \frac{L_i}{2}t^2$ is concave on $\{t : x + te_i \in \bar{X}\}$. We put $P = \{i : L_i > 0\}$, $n_P = |P|$, $L = \max_i L_i$, and

$$\|d\|_{\mathbf{L}}^2 = \sum_{i \in P} L_i d_i^2 \quad (d \in \mathbb{R}^n).$$

The condition concerns the sum F , not the individual factors, and it must hold on the continuous hull, including fractional points between integer values. If F is twice differentiable it says $\partial_{ii} F \leq L_i$ on $\bar{X}$. For a quadratic $F(x) = \frac{1}{2}x^\top Hx + b^\top x + c$ the smallest valid choice is $L_i = \max\{H_{ii}, 0\}$; the

off-diagonal entries of H are unrestricted, so F can be far from convex. For explicit polynomials, valid rational bounds follow from interval evaluation of $\partial_{ii}F$ over $\bar{X}$ (Lemma 6.20). A coordinate with $L_i = 0$ is concave along its own direction, so replacing x_i by the better of ℓ_i and u_i never increases F ; this classical endpoint property [53, 161] is the case of zero correction below.

3.3 Quadratic growth and conditioning

The running-time analysis uses a growth condition at a minimizer or at the set of minimizers.

Definition 3.2 (Growth and condition numbers). For $L' \geq 0$ and $g' > 0$ put $\kappa(L', g') = \max\{1, L'/g'\}$. Problem (3.1) has

- (a) *point growth* with constant $g > 0$ if there is $x^* \in X$ with

$$F(x) - F(x^*) \geq g\|x - x^*\|^2 \quad \text{for all } x \in X; \quad (3.2)$$

then x^* is the unique minimizer, and we put $\kappa = \kappa(L, g)$;

- (b) *weighted growth* with constant $\gamma > 0$ if there is a minimizer x^* with

$$F(x) - F^* \geq \gamma\|x - x^*\|_{\mathbf{L}}^2 \quad \text{for all } x \in X, \quad (3.3)$$

and we put $\bar{\kappa} = \kappa(1, \gamma) = \max\{1, 1/\gamma\}$;

- (c) *set growth* with constant $g_S > 0$ if

$$F(x) - F^* \geq g_S \text{dist}(x, \mathcal{S})^2 \quad \text{for all } x \in X, \quad (3.4)$$

and we put $\kappa_S = \kappa(L, g_S)$.

Quadratic growth refers to any of these conditions. The constants are not unique, and a larger constant never gives a larger condition number. Each condition number is defined from a valid constant: a result that assumes growth holds for every valid constant, and a statement that an instance has, for example, $\kappa \leq c$ means that some valid constant gives this bound. If a condition holds, its largest valid constant is the infimum of the ratio of $F(x) - F^*$ to $\|x - x^*\|^2$, $\|x - x^*\|_{\mathbf{L}}^2$ or $\text{dist}(x, \mathcal{S})^2$, respectively, over the $x \in X$ where this denominator is positive (every positive constant is valid if there is no such x); for weighted growth it does not depend on x^* , because (3.3) forces $x'_P = x^*_P$ for every minimizer x' . So each condition number has a smallest value, determined by the instance and the curvature bounds. As a parameter of the instance, $\bar{\kappa}$ denotes this value, $\max\{1, 1/\gamma^*\}$ with γ^* the largest valid γ ($\bar{\kappa} = 1$ if $P = \emptyset$).¹ Distances and $\|\cdot\|$ are Euclidean.

Point growth with constant g is set growth with constant g and $\mathcal{S} = \{x^*\}$, so $\kappa_S = \kappa$. Since $\|d\|_{\mathbf{L}}^2 \leq L\|d\|^2$, it also implies weighted growth, with constant g/L if $L > 0$ (then $\bar{\kappa} = \kappa$) and with every constant if $L = 0$. So $\bar{\kappa}$ can always be taken at most κ , and it has two further advantages. It does not change when a continuous coordinate is rescaled, $x_i = \alpha y_i$, because L_i is multiplied by α^2 and $L_i(x_i - x_i^*)^2$ is unchanged; for a separable quadratic $\sum_i \frac{L_i}{2}(x_i - c_i)^2$ with all $L_i > 0$ on a continuous box containing c , the largest constants give $\bar{\kappa} = 2$ and $\kappa = 2 \max_i L_i / \min_i L_i$. And coordinates with $L_i = 0$ need no growth at all: minimizers may differ outside P .

The conditions are global and imposed only on the feasible set X . They do not require convexity: F may have many local minima, provided each lies above F^* by at least a fixed multiple of its

¹Our κ is unrelated to the integer κ in [53, Lemma 20] (a result of [110]): the least number such that every principal submatrix of order $n - \kappa$ of the Hessian is positive semidefinite.

squared distance from x^* (Example 5.12); Remark 3.4 below states what growth restricts for box QPs. When integer coordinates are present, nearly tied integer assignments make the growth constants small even if F is strongly convex in the continuous coordinates. For quadratics, uniqueness of the minimizer implies point growth for some $g > 0$ (Lemma 6.12), and set growth always holds for some $g_S > 0$ (Remark 6.10). Neither fact bounds the condition number, which can be exponential in I even for $n = 2$. The function $F = (x_1 - x_2)^2 + 2^{-k}x_1^2$ on $[0, 1]^2$ has input length $O(k)$ and the unique minimizer 0. Valid curvature bounds satisfy $L_1 \geq 2 + 2^{1-k}$ and $L_2 \geq 2$, and $F(t, t) = 2^{-k}t^2$, so every valid constant satisfies $g \leq 2^{-k-1}$ and $\gamma \leq 2^{-k}/(L_1 + L_2) < 2^{-k-2}$. Hence $\kappa \geq L_2/g \geq 2^{k+2}$ and $\bar{\kappa} > 2^{k+2}$. Our algorithms use the curvature bounds explicitly; they never need g , γ , g_S or x^* , and the validity of every bound they return depends only on Definition 3.1.

For box QPs, the next lemma certifies growth from the gradient and the Hessian at a point, and shows what minimality and growth imply for the Hessian. For a point $x^* \in X$ let $J_0 = \{i \in I_C : \ell_i < x_i^* < u_i\}$ be its continuous coordinates strictly inside their bounds and $J_\partial = \{i : x_i^* \in \{\ell_i, u_i\}\}$ its coordinates at a bound.

Lemma 3.3 (Growth certificates for box QPs). *Let $F = \frac{1}{2}x^\top Hx + b^\top x + c$ be a quadratic on a mixed box X , $x^* \in X$ and $\zeta = \nabla F(x^*)$.*

- (a) *Suppose that $\zeta_i = 0$ for $i \in J_0$, and that for every $i \in I_C \setminus J_0$ the gradient points inward: $\zeta_i \geq 0$ if $x_i^* = \ell_i$ and $\zeta_i \leq 0$ if $x_i^* = u_i$. Let $M = \text{diag}(\mu)$ with $\mu_i = 0$ for $i \in J_0$, $\mu_i = |\zeta_i|/s_i$ for $i \in J_\partial$ with inward ζ_i , and $\mu_i = -|\zeta_i|$ for the remaining coordinates, which are integer coordinates. Then*

$$F(x) - F(x^*) \geq \frac{1}{2}(x - x^*)^\top (H + 2M)(x - x^*) \quad (x \in X). \quad (3.5)$$

Consequently, if $H + 2M \succeq 2g_0I$ for some $g_0 > 0$, then x^ is the unique minimizer and F has point growth with constant g_0 .*

- (b) *Conversely, let x^* be a minimizer. Then $\zeta_{J_0} = 0$ and $H_{J_0J_0} \succeq 0$. If F has point growth with constant g , then $H_{J_0J_0} \succeq 2gI$; if it has weighted growth with constant γ at x^* , then $H_{J_0J_0} \succeq 2\gamma \text{diag}(L_i)_{i \in J_0}$. In particular, under point growth $g \leq \frac{1}{2}v^\top H v / \|v\|^2$ for every nonzero v supported on J_0 , and if $J_0 \neq \emptyset$, every valid curvature bound satisfies $L \geq 2g$, so $\kappa \geq 2$. If $J_0 = [n]$, then F is convex, and strongly convex under point growth.*

Proof. (a) Let $x \in X$ and $d = x - x^*$. Since F is quadratic, $F(x) - F(x^*) = \zeta^\top d + \frac{1}{2}d^\top H d$ exactly, and it suffices to show $\zeta_i d_i \geq \mu_i d_i^2$ for every i . For $i \in J_0$ both sides vanish. If $x_i^* = \ell_i$ and $\zeta_i \geq 0$, then $0 \leq d_i \leq s_i$, so $\zeta_i d_i = |\zeta_i| d_i \geq |\zeta_i| d_i^2 / s_i = \mu_i d_i^2$; the case $x_i^* = u_i$, $\zeta_i \leq 0$ is symmetric. For the remaining $i \in I_Z$, $d_i \in \mathbb{Z}$ and $|d_i| \leq d_i^2$, which gives $\zeta_i d_i \geq -|\zeta_i| |d_i| \geq -|\zeta_i| d_i^2 = \mu_i d_i^2$. If $H + 2M \succeq 2g_0I$, the right side of (3.5) is at least $g_0 \|d\|^2$, which is positive for $d \neq 0$.

(b) For $i \in J_0$ both directions $\pm e_i$ are feasible from x^* , so minimality gives $\zeta_i = 0$. Let $v \neq 0$ be supported on J_0 . For all sufficiently small $|t|$, $x^* + tv \in X$ and $F(x^* + tv) - F^* = \frac{t^2}{2}v^\top H v$. This is nonnegative by minimality; point growth bounds it from below by $gt^2 \|v\|^2$, and weighted growth by $\gamma t^2 \|v\|_{\mathbf{L}}^2 = \gamma t^2 \sum_{i \in J_0} L_i v_i^2$; this gives the three matrix inequalities. Taking $v = e_i$ with $i \in J_0$ gives $H_{ii} \geq 2g$ under point growth, and every valid curvature bound satisfies $L \geq L_i \geq H_{ii}$, because $t \mapsto F(x + te_i) - L_i t^2 / 2$ has second derivative $H_{ii} - L_i$. If $J_0 = [n]$, then $H = H_{J_0J_0}$. $\square$

Remark 3.4 (What growth restricts). For a box QP, the block $H_{J_0J_0}$ at a minimizer is positive semidefinite (Lemma 3.3(b)), so at every minimizer, with or without growth, every direction of negative curvature involves active bounds or integer coordinates. Point growth with some $g > 0$ is equivalent to uniqueness of the minimizer (Lemma 6.12), so what restricts the instances is the

size of the constants. Point growth gives $H_{J_0 J_0} \succeq 2gI$, hence $\kappa \geq 2L/\lambda_{\min}(H_{J_0 J_0})$ if $J_0 \neq \emptyset$, and weighted growth gives $H_{J_0 J_0} \succeq 2\gamma \text{diag}(L_i)_{i \in J_0}$. For every $x' \in X$ with $F(x') > F^*$, (3.2) gives $\kappa \geq L\|x' - x^*\|^2/(F(x') - F^*)$ and (3.3) gives $\bar{\kappa} \geq \|x' - x^*\|_{\mathbf{L}}^2/(F(x') - F^*)$. Since $\|d\|_{\mathbf{L}}^2 \leq L\|d\|^2$, nearly optimal points far from x^* in the norm $\|\cdot\|_{\mathbf{L}}$ make both condition numbers large. Within these conditions an instance can have an indefinite Hessian, $\kappa \leq \frac{21}{4}$ and exponentially many strict local minima (Example 5.12). Part (a) certifies growth when the first-order terms $\zeta_i d_i$ at active bounds supply the growth that the Hessian lacks. At integer coordinates it uses integrality only to bound $\zeta_i d_i$ below by $-|\zeta_i|d_i^2$.

3.4 Notation

Table 2 lists the symbols used in several sections and the second meanings of a few of them; symbols local to one proof, example or algorithm are defined where they are used.

Table 2: Reserved symbols.

Symbol	Meaning	Defined in
$n, [n], I_C, I_Z$	number of variables, $\{1, \dots, n\}$, continuous and integer coordinates	§3
$X, \bar{X}, \ell_i, u_i$	domain, its continuous hull, bounds of coordinate i	§3
s_i, s	width $u_i - \ell_i$, largest width	§3
$F, \mathcal{A}, f_a, S_a$	objective, factors, factor, scope	(3.1)
$F^*, \mathcal{S}, \mathcal{S}_i$	optimal value, set of minimizers, its projection on coordinate i	§3
T, N, B_t, p	tree decomposition, number of bags, bag, maximum bag size	§3
S_{tu}	separator $B_t \cap B_u$ of adjacent tree nodes	§4.2
I, q	input length, target accuracy 2^{-q}	§3
$L_i, L, P, n_P, \ \cdot\ _{\mathbf{L}}$	upper coordinate curvatures, $\max_i L_i$, $\{i : L_i > 0\}$, $ P $, weighted norm	Def. 3.1
$\kappa(L', g')$	$\max\{1, L'/g'\}$	Def. 3.2
g, κ, x^*	point growth constant, $\kappa(L, g)$, minimizer	Def. 3.2
$\gamma, \bar{\kappa}$	weighted growth constant, $\max\{1, 1/\gamma\}$	Def. 3.2
g_S, κ_S	set growth constant, $\kappa(L, g_S)$	Def. 3.2
J_0, J_∂	continuous coordinates strictly inside, coordinates at a bound	Lemma 3.3
$X', X^{(j)}$	subbox, box of stage j	§4
$G = \prod_i G_i, K$	grid, largest number of nodes per coordinate	§4
K_S	bound on the number of nodes per coordinate under set growth	Thm. 9.4
$J, w(J), w_i(v)$	grid interval, its effective width, largest width at node v	§4
J	also the last level of CORE, TU-GRID and UC	§§7–9
d_i, D, Q, β, m_i	corrections, their sum, $F - D$, $\min_G Q$, min-marginals	Def. 4.1
U	filtering threshold; value of the best known feasible point	Def. 4.4
$\theta, h_{ij}, \eta_j, r_i$	grading, mesh $\eta_j r_i$ of coordinate i at stage j , scale 2^{E-j} , $2^{-\varpi_i}$	§5
$h_j, j_{\max}$	common mesh, stage limit of TRIAL and CT	§5
Δ, R, Ω, τ	denominator and height constants of exact output (radius R in §5)	§6
x°, J_f	a minimizer, free coordinates of REC	§6
$f(p, t)$	function in the time bound of CT (not a factor f_a)	Thm. 5.7
f_1, C_1	function and exponent in the time bound of EX	Thm. 6.14
$\mathcal{K}, \mathcal{R}, V$	retained and recourse coordinates, value function	§7
ν	$\max\{0, -\lambda_{\min}(\nabla^2 F)\}$, size of the negative curvature	§7.3
r	number of private blocks (§7); bound on all $ \mathcal{S}_i $ (§§8–10)	§§7, 8
$\bar{L}, \kappa_c, \eta$	directional curvature, $\kappa(\bar{L}, g_S)$, mesh unit under coupling constraints	§8

4 Corrected coordinate grids

This section shows how a finite dynamic program yields a valid lower bound for the continuous mixed-integer problem, and how its coordinate min-marginals remove parts of the domain without losing any global minimizer. Only Definition 3.1 is used: no convexity, uniqueness or growth.

4.1 Corrected grids and cell bounds

A *subbox* is a product $X' = \prod_i X'_i \subseteq X$ with $X'_i = [\ell'_i, u'_i]$ for $i \in I_C$ and $X'_i = [\ell'_i, u'_i] \cap \mathbb{Z}$ with integers ℓ'_i, u'_i for $i \in I_Z$, where $\ell'_i \leq u'_i$. A *grid* for X' is a product $G = \prod_i G_i$ of finite sets $G_i \subseteq X'_i$ that contain ℓ'_i and u'_i . The elements of G_i are *nodes*, and $|G_i|$ is the number of nodes per coordinate. Consecutive nodes $a < a'$ of G_i form a *grid interval* $J = [a, a']$; if $\ell'_i = u'_i$, the single node a forms the grid interval $[a, a]$. The *effective width* of a grid interval $J = [a, a']$ of coordinate i is $w(J) = 0$ if $i \in I_Z$ and $a' - a = 1$, and $w(J) = a' - a$ otherwise; an integer interval of length one has no feasible point in its interior.

Definition 4.1 (Corrections, bound and min-marginals). For $v \in G_i$ let $w_i(v)$ be the largest effective width of a grid interval with endpoint v . Put

$$d_i(v) = \frac{L_i}{8} w_i(v)^2, \quad D(y) = \sum_{i=1}^n d_i(y_i), \quad Q(y) = F(y) - D(y) \quad (y \in G), \quad (4.1)$$

$$\beta = \min_{y \in G} Q(y), \quad m_i(v) = \min\{Q(y) : y \in G, y_i = v\} \quad (v \in G_i). \quad (4.2)$$

The function m_i is the *min-marginal* of Q in coordinate i .

The corrections are unary, so Q has the same scopes as F and the same tree decomposition applies to it. A *cell* of G is a product $C = \prod_i J_i$ of grid intervals J_i of G_i ; its *cell bound* is

$$\beta(C) = \min_{v \in \text{vert}(C)} F(v) - \sum_i \frac{L_i}{8} w(J_i)^2, \quad \text{vert}(C) = \prod_i \{\text{endpoints of } J_i\}.$$

Subtracting $\frac{L_i}{2}(x_i - \text{mid } J_i)^2$ in every coordinate with $w(J_i) > 0$ makes F concave along each of these coordinates, so its minimum over $C \cap X$ is attained at a vertex, and $\beta(C)$ is a lower bound for F on $C \cap X$. For a single cell this is the vertex bound of Bajaj and Hasan [12], who evaluate at the vertices the edge-concave underestimator of Hasan [89], built from an upper bound on the diagonal Hessian entries; we use one bound L_i per coordinate. The term $\frac{L_i}{8} w(J_i)^2$ is the maximum separation of the α BB underestimator with $\alpha_i = L_i/2$ [2, 134], used with the opposite sign. The following observation shows that the unary correction (4.1) computes the best of these bounds over all cells of a product grid at once.

Proposition 4.2 (Cellwise form of the corrected grid). *Let G be a grid for a subbox X' .*

- (a) *For every cell C and every $x \in C \cap X$, $F(x) \geq \beta(C)$. Every $x \in X'$ lies in some cell.*
- (b) *$\beta = \min_C \beta(C)$, the minimum over all cells. In particular $\beta \leq \min_{X'} F$.*
- (c) *For a grid interval $J = [a, a']$ of coordinate i , put $\beta_i(J) = \min\{\beta(C) : C_i = J\}$. Then*

$$\beta_i(J) = \min_{v \in \{a, a'\}} (m_i(v) + d_i(v)) - \frac{L_i}{8} w(J)^2 \geq \min\{m_i(a), m_i(a')\},$$

and $F(x) \geq \beta_i(J)$ for every $x \in X'$ with $x_i \in J$. Moreover, $F(x) \geq m_i(v) + d_i(v)$ for every $x \in X'$ with $x_i = v \in G_i$.

Proof. (a) Let $C = \prod_i J_i$ with $J_i = [a_i, a'_i]$, let $W = \{i : w(J_i) > 0\}$, $c_i = (a_i + a'_i)/2$, and $\varphi(y) = F(y) - \sum_{i \in W} \frac{L_i}{2} (y_i - c_i)^2 \leq F(y)$. If $i \notin W$, then $a'_i = a_i$, or $i \in I_Z$ and $a'_i = a_i + 1$; in both cases $x_i \in \{a_i, a'_i\}$. If $i \in W$, then for fixed values of the other coordinates, $t \mapsto \varphi(\dots, t, \dots) = (F(\dots, t, \dots) - \frac{L_i}{2} t^2) + L_i c_i t + \text{const}$ is concave on J_i by Definition 3.1, so its minimum over J_i is attained at an endpoint. Replacing the coordinates in W one at a time by such endpoints keeps the point in $C \cap X$ and does not increase φ , and ends at a vertex v with $\varphi(v) \leq \varphi(x)$. Since $(v_i - c_i)^2 = w(J_i)^2/4$ for $i \in W$, $F(x) \geq \varphi(x) \geq \varphi(v) = F(v) - \sum_i \frac{L_i}{8} w(J_i)^2 \geq \beta(C)$. Every $x_i \in X'_i$ lies in a grid interval because G_i contains both endpoints of X'_i .

(b) For $v \in G_i$ let $\mathcal{J}_i(v)$ be the nonempty set of grid intervals with endpoint v . The pairs (C, y) with $y \in \text{vert}(C)$ are exactly the grid points y together with an independent choice of $J_i \in \mathcal{J}_i(y_i)$ for every i . Hence

$$\min_C \beta(C) = \min_{y \in G} \left[F(y) - \sum_i \max_{J \in \mathcal{J}_i(y_i)} \frac{L_i}{8} w(J)^2 \right] = \min_{y \in G} [F(y) - D(y)] = \beta,$$

and $\beta \leq \min_{X'} F$ follows from (a).

(c) The same computation with the i th interval fixed to J gives the formula, because $Q(y) = F(y) - \sum_{k \neq i} d_k(y_k) - d_i(v)$ when $y_i = v$. Since $J \in \mathcal{J}_i(v)$ for $v \in \{a, a'\}$, $d_i(v) \geq L_i w(J)^2/8$, which gives the inequality. The bound for $x_i \in J$ follows from (a) with $C_i = J$. If $x_i = v \in G_i$, choose $J_i \in \mathcal{J}_i(v)$ and grid intervals $J_k \ni x_k$ for $k \neq i$, and run the argument of (a) with W replaced by $W \setminus \{i\}$. Coordinate i stays at v , and the vertex y reached satisfies $F(x) \geq F(y) - \sum_{k \neq i} \frac{L_k}{8} w(J_k)^2 \geq F(y) - \sum_{k \neq i} d_k(y_k) = Q(y) + d_i(v) \geq m_i(v) + d_i(v)$. $\square$

Enumerating the cells and their up to 2^n vertices is the box-by-box computation of branch-and-bound. Since Q is factorized like F , Section 4.2 instead computes β and all min-marginals by dynamic programming on the given tree decomposition.

Families of intervals. The proof of Proposition 4.2 uses only that each grid interval has endpoints in G_i and that every node is an endpoint of one. The proposition therefore holds for finite sets $G_i \subseteq X_i$ and, in place of the grid intervals, finite families $\mathcal{J}_i$ of intervals with these two properties, with $w_i(v)$ the largest effective width of a member with endpoint v , cells the products of members, and X' replaced by $X \cap \prod_i \bigcup \mathcal{J}_i$. The bounds of Lemma 9.3 for unions of uniform cells are an instance.

Randomized rounding. The bound $\beta \leq F(x)$ for $x \in X'$ also follows from randomized rounding. Round each x_i independently to an endpoint of a grid interval $J_i \ni x_i$, with mean x_i , to obtain a grid point Y . Jensen's inequality for the concave functions $t \mapsto F(\dots, t, \dots) - \frac{L_i}{2} t^2$, one coordinate at a time, gives $\mathbb{E}F(Y) \leq F(x) + \sum_i \frac{L_i}{2} \text{Var}(Y_i)$, and $\frac{L_i}{2} \text{Var}(Y_i) \leq \frac{L_i}{8} w(J_i)^2 \leq d_i(Y_i)$ ($Y_i = x_i$ if $w(J_i) = 0$), so $\mathbb{E}Q(Y) \leq F(x)$. Section 8 uses a correlated version of this rounding that keeps coupling constraints satisfied (Lemma 8.3).

4.2 Dynamic programming and min-marginals

Let q_t be the sum of the factors assigned to bag t minus the corrections d_i of the coordinates whose home bag is t , regarded as a table on $G_{B_t} = \prod_{i \in B_t} G_i$; its entries are the *table entries* of bag t , and $Q = \sum_t q_t$ on G . For adjacent tree nodes t, u let $S_{tu} = B_t \cap B_u$ and define messages

$$M_{t \rightarrow u}(y_{S_{tu}}) = \min_{y_{B_t \setminus S_{tu}}} \left[q_t(y_{B_t}) + \sum_{v \in \mathcal{N}(t) \setminus \{u\}} M_{v \rightarrow t}(y_{S_{vt}}) \right], \quad (4.3)$$

where $\mathcal{N}(t)$ is the set of neighbors of t in T and an empty separator carries a single scalar. Put $A_t = q_t + \sum_{v \in \mathcal{N}(t)} M_{v \rightarrow t}$.

Lemma 4.3 (Two-pass dynamic programming). *For every bag t and every $y_{B_t} \in G_{B_t}$, $A_t(y_{B_t}) = \min\{Q(z) : z \in G, z_{B_t} = y_{B_t}\}$. Hence $\beta = \min A_t$, and $m_i(v)$ is the minimum of A_t over $y_i = v$ for any bag $t \ni i$. With $K = \max_i |G_i|$, all messages, β , a minimizer and all min-marginals are obtained with $O((|A| + n + N)K^p)$ additions and comparisons of table entries and $O(p)$ index work per table entry, hence $O(p(|A| + n + N)K^p)$ operations, plus the evaluation of every factor at the at most $K^{|S_a|}$ grid points of its scope.*

Proof. For an oriented edge $t \rightarrow u$ let $T_{t \rightarrow u}$ be the component of $T \setminus \{tu\}$ containing t , and $\mathcal{W}_{t \rightarrow u}$ the set of coordinates occurring in bags of $T_{t \rightarrow u}$ but not in S_{tu} . By induction on $|T_{t \rightarrow u}|$, $M_{t \rightarrow u}(y_{S_{tu}})$ is the minimum of $\sum_{t' \in T_{t \rightarrow u}} q_{t'}$ over $y_{\mathcal{W}_{t \rightarrow u}}$: the trees $T_{v \rightarrow t}$, $v \in \mathcal{N}(t) \setminus \{u\}$, partition $T_{t \rightarrow u} \setminus \{t\}$, and by running intersection a coordinate occurring both in $T_{v \rightarrow t}$ and outside it lies in S_{vt} , so the sets $\mathcal{W}_{v \rightarrow t}$ are disjoint from each other and from B_t and the minimization splits. The same argument with all neighbors of t gives the formula for A_t . Messages are computed toward a root and back; at bag t the sum of all incoming messages is formed once, and each outgoing message is obtained from $A_t - M_{u \rightarrow t}$ by one pass, with no product over children. A minimizer is recovered from the root outward. $\square$

Lemma 4.3 is classical nonserial dynamic programming, bucket elimination or junction-tree computation [20, 50, 123], and two passes give all min-marginals [113, 183]. The only new element is that its input is the corrected objective Q . Any exact method for β , a minimizer and the min-marginals can replace the messages; Section 7.5 uses minimum cuts for balanced quadratics without a width bound.

4.3 Filtering and certificates

Definition 4.4 (Filtering step). Let G be a grid for a subbox X' with corrected minimum β , and let $U \geq \beta$. For $i \in P$, retain a grid interval $[a, a']$ of coordinate i if

$$\min\{m_i(a), m_i(a')\} \leq U, \quad (4.4)$$

and let X''_i be the smallest interval containing the retained intervals (intersected with $\mathbb{Z}$ for $i \in I_Z$). For $i \notin P$ let $X''_i = X'_i$. The filtered subbox is $X'' = \prod_i X''_i$.

Proposition 4.5 (Safe filtering). *Every $x \in X' \setminus X''$ satisfies $F(x) > U$. Hence X'' contains every $x \in X'$ with $F(x) \leq U$, and every minimizer of F over X' if $U \geq \min_{X'} F$. Every minimizer y of Q over G lies in X'' ; in particular X'' is a subbox.*

Proof. If $x \in X' \setminus X''$, some $i \in P$ has $x_i \notin X''_i$, so every grid interval containing x_i was not retained, and Proposition 4.2(c) gives $F(x) \geq \min\{m_i(a), m_i(a')\} > U$ for such an interval. For a minimizer y of Q and $i \in P$, $m_i(y_i) \leq Q(y) = \beta \leq U$, and y_i is an endpoint of a grid interval of coordinate i , which is therefore retained. So $y_i \in X''_i$ for $i \in P$, and trivially for $i \notin P$. Hence each X''_i is nonempty, and for $i \in I_Z$ its endpoints are nodes, hence integers. $\square$

A method that filters repeatedly produces nested subboxes. The following certificate records only what is needed to check the result.

Definition 4.6 (Path certificate). A *path certificate* consists of grids $G^{(0)}, \dots, G^{(k)}$, a number β and a point $\hat{x}$. Let $X^{(j)}$ be the subbox spanned by $G^{(j)}$: with $\ell_i^{(j)} = \min G_i^{(j)}$ and $u_i^{(j)} = \max G_i^{(j)}$, $X_i^{(j)}$ is $[\ell_i^{(j)}, u_i^{(j)}]$ or its integer points. The certificate is *valid* if $\hat{x} \in X$, the nodes of integer coordinates are integers, $X^{(0)} = X$, $X^{(j+1)} \subseteq X^{(j)}$, and, with $Q^{(j)}$ and $m^{(j)}$ computed from $G^{(j)}$ and the curvature bounds,

(C1) for $j < k$, every grid interval $J = [a, a']$ of $G_i^{(j)}$ that is not contained in $[\ell_i^{(j+1)}, u_i^{(j+1)}]$ satisfies

$$\min\{m_i^{(j)}(a), m_i^{(j)}(a')\} \geq \beta,$$

or has $w(J) = 0$ and $m_i^{(j)}(v) \geq \beta$ for each endpoint v of J outside $[\ell_i^{(j+1)}, u_i^{(j+1)}]$;

(C2) $\min_{G^{(k)}} Q^{(k)} \geq \beta$.

The second alternative in (C1) covers filters that remove single integer nodes, such as the integer-label filter of Lemma B.1.

Theorem 4.7 (Path certificates). *If a path certificate is valid, then $\beta \leq F^* \leq F(\hat{x})$. Its validity is checked by at most $k + 1$ runs of the dynamic program of Lemma 4.3 in exact arithmetic, given the curvature bounds; for a quadratic the checker computes $L_i = \max\{H_{ii}, 0\}$ itself. No growth or uniqueness assumption is involved.*

Proof. Let $x \in X$ and let j be the largest index with $x \in X^{(j)}$. If $j = k$, Proposition 4.2(b) and (C2) give $F(x) \geq \beta$. Otherwise some $x_i \notin X_i^{(j+1)}$, so $x_i \notin [\ell_i^{(j+1)}, u_i^{(j+1)}]$, because x_i is an integer if $i \in I_Z$. A grid interval J of $G_i^{(j)}$ containing x_i is therefore not contained in $[\ell_i^{(j+1)}, u_i^{(j+1)}]$. If the first alternative of (C1) holds for J , Proposition 4.2(c) gives $F(x) \geq \beta$. Otherwise $w(J) = 0$, so the feasible value x_i is an endpoint of J outside $[\ell_i^{(j+1)}, u_i^{(j+1)}]$, and the last claim of Proposition 4.2(c) gives $F(x) \geq m_i^{(j)}(x_i) + d_i^{(j)}(x_i) \geq m_i^{(j)}(x_i) \geq \beta$. $\square$

Corollary 4.8 (Filtering runs give valid certificates). *Let $G^{(0)}$ be a grid for $X^{(0)} = X$ and, for $j < k$, let $G^{(j+1)}$ be a grid for the subbox $X^{(j+1)}$ obtained from $G^{(j)}$ by the filtering step with a threshold $U_j \geq F^*$, for example the value of a feasible point. Let β be the corrected minimum of $G^{(k)}$ and $\hat{x} \in X$. Then every threshold satisfies the condition of Definition 4.4, every minimizer of F lies in every $X^{(j)}$, and $(G^{(0)}, \dots, G^{(k)}, \beta, \hat{x})$ is a valid path certificate.*

Proof. Let β_j be the corrected minimum of $G^{(j)}$. If every minimizer lies in $X^{(j)}$, as it does in $X^{(0)} = X$, then Proposition 4.2(b) gives $\beta_j \leq F^* \leq U_j$, so the filtering step is defined, and Proposition 4.5 puts every minimizer in $X^{(j+1)}$. By induction this holds for all j , and $\beta = \beta_k \leq F^*$. Grids for subboxes have integer nodes in integer coordinates, $X^{(j+1)} \subseteq X^{(j)}$, and (C2) holds with equality. For (C1), let $J = [a, a']$ be a grid interval of $G_i^{(j)}$ not contained in $[\ell_i^{(j+1)}, u_i^{(j+1)}]$. Coordinates outside P are not filtered, so $i \in P$, and J was not retained, since $[\ell_i^{(j+1)}, u_i^{(j+1)}]$ contains every retained interval. Hence $\min\{m_i^{(j)}(a), m_i^{(j)}(a')\} > U_j \geq F^* \geq \beta$. $\square$

So a filtering run is certified by its stage grids (or the boxes with the centers, meshes and grading that generate them), one bound and one feasible point. Messages, min-marginal tables, intermediate incumbents and stages other than the last that remove nothing need not be stored; the stages that remove points are needed, because the last grid bounds F only on $X^{(k)}$. A checker runs the dynamic program once per recorded stage, the same table work as the filtering steps that produced the certificate; Section 11 reports measured replay times.

The certificate is a branch-and-bound proof whose tree is a path, in which filtering acts as optimality-based bound tightening (Section 2); all interval bounds of a stage come from one pair of message passes.

5 Accuracy-independent grids under quadratic growth

Theorem 4.7 holds for any choice of grids. This section chooses them so that, under weighted growth, every coordinate grid has $O(\sqrt{\kappa} \log(n+2))$ nodes at every refinement stage, independently of the accuracy reached. The growth constant is used only in the analysis.

5.1 Graded grids and one trial

Definition 5.1 (Graded grid). Let $X'_i = [\ell'_i, u'_i]$ (or its integer points if $i \in I_Z$), a center $c_i \in X'_i$, a mesh $h > 0$ and a grading $0 < \theta \leq 1/4$ be given. Put

$$\sigma(t) = h + \theta t \quad (i \in I_C), \quad \sigma(t) = \max\{1, \lfloor h + \theta t \rfloor\} \quad (i \in I_Z). \quad (5.1)$$

Starting from $t_0 = 0$, set $t_{k+1} = \min\{t_k + \sigma(t_k), u'_i - c_i\}$ until $t_{k+1} = u'_i - c_i$, and take the nodes $c_i + t_k$; proceed in the same way towards ℓ'_i . These nodes form the *graded grid* of X'_i .

Spacing grows linearly with the distance from the center, so a grid of radius R needs only $O(1 + \theta^{-1} \log(1 + \theta R/h))$ nodes.

Lemma 5.2 (Graded grids). (a) $w_i(v) \leq h + \theta|v - c_i|$ for every node v . (b) If an endpoint is at distance $R > 0$ from c_i , the number of nodes strictly on that side is at most $\lceil (4/\theta) \ln(1 + \theta R/\hat{h}) \rceil$, where $\hat{h} = h$ for $i \in I_C$ and $\hat{h} = \max\{h, 1\}$ for $i \in I_Z$. (c) If $h \geq u'_i - \ell'_i$, the grid has at most three nodes.

The proof is in Appendix A. The meshes are chosen per coordinate so that every coordinate sees the same correction scale. For $i \in P$ let $\varpi_i \in \mathbb{Z}$ satisfy $4^{\varpi_i-1} < L_i \leq 4^{\varpi_i}$ and put $r_i = 2^{-\varpi_i}$, so that $\frac{1}{4} < L_i r_i^2 \leq 1$. Let E be the least integer with $2^E r_i \geq s_i$ for all $i \in P$, and put $\eta_j = 2^{E-j}$ and $h_{ij} = \eta_j r_i$; every h_{ij} is a power of two.

Algorithm 5.3 (TRIAL: one trial). The parameters are a grading $\theta \in (0, 1/4]$, a cap K , an accuracy $\varepsilon > 0$ and a stage limit $j_{\max}$. Start with $X^{(0)} = X$, center $c^{(0)} = \ell$, and the incumbent $\hat{x} = \ell$ with value $U = F(\ell)$, or a better feasible point and its value if one is known. For $j = 0, 1, \dots, j_{\max}$:

- (i) for $i \in P$ let $G_i^{(j)}$ be the graded grid of $X_i^{(j)}$ with center $c_i^{(j)}$, mesh h_{ij} and grading θ ; for $i \notin P$ let $G_i^{(j)} = \{\ell_i, u_i\}$; if some grid would exceed K nodes, *abort* the trial before forming any table;
- (ii) compute β_j , a minimizer $y^{(j)}$ of Q over $G^{(j)}$ and all min-marginals by Lemma 4.3; if $F(y^{(j)}) < U$, set $\hat{x} = y^{(j)}$ and $U = F(y^{(j)})$;
- (iii) if $U - \beta_j \leq \varepsilon$, stop with *success*; otherwise filter with threshold U (Definition 4.4) to obtain $X^{(j+1)}$, and set $c^{(j+1)} = y^{(j)}$.

The trial *fails* if stage $j_{\max}$ ends without success.

By Proposition 4.5 the centers stay feasible and inside the current subbox, and by Corollary 4.8 every stage yields a valid path certificate. The analysis allows any first center in X (Lemma 5.4(ii) holds at $j = 0$ for every $c^{(0)} \in X$). Later centers need to be close to x^* : by Lemma 5.4(iii), the corrected minimizer $y^{(j)}$ is as close to x^* as Lemma 5.4(ii) requires of the center of stage $j + 1$. Because the centers are ℓ and grid points, denominators do not compound from stage to stage (Appendix A).

5.2 Contraction and localization

Lemma 5.4 (Stage invariants). *Assume weighted growth with constant γ at a minimizer x^* , let $k \geq \max\{1, 1/\gamma\}$ satisfy $8k\theta^2 \leq 1$, and write $a_j = n_P \eta_j^2$. At every stage j reached by TRIAL, with U the incumbent value after step (ii):*

- (i) $x^* \in X^{(j)}$ and $\beta_j \leq F^*$;
- (ii) $\|c^{(j)} - x^*\|_{\mathbf{L}}^2 \leq 4ka_j$;
- (iii) $\|y^{(j)} - x^*\|_{\mathbf{L}}^2 \leq ka_j$;
- (iv) $U - \beta_j \leq F(y^{(j)}) - \beta_j = D(y^{(j)}) \leq \frac{9}{16}a_j$;
- (v) every grid point z with $Q(z) \leq U$ satisfies $\|z - x^*\|_{\mathbf{L}}^2 \leq \frac{17}{15}ka_j$.

Proof. (i) By induction on j , starting from $x^* \in X^{(0)} = X$. If $x^* \in X^{(j)}$, then $\beta_j \leq \min_{X^{(j)}} F = F^*$ by Proposition 4.2(b), and filtering with threshold $U \geq F^* = F(x^*)$ keeps x^* by Proposition 4.5.

Next, every grid point z satisfies, with $c = c^{(j)}$,

$$D(z) \leq \frac{a_j}{4} + \frac{\theta^2}{2} (\|z - x^*\|_{\mathbf{L}}^2 + \|c - x^*\|_{\mathbf{L}}^2). \quad (5.2)$$

Indeed, for $i \in P$ Lemma 5.2(a) and $L_i h_{ij}^2 \leq \eta_j^2$ give $d_i(z_i) \leq \frac{L_i}{8} (h_{ij} + \theta|z_i - c_i|)^2 \leq \frac{\eta_j^2}{4} + \frac{\theta^2}{4} L_i (z_i - c_i)^2$, and $(z_i - c_i)^2 \leq 2(z_i - x_i^*)^2 + 2(c_i - x_i^*)^2$; for $i \notin P$, $d_i = 0$.

(ii) For $j = 0$ and any $c^{(0)} \in X$, $\|c^{(0)} - x^*\|_{\mathbf{L}}^2 \leq \sum_{i \in P} L_i s_i^2 \leq \sum_{i \in P} L_i r_i^2 \eta_0^2 \leq a_0$. For $j \geq 1$ it follows from (iii) at stage $j - 1$, because $c^{(j)} = y^{(j-1)}$ and $a_{j-1} = 4a_j$.

(iii) Write $Y = \|y^{(j)} - x^*\|_{\mathbf{L}}^2$ and $a = a_j$. Since $\theta^2/2 \leq 1/(16k) \leq \gamma/16$, growth, (i), $Q(y^{(j)}) = \beta_j$, (5.2) and (ii) give $\gamma Y \leq F(y^{(j)}) - F^* \leq F(y^{(j)}) - \beta_j = D(y^{(j)}) \leq \frac{a}{4} + \frac{\gamma}{16}(Y + 4ka)$, so $Y \leq \frac{4}{15}(\gamma^{-1} + k)a \leq \frac{8}{15}ka$.

(iv) By (5.2), (ii), (iii) and $\theta^2 k \leq \frac{1}{8}$, $D(y^{(j)}) \leq \frac{a}{4} + \frac{\theta^2}{2} \cdot 5ka \leq \frac{9}{16}a$. Moreover $U \leq F(y^{(j)})$ after step (ii), and $F(y^{(j)}) - \beta_j = D(y^{(j)})$ because $Q(y^{(j)}) = \beta_j$.

(v) Let $Z = \|z - x^*\|_{\mathbf{L}}^2$. Using $Q(z) \leq U \leq F(y^{(j)})$, $\beta_j \leq F^*$, (iv) and (5.2), $\gamma Z \leq F(z) - F^* = Q(z) + D(z) - F^* \leq D(y^{(j)}) + D(z) \leq \frac{13}{16}a + \frac{\gamma}{16}(Z + 4ka)$, so $Z \leq (\frac{13}{15}\gamma^{-1} + \frac{4}{15}k)a \leq \frac{17}{15}ka$. $\square$

The proof shows where filtering is essential. Part (v) bounds every grid point whose corrected value can keep an interval alive, and the retained subbox therefore shrinks at the same rate as the mesh.

Lemma 5.5 (Localization and grid size). *Under the hypotheses of Lemma 5.4, for every stage j followed by filtering and every $i \in P$,*

$$X_i^{(j+1)} \subseteq [y_i^{(j)} - R_{ij}, y_i^{(j)} + R_{ij}], \quad R_{ij} = 8\sqrt{kn_P} h_{ij} + [i \in I_Z],$$

where $[i \in I_Z]$ is 1 if $i \in I_Z$ and 0 otherwise. Consequently every grid of the trial has at most

$$K(\theta, n_P) = 10\theta^{-1} \lceil \log_2(n_P + 2) \rceil \quad (5.3)$$

nodes per coordinate, independently of the stage and of ε .

Proof. Write $y = y^{(j)}$, $c = c^{(j)}$, $h = h_{ij}$ and $\rho = \sqrt{kn_P}$. Let $[a, a']$ be a retained grid interval of coordinate i , let $v \in \{a, a'\}$ satisfy $m_i(v) \leq U$ (Definition 4.4), and let the grid point z attain $m_i(v)$, so $z_i = v$ and $Q(z) \leq U$. Because $\sqrt{L_i} > 1/(2r_i)$, a bound $\sqrt{L_i}|\delta| \leq \lambda\rho\eta_j$ implies $|\delta| < 2\lambda\rho h$; so Lemma 5.4(ii), (iii), (v) give $|v - y_i| < 2(1 + \sqrt{17/15})\rho h$ and $|v - c_i| < 2(2 + \sqrt{17/15})\rho h$. An integer interval of length one lies within $|v - y_i| + 1$ of y_i . Otherwise the interval has length at most $w_i(v) \leq h + \theta|v - c_i|$ (Lemma 5.2(a)), and $\theta\sqrt{k} \leq 1/\sqrt{8}$, $h \leq \rho h$ and $\sqrt{n_P} \leq \rho$ give that each of its points is within $(3 + 2\sqrt{17/15} + (2 + \sqrt{17/15})/\sqrt{2})\rho h < 7.3\rho h$ of y_i . So every retained interval, and hence their hull $X_i^{(j+1)}$, lies in $[y_i - R_{ij}, y_i + R_{ij}]$.

At stage $j+1$ the grid of coordinate i is centered at $y_i \in X_i^{(j+1)}$ with mesh $h/2$, and each side has length at most R_{ij} . With $\hat{h}$ as in Lemma 5.2(b) for mesh $h/2$, $\theta R_{ij}/\hat{h} \leq 16\theta\rho + \theta \leq 4\sqrt{2n_P} + \frac{1}{4}$, so by Lemma 5.2(b) the grid has at most $1 + 2\lceil \frac{4}{\theta} \ln(\frac{5}{4} + 4\sqrt{2n_P}) \rceil$ nodes, which is at most $K(\theta, n_P)$ by a calculus estimate in Appendix A. Stage 0 has at most three nodes per coordinate by Lemma 5.2(c), since $h_{i0} \geq s_i$, and coordinates outside P have two. $\square$

Section 2 relates Lemmas 5.4 and 5.5 to the cluster problem of branch-and-bound.

5.3 Unknown growth: capped trials

The algorithm cannot use $8\bar{\kappa}\theta^2 \leq 1$ directly because γ is unknown. It tries geometrically decreasing gradings and relies on the caps to bound the work of unsuitable trials.

Algorithm 5.6 (CT: capped trials). The input is an accuracy $\varepsilon > 0$. If $P = \emptyset$, minimize F over the grid $\prod_i \{\ell_i, u_i\}$ by Lemma 4.3 and return a minimizer $\hat{x}$, $\beta = F(\hat{x})$ and this grid. Otherwise let $j_{\max}$ be the least integer $j_{\max} \geq 0$ with $\frac{9}{16}n_P\eta_0^2 4^{-j_{\max}} \leq \varepsilon$, and for $\mu = 2, 3, \dots$ run TRIAL with grading $\theta = 2^{-\mu}$, cap $K_\mu = K(2^{-\mu}, n_P)$, accuracy ε and stage limit $j_{\max}$, keeping the incumbent from trial to trial, until a trial succeeds. Return the incumbent $\hat{x}$, the bound $\beta = \beta_j$ of the successful stage j and the grids of the successful trial.

Theorem 5.7 (Certified approximation). *Let (3.1) have explicit rational quadratic factors given as monomial lists, a tree decomposition of maximum bag size p , curvature bounds $L_i = \max\{0, \partial_{ii}F\}$, and let $\varepsilon = 2^{-q}$.*

- (a) *On every instance, $CT(\varepsilon)$ halts and returns a feasible rational $\hat{x}$ and a valid path certificate with $\beta \leq F^* \leq F(\hat{x}) \leq \beta + \varepsilon$.*
- (b) *If the instance has weighted growth with condition number $\bar{\kappa}$, $CT(\varepsilon)$ succeeds at a trial with $2^\mu \leq 6\sqrt{\bar{\kappa}}$ and uses at most*

$$f(p, \bar{\kappa})(I + q + 1)^5, \quad f(p, t) = c_0 (c_1 p \sqrt{t})^p t (1 + \log_2 t)^2,$$

bit operations, for absolute constants c_0, c_1 . The certificate has size, and is checked in time, within the same bound. Neither γ nor $\bar{\kappa}$ is an input. Since f is increasing in $t \geq 1$, the bound also holds with any larger number in place of $\bar{\kappa}$, such as the Euclidean condition number κ under point growth.

Proof. (a) If $P = \emptyset$, all corrections vanish and Proposition 4.2(b) gives $F(\hat{x}) = \beta \leq F^*$. Otherwise validity follows from Corollary 4.8, because every threshold is the value of a feasible point. For termination, take $\mu \geq 2$ with $\theta = 2^{-\mu} \leq \min_{i \in P} h_{i, j_{\max}}/s_i$, so that $\theta s_i \leq h_{ij}$ for all $j \leq j_{\max}$. In this trial every unclipped step at a stage $j \leq j_{\max}$ is at least h_{ij} : for $i \in I_C$ by (5.1), and for $i \in I_Z$ with $h_{ij} \geq 1$ because h_{ij} is a power of two, so $\lfloor h_{ij} \rfloor = h_{ij}$. A side of length at most $s_i \leq h_{ij}/\theta$

therefore takes at most $1/\theta$ steps, and the grid has at most $1 + 2/\theta < K_\mu$ nodes. For $i \in I_Z$ with $h_{ij} < 1$ the grid has at most $s_i + 1 < 1/\theta + 1$ nodes, because $s_i \leq h_{ij}/\theta < 1/\theta$. So the trial is not aborted. At stage $j_{\max}$, Lemma 5.2(a) gives $w_i(v) \leq h_{i,j_{\max}} + \theta s_i \leq 2h_{i,j_{\max}}$ for every node v , so $U - \beta_{j_{\max}} \leq D(y^{(j_{\max})}) \leq \sum_{i \in P} L_i h_{i,j_{\max}}^2 / 2 \leq a_{j_{\max}} / 2 \leq \frac{8}{9}\varepsilon$, and the trial succeeds.

(b) Let $\mu^* = \max\{2, \lceil \log_4(8\bar{\kappa}) \rceil\}$, so that $\theta = 2^{-\mu^*}$ satisfies $8\bar{\kappa}\theta^2 \leq 1$, and $2^{\mu^*} \leq 6\sqrt{\bar{\kappa}}$. Apply Lemmas 5.4 and 5.5 with the growth constant γ that defines $\bar{\kappa}$ and $k = \bar{\kappa}$: by Lemma 5.5 trial μ^* is never aborted, and by Lemma 5.4(iv) it succeeds at stage $j_{\max}$ at the latest, since $\frac{9}{16}a_{j_{\max}} \leq \varepsilon$. Every trial $\mu \leq \mu^*$ runs at most $j_{\max} + 1$ stages with at most K_μ nodes per coordinate, because an oversized grid is detected while it is generated. By Lemma 4.3 this costs $O((j_{\max} + 1)p(|\mathcal{A}| + n + N)K_\mu^p)$ table operations, and $\sum_{\mu \leq \mu^*} K_\mu^p \leq 2K_{\mu^*}^p$. Since $\sup_{t \geq 0} t^p e^{-t} = (p/e)^p$,

$$\lceil \log_2(n_P + 2) \rceil^p \leq 2p^p(n + 2), \quad (5.4)$$

so the table work is at most $c(j_{\max} + 1)pI^2(60p\sqrt{\bar{\kappa}})^p$ with $j_{\max} = O(I + q + 1)$ and an absolute constant c . Let Γ_X and Γ_F be the products of the denominators of the box endpoints and of the coefficients. Appendix A shows that all grid nodes of a trial have denominators dividing $\Gamma_X 2^\alpha$, where $\alpha = j_{\max} + O(I) + \mu K_\mu$, so one bound holds at all stages of a trial: denominators do not compound from stage to stage. It also shows that all table entries are integers of $O((1 + \mu 2^\mu)(I + q + 1))$ bits over the common denominator $8\Gamma_F \Gamma_X^2 4^\alpha$. Multiplying the counts proves the bound. $\square$

Remark 5.8 (In what sense the bound is fixed-parameter). As a parameter, $\bar{\kappa}$ is the smallest weighted condition number of the instance, which exists by Section 3. It is a real number, not part of the input, and CT never computes it. Since f is increasing in $t \geq 1$, the bound of Theorem 5.7(b) is at most $f(p, \lceil \bar{\kappa} \rceil)(I + q + 1)^5$, a fixed-parameter bound in the parameters p and $\lceil \bar{\kappa} \rceil$, which the algorithm does not need to know. The exponent 5 reflects schoolbook arithmetic and depends on neither parameter, and the accuracy enters only through $q = \log_2(1/\varepsilon)$. The factor $\lceil \log_2(n_P + 2) \rceil^p$ is absorbed by (5.4), and no bound on the number of bags containing a coordinate, on Hessian norms or on the number of local minima is used. If only a treewidth bound is known, the decomposition of [118] gives the same result with $p = 2\text{tw} + 2$. Theorem 1.2 states the lower bounds, proved in Section 10; since $\bar{\kappa} \leq \kappa$, the bounds stated for κ also apply to $\bar{\kappa}$.

A common mesh. The implementation of Section 11 and the exact-output results of Section 6.5 and Appendices B.1 and B.2 use one mesh for all coordinates and point growth with the Euclidean condition number κ .

Lemma 5.9 (CT with a common mesh). *Modify TRIAL by using the common mesh $h_j = s2^{-j}$ in place of h_{ij} for every $i \in P$; the corrections $L_i w_i(v)^2/8$ and the grids $\{\ell_i, u_i\}$ for $i \notin P$ are unchanged. Assume point growth with constant g at x^* , put $\kappa = \max\{1, L/g\}$, and let $8\kappa\theta^2 \leq 1$. At every stage j reached by such a trial, with center $c = c^{(j)}$, corrected minimizer $y_j = y^{(j)}$ and U the incumbent value after step (ii), we have $x^* \in X^{(j)}$, $\beta_j \leq F^*$, and:*

$$(G1) \quad D(z) \leq \frac{L}{4} n h_j^2 + \frac{L\theta^2}{2} (\|z - x^*\|^2 + \|c - x^*\|^2) \text{ for every grid point } z;$$

$$(G2) \quad \|c - x^*\|^2 \leq 4\kappa n h_j^2, \quad \|y_j - x^*\|^2 \leq \kappa n h_j^2, \text{ and } \|z - x^*\|^2 \leq \frac{17}{15} \kappa n h_j^2 \text{ for every grid point } z \text{ with } Q(z) \leq U;$$

$$(G3) \quad U - F^* \leq F(y_j) - F^* \leq F(y_j) - \beta_j = D(y_j) \leq \frac{9}{16} L n h_j^2;$$

$$(G4) \quad \text{if stage } j \text{ is followed by filtering, then for } i \in P \text{ the filtered interval } X_i^{(j+1)} \text{ lies within } 4.2\sqrt{n\kappa} h_j \text{ of } (y_j)_i \text{ if } i \in I_C, \text{ and within } 1 + 4.2\sqrt{n\kappa} h_j \text{ if } i \in I_Z;$$

(G5) every grid of the trial has at most $8\theta^{-1}\lceil\log_2(n+2)\rceil$ nodes per coordinate.

These statements also hold if the trial starts at any center $c^{(0)} \in X$ instead of ℓ . CT with a common mesh is CT with these trials, caps $8 \cdot 2^\mu \lceil\log_2(n+2)\rceil$, and the least stage limit $j_{\max} \geq 0$ with $\frac{9}{16} Lns^2 4^{-j_{\max}} \leq \varepsilon$. On every instance it halts with a valid path certificate and $F(\hat{x}) \leq \beta + \varepsilon$; under point growth it succeeds at the latest in the trial $\mu = \max\{2, \lceil\log_4(8\kappa)\rceil\}$.

The proof in Appendix A follows those of Lemmas 5.4 and 5.5 with $a_j = Lnh_j^2$ and $\|\cdot\|_{\mathbf{L}}^2$ replaced by $L\|\cdot\|^2$. Because (G2) bounds Euclidean distances in units of the common mesh, the coordinate bounds follow directly, without the conversion through $\sqrt{L_i} > 1/(2r_i)$ that costs Lemma 5.5 up to a factor 2; this gives the smaller radius and cap, at the price of $\kappa \geq \bar{\kappa}$. Items (G1)–(G4) and their proofs also hold for uniform grids, $\theta = 0$, with Lemma 5.2(a) read as $w_i(v) \leq h_j$; only (G5) needs $\theta > 0$.

5.4 Sharpness of the localization radius

The slack $U - \beta$ can be as large as a sum of n correction terms, while a single coordinate interval is removed only when its own excess exceeds the whole slack. The following family shows that the radius $\Theta(\sqrt{n\kappa}h_j)$ of Lemmas 5.5 and 5.9 is therefore attained, so that filtering alone leaves at least of order $\sqrt{n\kappa}$ nodes per coordinate on uniform grids; the graded grids reduce this to $O(\sqrt{\kappa} \log(n+2))$ (Lemma 5.5), which makes Theorem 5.7 a fixed-parameter bound (Remark 5.8).

Proposition 5.10 (Sharpness of localization). *Let $m \geq 1$, $n = 2m$, $\Lambda > 0$ and $0 < g \leq \Lambda/2$, and on $X = [-1, 1]^n$, with all coordinates continuous and named $(u_1, v_1, \dots, u_m, v_m)$, let*

$$F(x) = \sum_{k=1}^m \left[\frac{\Lambda}{2} (u_k^2 + v_k^2) + (\Lambda - 2g) u_k v_k \right].$$

Then $x^ = 0$, $F^* = 0$, the smallest valid curvature bounds are $L_i = \Lambda$, the largest growth constant is g , and $\kappa = \Lambda/g$ takes every value in $[2, \infty)$. Let G be a grid for a subbox that contains $[-R_0, R_0]^n$, and let every G_i contain 0 with $w_i(0) \geq h$. Then for every incumbent value $U \geq 0$, every coordinate i and every node $a \in G_i$ with*

$$|a| \leq \min\left\{R_0, h\sqrt{(n-2)\kappa/16}\right\}$$

we have $m_i(a) \leq U$. Hence every grid interval with such an endpoint is retained.

Corollary 5.11 (Filtered uniform grids need order $\sqrt{n\kappa}$ nodes). *Take the instance of Proposition 5.10 with $n \geq 4$, and run the stages of a trial with a common mesh (Lemma 5.9) without a cap, with $h_j = s2^{-j} = 2^{1-j}$, uniform grids (the construction of Definition 5.1 with $\theta = 0$) and initial center 0. Let $k_0 = \lfloor \sqrt{(n-2)\kappa/16} \rfloor$. At every stage the corrected minimizer is 0; at every stage $j \geq 1$ with $(k_0 + 1)h_j \leq 1$ that ends by filtering, the filtered box $X^{(j+1)}$ contains $[-(k_0 + 1)h_j, (k_0 + 1)h_j]^n$, and the grid of stage $j + 1$, if that stage is run, has at least $4k_0 + 5 \geq \sqrt{(n-2)\kappa} + 1$ nodes in every coordinate. With graded grids of grading $\theta \in (0, 1/4]$ instead: if a stage j ends by filtering, its box $X^{(j)}$ contains $[-R, R]^n$, where $R = h_j\sqrt{(n-2)\kappa/16}$, and its grids contain 0 with $w_i(0) \geq h_j$, then the grid of stage $j + 1$, if that stage is run, has at least $1 + \ln(1 + 2\theta R/h_j)/\ln(1 + \theta)$ nodes in every coordinate.*

Proofs are in Appendix A. Uniform grids also give accuracy-independent counts without any grading condition: under point growth, UC (Algorithm 9.2), which filters unions of uniform cells, has at most $12(2\sqrt{n\kappa} + 1)$ nodes per coordinate at every stage (Theorem 9.4 with $r = 1$ and $\kappa_S = \kappa$). The resulting running-time bound is polynomial in $I + q$ and κ for fixed p , but because of the factor $\sqrt{n}$ per coordinate it is not a fixed-parameter bound.

5.5 Examples

The first example shows that Theorem 5.7 covers nonconvex problems with many local minima and treewidth larger than one.

Example 5.12 (Coupled indefinite blocks). For a block b with variables $(u_b, v_b, \omega_b) \in [0, 1]^3$ let $\phi(u, v, \omega) = u^2 + v^2 - 4uv + \frac{1}{4}(u + v) + (\omega - \frac{u}{2})^2$, and for a graph $\mathcal{G}$ on the blocks with maximum degree $\Delta_{\mathcal{G}} \geq 1$ let

$$F = \sum_b \phi(u_b, v_b, \omega_b) + \frac{1}{16\Delta_{\mathcal{G}}} \sum_{bb' \in \mathcal{G}} (u_b - u_{b'})^2.$$

Then (a) $x^* = (1, 1, \frac{1}{2})$ in every block is the unique minimizer and $F(x) - F^* \geq \frac{1}{2}\|x - x^*\|^2$; (b) $L \leq \frac{21}{8}$, so $\kappa \leq \frac{21}{4}$; (c) each of the 2^m points with $(u_b, v_b, \omega_b) \in \{(0, 0, 0), (1, 1, \frac{1}{2})\}$ in all m blocks is a strict local minimizer; (d) a tree decomposition of $\mathcal{G}$ of width w yields one of F with $p = \max\{w + 1, 3\}$. Each block Hessian is indefinite and the minimizer is not a vertex. For $\mathcal{G}$ an $m_1 \times m_2$ grid graph with $m = m_1 m_2 \geq 2$, Theorem 5.7 applies with $p = \max\{m_1 + 1, 3\}$ and $\bar{\kappa} \leq \kappa \leq \frac{21}{4}$, although the instance has 2^m strict local minima. The proof is in Appendix A.

The minimizer of this family is easy to guess block by block; by (a), $F - F^*$ is at least $\frac{9}{8}$ times the number of blocks at $(0, 0, 0)$. The second example separates the grids from exact parametric dynamic programming.

Example 5.13 (Expanding-box chain). The family of Proposition 10.7, a chain of length m whose t th state has a box of width $2^t - 1$, has $p = 3$, a unique minimizer and $\kappa \leq 80$ for every m . By Theorem 5.7, a certified 2^{-q} -approximation takes time polynomial in m and q , whereas the exact Bellman message for the last state has at least 2^{m-1} quadratic pieces. Rescaling the t th state by 2^{-t} moves every coordinate into $[0, 1]$ and does not change $\bar{\kappa}$, which stays at most 80, but the rescaled instance has $L = 4^{m+1}$ and a largest growth constant between $\frac{1}{8}$ and 20, so every valid growth constant gives $\kappa \geq 4^m/5$, and the largest gives $\kappa \leq 32 \cdot 4^m$ (Appendix A). Section 11 reports the certified computation up to $m = 64$.

6 Exact output

A certified gap below 2^{-q} is not an exact solution. For rational quadratics, a global minimizer and the optimal value have bounded rational height, so a sufficiently small certified gap can be converted into an exact minimizer that is checked against the original data. This section proves a height bound for minimizers and for the optimal value (Section 6.2), gives an acceptance rule that needs neither uniqueness nor growth, and shows that an exact minimizer can be recovered from any feasible point close to the set of minimizers (Section 6.3). Under point growth this yields exact output in $f_1(p, \kappa) \text{poly}(I)$ bit operations (Theorem 6.14). Section 6.5 gives an acceptance test that uses the filtering history, and Section 6.6 treats explicit polynomial factors. Several minimizers are treated in Section 9.

6.1 Scaled data

In this section, except in Section 6.6, F is a quadratic with rational data,

$$F(x) = \frac{1}{2}x^T Hx + b^T x + c,$$

where H is the symmetric Hessian, on a bounded mixed box $X = \prod_i X_i$ with rational endpoints, after fixed coordinates have been substituted and integer endpoints rounded inward; thus $\ell_i < u_i$

for every i . The curvature bounds are $L_i = \max\{H_{ii}, 0\}$, so $P = \{i : H_{ii} > 0\}$. Let Δ be a positive common denominator of the monomial coefficients of F , namely $H_{ii}/2$, H_{ij} ($i < j$), b_i and c , and of all endpoints ℓ_i, u_i . Put

$$\hat{H} = \Delta H, \quad I_C^+ = I_C \cap P, \quad R = \Delta \prod_{i \in I_C^+} \hat{H}_{ii}, \quad \Omega = \Delta R^2, \quad \tau = \frac{1}{4nR}. \quad (6.1)$$

The matrix $\hat{H}$ is integral and symmetric, since $\hat{H}_{ii} = 2\Delta(H_{ii}/2)$ and $\hat{H}_{ij} = \Delta H_{ij}$; this choice of Δ absorbs the factor two of the off-diagonal entries. Each $\hat{H}_{ii}$ with $i \in I_C^+$ is a positive integer, and $\hat{H}_{ii} = \Delta L_i \leq \Delta L$, so $\log_2 R \leq \log_2 \Delta + |I_C^+| \log_2(\Delta L)$ is polynomial in I . Every positive H_{ii} is an integer multiple of $2/\Delta$, so $L \geq 2/\Delta$ if $L > 0$. The numbers R , Ω and τ are computed from the original data; refined boxes never enter them.

Two elementary facts are used repeatedly. If every coordinate of $x \in X$ lies in $\rho^{-1}\mathbb{Z}$ for a positive integer ρ , then $\Delta\rho^2 F(x) \in \mathbb{Z}$, because Δ clears every monomial coefficient. If $a/W \neq a'/W'$ are rationals with positive denominators, then $|a/W - a'/W'| \geq 1/(WW')$.

Lemma 6.1 (Hadamard's inequality [99, Theorem 7.8.1]). *If A is a real symmetric positive definite $m \times m$ matrix, then $\det A \leq \prod_{i=1}^m A_{ii}$.*

6.2 Stationary polytopes and rational height

The argument follows Vavasis [177] (see also [55]); we need its explicit constants. For $x \in X$ let $J_0(x) = \{i \in I_C : \ell_i < x_i < u_i\}$ be its interior coordinates.

Definition 6.2 (Stationary polytope). For $x^\circ \in \mathcal{S}$ let

$$\mathcal{P}(x^\circ) = \{x \in \mathbb{R}^n : x_i = x_i^\circ \ (i \notin J_0(x^\circ)), \ \ell_i \leq x_i \leq u_i \text{ and } \partial_i F(x) = 0 \ (i \in J_0(x^\circ))\}.$$

Lemma 6.3 (Stationary polytopes). *Let $x^\circ \in \mathcal{S}$ and $J_0 = J_0(x^\circ)$.*

- (a) $\nabla_{J_0} F(x^\circ) = 0$ and $H_{J_0 J_0} \succeq 0$.
- (b) $\mathcal{P}(x^\circ)$ is a nonempty polytope contained in X , and $\mathcal{P}(x^\circ) \subseteq \mathcal{S}$.
- (c) Let v be a vertex of $\mathcal{P}(x^\circ)$ and $J_v = J_0(v)$. Then $J_v \subseteq J_0 \cap I_C^+$, $\hat{H}_{J_v J_v} \succ 0$, every coordinate of v lies in $\rho^{-1}\mathbb{Z}$ for $\rho = \Delta \det \hat{H}_{J_v J_v}$, and $\Delta \leq \rho \leq R$ (with $\det \hat{H}_{J_v J_v} = 1$ if $J_v = \emptyset$).

Proof. (a) For $w \in \mathbb{R}^n$ supported on J_0 , the point $x^\circ + tw$ lies in X for all small $|t|$, because the coordinates in J_0 are continuous and strictly inside their bounds. Since $F(x^\circ + tw) = F^* + t\nabla F(x^\circ)^\top w + \frac{t^2}{2} w^\top H w$ has a minimum at $t = 0$, we get $\nabla F(x^\circ)^\top w = 0$ and $w^\top H w \geq 0$.

(b) By (a), $x^\circ \in \mathcal{P}(x^\circ)$. The set is defined by finitely many linear equations and inequalities inside a bounded box, so it is a polytope. Its points keep the integer coordinates of x° and satisfy all bounds, so $\mathcal{P}(x^\circ) \subseteq X$. For $x \in \mathcal{P}(x^\circ)$, $d = x - x^\circ$ is supported on J_0 , and $H_{J_0 J_0} d_{J_0} = \nabla_{J_0} F(x) - \nabla_{J_0} F(x^\circ) = 0$. Hence

$$F(x) - F(x^\circ) = \nabla F(x^\circ)^\top d + \frac{1}{2} d^\top H d = \nabla_{J_0} F(x^\circ)^\top d_{J_0} + \frac{1}{2} d_{J_0}^\top H_{J_0 J_0} d_{J_0} = 0,$$

so $x \in \mathcal{S}$.

(c) A continuous coordinate outside J_0 is fixed at $x_i^\circ \in \{\ell_i, u_i\}$ in $\mathcal{P}(x^\circ)$, so $J_v \subseteq J_0$. Suppose $w \in \mathbb{R}^{J_v}$ is nonzero and $H_{J_0 J_v} w = 0$. Extend w by zero to $\tilde{w} \in \mathbb{R}^n$. Then $v \pm t\tilde{w} \in \mathcal{P}(x^\circ)$ for small $t > 0$: the equations are preserved, the coordinates in J_v have slack, and all other coordinates are

unchanged. This contradicts that v is a vertex, so $H_{J_0 J_v}$ has trivial kernel. The matrix $H_{J_v J_v}$ is a principal submatrix of $H_{J_0 J_0} \succeq 0$; if $w \in \mathbb{R}^{J_v} \setminus \{0\}$ had $w^\top H_{J_v J_v} w = \tilde{w}^\top H \tilde{w} = 0$, then $H_{J_0 J_0} \tilde{w} = 0$, that is $H_{J_0 J_v} w = 0$. Hence $H_{J_v J_v} \succ 0$ and $\hat{H}_{J_v J_v} = \Delta H_{J_v J_v} \succ 0$. In particular $H_{ii} > 0$ for $i \in J_v$, so $J_v \subseteq I_C^+$.

The coordinates of v outside J_v are integers or endpoints, so they lie in $\Delta^{-1}\mathbb{Z}$. The rows $i \in J_v$ of $\nabla F(v) = 0$, multiplied by Δ^2 , read

$$\hat{H}_{J_v J_v}(\Delta v_{J_v}) = -\Delta(\Delta b_{J_v}) - \sum_{j \notin J_v} \hat{H}_{J_v j}(\Delta v_j),$$

whose right-hand side is integral. Cramer's rule gives $\Delta v_{J_v} \in (\det \hat{H}_{J_v J_v})^{-1} \mathbb{Z}^{J_v}$, so every coordinate of v lies in $\rho^{-1}\mathbb{Z}$. Finally $\det \hat{H}_{J_v J_v}$ is a positive integer, and by Lemma 6.1 and $J_v \subseteq I_C^+$, $\det \hat{H}_{J_v J_v} \leq \prod_{i \in J_v} \hat{H}_{ii} \leq \prod_{i \in I_C^+} \hat{H}_{ii}$. $\square$

Corollary 6.4 (Rational height). *(a) Some global minimizer has all coordinates in $\rho^{-1}\mathbb{Z}$ for an integer ρ with $\Delta \mid \rho$ and $\rho \leq R$; every isolated global minimizer has this property. The endpoints ℓ_i, u_i then also lie in $\rho^{-1}\mathbb{Z}$. (b) $F^* = a/W$ in lowest terms with $1 \leq W \leq \Omega$.*

Proof. (a) Take any $x^\circ \in \mathcal{S}$. The nonempty polytope $\mathcal{P}(x^\circ)$ has a vertex, which is a global minimizer by Lemma 6.3(b) and has the stated height by Lemma 6.3(c). If x° is isolated in $\mathcal{S}$, then the convex set $\mathcal{P}(x^\circ) \subseteq \mathcal{S}$ contains no other point near x° , hence equals $\{x^\circ\}$, and x° is its vertex. The endpoints lie in $\Delta^{-1}\mathbb{Z} \subseteq \rho^{-1}\mathbb{Z}$. (b) Evaluate F at the minimizer of (a): $\Delta \rho^2 F^* \in \mathbb{Z}$, so $W \leq \Delta \rho^2 \leq \Omega$. $\square$

Remark 6.5 (Other admissible height constants). Any bound on the minors of $\hat{H}$ restricted to I_C also works in Lemma 6.3(c) and Corollary 6.4, for example $|I_C|! \max\{1, \max_{ij} |\hat{H}_{ij}|\}^{|I_C|}$ by the Leibniz expansion, or $\prod_{i \in I_C} \max\{1, \sum_{j \in I_C} |\hat{H}_{ij}|\}$ by Hadamard's inequality for rows. Both also bound *nonprincipal* minors, which arise if a vertex of $\mathcal{P}(x^\circ)$ is described by a nonprincipal set of stationarity rows. Part (c) shows that the principal system $\hat{H}_{J_v J_v}$ always suffices, and Lemma 6.1 then gives the diagonal bound in (6.1), which is never larger than either alternative and depends only on the coordinate curvatures. All these constants have polynomial bit length; only the threshold values below change.

6.3 Acceptance and recovery

Proposition 6.6 (Exact acceptance). *Let F attain its minimum F^* over the feasible set X , and let F^* be a rational number with denominator at most Ω' . Let $\beta \leq F^*$ and let $x \in X$ with $F(x) = a/W$ in lowest terms. If $F(x) - \beta < 1/(\Omega'W)$, then x is a global minimizer. For the quadratics of this section, $\Omega' = \Omega$ is admissible by Corollary 6.4(b).*

Proof. If $F(x) > F^*$, then $F(x) - F^* \geq 1/(\Omega'W)$ by the second elementary fact, because the denominator of F^* is at most Ω' . With $\beta \leq F^*$ this contradicts $F(x) - \beta < 1/(\Omega'W)$. $\square$

The rule uses the denominator of the candidate itself. It does not assume that feasible values lie on a lattice, which is false when continuous coordinates vary. The proof uses only the rationality of $F(x)$ and the denominator bound for F^* ; Sections 7 and 8 apply the proposition with other feasible sets and other values of Ω' .

Algorithm 6.7 (REC: snapping recovery). The input is a point $y \in X$. Fix every integer coordinate at y_i . For every continuous coordinate, fix it at ℓ_i if $y_i - \ell_i \leq \tau$, else at u_i if $u_i - y_i \leq \tau$, and otherwise call it free; let J_f be the set of free coordinates. Find by exact linear programming a point $\tilde{x}$ that agrees with the fixed values and satisfies $\ell_{J_f} \leq \tilde{x}_{J_f} \leq u_{J_f}$ and $\nabla_{J_f} F(\tilde{x}) = 0$, or report failure.

The data of this linear system are original coefficients, endpoints and integer values of y , so it is solved in time polynomial in I [82, Theorem 6.4.12]. If $H_{J_f J_f}$ is nonsingular, Gaussian elimination suffices.

Lemma 6.8 (Snapping recovery). *If $y \in X$ and $\text{dist}(y, \mathcal{S}) \leq \tau/2$, then $\text{REC}(y)$ does not fail, and every point it can return lies in $\mathcal{S}$.*

Proof. Let $x^\circ \in \mathcal{S}$ be nearest to y ; it exists since $\mathcal{S}$ is compact. As $\|y - x^\circ\| \leq \tau/2 < 1$, the integer coordinates of y and x° agree. The two endpoints of a continuous coordinate differ by at least $1/\Delta \geq 1/R = 4n\tau > 2\tau$, so no coordinate is within τ of both endpoints. If $x_i^\circ = \ell_i$ then $y_i - \ell_i \leq \tau/2$ and coordinate i is fixed at ℓ_i ; similarly for $x_i^\circ = u_i$. Hence every continuous coordinate at a bound in x° is fixed at its value in x° , and $J_f \subseteq J_0 = J_0(x^\circ)$. Let $J' = J_0 \setminus J_f$ be the remaining fixed coordinates, each fixed at an endpoint $\xi_i \in \{\ell_i, u_i\}$, and define on $\mathcal{P}(x^\circ)$ the linear function

$$\phi(x) = \sum_{i \in J'} |x_i - \xi_i| = \sum_{i \in J', \xi_i = \ell_i} (x_i - \ell_i) + \sum_{i \in J', \xi_i = u_i} (u_i - x_i) \geq 0.$$

For $i \in J'$ we have $|y_i - \xi_i| \leq \tau$ and $|x_i^\circ - y_i| \leq \tau/2$, so $\phi(x^\circ) \leq \frac{3}{2}|J'|\tau \leq \frac{3}{8R} < \frac{1}{R}$. The minimum of ϕ over the polytope $\mathcal{P}(x^\circ)$ is attained at a vertex v . By Lemma 6.3(c) the coordinates of v and all endpoints lie in $\rho^{-1}\mathbb{Z}$ for an integer $\rho \leq R$, so $\phi(v) \in \rho^{-1}\mathbb{Z}$. Since $0 \leq \phi(v) \leq \phi(x^\circ) < 1/R \leq 1/\rho$, we get $\phi(v) = 0$.

Thus $v \in \mathcal{P}(x^\circ) \subseteq \mathcal{S}$ satisfies $v_i = \xi_i$ for $i \in J'$, agrees with x° (hence with the fixed values of REC) outside J_0 , and has $\nabla_{J_f} F(v) = 0$ because $J_f \subseteq J_0$. So the system solved by REC is feasible. If $\tilde{x}$ is any solution, then $d = \tilde{x} - v$ is supported on J_f and $H_{J_f J_f} d_{J_f} = \nabla_{J_f} F(\tilde{x}) - \nabla_{J_f} F(v) = 0$, so $F(\tilde{x}) - F(v) = \nabla_{J_f} F(v)^\top d_{J_f} + \frac{1}{2} d_{J_f}^\top H_{J_f J_f} d_{J_f} = 0$. Since $\tilde{x} \in X$, it is a global minimizer. Nonsingularity of $H_{J_f J_f}$ is used. $\square$

Theorem 6.9 (From certified gaps to exact minimizers). *Let $\beta \leq F^*$ be certified and $y \in X$.*

(a) *If F has set growth (3.4) with constant g_S and*

$$F(y) - \beta \leq \varepsilon_S := \min\left\{\frac{g_S \tau^2}{4}, \frac{1}{2\Omega^2}\right\} = \min\left\{\frac{g_S}{64n^2 R^2}, \frac{1}{2\Omega^2}\right\}, \quad (6.2)$$

then $\text{REC}(y)$ returns a point $\tilde{x}$ that passes the test of Proposition 6.6 with $\Omega' = \Omega$.

(b) *Without any growth assumption there is $\varepsilon_0 > 0$, depending on the instance, such that the conclusion of (a) holds whenever $F(y) - \beta \leq \varepsilon_0$.*

Proof. (a) $g_S \text{dist}(y, \mathcal{S})^2 \leq F(y) - F^* \leq F(y) - \beta \leq g_S \tau^2/4$, so Lemma 6.8 applies and $\tilde{x} \in \mathcal{S}$. Then $F(\tilde{x}) = F^* = a/W$ with $W \leq \Omega$, and $F(\tilde{x}) - \beta \leq F(y) - \beta \leq 1/(2\Omega^2) < 1/(\Omega W)$. (b) The set $Y = \{x \in X : \text{dist}(x, \mathcal{S}) \geq \tau/2\}$ is compact and $F > F^*$ on it, so $\varepsilon_Y = \min_Y F - F^* > 0$ (put $\varepsilon_Y = \infty$ if $Y = \emptyset$). With $\varepsilon_0 = \min\{\varepsilon_Y/2, 1/(2\Omega^2)\}$, $F(y) - \beta \leq \varepsilon_0$ gives $\text{dist}(y, \mathcal{S}) < \tau/2$, and the argument of (a) applies. $\square$

Remark 6.10 (Set growth is automatic). Every rational mixed box QP has set growth with some constant $g_S > 0$. On an integer slice whose continuous box contains a minimizer, Luo and Sturm's error bound for a quadratic on a polytope [129, Theorem 3.3], applied to $F - F^*$, gives $F(x) - F^* \geq c \text{dist}(x, \mathcal{S} \cap \text{slice})^2 \geq c \text{dist}(x, \mathcal{S})^2$; on the finitely many other slices $F - F^*$ is bounded below by a positive constant while $\text{dist}(x, \mathcal{S})$ is bounded by the diameter of $\bar{X}$. As for point growth, no useful lower bound on g_S follows. Theorem 6.9 shows that exactness costs only $\log_2(1/\varepsilon_S) \leq \text{poly}(I) + \max\{0, \log_2(1/g_S)\}$ bits of certified accuracy; the cost of reaching that accuracy is a property of the approximation algorithm.

Remark 6.11 (Relation to rational-witness results). Corollary 6.4 is the box case, with explicit constants, of a classical fact: a quadratic program with rational data that attains its minimum has a global minimizer of polynomial encoding length [177]. Vavasis used it to show that the decision version of quadratic programming is in NP, and Del Pia, Dey and Molinaro extended NP membership to mixed-integer quadratic programs with unbounded integer variables [55]; on a bounded mixed box the integer coordinates have polynomial length, so that extension is not needed. These results certify $F^* \leq t$. Deciding $F^* \leq t$ for box QP is NP-complete, and NP-hard already at treewidth two [53], so certificates of $F^* > t$ of polynomial size, checkable in polynomial time, for all instances would place an NP-complete problem in coNP. The certificates of Theorem 6.14 below have size $f_1(p, \kappa) \text{poly}(I)$: they exist for bounded (p, κ) , and they are checked without trusting growth. The explicit constants R and Ω are what make the stopping test computable.

6.4 Exact output under point growth

Lemma 6.12 (Uniqueness implies growth). *If $\mathcal{S} = \{x^*\}$, then F has point growth (3.2) with some constant $g > 0$.*

For quadratic programs over polyhedra this is part of the second-order theory of [39]; the proof below also covers integer coordinates.

Proof. Otherwise there are $x^k \in X \setminus \{x^*\}$ with $(F(x^k) - F^*)/\|x^k - x^*\|^2 \rightarrow 0$. As $\bar{X}$ is bounded, $F(x^k) \rightarrow F^*$, and compactness of X with uniqueness gives $x^k \rightarrow x^*$; so the integer coordinates of x^k equal those of x^* for large k , and since $x^k \neq x^*$, they differ in some continuous coordinate. Put $t_k = \|x^k - x^*\|$ and $u^k = (x^k - x^*)/t_k$; pass to a subsequence with $u^k \rightarrow u$, $\|u\| = 1$, $u_{I_Z} = 0$. Then

$$\frac{F(x^k) - F^*}{t_k^2} = \frac{\nabla F(x^*)^\top u^k}{t_k} + \frac{1}{2}(u^k)^\top H u^k.$$

Since x^* minimizes F over the convex continuous box of its integer slice and x^k lies in that box, the first term is nonnegative. The left side tends to zero and the second term is bounded, so $\nabla F(x^*)^\top u = 0$ and $u^\top H u \leq 0$. Each u^k satisfies $u_i^k \geq 0$ when $x_i^* = \ell_i$ and $u_i^k \leq 0$ when $x_i^* = u_i$; so does u , and hence $x^* + tu \in X$ for small $t > 0$. Then $F(x^* + tu) - F^* = \frac{t^2}{2} u^\top H u \leq 0$, contradicting uniqueness. $\square$

As noted after Definition 3.2, the lemma gives no bound on the condition number, which can be exponential in I .

Algorithm 6.13 (EX: exact output). For $q = 1, 2, 4, 8, \dots$: run CT(2^{-q}) (Algorithm 5.6) and obtain a feasible $\hat{x}$ and a certified β ; run REC($\hat{x}$); if it returns $\tilde{x}$ with $F(\tilde{x}) - \beta < 1/(\Omega W)$, where W is the reduced denominator of $F(\tilde{x})$, stop and return $\tilde{x}$, $F(\tilde{x})$ and the path certificate of CT.

If $L = 0$, then $P = \emptyset$, CT is exact and returns a vertex $\hat{x}$ of the box with $F(\hat{x}) = \beta$; REC fixes every coordinate of a vertex, so $\tilde{x} = \hat{x}$ and the test passes at $q = 1$.

Theorem 6.14 (Exact rational output). *For every rational mixed box QP with a supplied tree decomposition, EX stops, and its output is a global minimizer and F^* , with a certificate whose validity uses neither growth nor uniqueness. If the instance has point growth (3.2) with condition number κ , EX uses at most $f_1(p, \kappa)(I + 1)^{C_1}$ bit operations, where f_1 is a computable function and C_1 an absolute constant; neither g nor κ is an input. This bound remains valid if REC skips every call whose matrix $H_{J_f J_f}$ is singular, so under point growth no linear programming is needed.*

Proof. Validity. Each returned point passes Proposition 6.6 with a certified β (Theorem 5.7), so it is a global minimizer.

Termination. CT stops on every instance (Theorem 5.7(a)). Once $2^{-q} \leq \varepsilon_0$ of Theorem 6.9(b), the test passes.

Bound under growth. If $L = 0$, EX stops after one call of CT. Otherwise $L \geq 2/\Delta$. Point growth is set growth with $\mathcal{S} = \{x^*\}$ and $g_S = g$. By Theorem 6.9(a), EX stops at the latest at the first q with $2^{-q} \leq \varepsilon_S$, so the largest q used is less than $2q^*$ with $q^* = \lceil \log_2(1/\varepsilon_S) \rceil \geq 1$. Since $g \geq L/\kappa$ and $\log R, \log \Omega, \log n$ and $\log(1/L)$ are at most polynomial in I , $q^* \leq \text{poly}(I) + \log_2 \kappa$. There are $O(\log q^*)$ calls of CT, each within $f(p, \kappa)(I + 2q^* + 1)^5$ bit operations by Theorem 5.7, whose bound holds with $\kappa \geq \bar{\kappa}$, and $(I + 2q^* + 1)^5 \leq (1 + \log_2 \kappa)^5 \text{poly}(I)$. REC, the evaluation of $F(\tilde{x})$ and the comparison cost a polynomial in I and in the bit lengths of $\hat{x}$ and β . By Appendix A these bit lengths are $O((1 + \mu 2^\mu)(I + 2q^* + 1))$, where CT succeeds at trial μ (an incumbent kept from an earlier trial satisfies the smaller bound of that trial), and $2^\mu \leq 6\sqrt{\bar{\kappa}} \leq 6\sqrt{\kappa}$ by Theorem 5.7(b). Absorbing all factors that depend only on κ into f_1 proves the bound.

Finally, let $J_0 = J_0(x^*)$. By Lemma 6.3(a), $\nabla_{J_0} F(x^*) = 0$, so a nonzero w supported on J_0 with $H_{J_0 J_0} w_{J_0} = 0$ would make F constant on a short segment through x^* ; hence $H_{J_0 J_0} \succ 0$. If $\text{dist}(\hat{x}, x^*) \leq \tau/2$, then $J_f \subseteq J_0$ as in the proof of Lemma 6.8, so $H_{J_f J_f}$ is nonsingular. This holds at every call that the bound uses, so skipping singular calls does not change it. $\square$

Remark 6.15 (Coordinatewise reconstruction). Under point growth one may instead round each coordinate of $\hat{x}$ to the unique rational of denominator at most R within $1/(4R^2)$ (continued fractions, [82, Theorem 5.1.9]); this succeeds once $2^{-q} \leq g/(32R^4)$, by Corollary 6.4(a) and the $1/R^2$ separation of such rationals. Snapping recovery succeeds once $2^{-q} \leq g/(64n^2 R^2)$ (Lemma 6.8), which is the weaker requirement when $R^2 \geq 2n^2$; it does not need uniqueness, and it returns a minimizer even when $\mathcal{S}$ is a continuum. In both routes, $2^{-q} \leq 1/(2\Omega^2)$ makes the recovered minimizer pass the acceptance test.

6.5 Localized acceptance

The acceptance rule of Proposition 6.6 needs a certified gap below $1/(\Omega W)$, as small as about 2^{-315} on the instances of experiment E4 (Section 11.4), which have at most six variables. The filtering history permits a different test that accepts a candidate as soon as the retained box is small compared with a first-order margin.

Consider stage j of a filtering run, such as a trial of CT: a grid G for the current box $X^{(j)}$ with corrections $d_i(v) = L_i w_i(v)^2/8$ and the grids $G_i = \{\ell_i, u_i\}$ for $i \notin P$, min-marginals m_i , a corrected minimizer y_j , the incumbent value U , and the filtered box $X^{(j+1)}$. The boxes are reached from X by filtering steps whose thresholds are values of feasible points and at least U , as in a trial, where the incumbent value never increases. So by Proposition 4.5 every $x \in X$ with $F(x) \leq U$ lies in $X^{(j+1)}$, and $y_j \in X^{(j+1)}$. Filtering never narrows a coordinate $i \notin P$, so $X_i^{(j+1)} = X_i^{(j)} = X_i$.

Lemma 6.16 (Node exclusion). *Let $x \in X^{(j)}$ with $F(x) \leq U$.*

- (a) *If $i \in I_Z$, then x_i lies in the set Z_i of nodes v with $m_i(v) \leq U$ and of integers strictly inside grid intervals $[a, a']$ with $a' - a \geq 2$ and $\min\{m_i(a), m_i(a')\} \leq U$.*
- (b) *If $i \notin P$, $G_i = \{a, a'\}$ and $m_i(a) > U$, then setting $x_i = a'$ gives a point of $X^{(j)}$ with value at most $F(x)$.*

Proof. (a) If x_i is a node v , Proposition 4.2(c) gives $m_i(v) \leq m_i(v) + d_i(v) \leq F(x) \leq U$. Otherwise x_i is an integer strictly inside a grid interval $[a, a']$, which then has $a' - a \geq 2$, and Proposition 4.2(c)

gives $\min\{m_i(a), m_i(a')\} \leq F(x) \leq U$. (b) Since $L_i = 0$, F is concave along coordinate i , so $F(x)$ is at least the smaller of its values at $x_i = a$ and $x_i = a'$. By Proposition 4.2(c) the first is at least $m_i(a) > U \geq F(x)$, so the second is at most $F(x)$. $\square$

Proposition 6.17 (Localized exact acceptance). *Define the narrowed box $\tilde{X} \subseteq X^{(j+1)}$ coordinate-wise: for $i \in I_Z \cap P$ let $\tilde{X}_i$ be the integer hull of $Z_i \cap X_i^{(j+1)}$; for $i \notin P$ and $G_i = \{a, a'\}$ let $\tilde{X}_i = \{a'\}$ if $m_i(a) > U \geq m_i(a')$, $\tilde{X}_i = \{a\}$ if $m_i(a') > U \geq m_i(a)$, and $\tilde{X}_i = X_i$ otherwise; and let $\tilde{X}_i = X_i^{(j+1)}$ for $i \in I_C \cap P$. Let $\tilde{x} \in \tilde{X}$ with $F(\tilde{x}) \leq U$, $\zeta = \nabla F(\tilde{x})$, $J_+ = \{i : \tilde{X}_i \text{ has positive length}\}$, and $\tilde{X}_i = [\alpha_i, \alpha'_i]$ (or its integer points) for $i \in J_+$. Suppose every $i \in J_+$ falls under one of the cases*

- (i) $\tilde{x}_i = \alpha_i, \zeta_i \geq 0 : \mu_i = \zeta_i/(\alpha'_i - \alpha_i);$
- (ii) $\tilde{x}_i = \alpha'_i, \zeta_i \leq 0 : \mu_i = -\zeta_i/(\alpha'_i - \alpha_i);$
- (iii) $\alpha_i < \tilde{x}_i < \alpha'_i, \zeta_i = 0 : \mu_i = 0;$
- (iv) $i \in I_Z \text{ otherwise} : \mu_i = -|\zeta_i|,$

and that $H_{J_+J_+} + 2 \operatorname{diag}(\mu_{J_+}) \succeq 0$. Then $\tilde{x}$ is a global minimizer.

Proof. Let $x \in X$. If $F(x) > U$, then $F(x) > F(\tilde{x})$. Otherwise $x \in X^{(j+1)}$. Applying Lemma 6.16(b) to the coordinates $i \notin P$ whose $\tilde{X}_i$ is a single endpoint keeps the point in $X^{(j+1)}$ and does not increase F , and gives a point x'' that, by Lemma 6.16(a), lies in $\tilde{X}$. Let $d = x'' - \tilde{x}$; then $d_i = 0$ for $i \notin J_+$ and $\alpha_i - \tilde{x}_i \leq d_i \leq \alpha'_i - \tilde{x}_i$ for $i \in J_+$. In cases (i) and (ii), $\zeta_i d_i = |\zeta_i| |d_i| \geq \mu_i d_i^2$; in case (iii) both sides vanish; in case (iv) $d_i \in \mathbb{Z}$, so $\zeta_i d_i \geq -|\zeta_i| |d_i| \geq -|\zeta_i| d_i^2$. Hence $F(x'') - F(\tilde{x}) = \zeta^\top d + \frac{1}{2} d^\top H d \geq \frac{1}{2} d^\top (H_{J_+J_+} + 2 \operatorname{diag}(\mu_{J_+})) d_{J_+} \geq 0$, and $F(x) \geq F(x'') \geq F(\tilde{x})$. $\square$

The test needs one exact $LDL^\top$ factorization of a $|J_+| \times |J_+|$ matrix, which a checker repeats after replaying the filtering history. Its validity needs neither uniqueness nor growth, and the candidate $\tilde{x}$ is arbitrary. The corrected minimizer lies in $\tilde{X}$: for $i \in I_Z \cap P$, $(y_j)_i$ is a node with $m_i((y_j)_i) \leq Q(y_j) = \beta_j \leq F^* \leq U$, and for $i \notin P$ the same inequality excludes the endpoint whose min-marginal exceeds U . The following rule computes a candidate from $\tilde{X}$ and y_j .

Definition 6.18 (Face candidate). Put $\tilde{x}_i = (y_j)_i$ for $i \in I_Z$ and for $i \notin J_+$. For $i \in I_C \cap J_+$, put $\tilde{x}_i = \ell_i$ if $\ell_i \in \tilde{X}_i$, else $\tilde{x}_i = u_i$ if $u_i \in \tilde{X}_i$, and otherwise call i free. If the set J_f of free coordinates has $H_{J_f J_f}$ nonsingular, the *face candidate* is the point $\tilde{x}$ whose free part solves $\nabla_{J_f} F(\tilde{x}) = 0$; otherwise stage j has no face candidate.

The face candidate costs one exact solve of a $|J_f| \times |J_f|$ linear system. It is tested by Proposition 6.17, whose hypotheses include $\tilde{x} \in \tilde{X}$ and $F(\tilde{x}) \leq U$.

Corollary 6.19 (Acceptance of the face candidate). *Let F have point growth with constant g at x^* . Let $J_0 = J_0(x^*)$ and $A = J_\partial \cap I_C^+$, the continuous coordinates with $L_i > 0$ at which x^* is at a bound, and assume strict complementarity, $\partial_i F(x^*) \neq 0$ for $i \in A$. Put $\lambda_A = \min_{i \in A} |\partial_i F(x^*)|$ and $\Gamma = \|H_{AA}\|_2 + \|H_{J_0 A}\|_2^2/(2g)$, and let δ_X be the smallest distance $|x_i^* - e|$ over coordinates $i \in I_C \cup (I_Z \setminus P)$ and endpoints $e \in \{\ell_i, u_i\}$ with $e \neq x_i^*$. Put $h^* = \omega^*/\sqrt{n\kappa}$, where ω^* is the smallest of the numbers $\frac{1}{2}$ (if $I_Z \cap P \neq \emptyset$), $\delta_X/6$ (if $I_C \cup (I_Z \setminus P) \neq \emptyset$) and $\lambda_A/(5\Gamma)$ (if $A \neq \emptyset$); at least one of the first two conditions holds, because $I_Z \cap P$ and $I_C \cup (I_Z \setminus P)$ cover $[n]$. Consider a trial of CT with a common mesh (Lemma 5.9) with $8\kappa\theta^2 \leq 1$. At every stage j with $h_j \leq h^*$ that the trial reaches and ends by filtering, the face candidate exists, equals x^* , and passes the test of Proposition 6.17. Since $\delta_X \geq 1/R$, $\lambda_A \geq 1/(\Delta R)$ and $\log_2 \Gamma \leq \operatorname{poly}(I) + \log_2 \kappa$, the least j with $h_j \leq h^*$ is at most $\operatorname{poly}(I) + O(\log \kappa)$.*

The proof is in Appendix B.1. It also shows that at these stages $\tilde{X}_i = \{x_i^*\}$ for every integer coordinate and every coordinate with $L_i = 0$, and that $J_f = J_0$. With several minimizers the test can fail, for example when two optimal vertices differ in a coordinate $i \notin P$.

In experiment S1 (Section 11.4) the test accepted the face candidate within nine stages on 29 of 30 random mixed-integer instances; on the five of them on which EX used the most stages, one CT run needed 40 to 72 stages to reach the threshold of Proposition 6.6 with the constants of (6.1). The idea of certifying a neighborhood of a candidate by local information is that of exclusion boxes in interval branch-and-bound [148]; here it is carried out exactly on the box produced by the filtering history.

6.6 Explicit polynomial factors

Now let each factor be an explicit list of rational monomials of degree at most $d \geq 2$, with d fixed (or encoded in unary).

Lemma 6.20 (Interval curvature certificate). *For each i write $\partial_{ii}F = \sum_m c_m x^{\alpha_m}$ and let $[\underline{\mu}_m, \bar{\mu}_m]$ be the range of the monomial x^{α_m} on $\bar{X}$. Put $\hat{L}_i = \sum_m \max\{c_m \bar{\mu}_m, c_m \underline{\mu}_m\}$ and $L_i = \max\{\hat{L}_i, 0\}$. Then the L_i are upper coordinate curvatures, and they are computed in time polynomial in I .*

Proof. The monomial x^α is a product of functions $x_k^{\alpha_k}$ of distinct coordinates, so its range on the box $\bar{X}$ is the product of the intervals $\{t^{\alpha_k} : t \in [\ell_k, u_k]\}$, each of which is computed exactly from $\ell_k^{\alpha_k}, u_k^{\alpha_k}$ and, for even α_k with $\ell_k < 0 < u_k$, the value 0. Hence $c_m x^{\alpha_m} \leq \max\{c_m \bar{\mu}_m, c_m \underline{\mu}_m\}$ on $\bar{X}$, and $\partial_{ii}F \leq \hat{L}_i \leq L_i$. For every $x \in \bar{X}$, $t \mapsto F(x + te_i) - \frac{L_i}{2}t^2$ has second derivative $\partial_{ii}F - L_i \leq 0$ on its interval, so it is concave. The endpoints are powers of degree at most d of input rationals, so their bit length is $O(dI)$, and there are at most as many terms as input monomials. $\square$

The bound can overestimate when monomials cancel; the condition number in the results below is computed with the accepted L . Any sharper bound with a certificate checkable in polynomial time may be used instead.

Proposition 6.21 (Lattice stopping rule). *Suppose every coordinate is integer, and let Γ_F be the product of the denominators of the coefficients of the objective (after substituting fixed coordinates). If $\hat{x} \in X$, $\beta \leq F^*$ and $F(\hat{x}) - \beta < 1/\Gamma_F$, then $\hat{x} \in \mathcal{S}$. It suffices that $F(\hat{x}) - \beta \leq 2^{-q}$ with $q = 1 + \lceil \log_2 \Gamma_F \rceil = O(I)$.*

Proof. At integer points every monomial value lies in $\Gamma_F^{-1}\mathbb{Z}$, so $F(X) \subseteq \Gamma_F^{-1}\mathbb{Z}$. If $F(\hat{x}) > F^*$ then $F(\hat{x}) - F^* \geq 1/\Gamma_F > F(\hat{x}) - \beta$, contradicting $\beta \leq F^*$. $\square$

Corollary 6.22 (Polynomial factors). *For fixed d , under point growth (3.2) with the accepted curvature bounds, $CT(2^{-q})$ returns a certified 2^{-q} -approximation in $f_d(p, \kappa)(I + q + 1)^C$ bit operations, where f_d is a computable function and C an absolute constant. If all coordinates are integer, $CT(2^{-(1 + \lceil \log_2 \Gamma_F \rceil)})$ returns an exact minimizer, certified by Proposition 6.21, in $f_d(p, \kappa)(I + 1)^C$ bit operations.*

Proof. The interpolation, filtering, contraction, localization and grid-size arguments of Sections 4 and 5 use only Definition 3.1 and growth, not the quadratic form. In the bit analysis of Theorem 5.7 (Appendix A) only the exponents that define the meshes h_{ij} (Section 5) and the factor evaluation change. The L_i of Lemma 6.20 have $O(dI)$ bits and lie in $(\Gamma_F \Gamma_X^{d-2})^{-1}\mathbb{Z}$, because the coefficients of $\partial_{ii}F$ are integer multiples of coefficients of F and the endpoints of the monomial ranges are products of at most $d - 2$ box endpoints. So these exponents are $O(dI)$ in absolute value, and

every node has a denominator dividing $\Gamma_X 2^\alpha$, with α as in Appendix A but with $O(dI)$ in place of $O(I)$. A monomial of degree at most d at such nodes has a value with denominator dividing $\Gamma_F(\Gamma_X 2^\alpha)^d$, and since $d \geq 2$, so has 8 times a correction $L_i w^2/8$. Hence factor values, corrections, table entries and messages have the common denominator $8\Gamma_F(\Gamma_X 2^\alpha)^d$, and over it they are integers of $O(d(I + \alpha))$ bits. For fixed d these bit lengths exceed those of Appendix A by a constant factor, and the evaluation of an explicit factor costs a polynomial in its length and these bit lengths. The second statement follows from Proposition 6.21 with $q = O(I)$. $\square$

Example 6.23 (Limits of the polynomial extension). (a) $F(x) = x^3 - 6x$ on $[1, 2]$ has $\partial_{xx}F = 6x \leq 12 = \hat{L}$ and $F(x) - F^* = (x - \sqrt{2})^2(x + 2\sqrt{2}) \geq 3(x - \sqrt{2})^2$, so $g = 3$ is a growth constant and $\kappa \leq 4$; but its minimizer $\sqrt{2}$ and value $-4\sqrt{2}$ are irrational, so no rational exact output exists for continuous polynomial coordinates. (b) $F(x) = x^4$ on $[-1, 1]$ has a unique minimizer and no $g > 0$, so Lemma 6.12 fails for polynomials. (c) For $k \geq 1$, $F(x, y) = x^2 - \frac{1}{2}x^{2^k} + y^2$ on $[0, 1]^2$ has $\partial_{xx}F \leq 2$, $\partial_{yy}F = 2$ and growth constant $\frac{1}{2}$, since $x^{2^k} \leq x^2$; its input has $O(k)$ bits if exponents are written in binary, but $F(\frac{1}{2}, 0)$ has denominator 2^{2^k+1} . Binary-encoded exponents therefore need a separate evaluation analysis.

For continuous polynomial objectives an exact answer must therefore be implicit. Appendix B.2 gives one form: under point growth and strict complementarity at the active bounds, sign tests on the certified retained box identify the active face, and a uniform convexity test on the remaining coordinates yields a strongly convex polynomial subproblem whose unique minimizer is the global one. Besides p , κ and I , the cost bound depends polynomially on $\log_2 \max\{2, C_2/\lambda_A\}$, where C_2 bounds the absolute row sums of the second derivatives in the continuous coordinates on $\bar{X}$ and λ_A is the smallest inward derivative at the active bounds (Theorem B.4); Propositions B.5 and B.6 show that this term is intrinsic to certificates of this form and that weak complementarity defeats them.

7 Conditional recourse

The condition numbers of Section 5 are built from the coordinate curvatures L_i of F and a growth constant. Stiff convex terms, which are easy to optimize, can make the curvatures large. A residual problem with combinatorial structure causes two other difficulties: nearly tied endpoint assignments of its coordinates make the point-growth constant small, and dense interactions among them make the bags large. This section minimizes exactly over part of the variables for fixed values of the others; we call this *recourse*, as for the second stage of a two-stage program, although nothing here is random. The coordinates split into a *retained* set $\mathcal{K}$ and a *recourse* set $\mathcal{R} = [n] \setminus \mathcal{K}$, with $v = x_{\mathcal{K}}$ and $y = x_{\mathcal{R}}$. For given v , y is chosen optimally, and CT (Algorithm 5.6) runs on the value function V of (7.1).

Section 7.1 shows that V inherits coordinate curvature and growth from F , so that Sections 4–6 apply to V (Theorem 7.2). Section 7.3 eliminates private convex quadratic blocks and certifies the curvature of their value factors (Theorem 7.9). Section 7.4 eliminates residual problems that are concave in each coordinate and submodular after sign changes, whose conditional values are minimum cuts (Theorem 7.15). These are the main results of this section. Section 7.2 shows that a correction charged to one bag is valid with exact or certified recourse (Theorem 7.4), but not with grid min-marginals (Proposition C.2 in Appendix C); Section 7.4 uses its Lemma 7.3 and Theorem 7.4. Remark 7.21 extends the cut oracle to concave–convex residuals. Section 7.5 does not use recourse: for balanced quadratics the grid problem of CT is itself a minimum cut, so no tree decomposition is needed (Theorem 7.24).

7.1 Value functions inherit curvature and growth

For the retained set $\mathcal{K}$ and the recourse set $\mathcal{R} = [n] \setminus \mathcal{K}$ define

$$V(v) = \min_{y \in X_{\mathcal{R}}} F(v, y), \quad v \in \bar{X}_{\mathcal{K}}, \quad (7.1)$$

where $\bar{X}_{\mathcal{K}} = \prod_{i \in \mathcal{K}} [\ell_i, u_i]$ and the feasible set $X_{\mathcal{R}}$ of the recourse problem is compact and does not depend on v ; this is the only structural requirement. For model (3.1), $X_{\mathcal{R}} = \prod_{i \in \mathcal{R}} X_i$; in Section 7.3 it is a product of polytopes. We have $\min_{X_{\mathcal{K}}} V = F^*$, and a retained point $v \in X_{\mathcal{K}}$ together with a minimizer $y(v)$ of the recourse problem is a feasible point with value $V(v)$.

Lemma 7.1 (Partial minimization). *Let V be defined by (7.1).*

- (a) *Let $i \in \mathcal{K}$ and $L_i \geq 0$. If $t \mapsto F(v + te_i, y) - \frac{L_i}{2}t^2$ is concave on $\{t : v + te_i \in \bar{X}_{\mathcal{K}}\}$ for every $v \in \bar{X}_{\mathcal{K}}$ and every $y \in X_{\mathcal{R}}$, then so is $t \mapsto V(v + te_i) - \frac{L_i}{2}t^2$ for every $v \in \bar{X}_{\mathcal{K}}$. In particular, upper coordinate curvatures of F for the coordinates in $\mathcal{K}$ are upper coordinate curvatures of V on $\bar{X}_{\mathcal{K}}$.*
- (b) *If F has point growth with constant g at (v^*, y^*) , then V has point growth with constant g at v^* on $X_{\mathcal{K}}$.*

Proof. (a) For fixed v , the function $t \mapsto V(v + te_i) - \frac{L_i}{2}t^2$ is the pointwise minimum over $y \in X_{\mathcal{R}}$ of the concave functions $t \mapsto F(v + te_i, y) - \frac{L_i}{2}t^2$. It is finite, and it is concave because its hypograph is the intersection of their hypographs. (b) Since $F^* \leq V(v^*) \leq F(v^*, y^*) = F^*$, we have $V(v^*) = F^*$. For $v \in X_{\mathcal{K}}$ let $y(v)$ attain the minimum. Then $V(v) - V(v^*) = F(v, y(v)) - F^* \geq g(\|v - v^*\|^2 + \|y(v) - y^*\|^2) \geq g\|v - v^*\|^2$. $\square$

Part (a) is the classical stability of semiconcavity under infima [34]. Lemma 7.1(a) explains why the corrected-grid bound survives partial minimization, and Lemma 7.1(b) why the growth analysis does. The curvature of V can be much smaller than that of F . If $F(v, y) = M(y - v)^2$ with $v, y \in [0, 1]$, the diagonal curvature in v is $2M$, whereas $V \equiv 0$. Section 7.3 evaluates V exactly and certifies curvature bounds that can cancel such stiff convex terms (Theorem 7.9), though not always (Proposition 7.11); Section 7.4 evaluates V exactly when the eliminated problem is combinatorial.

The factor structure survives the elimination as follows. Suppose the recourse coordinates form disjoint *private blocks* $y^{(1)}, \dots, y^{(r)}$ with domains $Y_1, \dots, Y_r$, so that $X_{\mathcal{R}} = \prod_t Y_t$, and

$$F(v, y) = F_0(v) + \sum_{t=1}^r \phi_t(v_{E_t}, y^{(t)}), \quad (7.2)$$

where F_0 is a sum of factors in v and each block interacts with the retained coordinates only through its *scope* $E_t \subseteq \mathcal{K}$. Then

$$V(v) = F_0(v) + \sum_{t=1}^r \psi_t(v_{E_t}), \quad \psi_t(w) = \min_{y^{(t)} \in Y_t} \phi_t(w, y^{(t)}),$$

is again a sum of factors, the *value factors* ψ_t , with scopes E_t . Any tree decomposition of the scopes of F_0 and the sets E_t serves for V . Its width is the width after elimination of the blocks, which can differ from the width of the original decomposition.

Theorem 7.2 (Grids over value functions). *Suppose V has the form above, with a tree decomposition of the scopes of F_0 and the sets E_t of maximum bag size p , and with rational upper coordinate curvatures L_i^V ($i \in K$) of V whose encoding lengths are bounded by a polynomial in I . Suppose that, for every rational point w , the factors of F_0 can be evaluated exactly, and an oracle returns $\psi_t(w)$ and a minimizer $y^{(t)}(w)$ exactly for every block t , both in time bounded by a polynomial in I and the encoding length of w . Put $L^V = \max_i L_i^V$.*

- (a) *Theorem 4.7 holds for V , and the incumbent of CT applied to V lifts to a feasible point $(\hat{v}, y(\hat{v}))$ of value $V(\hat{v})$.*
- (b) *If V has point growth with constant g (by Lemma 7.1(b), this holds if F has), then Theorem 5.7 holds for V with $\kappa_V = \kappa(L^V, g)$ in place of $\bar{\kappa}$ and with the bound $f(p, \kappa_V) \kappa_V^C (I + q + 1)^C$ in place of $f(p, \bar{\kappa})(I + q + 1)^5$, where C depends only on the polynomials in the hypotheses.*
- (c) *If F is a rational quadratic on the mixed box X of (3.1), Theorem 6.14 holds for EX (Algorithm 6.13) with CT applied to V and with REC (Algorithm 6.7) and the acceptance test applied to F at the lifted incumbent. If F has point growth with constant g , the bound becomes $f(p, \kappa_V) \kappa_V^{C_1} (I + 1)^{C_1}$, where C_1 depends only on the polynomials in the hypotheses.*

Proof. (a) Sections 4 and 5 use the objective only through Definition 3.1, exact factor values at grid points, and, for the running time, growth. The L_i^V satisfy Definition 3.1 for V , the oracle gives exact values and $y(\hat{v})$, and $\min_{X_K} V = F^*$.

(b), (c) The accounting of bit lengths, which replaces the common denominators of Theorem 5.7 by the bounds of the oracle, and of the accuracy needed for exact output is in Appendix C. $\square$

The theorem charges the work of the recourse oracle to the polynomial factor, so it is useful when the recourse problems are easy and their elimination does not increase the width. Sections 7.3 and 7.4 give classes in which the recourse problems are solved exactly and with certificates.

7.2 Bag-local corrections with exact or certified recourse

The correction of Section 4 is global: every coordinate is rounded, and the allowance is the sum of all coordinate corrections. A bag-local allowance is valid when the coordinates outside the bag are minimized exactly or with a certified error.

Bag cells. Let B be a nonempty set of coordinates, for example a bag or the core of Section 7.4. We write V_B for the value function (7.1) with $K = B$, and $\bar{X}_B = \prod_{i \in B} [\ell_i, u_i]$. Let L_i ($i \in B$) be upper coordinate curvatures of F for the coordinates in B . A *bag cell* is a product $C = \prod_{i \in B} [a_i, b_i]$ with $a_i \leq b_i$ and $a_i, b_i \in X_i$. Its *corners* are the points v with $v_i \in \{a_i, b_i\}$; they lie in X_B . With the effective width w of Section 4, put

$$e_B(C) = \sum_{i \in B} \frac{L_i}{8} w([a_i, b_i])^2.$$

If every side of C is at most h , then $e_B(C) \leq \frac{h^2}{8} \sum_{i \in B} L_i$.

Lemma 7.3 (Bag-cell interpolation). *Let C be a bag cell. Every $x \in X$ with $x_B \in C$ satisfies*

$$F(x) \geq \min_{v \text{ a corner of } C} V_B(v) - e_B(C).$$

Proof. By Lemma 7.1(a), the L_i satisfy Definition 3.1 for V_B on $\bar{X}_B$. The proof of Proposition 4.2(a) uses only Definition 3.1; applied to V_B , to the subbox spanned by C and to the grid of its corners, it gives $V_B(x_B) \geq \min_{\text{corners}} V_B - e_B(C)$. Finally $F(x) \geq V_B(x_B)$ because $x_{[n] \setminus B} \in X_{[n] \setminus B}$. $\square$

No coordinate outside B is rounded, so the error $e_B(C)$ does not grow with n . The price is that V_B is a global optimization problem over the original outside domain.

Theorem 7.4 (Conditional-recourse filter). *Let $\mathcal{Q}$ be a finite family of bag cells, $e \geq \max_{C \in \mathcal{Q}} e_B(C)$ and $\varepsilon_{\text{or}} \geq 0$. Suppose that for every corner v of every cell of $\mathcal{Q}$ an oracle supplies a number $\underline{V}(v)$ and a point $x^v \in X$ with $x_B^v = v$ such that*

$$\underline{V}(v) \leq V_B(v), \quad F(x^v) \leq \underline{V}(v) + \varepsilon_{\text{or}}. \quad (7.3)$$

Let U be the value of a feasible point with $U \leq \min_v F(x^v)$, the minimum taken over all supplied completions. Retain $C \in \mathcal{Q}$ if and only if $\min_v \text{corner of } C \underline{V}(v) - e \leq U$. Then:

- (a) *every $C \in \mathcal{Q}$ that contains x_B^* for some global minimizer x^* is retained;*
- (b) *if some $C \in \mathcal{Q}$ contains x_B^* for a global minimizer x^* , then $U \leq F^* + e + \varepsilon_{\text{or}}$, and every retained cell has a corner v with $V_B(v) \leq F^* + 2e + 2\varepsilon_{\text{or}}$;*
- (c) *every $x \in X$ with $x_B \in \bigcup \mathcal{Q}$ satisfies*

$$F(x) \geq \lambda := \min_{C \in \mathcal{Q}} \left(\min_{\text{corners of } C} \underline{V} - e \right),$$

$$\text{and } U - \lambda \leq e + \varepsilon_{\text{or}}.$$

The proof is in Appendix C.

Appendix C applies the theorem level by level to multilevel schemes and shows that filtering alone needs only values whose error varies by at most ε_{or} over the corners (Proposition C.1). Grid min-marginals, which minimize over the outside coordinates on the grid instead of their domain, do not suffice: a constant correction charged to one bag then keeps a cell containing the bag projection of the minimizer only if the constant is at least a quantity that grows with the number of outside coordinates, already for bag size two and $\kappa < 11$ (Proposition C.2 in Appendix C).

7.3 Certified convex recourse

Large positive curvature often sits in variables that enter the objective convexly once the other variables are fixed. This subsection eliminates such variables exactly, as convex value factors, and shows that CT then depends only on the coordinate curvature and the growth of the reduced objective. Proposition 7.8 covers three cases (no certificate, a globally affine response, a piecewise-affine response), and Proposition 7.11 shows the limits of this approach.

Definition 7.5 (Convex recourse model). The coordinates split into retained coordinates $\mathcal{K}$, with $v = x_{\mathcal{K}} \in X_{\mathcal{K}} = \prod_{i \in \mathcal{K}} X_i$ and X_i as in (3.1), and r private blocks $y^{(t)} \in Y_t$, $t = 1, \dots, r$, where $Y_t = \{y \in \mathbb{R}^{n_t} : G_t y \leq b_t\}$ is a nonempty bounded rational polytope. Private variables are continuous. Block t has an ordered scope $E_t \subseteq \mathcal{K}$ of k_t retained coordinates; for $i \in E_t$ let $j(t, i)$ be its position in E_t , and let $\Pi_t = \prod_{i \in E_t} [\ell_i, u_i]$ be the *scope box*. The objective has the form (7.2),

$$F(v, y) = F_0(v) + \sum_{t=1}^r \phi_t(v_{E_t}, y^{(t)}), \quad \phi_t(w, y) = \frac{1}{2} y^{\top} C_t y + y^{\top} \Gamma_t w + c_t^{\top} y, \quad (7.4)$$

with rational data, $C_t \succeq 0$, and a rational quadratic $F_0(v) = \sum_{a \in \mathcal{A}} f_a(v_{S_a})$ with Hessian H_0 . All terms that involve retained variables only are placed in F_0 ; thus $(H_0)_{ii}$ is the diagonal entry of the full Hessian of F at v_i . A tree decomposition of the retained coordinates is supplied, with maximum bag size p , such that every S_a and every E_t lies in a bag.

The full Hessian of F need not be positive semidefinite, and C_t may be singular. The value factors and the reduced objective are

$$\psi_t(w) = \min_{y \in Y_t} \phi_t(w, y) \quad (w \in \mathbb{R}^{k_t}), \quad V(v) = F_0(v) + \sum_t \psi_t(v_{E_t}). \quad (7.5)$$

Since the blocks are independent once v is fixed, V is the value function (7.1) with $X_{\mathcal{R}} = \prod_t Y_t$, and $\min_{X_{\mathcal{K}}} V = F^*$. Every v lifts to $(v, y(v))$ with $y^{(t)}(v) \in \arg \min_{Y_t} \phi_t(v_{E_t}, \cdot)$ and $F(v, y(v)) = V(v)$.

Leaf certificates and the curvature of a value factor. Fix one block, write $k = k_t$, and drop the index t : $\phi(w, y) = \frac{1}{2}y^{\top}Cy + y^{\top}\Gamma w + c^{\top}y$ for $w \in \mathbb{R}^k$, $Y = \{y : Gy \leq b\}$ with m rows, and $\psi(w) = \min_{y \in Y} \phi(w, y)$.

Definition 7.6 (Leaf certificate). Let $\Pi' \subseteq \mathbb{R}^k$ be a polytope with nonempty interior. A *certificate* on Π' consists of affine maps $\bar{y}(w) = \bar{y}_0 + Bw$ and $\lambda(w) = \lambda_0 + \Lambda w \in \mathbb{R}^m$ such that, for all $w \in \Pi'$,

$$G\bar{y}(w) \leq b, \quad \lambda(w) \geq 0, \quad (7.6)$$

$$C\bar{y}(w) + \Gamma w + c + G^{\top}\lambda(w) = 0, \quad (7.7)$$

$$\lambda_s(w)(b_s - G_s\bar{y}(w)) = 0 \quad (s = 1, \dots, m). \quad (7.8)$$

A *certificate on a rational box* $\Pi \subseteq \mathbb{R}^k$ is a rational binary space partition of Π , in which each internal node splits its polytope by one rational hyperplane into two closed halves with nonempty interiors, together with a certificate on each of its leaves $\Pi'_1, \dots, \Pi'_{N_{\Pi}}$. For a private box $Y = [\ell^Y, u^Y]$ the rows are $y \leq u^Y$ and $-y \leq -\ell^Y$, and λ splits into upper and lower bound multipliers.

A certificate is checked exactly in time polynomial in its encoding length. The leaves of a binary space partition cover its root, each leaf is described by the inequalities of Π and of the halfspaces on its root path, and one linear program per node checks that both halves have nonempty interiors. Stationarity (7.7) is an identity between affine maps, and complementarity (7.8) says that a quadratic polynomial vanishes on a set with nonempty interior, that is, all its coefficients vanish. By linear programming duality, an affine inequality $\alpha_0 + \alpha^{\top}w \geq 0$ holds on a nonempty polytope $\{w : Aw \leq a\}$ if and only if some $\pi \geq 0$ has $A^{\top}\pi = -\alpha$ and $a^{\top}\pi \leq \alpha_0$; such rational multipliers are part of the certificate.

Lemma 7.7 (Certificate identity and piece Hessian). *Let $(\bar{y}, \lambda)$ be a certificate on Π' . Then:*

(a) *For $w \in \Pi'$ and $y \in \mathbb{R}^{n_t}$,*

$$\phi(w, y) - \phi(w, \bar{y}(w)) = \frac{1}{2}(y - \bar{y}(w))^{\top}C(y - \bar{y}(w)) + \lambda(w)^{\top}(b - Gy). \quad (7.9)$$

The right side is nonnegative for $y \in Y$. Hence $\psi = q_{\Pi'}$ on Π' , where $q_{\Pi'}(w) = \phi(w, \bar{y}(w))$.

(b) *Let $\mathcal{E} = \{s : \lambda_s \neq 0\}$. Then $G_s B = 0$ for $s \in \mathcal{E}$, and*

$$\nabla^2 q_{\Pi'} = B^{\top}CB + B^{\top}\Gamma + \Gamma^{\top}B = -B^{\top}CB \preceq 0.$$

Proof. (a) The function $\phi(w, \cdot)$ is quadratic with Hessian C , and by (7.7) its gradient at $\bar{y} = \bar{y}(w)$ is $-G^\top \lambda$. Exact expansion gives

$$\phi(w, y) - \phi(w, \bar{y}) = \lambda^\top (G\bar{y} - Gy) + \frac{1}{2}(y - \bar{y})^\top C(y - \bar{y}),$$

and $\lambda^\top (G\bar{y} - Gy) = \lambda^\top (b - Gy) - \lambda^\top (b - G\bar{y}) = \lambda^\top (b - Gy)$ by (7.8). For $y \in Y$ both terms on the right of (7.9) are nonnegative, since $C \succeq 0$ and $\lambda(w) \geq 0$. Equality holds at $y = \bar{y}(w) \in Y$.

(b) For each s , the polynomial $\lambda_s(w)(b_s - G_s \bar{y}_0 - G_s Bw)$ vanishes on Π' , which has nonempty interior, so it is the zero polynomial. The polynomial ring is an integral domain; for $s \in \mathcal{E}$ the first factor is a nonzero polynomial, so the second is zero and $G_s B = 0$. The w -linear part of (7.7) reads $CB + \Gamma + G^\top \Lambda = 0$, and row s of Λ vanishes for $s \notin \mathcal{E}$. Hence

$$B^\top (CB + \Gamma) = - \sum_{s \in \mathcal{E}} (G_s B)^\top \Lambda_s = 0,$$

so $B^\top \Gamma = -B^\top CB$, which is symmetric; therefore also $\Gamma^\top B = -B^\top CB$. Substituting $y = \bar{y}_0 + Bw$ in ϕ gives $\nabla^2 q_{\Pi'} = B^\top CB + B^\top \Gamma + \Gamma^\top B = -B^\top CB$. $\square$

By (C.1) in Appendix C, $(B^\top CB)_{jj}$ is the energy that the private response spends to follow a unit change of w_j .

Proposition 7.8 (Coordinate curvature of a value factor). *(a) ψ is finite and concave on $\mathbb{R}^k$.*

(b) Let a certificate on a box Π be given, with linear part $B_{\Pi'}$ of the response on the leaf Π' , and put

$$\sigma_j = \min_{\Pi'} (B_{\Pi'}^\top C B_{\Pi'})_{jj} \geq 0,$$

the minimum over all leaves. For every $w \in \Pi$ and every interval J with $w + J e_j \subseteq \Pi$, the function $t \mapsto \psi(w + t e_j) + \sigma_j t^2 / 2$ is concave on J .

(c) If a number σ' has the property in (b), then $\sigma' \leq \sigma_j$. In particular σ_j is the same for every certificate on Π .

Proof. (a) For fixed y , $\phi(w, y)$ is affine in w . Thus ψ is a pointwise minimum of affine functions over the compact nonempty set Y , hence finite and concave.

(b) On each leaf, $\psi = q_{\Pi'}$ has second derivative $-(B_{\Pi'}^\top C B_{\Pi'})_{jj} \leq -\sigma_j$ along e_j (Lemma 7.7), and at the finitely many points where a segment changes leaf, the one-sided derivatives of the concave function ψ can only decrease. Appendix C gives the details.

(c) Take any leaf Π' and an interior point w' of Π' . For small $|t|$, $\psi(w' + t e_j) = q_{\Pi'}(w' + t e_j)$, whose second derivative is $-(B_{\Pi'}^\top C B_{\Pi'})_{jj}$. Concavity of $\psi(w' + t e_j) + \sigma' t^2 / 2$ forces $\sigma' \leq (B_{\Pi'}^\top C B_{\Pi'})_{jj}$. Minimizing over the leaves gives $\sigma' \leq \sigma_j$. Applying this to two certificates shows that each σ_j is at most the other. $\square$

Part (a) alone certifies $\sigma_j = 0$ for every block; this is the direct bound. Part (c) shows that a piecewise certificate loses nothing: the cancellation it certifies is the largest one possible for coordinatewise semiconcavity of this factor. For a private box, a certificate on every rational box always exists, so this largest cancellation is always certifiable (Proposition C.4 in Appendix C). The construction can be exponential; no bound on the number of leaves is claimed.

Theorem 7.9 (Certified convex recourse). *In the model of Definition 7.5, let each block carry a certificate on its scope box Π_t , or none. Put $\sigma_t = 0$ for a block without certificate and otherwise let σ_t be the vector of Proposition 7.8(b). Define*

$$L_i = (H_0)_{ii} - \sum_{t: i \in E_t} \sigma_{t,j(t,i)}, \quad L_i^+ = \max\{L_i, 0\}, \quad L^+ = \max_i L_i^+, \quad (7.10)$$

and let I be the total encoding length of data, decomposition and certificates.

- (i) *For each $i \in \mathcal{K}$, the function $v \mapsto V(v) - L_i v_i^2/2$ is concave on every segment of $\bar{X}_{\mathcal{K}}$ parallel to e_i . All certificates are checked in $\text{poly}(I)$ time.*
- (ii) *If $L^+ = 0$, that is, $L_i \leq 0$ for all i , endpoint dynamic programming with two nodes per coordinate returns an exact minimizer of F in $2^{O(p)} \text{poly}(I)$ bit operations. No growth or uniqueness is needed.*
- (iii) *Suppose that $L^+ > 0$ and that V has point growth with constant g , as it has if F has (Lemma 7.1(b)). With $\kappa_V = \kappa(L^+, g)$, CT applied to V with the curvatures L_i^+ returns a feasible original point with a certified gap at most 2^{-q} in $f(p, \kappa_V) \kappa_V^C (I + q + 1)^C$ bit operations. If F has point growth with constant g , an exact rational minimizer is found in $f(p, \kappa_V) \kappa_V^{C_1} (I + 1)^{C_1}$ bit operations. Here f is the function of Theorem 5.7, C and C_1 are absolute constants, and neither g nor κ_V is an input. Every certified lower bound and every returned exact minimizer is valid without growth or uniqueness. Without point growth of F , the exact-output procedure has no termination guarantee; if every Y_t is a box, Theorem 7.2(c) applies instead, and EX stops on every instance.*
- (iv) *$L_i \leq (H_0)_{ii}$ for every i . Hence, if F has point growth with constant g and L' is a common upper coordinate curvature of F for all coordinates, retained and private, then $\kappa(L^+, g) \leq \kappa(L', g)$: the parameter κ_V of (iii) never exceeds the Euclidean condition number of F with the private variables kept as ordinary coordinates.*

The proof is in Appendix C.4.

Remark 7.10 (Three instances and prior work). Without certificates, $L_i = (H_0)_{ii}$: changing active sets are handled exactly, but direct stiffness is not cancelled. A one-leaf certificate is a globally affine response; then ψ_t is an explicit quadratic with Hessian $-B_t^T C_t B_t$ that can be merged into F_0 (for $M(y - v)^2$ on $[0, 1]^2$ it gives $\sigma = 2M$ and $L_v = 0$). A certificate with several leaves is a piecewise-affine response and certifies the largest possible σ_j . The partitions of different blocks are never overlaid; their common refinement can have exponentially many cells. Parametric-QP critical regions are classical [18]; the contribution here is their curvature accounting inside CT. Khajavirad [110] eliminates the components formed by variables with positive diagonal into value functions of their neighbors; the neighbors have nonpositive diagonal, can be taken binary, and their values are enumerated. Here the neighbors v_{E_t} are gridded continuous or integer coordinates, and what is certified is the coordinate curvature of each value factor.

Appendix C derives an affine condensation (Corollary C.9), the bound $\kappa_V \leq \max\{1, C_0\} \cdot \max\{1, \nu/g\}$ when $L^+ \leq C_0 \nu$ with $\nu = \max\{0, -\lambda_{\min}(\nabla^2 F)\}$ (Corollary C.10), a block whose certified curvature is 8 while its direct curvature is $2M + 8$ (Example C.11), and a ladder with m negative eigenvalues whose certified ratio L^+/g is at most 123 for all m and M (Proposition C.12). For a block whose domain is a box, one exact convex quadratic program and one linear program decide whether some affine map is an optimal response on the whole scope box, and if so return it with a one-leaf certificate (Theorem C.7 in Appendix C).

Limits of global recourse curvature. Call a retained set $\mathcal{K} \subseteq [n]$ *admissible* if the principal Hessian block of the coordinates in $\mathcal{R} = [n] \setminus \mathcal{K}$ is positive semidefinite, so that they form convex private blocks. For admissible $\mathcal{K}$ the value function $V_{\mathcal{K}}$ is intrinsic: it does not depend on how the private part is grouped or certified. Let $L_{\mathcal{K}}$ be the least L for which Definition 3.1 holds for $V_{\mathcal{K}}$ in every retained coordinate, and $g_{\mathcal{K}}$ the largest growth constant of $V_{\mathcal{K}}$. Since every certified L_i is valid (Theorem 7.9(i)), $L^+ \geq L_{\mathcal{K}}$, and every growth constant g of $V_{\mathcal{K}}$, in particular every growth constant of F (Lemma 7.1(b)), satisfies $g \leq g_{\mathcal{K}}$; so no certificate yields a parameter below $L_{\mathcal{K}}/g_{\mathcal{K}}$.

Proposition 7.11 (Recourse cannot replace L/g by ν/g in general). *For $M \geq 1$ let $x \in [0, 2]$, $y \in [1, 3]$ and*

$$F_M(x, y) = M(x - y)^2 - \frac{1}{8}(x + y - 2)^2 + 2(y - 1).$$

Then $(1, 1)$ is the unique minimizer, $F^ = 0$, $F_M(x, y) - F^* \geq \frac{1}{8}((x - 1)^2 + (y - 1)^2)$, and $\nu = \frac{1}{2}$; thus $\nu/g \leq 4$ with $g = \frac{1}{8}$. Nevertheless, for every admissible retained set $\mathcal{K}$ and every positive diagonal rescaling of the retained coordinates,*

$$L_{\mathcal{K}}/g_{\mathcal{K}} \geq (4M - \frac{1}{2})/3 \geq M.$$

The proof is in Appendix C, which also locates the obstruction in a clipped piece of the private response. Disjoint copies give arbitrarily many variables with the same constants and bag size two. The instance is trivial to solve; the proposition limits the parameter of Theorem 7.9, not the problem.

7.4 Separately concave residuals and minimum cuts

In this subsection $F(x) = \frac{1}{2}x^T Hx + b^T x + c$ is a rational quadratic, the retained set $\mathcal{K}$ is a *core* of k continuous coordinates, and the recourse set $\mathcal{R}$ is called the *residual*. Assume

$$(R1) \ H_{ii} \leq 0 \ (i \in \mathcal{R}), \quad (R2) \ o_i o_j H_{ij} \leq 0 \ (i \neq j, \ i, j \in \mathcal{R}) \text{ for some } o \in \{-1, 1\}^{\mathcal{R}}. \quad (7.11)$$

Nothing is assumed about the entries of H in core rows, and the residual interaction graph may be dense. A residual coordinate may be continuous or integer; integer bounds are rounded inward first, and an empty domain is rejected. Signs o as in (R2) are found, or shown not to exist, by the following lemma applied to $H_{\mathcal{R}\mathcal{R}}$.

Lemma 7.12 (Recognizing balance). *For a symmetric $m \times m$ matrix H , signs $o \in \{-1, 1\}^m$ with $o_i o_j H_{ij} \leq 0$ for all $i \neq j$ can be found, or shown not to exist, in time linear in m plus the number of nonzero off-diagonal entries.*

Proof. The conditions are $o_i = o_j$ when $H_{ij} < 0$ and $o_i = -o_j$ when $H_{ij} > 0$. Breadth-first search fixes one sign in each connected component of the graph of nonzero off-diagonal entries, propagates the conditions along a spanning tree and checks every other edge. If an edge violates its condition, it closes a cycle with an odd number of positive entries, around which the conditions are inconsistent [88]; otherwise all conditions hold. $\square$

Lemma 7.13 (Endpoint reduction). *Fix $v \in \bar{X}_{\mathcal{K}}$ and a box $\prod_{i \in \mathcal{R}} [\ell_i, u_i]$ whose bounds are integers for integer coordinates. Under (R1), the minimum of $F(v, \cdot)$ over this continuous box is attained at a point with $y_i \in \{\ell_i, u_i\}$ for all $i \in \mathcal{R}$, and it equals the minimum over the mixed residual domain.*

Proof. Start from a minimizer, which exists by compactness. For each residual coordinate in turn, F as a function of that coordinate alone is $\frac{1}{2}H_{ii}t^2$ plus an affine function of t , which is concave by (R1); replace the coordinate by an endpoint with no larger value. The result is a minimizer whose residual coordinates are endpoints. Its integer coordinates are integral, so it lies in the mixed domain, which is contained in the continuous box. $\square$

For $i \in \mathcal{R}$ put $\alpha_i = \ell_i$, $\delta_i = u_i - \ell_i$ if $o_i = 1$, and $\alpha_i = u_i$, $\delta_i = -(u_i - \ell_i)$ if $o_i = -1$. For a label $z \in \{0, 1\}^{\mathcal{R}}$ let $y(z)$ have coordinates $y(z)_i = \alpha_i + \delta_i z_i$; as z ranges over $\{0, 1\}^{\mathcal{R}}$, $y(z)$ ranges over all residual endpoint vectors. In a network with source $\mathbf{s}$ and sink $\mathbf{t}$, $\text{cap}(S)$ denotes the total capacity of the arcs leaving a node set S that contains $\mathbf{s}$ but not $\mathbf{t}$.

Proposition 7.14 (Signed cut representation). (a) Let $\Phi(z) = \Phi_0 + \sum_i \mu_i z_i + \sum_{i < j} \omega_{ij} z_i z_j$ on $\{0, 1\}^m$ with $\omega_{ij} \leq 0$ for all $i < j$, and put $\omega_{ji} = \omega_{ij}$ and $\rho_i = \mu_i + \frac{1}{2} \sum_{j \neq i} \omega_{ij}$. Build a network on $\{1, \dots, m, \mathbf{s}, \mathbf{t}\}$ with arcs $i \rightarrow \mathbf{t}$ of capacity $\max\{\rho_i, 0\}$ and $\mathbf{s} \rightarrow i$ of capacity $\max\{-\rho_i, 0\}$ for each i , and arcs $i \rightarrow j$ and $j \rightarrow i$ of capacity $-\omega_{ij}/2 \geq 0$ for each pair with $\omega_{ij} \neq 0$. With $S_z = \{\mathbf{s}\} \cup \{i : z_i = 1\}$,

$$\Phi(z) = \Phi_0 + \sum_i \min\{0, \rho_i\} + \text{cap}(S_z) \quad (z \in \{0, 1\}^m). \quad (7.12)$$

(b) Under (7.11) and for a rational core point v , the function $z \mapsto F(v, y(z))$ on $\{0, 1\}^{\mathcal{R}}$ has the form of (a) with rational coefficients obtained by exact evaluation of F , and $\omega_{ij} = H_{ij} \delta_i \delta_j \leq 0$.

Proof. (a) For binary z , $\omega_{ij} z_i z_j = \frac{\omega_{ij}}{2} (z_i + z_j) - \frac{\omega_{ij}}{2} |z_i - z_j|$, so $\Phi(z) = \Phi_0 + \sum_i \rho_i z_i - \frac{1}{2} \sum_{i < j} \omega_{ij} |z_i - z_j|$. In the cut S_z , an arc $i \rightarrow \mathbf{t}$ is cut exactly when $z_i = 1$, an arc $\mathbf{s} \rightarrow i$ exactly when $z_i = 0$, and exactly one of the arcs $i \rightarrow j$, $j \rightarrow i$ when $z_i \neq z_j$, none otherwise. Hence $\text{cap}(S_z) = \sum_i [\max\{\rho_i, 0\} z_i + \max\{-\rho_i, 0\} (1 - z_i)] - \frac{1}{2} \sum_{i < j} \omega_{ij} |z_i - z_j|$, and $\rho_i z_i = \max\{\rho_i, 0\} z_i + \max\{-\rho_i, 0\} (1 - z_i) + \min\{0, \rho_i\}$.

(b) Expand $F(v, \alpha + \delta \circ z)$ and use $z_i^2 = z_i$. Terms without residual coordinates are constant. The terms $\frac{1}{2} H_{ii} y_i^2$, $b_i y_i$ and $H_{ij} v_j y_i$ with $j \in \mathcal{K}$ are affine in z_i , and $H_{ij} y_i y_j$ with $i \neq j$ in $\mathcal{R}$ gives $H_{ij} \delta_i \delta_j z_i z_j$ plus affine terms. By (R2), $\omega_{ij} = H_{ij} o_i o_j (u_i - \ell_i)(u_j - \ell_j) \leq 0$. $\square$

Every node set that contains $\mathbf{s}$ but not $\mathbf{t}$ is S_z for exactly one z , so by (7.12) minimizing Φ is a minimum $\mathbf{s}$ - $\mathbf{t}$ cut problem with nonnegative capacities.

Theorem 7.15 (Certified cut recourse). Assume (7.11). For every rational core point v , one maximum-flow computation returns a minimizer $y(v)$ of $F(v, \cdot)$ over $X_{\mathcal{R}}$, which is a residual endpoint vector, the exact value $V(v)$, and a certificate: a label z with $y(v) = y(z)$ and a feasible flow in the network of Proposition 7.14 whose value equals $\text{cap}(S_z)$. The bit complexity is polynomial in the encoding length of F and v , with an absolute exponent. Given the network, the certificate is checked with a number of rational operations linear in $|\mathcal{R}|$ plus the number of nonzero entries H_{ij} with $i \neq j$ in $\mathcal{R}$.

The proof is in Appendix D.1.

Remark 7.16 (Scope and prior work). Lemmas 7.12 and 7.13 and Proposition 7.14 are classical (Section 2). With an empty core, Theorem 7.15 is the polynomial solvability of box QPs that are submodular after sign changes and have nonpositive diagonal [32, 150]. What we use is the conditional form: the core and the core-residual couplings are arbitrary, and each conditional value comes with a flow certificate. Del Pia and Khajavirad also fix the coordinates with nonpositive diagonal at endpoints; their polynomial classes [53, Theorem 4] restrict the structure of these coordinates (logarithmic treewidth and logarithmic interfaces to the others) and the rank of their coupling to the others, whereas (R2) restricts only signs and allows a dense residual. Condition (R1) is what reduces the residual to endpoint vectors. Without it, and with continuous residual coordinates, the conditional problem is a box QP that is submodular after sign changes, and the complexity of that class is open [32, footnote 2]; Section 7.5 grids such coordinates instead (Corollary 7.25).

Search over a continuous core. Let the core domain be $X_K = [0, 1]^k$ with $k \geq 1$, and let the residual domain X_R be any compact mixed box. Let $L \geq 0$ be an upper coordinate curvature of F for every core coordinate (for a quadratic, any $L \geq \max_{i \in K} H_{ii}$); by Lemma 7.1(a) it is one for V . An *exact oracle* returns, for every dyadic $v \in [0, 1]^k$, the value $V(v)$ and a completion $y(v) \in X_R$ with $F(v, y(v)) = V(v)$, together with a certificate. Theorem 7.15 supplies one under (7.11), and Remark 7.21 supplies one when the residual has a convex part. Level j of the search uses dyadic cells of side $h_j = 2^{-j}$, the common mesh of Section 5 with $s = 1$, and

$$e_j = \frac{kLh_j^2}{8}, \quad \beta_V(C) = \min\{V(v) : v \text{ a corner of } C\} - e_j.$$

Algorithm 7.17 (CORE: core search). Given an exact oracle, L , an accuracy $\varepsilon > 0$ and a level limit J , start with $\mathcal{Q}_0 = \{[0, 1]^k\}$ and $U = +\infty$. For $j = 0, 1, \dots, J$:

- (i) query the oracle at every corner of a cell of $\mathcal{Q}_j$ that has not been queried before, and let U be the least value $V(v)$ of all queried corners, attained by the incumbent $(v_U, y(v_U))$;
- (ii) put $\lambda_j = \min_{C \in \mathcal{Q}_j} \beta_V(C)$ and retain $\mathcal{Q}'_j = \{C \in \mathcal{Q}_j : \beta_V(C) \leq U\}$;
- (iii) if $U - \min\{\lambda_j, U\} \leq \varepsilon$, stop; otherwise let $\mathcal{Q}_{j+1}$ consist of the 2^k dyadic children of each cell of $\mathcal{Q}'_j$.

Return the incumbent, U and the lower bound $\min\{\lambda_j, U\}$.

Theorem 7.18 (Core search). *At every level j of CORE:*

- (a) every cell of $\mathcal{Q}_j$ that contains x_K^* for a global minimizer x^* is retained; in particular $\mathcal{Q}'_j \neq \emptyset$;
- (b) $\lambda_j \leq F^* \leq U \leq \lambda_j + e_j$;
- (c) every retained cell has a corner v with $V(v) \leq F^* + 2e_j$;
- (d) every $x \in X$ satisfies $F(x) \geq \min\{\lambda_j, U\}$, and this can be verified from the list of cells, the certified queried values and the retention decisions, without any optimization.

The proof is in Appendix D.1.

By Theorem 7.18(b), CORE stops at the latest at the first level j with $e_j \leq \varepsilon$, provided that J is at least this level. Its cost is controlled by the number of near-optimal corners.

Proposition 7.19 (Query count under core growth). *Suppose $V(v) - F^* \geq g\|v - v^*\|^2$ for all $v \in [0, 1]^k$, some v^* and some $g > 0$; by Lemma 7.1(b) this holds with the same g if F has point growth with constant g . Let $\kappa_K = \kappa(L, g) = \max\{1, L/g\}$. Then levels $0, \dots, J$ of CORE make at most*

$$2^k + 8^k J(1 + \sqrt{k\kappa_K})^k$$

oracle queries. The constant g is not used by the algorithm.

The proof is in Appendix D.1.

The bound is linear in the number of levels, and the residual enters only the cost of each query. Without growth the count can be exponential in the number of levels.

Example 7.20 (Flat directions). Take k even, no residual, and $V(v) = \sum_{t=1}^{k/2} (v_{2t-1} - v_{2t})^2$, so $L = 2$ and $F^* = 0$. The corner $v = 0$ gives $U = 0$ at level 0. Every product of diagonal cells $\prod_t ([a_t h_j, (a_t + 1)h_j] \times [a_t h_j, (a_t + 1)h_j])$ with integers $0 \leq a_t < 2^j$ has a corner with $V = 0$, hence is retained, and its parent is again such a product. So level j retains at least $2^{j k/2}$ cells, while $U - \lambda_j = e_j$; stopping at accuracy ε therefore takes $\Omega(\varepsilon^{-k/4})$ queries for fixed k .

Exact output. Under (7.11) and for rational data, F^* and every value $\min_{v \in [0,1]^k} F(v, y)$ at a residual endpoint vector y have denominators at most a constant Ω_K of bit length $O(I^2)$ (Lemma D.1). Run CORE with the oracle of Theorem 7.15 and accuracy $\varepsilon = \Omega_K^{-2}/2$, and minimize $F(\cdot, y)$ exactly over the faces of the core cube, where y is the residual part of the incumbent. The result is a global minimizer that passes the acceptance test of Proposition 6.6 with the denominator bound Ω_K . Under core growth the total cost is $8^k(1 + \sqrt{k\kappa_K})^k \text{poly}(I)$ bit operations with an absolute exponent, and g is not an input (Theorem D.2 in Appendix D).

Remark 7.21 (Concave–convex residuals). The oracle extends to residuals with a convex part. Split $\mathcal{R} = \mathcal{R}_- \cup \mathcal{R}_+$, require (R1) only on $\mathcal{R}_-$ and (R2) on all of $\mathcal{R}$, and let the coordinates in $\mathcal{R}_+$ be continuous with $H_{\mathcal{R}_+ \mathcal{R}_+} \succeq 0$. For a fixed core point, the coordinates in $\mathcal{R}_-$ can be moved to endpoints without increasing F , as in Lemma 7.13. For each set $S \subseteq \mathcal{R}_-$ of coordinates with label $z_i = 1$, what remains is a convex box QP in $y_{\mathcal{R}_+}$, which is solved exactly in polynomial time [119]. Its value $\Phi(S)$ is a submodular function of S , because partial minimization preserves submodularity [173], and the conditional minimum is $\min_S \Phi(S)$. Submodular minimization by the ellipsoid method [82] and Edmonds’ min–max theorem [63] then give the exact conditional value, a minimizer, and a certificate: a convex combination of at most $|\mathcal{R}_-| + 1$ greedy vectors of Φ . All these ingredients are classical, and the same composition underlies the mixed continuous–discrete methods cited in Section 2. Proposition D.4 in Appendix D states the result precisely. It supplies, for concave–convex residuals, the exact oracle required by CORE (Theorem 7.18 and Proposition 7.19), with a certificate that can be checked without optimization (Proposition D.4(d)). For $\mathcal{R}_+ = \emptyset$ this is the class (7.11), where Theorem 7.15 needs only one maximum flow.

7.5 Corrected grids without a tree decomposition

CT (Algorithm 5.6) uses its tree decomposition only to compute, on the current product grid $G = \prod_i G_i$, the corrected minimum $\beta = \min_G Q$, one minimizer and the min-marginals $m_i(v)$ of $Q = F - D$ (Lemma 4.3). Any exact method for these quantities can replace the dynamic program. For the following sign structure, minimum cuts are such a method, whatever the interaction graph.

Definition 7.22 (Balanced quadratic). A quadratic $F(x) = \frac{1}{2}x^\top Hx + b^\top x + c$ is *balanced* if there are signs $o \in \{-1, 1\}^n$ with $o_i o_j H_{ij} \leq 0$ for all $i \neq j$.

Equivalently, after the substitution $x_i \mapsto o_i x_i$ all off-diagonal Hessian entries are nonpositive, so F becomes submodular on $\mathbb{R}^n$. Condition (R2) of Section 7.4 says that F becomes balanced once the core coordinates are fixed, and Lemma 7.12 recognizes balance in linear time. No sign condition is imposed on the diagonal or on the linear terms. If $H_{ii} \leq 0$ for all i , then F satisfies (7.11) with an empty core, and Theorem 7.15 minimizes it exactly with one maximum flow.

Lemma 7.23 (Grid minimization by one minimum cut). *Let F be balanced with signs o , let $G_i \subset \mathbb{Q}$ be finite and nonempty with $|G_i| = k_i + 1$, and let $\chi_i : G_i \rightarrow \mathbb{Q}$ be arbitrary. Put $F_\chi(y) = F(y) + \sum_i \chi_i(y_i)$ for $y \in \prod_i G_i$. Then $\min F_\chi$ and a minimizer are obtained from one minimum s–t cut in a network with $\sum_i k_i + 2$ nodes and at most $3 \sum_i k_i + 2 \sum_{i < j: H_{ij} \neq 0} k_i k_j$ arcs, whose capacities are computed from H , b , the grid nodes and the values of the χ_i by polynomially many rational operations. A feasible flow of equal value certifies the minimum. The same holds for each min-marginal $\min\{F_\chi(y) : y_i = v\}$, with G_i replaced by $\{v\}$.*

The proof (Appendix D) encodes each coordinate by threshold variables and applies Proposition 7.14(a); arcs of infinite capacity exclude the binary vectors that encode no grid node. Threshold encodings, and the reduction of pairwise problems that are submodular with respect to label orders

to one minimum cut, are classical [164, Sections 3–4], [102]. Lemma 7.23 is this construction for quadratic pair terms; what matters here is that the unary terms χ_i are arbitrary, so that they can be the negative corrections $-d_i$ on nonuniform grids.

Theorem 7.24 (Balanced box quadratics without a width parameter). *Let F be a balanced rational quadratic on a mixed box X with input length I . Run CT and EX (Algorithm 6.13) with β , a minimizer and all min-marginals computed by Lemma 7.23 instead of Lemma 4.3.*

- (a) *On every instance, $CT(2^{-q})$ halts with a feasible rational point and a valid path certificate of gap at most 2^{-q} , in which the dynamic programs are replaced by minimum cuts with flows of equal value; and EX halts with a global minimizer and F^* .*
- (b) *If F has point growth with condition number κ , then $CT(2^{-q})$ uses $(n\kappa(I + q + 1))^{O(1)}$ bit operations and EX uses $(n\kappa(I + 1))^{O(1)}$ bit operations.*

No decomposition is used, g and κ are not inputs, and the validity of every certificate uses neither growth nor uniqueness.

Proof. (a) In CT and EX, and in the proofs of Propositions 4.2 and 4.5, Theorem 4.7, Lemmas 5.4 and 5.5 and Theorems 5.7, 6.9 and 6.14, the tree decomposition enters only through Lemma 4.3, which computes β , a minimizer of Q and the min-marginals $m_i(v)$ on the current grid (if $P = \emptyset$, CT minimizes F over the grid $\prod_i \{\ell_i, u_i\}$, on which $Q = F$). Everything else uses the data of F directly: the condition of Definition 3.1 holds with $L_i = \max\{H_{ii}, 0\}$, because the restriction of F to a coordinate line has second derivative H_{ii} , and REC (Algorithm 6.7) and Proposition 6.6 work with F and its gradient. Since D is a sum of unary terms, $Q = F_\chi$ with $\chi_i = -d_i$, so Lemma 7.23 computes β and a minimizer with one cut and each $m_i(v)$ with one further cut. A flow of equal value certifies each of these values from below by weak duality, which is what conditions (C1) and (C2) of a path certificate require. Hence the validity and termination statements of these results hold verbatim, including the abort rule of CT and, under weighted growth, the success of a trial with $2^\mu \leq 6\sqrt{\kappa}$; only the operation counts change.

(b) It remains to count cut computations and bit lengths in place of table operations; the count is in Appendix D. $\square$

Corollary 7.25 (Balanced after deleting a supplied core). *Let F and X be as in Theorem 7.24, except that F need not be balanced. Suppose a set $\mathcal{K}$ of k coordinates is supplied such that $o_i o_j H_{ij} \leq 0$ for some signs o and all $i \neq j$ outside $\mathcal{K}$. Then Theorem 7.24 holds for F , with the work bounds of part (b) multiplied by K^k , where $K = O(\sqrt{\kappa} \log n)$ bounds the nodes per coordinate in every trial that CT runs.*

Proof. Fixing the core coordinates at grid nodes changes only unary and constant terms of the remaining quadratic, which is balanced. Hence β , a minimizer and each min-marginal are minima, over at most K^k vectors of core nodes, of problems solved by Lemma 7.23; a min-marginal of a core coordinate fixes that coordinate. Every trial that CT runs has at most $K_\mu \leq K_{\mu^*}$ nodes per coordinate, where $2^{\mu^*} \leq 6\sqrt{\kappa} \leq 6\sqrt{\kappa}$ (proof of Theorem 5.7(b) and Appendix D), so $K = K_{\mu^*} = O(\sqrt{\kappa} \log n)$. The rest of the proof of Theorem 7.24 is unchanged. $\square$

Corollary 7.25 complements Section 7.4: residual coordinates may now have positive diagonal, but they are gridded and filtered like the core, so their curvature enters κ , and point growth of F is needed instead of growth of V in the core coordinates; the residual graph may still be dense. Section 2 compares Theorem 7.24 with methods for continuous submodular minimization.

8 Coupling constraints

The corrected-grid bound of Proposition 4.2 follows from rounding each coordinate independently to an endpoint of its grid interval, and every rounded point is feasible because the domain is a product. A coupling constraint, even one as sparse as $x_1 = x_2$, destroys this: independent rounding leaves the feasible set, and the minimum over feasible grid points need not be a lower bound. This section shows that totally unimodular (TU) coupling of the continuous variables, with data aligned to a uniform mesh, admits a *feasible* mean-preserving rounding. Filtering, certificates, accuracy-independent node counts and exact output then carry over. The price is visible in two places. The rounding is correlated, so a bound on the full continuous Hessian replaces coordinate curvature; Example 8.5 shows that this is necessary. The analysis uses a uniform aligned mesh, so the number of nodes per coordinate is of order $\sqrt{n_c \kappa_c}$ instead of logarithmic in n ; Section 8.7 shows that the algorithm incurs this cost, and that two natural nonuniform alternatives, misaligned coordinate grids and a common graded pattern, give invalid bounds with curvature-only corrections. Section 8.5 states when the algorithm is polynomial and why it is not fixed-parameter tractable. Its levels resemble the scaling phases of Hochbaum and Shanthikumar [96] for separable convex objectives; Section 2 compares the two.

8.1 Model

Definition 8.1 (Aligned TU model). Let $n_c \geq 1$ and $n_d \geq 0$. The feasible set is

$$X = \{(x, z) \in \mathbb{R}^{n_c} \times \mathbb{Z}^{n_d} : \ell \leq x \leq u, z \in \mathcal{Z}, Ax + Bz \leq b\}, \quad \mathcal{Z} = Z_1 \times \cdots \times Z_{n_d},$$

with the following data.

- (i) $A \in \{0, \pm 1\}^{m \times n_c}$ is totally unimodular; B and b are rational.
- (ii) $\ell < u$ componentwise (fixed continuous coordinates have been substituted), and every Z_k is a nonempty, explicitly listed finite set of integers.
- (iii) A rational *mesh unit* $\eta > 0$ is given such that

$$\ell/\eta \in \mathbb{Z}^{n_c}, \quad u/\eta \in \mathbb{Z}^{n_c}, \quad (b - Bz)/\eta \in \mathbb{Z}^m \quad \text{for every } z \in \mathcal{Z}. \quad (8.1)$$

For integral ℓ, u, B, b one may take $\eta = 1$. Write Δ_η for the denominator of η in lowest terms.

- (iv) $F(x, z) = \sum_{a \in \mathcal{A}} f_a(v_{S_a})$ with $v = (x, z)$, and a tree decomposition with N nodes and bags of size at most p is given such that every scope S_a and every row support $\{i : A_{ri} \neq 0\} \cup \{k : B_{rk} \neq 0\}$ is contained in some bag.
- (v) (Directional curvature.) A linear subspace $\mathcal{W} \subseteq \mathbb{R}^{n_c}$ and a rational $\bar{L} > 0$ are given such that, for every $z \in \mathcal{Z}$, $F(\cdot, z)$ is twice continuously differentiable on an open set containing $[\ell, u]$ and

$$w^\top \nabla_{xx}^2 F(x, z) w \leq \bar{L} \|w\|^2 \quad (x \in [\ell, u], w \in \mathcal{W}). \quad (8.2)$$

Either $\mathcal{W} = \mathbb{R}^{n_c}$, or $\mathcal{W} = \ker C$, where the rows of C are the continuous parts A_r of some *equality pairs*, that is, pairs of rows r, r' with $(A_{r'}, B_{r'}, b_{r'}) = -(A_r, B_r, b_r)$.

Equalities are thus written as two opposite inequalities; this preserves total unimodularity. Only the continuous columns A must be TU: the integer columns B are arbitrary, because the rounding below never changes z . Throughout we write $n = n_c + n_d$,

$$s = \max_{i \leq n_c} (u_i - \ell_i), \quad K_Z = \max \left\{ 1, \max_k |Z_k| \right\}, \quad h_j = \eta 2^{-j}, \quad E_j = \frac{n_c \bar{L} h_j^2}{8},$$

$\mathcal{S} = \arg \min_X F$, $F^* = \min_X F$, and $\mathcal{S}_i = \{v_i : v \in \mathcal{S}\}$ for the projection of $\mathcal{S}$ on coordinate i (continuous or discrete). Since the sets Z_k are listed, $K_Z \leq I$. As in CT with a common mesh (Lemma 5.9), h_j is the same for all continuous coordinates; its base is the mesh unit η instead of s , so that every grid is aligned with the data. When a growth constant is discussed, it is a set-growth constant g_S as in Definition 3.2, for the Euclidean distance on $\mathbb{R}^{n_c+n_d}$:

$$F(v) - F^* \geq g_S \text{dist}(v, \mathcal{S})^2 \quad (v \in X), \quad \kappa_c = \kappa(\bar{L}, g_S) = \max\{1, \bar{L}/g_S\}. \quad (8.3)$$

Point growth at a unique minimizer is the case $\mathcal{S} = \{v^*\}$. Neither g_S nor $\mathcal{S}$ is used by any algorithm below.

Conditions (8.1) and (8.2) can be certified in time polynomial in I (Appendix E.1): alignment by $1 + \sum_k |Z_k|$ vector tests, without enumerating $\mathcal{Z}$ (Proposition E.2), and directional curvature, for quadratic and fixed-degree polynomial factors, by rational sufficient tests such as a bound on the absolute row sums of the continuous Hessian (Remark E.1).

8.2 Feasible rounding on aligned cells

Definition 8.2 (Level- j domains). A *level- j domain* is a product $\mathcal{D} = \mathcal{D}_1 \times \cdots \times \mathcal{D}_{n_c} \times \Lambda_1 \times \cdots \times \Lambda_{n_d}$, where each $\mathcal{D}_i \subseteq [\ell_i, u_i]$ is a nonempty union of finitely many closed intervals $[a, a']$, $a \leq a'$, with $a, a' \in h_j \mathbb{Z}$, and each $\Lambda_k \subseteq Z_k$ is nonempty. Its grid in coordinate i is $G_j(\mathcal{D}_i) = \mathcal{D}_i \cap h_j \mathbb{Z}$, its *cells* are the intervals $[t, t + h_j] \subseteq \mathcal{D}_i$ with $t \in h_j \mathbb{Z}$, and its grid is $G_j(\mathcal{D}) = \prod_i G_j(\mathcal{D}_i) \times \prod_k \Lambda_k$.

Since $\ell, u \in \eta \mathbb{Z}^{n_c} \subseteq h_j \mathbb{Z}^{n_c}$, the box $[\ell, u] \times \mathcal{Z}$ is a level-0 domain, and a level- j domain is also a level- $(j+1)$ domain. Components of $\mathcal{D}_i$ may be single points.

Lemma 8.3 (Feasible corner rounding). *Let $\mathcal{D}$ be a level- j domain and $(x, z) \in X \cap \mathcal{D}$. There is a random vector $Y \in \mathbb{R}^{n_c}$ with finitely many values such that, almost surely, $(Y, z) \in X \cap G_j(\mathcal{D})$ and $Y - x \in \mathcal{W}$, and*

- (a) $\mathbb{E}Y = x$;
- (b) if $x_i \in h_j \mathbb{Z}$ then $Y_i = x_i$; otherwise, with $t_i = h_j \lfloor x_i/h_j \rfloor$, the cell $[t_i, t_i + h_j]$ lies in $\mathcal{D}_i$ and $Y_i \in \{t_i, t_i + h_j\}$;
- (c) $\mathbb{E}\|Y - x\|^2 \leq n_c h_j^2/4$.

Proof. Write $h = h_j$ and $t_i = h \lfloor x_i/h \rfloor$, so $x = t + h\vartheta$ with $\vartheta \in [0, 1)^{n_c}$. Let $\mathcal{I}_0 = \{i : \vartheta_i = 0\} = \{i : x_i \in h\mathbb{Z}\}$. If $i \notin \mathcal{I}_0$, then x_i lies in a component $[a, a']$ of $\mathcal{D}_i$ with $a, a' \in h\mathbb{Z}$ and $a < x_i < a'$, so $a \leq t_i$ and $t_i + h \leq a'$; the cell $[t_i, t_i + h]$ lies in $\mathcal{D}_i$.

Consider the polytope

$$\mathcal{Q} = \left\{ \vartheta' \in \mathbb{R}^{n_c} : A\vartheta' \leq \frac{b - Bz - At}{h}, 0 \leq \vartheta' \leq \mathbf{1}, \vartheta'_i \leq 0 \ (i \in \mathcal{I}_0) \right\}.$$

The right-hand side $(b - Bz - At)/h$ is integral: $(b - Bz)/h = 2^j(b - Bz)/\eta \in \mathbb{Z}^m$ by (8.1), and $At/h \in \mathbb{Z}^m$ because A is integral and $t/h \in \mathbb{Z}^{n_c}$. Appending unit rows to a TU matrix gives a TU matrix [165, Section 19.1], so by the Hoffman–Kruskal theorem [97] every vertex of $\mathcal{Q}$ is integral, hence lies in $\{0, 1\}^{n_c}$ and vanishes on $\mathcal{I}_0$. Since $(x, z) \in X$, $\vartheta \in \mathcal{Q}$, and the polytope $\mathcal{Q}$ is the convex hull of its vertices, so $\vartheta = \sum_k \lambda_k \vartheta^k$ with vertices ϑ^k and convex weights λ_k . Let $Y = t + h\vartheta^k$ with probability λ_k .

Then $\mathbb{E}Y = t + h\vartheta = x$. For each outcome, $AY + Bz = At + hA\vartheta^k + Bz \leq b$. For $i \in \mathcal{I}_0$, $Y_i = t_i = x_i$; for $i \notin \mathcal{I}_0$, $Y_i \in \{t_i, t_i + h\}$, which lies in the cell $[t_i, t_i + h] \subseteq \mathcal{D}_i \subseteq [\ell_i, u_i]$. Hence

$(Y, z) \in X \cap G_j(\mathcal{D})$ and (b) holds. If $\mathcal{W} = \ker C$, each row A_r of C belongs to an equality pair, so $A_r Y + B_r z = b_r = A_r x + B_r z$ and $A_r(Y - x) = 0$. Finally, for $i \notin \mathcal{I}_0$ the variable Y_i takes two values with mean x_i , so $\mathbb{E}(Y_i - x_i)^2 = (x_i - t_i)(t_i + h - x_i) \leq h^2/4$, and coordinates in $\mathcal{I}_0$ contribute zero. This proves (c). $\square$

The rounded coordinates are dependent: they follow a vertex decomposition of the polytope $\mathcal{Q}$, not a product law. This is why the next lemma needs (8.2) rather than coordinate curvature.

Lemma 8.4 (Rounding allowance). *In the situation of Lemma 8.3,*

$$\mathbb{E}F(Y, z) \leq F(x, z) + \frac{\bar{L}}{2} \mathbb{E}\|Y - x\|^2 \leq F(x, z) + E_j.$$

Proof. Fix an outcome and put $w = Y - x \in \mathcal{W}$. The segment from x to Y lies in the box $[\ell, u]$, and $\phi(t) = F(x + tw, z)$ satisfies $\phi''(t) = w^\top \nabla_{xx}^2 F(x + tw, z) w \leq \bar{L} \|w\|^2$ by (8.2). Taylor's formula with integral remainder gives $\phi(1) \leq \phi(0) + \phi'(0) + \bar{L} \|w\|^2/2$, that is, $F(Y, z) \leq F(x, z) + \nabla_x F(x, z)^\top (Y - x) + \frac{\bar{L}}{2} \|Y - x\|^2$. Take expectations and use $\mathbb{E}(Y - x) = 0$ and Lemma 8.3(c). $\square$

Example 8.5 (Coordinate curvature is not enough). Let $0 < h \leq 1/2$, $X = \{x \in [0, 1]^2 : x_1 = x_2\}$ and $F(x) = 2x_1x_2 - h(x_1 + x_2)$. Both coordinate curvatures are zero, so the corrections of Proposition 4.2 vanish. On X , $F(t, t) = 2t^2 - 2ht$ has minimum $F^* = -h^2/2$ at $t = h/2$, whereas every feasible point of the mesh- h grid has value $F(kh, kh) = 2h^2k(k-1) \geq 0$. Corrections computed from coordinate curvature would therefore certify the false bound 0. The Hessian has eigenvalues ± 2 , so $\bar{L} = 2$ is valid with $\mathcal{W} = \mathbb{R}^2$, and the allowance $n_c \bar{L} h^2/8 = h^2/2$ gives the correct bound $0 - h^2/2 = F^*$; the allowance is attained.

8.3 Filtering and certificates

The tree computation of Lemma 4.3 applies to F on a grid $G_j(\mathcal{D})$ after each row r is assigned to a bag containing its support, as the table $\iota_r(y) = 0$ if $A_r y_x + B_r y_z \leq b_r$ and $\iota_r(y) = +\infty$ otherwise. The proof of Lemma 4.3 only adds and minimizes tables, so it holds for values in $\mathbb{R} \cup \{+\infty\}$; when outgoing messages are formed by subtracting one incoming term from a sum, one stores the sum of the finite terms and the number of infinite ones. For a level- j domain $\mathcal{D}$ it returns

$$\mu = \min\{F(y) : y \in X \cap G_j(\mathcal{D})\}, \quad m_i(t) = \min\{F(y) : y \in X \cap G_j(\mathcal{D}), y_i = t\} - E_j$$

for every coordinate i and every node t of that coordinate (for the k th discrete coordinate we write m_k), and a minimizing grid point, with $\min \emptyset = +\infty$, using $O(p(|\mathcal{A}| + n + N + m)K^p)$ table operations, where K bounds the number of nodes per coordinate. The correction is the constant E_j , so m_i is the min-marginal of $Q = F - E_j$ over the feasible grid points, in the notation of Definition 4.1, and $\mu - E_j$ is the minimum of Q .

Algorithm 8.6 (TU-GRID: grids under TU coupling). Input: the data of Definition 8.1 and an accuracy $\varepsilon > 0$. Set $\mathcal{D}^{(0)} = [\ell, u] \times \mathcal{Z}$. For $j = 0, 1, 2, \dots$:

- (1) Compute μ_j , a minimizer $y^{(j)} \in X \cap G_j(\mathcal{D}^{(j)})$, and the min-marginals m_i on $G_j(\mathcal{D}^{(j)})$. If $\mu_0 = +\infty$, stop and report $X = \emptyset$.
- (2) Put $U_j = \mu_j$ and $\beta_j = \mu_j - E_j$. If $E_j \leq \varepsilon$, return $y^{(j)}$ and $[\beta_j, U_j]$.
- (3) (Union filter.) For each continuous i , let $\mathcal{D}_i^{(j+1)}$ be the union of all cells $[t, t + h_j]$ of $\mathcal{D}_i^{(j)}$ with $\min\{m_i(t), m_i(t + h_j)\} \leq U_j$ and of all single-point components $\{t\}$ of $\mathcal{D}_i^{(j)}$ with $m_i(t) \leq U_j$. For each k , let $\Lambda_k^{(j+1)} = \{t \in \Lambda_k^{(j)} : m_k(t) \leq U_j\}$.

The *hull variant* replaces each $\mathcal{D}_i^{(j+1)}$ by its convex hull.

Proposition 8.7 (Soundness). *Consider TU-GRID or its hull variant, with or without the stopping test of step (2). For every level j that is reached, (a), (b) and (e) hold, and (c) and (d) hold if step (3) is executed at level j :*

- (a) if $X \neq \emptyset$ then $\mu_0 < \infty$;
- (b) $\beta_j \leq F(v)$ for every $v \in X \cap \mathcal{D}^{(j)}$;
- (c) every $v \in (X \cap \mathcal{D}^{(j)}) \setminus \mathcal{D}^{(j+1)}$ has $F(v) > U_j$;
- (d) $\mathcal{D}^{(j+1)}$ is a level- $(j+1)$ domain that contains $\mathcal{S}$ and $y^{(j)}$; hence $y^{(j)} \in X \cap G_{j+1}(\mathcal{D}^{(j+1)})$ and $U_{j+1} \leq U_j$;
- (e) $\beta_j \leq F^* \leq U_j \leq F^* + E_j$ and $U_j - \beta_j = E_j$.

Moreover, a checker that recomputes every μ_j and m_i from the stored domains and verifies every filtering decision obtains $\beta_J \leq F^*$ for the last level J from (b) and (c) alone, without using any minimizer, uniqueness or growth.

The proof is in Appendix E.2. The certificate record consists of the data, a TU certificate for A (a structural certificate such as a network or interval representation, or a run of a polynomial-time TU recognition algorithm [165, Chapter 20]), the curvature certificate for (8.2) (Remark E.1), η , the domains, the tables, and the points $y^{(j)}$. As in Theorem 4.7, the whole history is needed: the last grid bounds F only on $X \cap \mathcal{D}^{(J)}$.

8.4 Accuracy-independent node counts

Theorem 8.8 (Node counts under set growth). *Assume (8.3) and put $a_j = \sqrt{n_c \bar{L}/(4g_S)} h_j$. Run TU-GRID with or without the stopping test of step (2). For every level j at which step (3) is executed:*

- (a) every point of $\mathcal{D}_i^{(j+1)}$ lies within $a_j + h_j$ of $\mathcal{S}_i$, and every value in $\Lambda_k^{(j+1)}$ lies within a_j of $\mathcal{S}_k$;
- (b) if a continuous projection $\mathcal{S}_i$ is finite, then

$$|G_{j+1}(\mathcal{D}_i^{(j+1)})| \leq |\mathcal{S}_i| \left(5 + \lfloor 2\sqrt{n_c \bar{L}/g_S} \rfloor \right) \leq |\mathcal{S}_i| (5 + \lfloor 2\sqrt{n_c \kappa_c} \rfloor);$$

- (c) in the hull variant the statement on values in (a) holds; if moreover $\mathcal{S} = \{v^*\}$, every point of $\mathcal{D}_i^{(j+1)}$ lies within $a_j + h_j$ of v_i^* , and the bound in (b) holds.

The proof is in Appendix E.2. The bound is per optimal coordinate value: different coordinates may be near different minimizers, and no optimal combination is enumerated; $\mathcal{S}$ can have up to $\prod_i |\mathcal{S}_i| \prod_k |Z_k|$ points. Step (3) filters by unions of cells because one coordinate can have optimal values far apart: on independent copies of a block with two minimizers, the hull variant keeps $2^j + 1$ nodes in one coordinate of every block at level j , whereas the union variant keeps a number of nodes bounded independently of j (Example E.3 in Appendix E.3). The finite-projection hypothesis cannot be dropped for TU-GRID: if $\mathcal{S}_i$ contains an interval $[a, a']$ with $a < a'$, then $[a, a'] \subseteq \mathcal{D}_i^{(j)}$ for every j by Proposition 8.7(d), so the level- j grid has at least $(a' - a)/h_j - 1$ nodes in coordinate i , which grows with the accuracy.

8.5 Complexity of certified approximation

Theorem 8.9 (Certified approximation with TU coupling). *Assume Definition 8.1 with factors that are explicit rational quadratics or explicitly encoded rational polynomials of fixed degree, and let I be the binary input length, including η , $\bar{L}$ and the decomposition. For $\varepsilon > 0$, TU-GRID(ε) either certifies $X = \emptyset$ at level 0, or stops at*

$$J = \min\{j \geq 0 : E_j \leq \varepsilon\} \leq \max\left\{0, \left\lceil \frac{1}{2} \log_2 \frac{n_c \bar{L} \eta^2}{8\varepsilon} \right\rceil \right\}$$

with a feasible rational point and an interval $[\beta_J, U_J] \ni F^$ of width $E_J \leq \varepsilon$, certified as in Proposition 8.7. All numbers have bit length polynomial in $I + J$, and the number of table operations is*

$$O\left(p(|\mathcal{A}| + n + N + m) \sum_{j=0}^J K_j^p\right), \quad K_0 \leq \max\{K_Z, s/\eta + 1\},$$

where K_j is the largest number of nodes per coordinate at level j . Under (8.3) with $|\mathcal{S}_i| \leq r$ for every continuous i , $K_j \leq K = \max\{K_Z, r(5 + \lfloor 2\sqrt{n_c \kappa_c} \rfloor)\}$ for every $j \geq 1$. The algorithm uses neither g_S nor r .

The proof is in Appendix E.2. Without growth the same certificate is produced, but the node counts depend on the accuracy: a continuous coordinate has at most $s2^j/\eta + 1$ nodes at level j , and if $J \geq 1$, minimality of J gives $2^J < \eta\sqrt{n_c \bar{L}/(2\varepsilon)}$. A bag of p continuous coordinates then has fewer than $(s\sqrt{n_c \bar{L}/(2\varepsilon)} + 1)^p$ table entries at the last level, which is polynomial in $1/\varepsilon$, not in $\log(1/\varepsilon)$.

Under set growth with finitely many optimal values per coordinate, TU-GRID runs in time polynomial in $I + \log(1/\varepsilon)$ for fixed p when the condition number κ_c , the box widths measured in mesh units (s/η), and r are polynomially bounded; recall that $K_Z \leq I$. It is not fixed-parameter tractable: its running time is not bounded by $f(p, \kappa_c) \text{poly}(I + \log(1/\varepsilon))$, because K^p contains the factor $n_c^{p/2}$ and Section 8.7 shows that this factor is incurred. The level-0 grid has up to $s/\eta + 1$ nodes per coordinate, which is pseudopolynomial in the data. A larger common unit η , checked by Proposition E.2, makes this smaller, but unless $P = NP$ the dependence on s/η cannot be removed. Indeed, Proposition 10.10 (Section 10.6) gives instances of Definition 8.1 with $p = 3$, a unique minimizer and $K_Z = 2$ on which approximating F^* within $\frac{1}{2}$ is NP-hard. Their objective is affine, so every $\bar{L} > 0$ is valid, and $\bar{L} \leq g_S$ gives $\kappa_c = 1$; of the quantities above, only s/η is not polynomially bounded on them.

8.6 Exact output for quadratic objectives

In this subsection $F(v) = \frac{1}{2}v^\top H v + \nabla F(0)^\top v + F(0)$ is a quadratic with rational data on the model of Definition 8.1. Let Δ be a positive common denominator of its monomial coefficients $H_{ii}/2$, H_{ik} ($i < k$), the entries of $\nabla F(0)$, and $F(0)$, and put $\hat{H} = \Delta H$, which is integral and symmetric, with continuous block $\hat{H}_{xx}$ and mixed block $\hat{H}_{xz}$. For $z \in \mathcal{Z}$ the continuous part of X is the polytope

$$\mathcal{P}_z = \{x \in \mathbb{R}^{n_c} : \hat{A}x \leq \hat{b}_z\}, \quad \hat{A} = \begin{pmatrix} A \\ I \\ -I \end{pmatrix} \in \{0, \pm 1\}^{m' \times n_c}, \quad \hat{b}_z = \begin{pmatrix} b - Bz \\ u \\ -\ell \end{pmatrix} \in \Delta_\eta^{-1} \mathbb{Z}^{m'},$$

with $m' = m + 2n_c$, so that $\{x : (x, z) \in X\} = \mathcal{P}_z$. Put $c_z = \hat{H}_{xz}z + \Delta \nabla_x F(0) \in \mathbb{Z}^{n_c}$, so that $\Delta \nabla_x F(x, z) = \hat{H}_{xx}x + c_z$. The *TU height constants* are

$$\begin{aligned} C_{\hat{H}} &= \max \left\{ 1, \max_{i,k \leq n_c} |(\hat{H}_{xx})_{ik}| \right\}, & R_{\text{TU}} &= (2n_c C_{\hat{H}})^{n_c}, \\ \Omega_{\text{TU}} &= \Delta (\Delta_\eta R_{\text{TU}})^2, & \tau_{\text{TU}} &= \frac{1}{4m' \Delta_\eta R_{\text{TU}}}. \end{aligned} \tag{8.4}$$

They are computed from the original data and have bit length polynomial in I . They take the place of the constants R , Ω and τ of Section 6, where all rows are bound rows and the stationarity systems are principal submatrices of the Hessian; with TU rows a Hadamard bound on the rows of a saddle-point matrix replaces the diagonal bound. Appendix E.4 proves the analogues of Lemma 6.3 and Corollary 6.4: some global minimizer has continuous coordinates in $\rho^{-1}\mathbb{Z}$ for an integer $1 \leq \rho \leq \Delta_\eta R_{\text{TU}}$, every isolated global minimizer has this property, and $F^* = a/W$ in lowest terms with $1 \leq W \leq \Omega_{\text{TU}}$ (Corollary E.6). Proposition 6.6 therefore applies with $\Omega' = \Omega_{\text{TU}}$: if $\beta \leq F^*$, $v \in X$ and $F(v) = a/W$ in lowest terms with $F(v) - \beta < 1/(\Omega_{\text{TU}}W)$, then v is a global minimizer.

TU-EXACT (Algorithm E.7 in Appendix E.5) runs TU-GRID without the stopping test of step (2) and snaps rows instead of coordinates. At every level j it makes active the rows of $\hat{A}$ with slack at most τ_{TU} at the grid minimizer $y^{(j)}$, imposes stationarity along the remaining face by linear equations, finds a point of the resulting polytope by exact linear programming, and returns that point if it passes the acceptance test above with $\beta = \beta_j$.

Theorem 8.10 (Exact output with TU coupling). *Let F be a rational quadratic on a nonempty set X as in Definition 8.1.*

- (a) *Every point returned by TU-EXACT is a global minimizer, and its value is F^* . This uses neither uniqueness nor growth.*
- (b) *TU-EXACT terminates.*
- (c) *Under (8.3) it terminates by the first level j with $E_j \leq \min\{g_S \tau_{\text{TU}}^2/(4n_c), 1/(2\Omega_{\text{TU}}^2)\}$, hence at the latest at level*

$$J_{\text{ex}} = \max \left\{ 0, \left\lceil \log_2 \frac{\eta n_c \sqrt{\kappa_c}}{\tau_{\text{TU}}} \right\rceil, \left\lceil \log_2 (\eta \Omega_{\text{TU}} \sqrt{n_c \bar{L}}) \right\rceil \right\} = \text{poly}(I) + O(\log \kappa_c).$$

If moreover every continuous projection $\mathcal{S}_i$ has at most r points, the table operations are as in Theorem 8.9 with J replaced by J_{ex} , and recovery and the acceptance test add a cost polynomial in $I + j$ at level j .

The proof (Appendix E.6) follows the proofs of Theorems 6.9 and 6.14, with a snapping lemma for rows of $\hat{A}$ (Lemma E.8) in place of Lemma 6.8.

8.7 The uniform mesh: cost and nonuniform alternatives

The running time of CT (Algorithm 5.6) is bounded by $f(p, \bar{\kappa}) \text{poly}(I + q)$ because its graded grids cover the localization radius with $O(\sqrt{\bar{\kappa}} \log(n + 2))$ nodes per coordinate (Section 5). With TU coupling the radius is $a_j + h_j$, of order $\sqrt{n_c \kappa_c} h_j$ (Theorem 8.8), but the mesh is uniform. The factor $n_c^{p/2}$ in K^p is incurred: Corollary 5.11 shows that filtered uniform grids keep at least $\sqrt{(n - 2)\kappa} + 1$ nodes per coordinate on the box instance of Proposition 5.10, and its proof applies to TU-GRID,

whose allowance E_j is at least the sum of the coordinate corrections of a uniform grid. With a redundant TU row that forces a bag of size p , TU-GRID(ε) then builds a table with at least $n_c^{p/2}$ entries for every $\varepsilon \leq \bar{L}/64$, while $\kappa_c = 2$ (Lemma E.10 in Appendix E.7).

Two natural nonuniform alternatives, misaligned coordinate grids and a common graded pattern, give invalid bounds with curvature-only corrections (Proposition E.11 and Example E.12 in Appendix E.7). With misaligned grids, the multiplier of an order constraint $x_1 \leq x_2$ is invisible to the curvature and to the growth constant; with a common graded pattern and a sum row, a minimizer need not be a convex combination of feasible grid points of its cell.

Remark 8.11 (Comparison with Bienstock and Muñoz). Bienstock and Muñoz [24, Theorems 4 and 15] construct, for mixed-binary polynomial problems with variables scaled to $[0, 1]$, a linear objective c , constraints of degree at most π and coefficient 1-norm at most $\mathfrak{F}$, a constraint intersection graph of treewidth ω and a tolerance ϵ (we keep their symbol, distinct from our accuracy ε), a linear program with $O((2\pi/\epsilon)^{\omega+1} n \log(\pi/\epsilon))$ variables and constraints whose solutions violate each constraint by at most $\mathfrak{F}\epsilon$ and are $\|c\|_1\epsilon$ -optimal. They also argue that, unless $P = NP$, neither the pseudopolynomial dependence on $1/\epsilon$ nor the coefficient scaling of the violation can be removed for that class. The results here differ in four respects. (i) Feasibility is exact, but only for TU continuous columns with aligned data; their constraints are arbitrary polynomials. (ii) A nonlinear objective is handled directly through $\bar{L}$, whereas their objective is linear and nonlinear objectives are encoded by additional constraints, which may increase the width. (iii) Under growth with finitely many optimal coordinate values, the per-level tables of TU-GRID do not depend on its accuracy ε , and ε enters only through $O(\log(1/\varepsilon))$ levels; this does not conflict with their lower bounds, which concern their whole class and hence instances without growth or TU structure, and the factor $(n_c\kappa_c)^{p/2}$ can itself be exponential in I because κ_c can be. (iv) Without growth, if $J \geq 1$, a last-level table of TU-GRID for a bag of p continuous coordinates has fewer than $(s\sqrt{n_c\bar{L}/(2\varepsilon)} + 1)^p$ entries (Section 8.5), an exponent $p/2$ in $1/\varepsilon$ against their $\omega + 1$ in $1/\epsilon$ for comparable decompositions; the halving comes from second-order rounding error with exact feasibility, and the comparison is only indicative because the tolerances are normalized differently. Neither result contains the other.

9 Several minimizers

The growth analysis of Section 5 localizes every retained coordinate near one minimizer. When the optimal set $\mathcal{S}$ is not a single point, this localization fails, although Theorem 6.9 shows that exactness itself costs only $\text{poly}(I) + \max\{0, \log_2(1/g_S)\}$ bits of accuracy under set growth with constant g_S . This section treats three situations. With finitely many minimizers, unions of uniform cells give grids with $O(r\sqrt{n\kappa_S})$ nodes per coordinate, where r is the largest number of optimal values of one coordinate (Section 9.1). For coordinatewise concave quadratics, the decomposition describes the entire optimal set (Section 9.2). For continuous box quadratics certified by a diagonal Lagrangian multiplier, an exact minimizer and an exact description of $\mathcal{S}$ are found in time $f'(p, \kappa_S) \text{poly}(I)$ for a computable function f' (Section 9.3). Throughout, $F = \frac{1}{2}x^\top Hx + b^\top x + c$ is a rational quadratic on a mixed box, and set growth, g_S and $\kappa_S = \kappa(L, g_S)$ are as in Definition 3.2. Some $g_S > 0$ always exists (Remark 6.10), but its size is not controlled, and no algorithm below computes it.

9.1 One center cannot serve several minimizers

Theorem 6.9 reduces exact output to reaching accuracy ε_S . CT (Algorithm 5.6) does not reach it at a rate controlled by p and κ_S when the minimizer is not unique, even with two isolated minimizers:

its grids are graded around one center, and coarse cells near a distant minimizer keep the corrected minimum low.

Proposition 9.1 (One center cannot certify two distant minimizers). *For $M > 0$ let*

$$F_M(x, z) = x^2 - 2xz + Mz \quad \text{on } X = [0, M]^2 \text{ (both coordinates continuous).}$$

Then $\mathcal{S} = \{(0, 0), (M, M)\}$, $F^ = 0$, $L_x = 2$, $L_z = 0$, and $F_M(v) \geq \frac{1}{20} \text{dist}(v, \mathcal{S})^2$ on X , so $\kappa_S \leq 40$; one bag $\{x, z\}$ covers both scopes ($p = 2$). Let $\theta \in (0, 1/4]$, $h \in (0, M]$ and $c \in X$. Let G_x be the graded grid (Definition 5.1) of $[0, M]$ with center c_x , mesh h and grading θ , let $G_z \subseteq [0, M]$ be a finite set containing 0 and M , and use the corrections $d_x = L_x w_x^2/8$ with $L_x \geq 2$ and any $d_z \geq 0$. Then*

$$\beta = \min_{G_x \times G_z} Q \leq -\max\left\{\frac{\theta^2 M^2}{379}, \frac{h^2}{20}\right\}. \quad (9.1)$$

Consequently:

- (a) *Every stage of TRIAL (Algorithm 5.3), and hence every stage of CT, satisfies (9.1) with its grading θ and its mesh $h = h_{xj}$ of coordinate x . Since $U \geq F^* = 0$, its certified gap $U - \beta$ is at least $\max\{\theta^2 M^2/379, h^2/20\}$. The same holds for CT with a common mesh (Lemma 5.9).*
- (b) *If such a stage has $\beta \geq -\varepsilon$, in particular if it certifies a gap $U - \beta \leq \varepsilon$, then its x -grid has more than $M/(48\sqrt{\varepsilon})$ nodes and its table for the bag $\{x, z\}$ has more than $M/(24\sqrt{\varepsilon})$ entries.*
- (c) *Any procedure that stops only after such a stage with $\beta > -1$, such as EX (Algorithm 6.13), builds a table with more than $M/24$ entries. For integer M the input length is $I = O(\log M)$, so this work is $2^{\Omega(I)}$, although $\kappa_S \leq 40$ and $p = 2$.*

The proof is in Appendix F. Other methods solve this instance easily; the proposition shows that the failure comes from the single center, not from conditioning. Replacing hulls by unions of retained cells, on uniform meshes, removes it, at the price of $O(r\sqrt{n\kappa_S})$ instead of $O(\sqrt{\kappa} \log(n+2))$ nodes per coordinate.

Algorithm 9.2 (UC: uniform cells). The input is an accuracy $\varepsilon > 0$. For each coordinate keep a finite set $\mathcal{C}_i$ of *retained cells*, closed intervals of positive length; initially $\mathcal{C}_i = \{[\ell_i, u_i]\}$. Let J be the least $j \geq 0$ with $\frac{1}{2}nLs^24^{-j} \leq \varepsilon$, start with the incumbent $\hat{x} = \ell$ and $U = F(\ell)$, and for $j = 0, 1, \dots, J$ with $h_j = s2^{-j}$:

- (i) *Refine.* Let $\mathcal{G}_{ij} = \{\ell_i + kh_j : k \in \mathbb{Z}_{\geq 0}\}$ for $i \in I_C$ and $\mathcal{G}_{ij} = \{\ell_i + \lceil kh_j \rceil : k \in \mathbb{Z}_{\geq 0}\}$ for $i \in I_Z$. Split every retained cell $[a, a']$ at the points of $\mathcal{G}_{ij} \cap (a, a')$; the resulting intervals are the *stage cells*, and G_i is the set of their endpoints. Let $w_i(v)$ be the largest effective width of a stage cell with endpoint v , and define d_i , D , Q , β_j and m_i by (4.1)–(4.2) on $G = \prod_i G_i$.
- (ii) *Solve.* Compute β_j , a minimizer $y^{(j)}$ of Q over G and all m_i by Lemma 4.3; if $F(y^{(j)}) < U$, set $\hat{x} = y^{(j)}$ and $U = F(y^{(j)})$.

(iii) *Filter.* Let $\mathcal{C}_i$ be the set of stage cells $[a, a']$ with $\min\{m_i(a), m_i(a')\} \leq U$. Do not take hulls. Return $\hat{x}$, U and β_J . The *domain* of stage j is $X_j = X \cap \prod_i \mathcal{C}_i$, with $\mathcal{C}_i$ as at the start of stage j .

Since $\mathcal{G}_{i,j-1} \subseteq \mathcal{G}_{ij}$ (for integers, $\lceil kh_{j-1} \rceil = \lceil 2kh_j \rceil$), every stage cell is either a segment between consecutive points of $\mathcal{G}_{ij} \cap [\ell_i, u_i]$ or such a segment clipped at u_i . Hence a continuous stage cell has length at most h_j ; an integer stage cell has length at most $\lceil h_j \rceil$, and if that length is at least two then $h_j > 1$ and $\lceil h_j \rceil < 2h_j$. Consecutive points of $\mathcal{G}_{ij}$ are at least h_j apart for $i \in I_C$ and at least $\max\{1, \lceil h_j \rceil\} \geq h_j/2$ apart for $i \in I_Z$. Each stage cell contains at most one point of $\mathcal{G}_{i,j+1}$ in its interior.

Lemma 9.3 (Uniform cells: bounds and safe filtering). *Assume only Definition 3.1. At every stage j of UC, $\beta_j \leq \min_{X_j} F$, and $F(x) \geq \min\{m_i(a), m_i(a')\}$ for every $x \in X_j$ and every stage cell $[a, a']$ of a coordinate i with $x_i \in [a, a']$. Every global minimizer lies in every X_j , and $\beta_J \leq F^* \leq U$.*

Proof. The stage cells of coordinate i form a finite family of intervals with endpoints in G_i , every node is an endpoint of one of them, w_i is defined from this family, and X_j is the intersection of X with the product of their unions. The first two claims are therefore Proposition 4.2(b) and (c) in the form for families of intervals (Section 4). If step (iii) of stage j removes a point $x \in X_j$, then for some i every stage cell containing x_i is discarded, so $F(x) > U$ for the threshold U used at stage j . As $U \geq F^*$, global minimizers are never removed; by induction every X_j contains $\mathcal{S}$, and $\beta_J \leq \min_{X_J} F = F^* \leq U$. $\square$

Theorem 9.4 (Uniform cells). *Assume Definition 3.1. For every $\varepsilon > 0$, $UC(\varepsilon)$ returns a feasible $\hat{x}$ with $\beta_J \leq F^* \leq F(\hat{x}) \leq \beta_J + \varepsilon$, together with a certificate (all stage tables and filtering decisions) whose validity uses only Definition 3.1. Suppose moreover that $\mathcal{S}$ is finite, $r = \max_i |\mathcal{S}_i|$, and F has set growth with constant g_S . Then every grid of every stage has at most*

$$K_S = 12r(2\sqrt{n\kappa_S} + 1) \quad (9.2)$$

nodes per coordinate. For an explicit rational quadratic, $UC(2^{-q})$ uses $O(p(|\mathcal{A}| + n + N)K_S^p) \text{poly}(I + q)$ bit operations, and EX with UC in place of CT returns an exact minimizer in $O(p(|\mathcal{A}| + n + N)K_S^p) \text{poly}(I + \log \kappa_S)$ bit operations. None of g_S , $\mathcal{S}$ and r is an input.

The proof is in Appendix F.2.

Remark 9.5 (What grading buys). With $r = 1$ the minimizer is unique and $\kappa_S = \kappa$, and Theorem 9.4 gives accuracy-independent grids with $O(\sqrt{n\kappa})$ nodes per coordinate without gradings or caps. The graded single-center grids of Section 5 need only $O(\sqrt{\kappa} \log(n+2))$ nodes per coordinate (Lemma 5.5), which gives the $f(p, \bar{\kappa}) \text{poly}(I + q)$ bound of Theorem 5.7; Proposition 9.1 shows that they lose this bound when the minimizer is not unique. Whether exact output admits an $f'(p, \kappa_S, r) \text{poly}(I)$ bound when $\mathcal{S}$ is finite is open. When $\mathcal{S}$ is a continuum, Proposition 10.8 shows that runs of filtering with thresholds $U_j \geq F^*$, in particular TRIAL and CT, can need $1 + 1/(2\sqrt{\varepsilon})$ nodes per coordinate to certify accuracy ε ; whether other certificates avoid this is open.

9.2 Coordinatewise concave quadratics: every minimizer at once

Write $E_i = \{\ell_i, u_i\}$ and $E(B) = \prod_{i \in B} E_i$. For $x \in \bar{X}$ let Y be the random endpoint vector with independent coordinates, $\Pr(Y_i = \ell_i) = (u_i - x_i)/s_i$ and $\Pr(Y_i = u_i) = (x_i - \ell_i)/s_i$, so that $\mathbb{E}Y_i = x_i$. For a bag t and $v \in E(B_t)$ put $\pi_t(v; x) = \Pr(Y_{B_t} = v) = \prod_{i \in B_t} \Pr(Y_i = v_i)$, a product of at most p affine functions of x . Root T at t_0 , let $\text{ch}(t)$ be the children of t , and write $B_t^\uparrow = B_t \cap B_{\text{pa}(t)}$, with $B_{t_0}^\uparrow = \emptyset$. Let q_t be the sum of the factors assigned to bag t .

Lemma 9.6 (Endpoint Bellman residual identity). *Assume $H_{ii} \leq 0$ for every i . Compute, in post-order and with all arguments ranging over endpoint labels,*

$$\begin{aligned} M_t(v_{B_t^\uparrow}) &= \min_{v_{B_t \setminus B_t^\uparrow}} \left[q_t(v_{B_t}) + \sum_{t' \in \text{ch}(t)} M_{t'}(v_{B_{t'}^\uparrow}) \right], \\ r_t(v_{B_t}) &= q_t(v_{B_t}) + \sum_{t' \in \text{ch}(t)} M_{t'}(v_{B_{t'}^\uparrow}) - M_t(v_{B_t^\uparrow}) \geq 0. \end{aligned}$$

Then the scalar M_{t_0} equals F^* , it is attained at an endpoint assignment, and for every $x \in \bar{X}$

$$F(x) - F^* = \sum_t \sum_{v \in E(B_t)} r_t(v) \pi_t(v; x) + \frac{1}{2} \sum_{i=1}^n (-H_{ii})(x_i - \ell_i)(u_i - x_i). \quad (9.3)$$

All tables and residuals are computed in $O(p2^p(N + |\mathcal{A}|))$ table operations on rationals of bit length polynomial in I .

The proof is in Appendix F. It uses only the nonnegativity of the residuals and one endpoint assignment with zero residuals, so a checker needs the tables M_t and that assignment, not the minimization that produced them.

Theorem 9.7 (All minimizers of a coordinatewise concave quadratic). *Under the hypothesis of Lemma 9.6, a point $x \in X$ is a global minimizer if and only if*

- (a) $x_i \in \{\ell_i, u_i\}$ for every i with $H_{ii} < 0$, and
- (b) $\pi_t(v; x) = 0$ for every bag t and every $v \in E(B_t)$ with $r_t(v) > 0$.

These are at most $n + N2^p$ factored polynomial equations of degree at most p , together with the original bounds and integrality. They are computed and checked in $2^{O(p)} \text{poly}(I)$ bit operations, without a growth assumption and without enumerating interior integer labels or optimal components.

Proof. By (9.3), x is optimal exactly when every nonnegative term vanishes. The diagonal terms vanish if and only if (a) holds, and the residual terms $r_t(v)\pi_t(v; x)$ vanish if and only if (b) holds. $\square$

Condition (b) is a support condition, not a condition on coordinate projections. For $F = x + y - 2xy = x(1 - y) + y(1 - x)$ on $[0, 1]^2$, both optimal corners $(0, 0)$ and $(1, 1)$ use every endpoint of every coordinate, yet the two positive residuals give $x(1 - y) = 0$ and $y(1 - x) = 0$, which exclude every other point. A sum of m disjoint copies has 2^m isolated minimizers and a description with $2m$ equations.

The equations have a finite combinatorial form. For $x \in \bar{X}$ define its *face pattern* $\chi(x) \in \{\mathsf{L}, \mathsf{U}, *\}^n$ by $\chi_i(x) = \mathsf{L}$ if $x_i = \ell_i$, $\chi_i(x) = \mathsf{U}$ if $x_i = u_i$, and $\chi_i(x) = *$ otherwise. For a pattern χ on $B \subseteq [n]$ let $\text{cube}(\chi) \subseteq E(B)$ consist of the v with $v_i = \ell_i$ where $\chi_i = \mathsf{L}$ and $v_i = u_i$ where $\chi_i = \mathsf{U}$. For a pattern χ on $[n]$, the closed face $\text{face}(\chi) = \{x \in \bar{X} : x_i = \ell_i \text{ if } \chi_i = \mathsf{L}, x_i = u_i \text{ if } \chi_i = \mathsf{U}\}$ of $\bar{X}$ has the vertex set $\text{cube}(\chi)$. Then $\pi_t(v; x) > 0$ if and only if $v \in \text{cube}(\chi(x)_{B_t})$. Let $\Sigma_i = \{\mathsf{L}, \mathsf{U}\}$, with $*$ added when $H_{ii} = 0$ and X_i has a point strictly between ℓ_i and u_i . Call $\chi \in \prod_i \Sigma_i$ *admissible* if, for every bag, $\text{cube}(\chi_{B_t}) \subseteq Z_t := \{v \in E(B_t) : r_t(v) = 0\}$.

Corollary 9.8 (Optimal faces form a tree-structured constraint problem). *Under the hypothesis of Lemma 9.6:*

- (i) $x \in X$ is optimal if and only if $\chi(x) \in \prod_i \Sigma_i$ and $\chi(x)$ is admissible.
- (ii) Replacing any $*$ of an admissible pattern by L or U keeps it admissible, and $\mathcal{S} = \bigcup_{\chi \text{ admissible}} (X \cap \text{face}(\chi))$. Distinct admissible patterns give disjoint nonempty sets $\{x \in X : \chi(x) = \chi\}$.
- (iii) Admissible patterns are the solutions of a constraint problem on the same tree decomposition, with at most three labels per variable and bag relations $\text{Adm}_t = \{\chi \in \prod_{i \in B_t} \Sigma_i : \text{cube}(\chi) \subseteq Z_t\}$. Consequently, in $2^{O(p)}(N + n) \text{poly}(I)$ bit operations one can decide whether the minimizer is unique, compute the dimension of $\mathcal{S}$, count the admissible patterns, count $|\mathcal{S}|$ when $\mathcal{S}$ is finite, compute $\text{dist}(y, \mathcal{S})^2$ and a nearest minimizer for a rational y , and minimize a linear function over $\mathcal{S}$.

The proof is in Appendix F.

Remark 9.9 (Two-label integer coordinates and multilinear objectives). If $X_i = \{\ell_i, \ell_i + 1\}$, replacing $\frac{1}{2}H_{ii}x_i^2$ by $\frac{1}{2}H_{ii}((2\ell_i + 1)x_i - \ell_i(\ell_i + 1))$ does not change F on X . Hence the sign hypothesis is needed only for coordinates with at least three feasible values; all binary quadratic programs are covered. The proofs also apply verbatim to $F = \sum_a f_a + \sum_i \psi_i(x_i)$ in which each f_a is affine in each of its variables separately and each ψ_i is concave on $[\ell_i, u_i]$. Then $\mathbb{E}f_a(Y) = f_a(x)$, and the diagonal term of (9.3) becomes $\psi_i(x_i)$ minus its chord at x_i . That difference vanishes at an interior point exactly when ψ_i is affine on $[\ell_i, u_i]$, which replaces the condition $H_{ii} < 0$ in (a).

Endpoint optimality for nonpositive diagonals is classical. For multilinear functions on a box, Rosenberg characterized the optimal set as the union of the faces all of whose vertices are optimal [161]. For a quadratic on a continuous box with $H_{ii} = 0$ for all i , this is Corollary 9.8(ii), because by (i) a pattern is admissible exactly when all vertices of its face are optimal. Del Pia and Khajavirad use endpoint optimality to separate binary-valued from continuous variables [53]. Min-sum messages are standard [50], and nonnegative reparameterizations whose zero entries describe all optimal labelings of a discrete problem are known from max-sum inference [183, 187]. Theorem 9.7 combines these facts and extends them to strictly concave coordinates and mixed boxes. The new part is the factored form on the decomposition: Corollary 9.8 turns the optimal set into a tree-structured pattern problem, which gives uniqueness, dimension, counts, distances and linear optimization over $\mathcal{S}$ with $2^{O(p)}$ operations per bag.

9.3 The diagonal Lagrangian certificate and unknown growth

In this subsection $X = \bar{X}$ is a continuous box and F is a rational quadratic. A point $\bar{x} \in X$ is a *box KKT point* if $\partial_i F(\bar{x}) \geq 0$ when $\bar{x}_i = \ell_i$, $\partial_i F(\bar{x}) \leq 0$ when $\bar{x}_i = u_i$, and $\partial_i F(\bar{x}) = 0$ otherwise; every minimizer is one. For such a point put

$$\lambda_i(\bar{x}) = \frac{2|\partial_i F(\bar{x})|}{u_i - \ell_i}, \quad \Lambda(\bar{x}) = \text{diag}(\lambda_1(\bar{x}), \dots, \lambda_n(\bar{x})).$$

For $\lambda \in \mathbb{R}_{\geq 0}^n$ define the Lagrangian dual function

$$\Psi(\lambda) = \inf_{x \in \mathbb{R}^n} \left[F(x) - \frac{1}{2} \sum_i \lambda_i (x_i - \ell_i)(u_i - x_i) \right].$$

Lemma 9.10 (Diagonal certificate, full optimal set and invariance). *Let X be a continuous box and F a quadratic.*

(i) *For every box KKT point $\bar{x}$ and every $x \in \mathbb{R}^n$,*

$$\begin{aligned} F(x) - F(\bar{x}) &= \frac{1}{2} (x - \bar{x})^\top (H + \Lambda(\bar{x})) (x - \bar{x}) \\ &\quad + \frac{1}{2} \sum_i \lambda_i(\bar{x}) (x_i - \ell_i)(u_i - x_i). \end{aligned} \tag{9.4}$$

(ii) *If $\bar{x}$ is a box KKT point and $H + \Lambda(\bar{x}) \succeq 0$, then $\bar{x} \in \mathcal{S}$ and*

$$\mathcal{S} = \{x \in X : (H + \Lambda(\bar{x}))(x - \bar{x}) = 0, \quad \lambda_i(\bar{x})(x_i - \ell_i)(u_i - x_i) = 0 \text{ for all } i\}. \tag{9.5}$$

(iii) $\Psi(\lambda) \leq F^*$ for every $\lambda \geq 0$, and the following are equivalent: (a) $H + \Lambda(\bar{x}) \succeq 0$ for some $\bar{x} \in \mathcal{S}$; (b) $H + \Lambda(\bar{x}) \succeq 0$ for every $\bar{x} \in \mathcal{S}$; (c) $\Psi(\lambda) = F^*$ for some $\lambda \geq 0$. When they hold, the vector $\lambda(\bar{x})$ is the same for every $\bar{x} \in \mathcal{S}$, and it is the unique maximizer of Ψ .

The proof is in Appendix F.

We write $\mathfrak{D}$ for the class of continuous box QPs that satisfy the equivalent conditions (a)–(c) of Lemma 9.10(iii), and call it the *diagonal certificate class*.

Remark 9.11 (The diagonal certificate class). Convex box QPs belong to $\mathfrak{D}$, since $\Lambda(\bar{x}) \succeq 0$. The class also contains indefinite instances with tilted and disconnected optimal sets (Example F.1 in Appendix F). The function Ψ is the Lagrangian dual for the constraints $(x_i - \ell_i)(u_i - x_i) \geq 0$, so condition (c) says that this dual, equivalently the basic Shor semidefinite relaxation, is exact. Sufficient global optimality conditions of this box-multiplier form are classical [106, 125], and stronger relaxations with complete exactness characterizations are studied in [159]. Lemma 9.10 is therefore not a new certificate. What we use is its invariance: whichever minimizer a search finds, the same rational test accepts it.

Lemma 9.10 checks a candidate but does not produce one. Candidates come from proximal stages: graded grids centered at the previous output, with a proximal term instead of filtering, and a guess $\hat{\kappa}$ in place of the unknown κ_S .

Algorithm 9.12 (PROX: proximal stages with guess $\hat{\kappa}$). The input is a rational $\hat{\kappa} \geq 1$. Let $\mu \geq 2$ be the least integer with $4^{-\mu}\hat{\kappa} \leq 1/8$, and put $\theta = 2^{-\mu}$, $\omega = L\theta^2/4$ and $\varrho = 2\lceil\sqrt{\hat{\kappa}n}\rceil$. A *proximal stage* with center $c \in X$ and mesh $h > 0$ takes for G_i the graded grid of X'_i with center c_i , mesh h and grading θ , where $X' = X \cap \prod_i [c_i - \varrho h, c_i + \varrho h]$. It returns a minimizer y^+ over $G = \prod_i G_i$ of $Q_\omega(y) = F(y) - D(y) + \omega\|y - c\|^2$, with D from (4.1) on G , computed by Lemma 4.3 (the added terms are unary). PROX($\hat{\kappa}$) runs proximal stages $j = 0, 1, \dots$ with meshes $h_j = s2^{-j}$ and centers $c^{(0)} = \ell$ and $c^{(j+1)} = y^{(j)}$, where $y^{(j)}$ is the output of stage j . There is no filtering.

Let $K_\theta = 8\theta^{-1}\lceil\log_2(n+2)\rceil$. Every grid of PROX($\hat{\kappa}$) has at most K_θ nodes per coordinate, and stage j costs $O(p(|\mathcal{A}| + n + N)K_\theta^p)$ table operations on rationals of bit length $O(I + j + \mu K_\theta)$. If F has set growth with a constant g_S such that $\kappa_S \leq \hat{\kappa}$, then $\text{dist}(y^{(j)}, \mathcal{S})^2 \leq \frac{99}{256}\hat{\kappa}nh_j^2$ for every $j \geq 0$ (Lemma F.2 in Appendix F).

Algorithm 9.13 (DISC: discovery under unknown growth). For $\hat{\kappa} = 1, 2, 4, \dots$: let $J_{\hat{\kappa}}$ be the least $j \geq 0$ with $4^j\tau^2 \geq 4\hat{\kappa}ns^2$ (τ as in REC, Algorithm 6.7); run PROX($\hat{\kappa}$) for stages $0, \dots, J_{\hat{\kappa}}$ and let $y = y^{(J_{\hat{\kappa}})}$. Run REC(y); if it does not fail, let $\bar{x}$ be a vertex of the polytope of solutions of its linear system, as returned by exact linear programming. If $\bar{x}$ is a box KKT point and $H + \Lambda(\bar{x}) \succeq 0$, stop and output $\bar{x}$, $F^* = F(\bar{x})$, $\Lambda(\bar{x})$ and the description (9.5) of $\mathcal{S}$; otherwise continue with the next guess.

Theorem 9.14 (Discovery of a certified optimal set under unknown growth). *Let X be a continuous box and F a rational quadratic.*

- (a) *Every output of DISC is correct, with no assumption on the instance. If $F \notin \mathfrak{D}$, DISC does not stop.*
- (b) *If $F \in \mathfrak{D}$, then F has set growth with some constant $g_S > 0$, and for every such g_S , DISC stops no later than at the first guess $\hat{\kappa} \geq \kappa_S$, so with $\hat{\kappa} \leq 2\kappa_S$.*
- (c) *Under the hypotheses of (b), DISC uses at most $f'(p, \kappa_S) \text{poly}(I)$ bit operations, where f' is a computable function that is nondecreasing and polynomial in κ_S for fixed p , and the polynomial in I has an absolute exponent. The output has bit length $\text{poly}(I)$.*

Neither $\mathcal{S}$, nor g_S , nor a bound on κ_S is supplied.

The proof is in Appendix F. Since DISC does not stop outside $\mathfrak{D}$, an implementation needs a work limit, and an unsuccessful guess certifies nothing about membership (Remark F.3).

The optimal value of an instance of $\mathfrak{D}$ equals $\max \Psi$, the value of a convex relaxation; the difficulty is in locating a minimizer. The Subset Sum reduction of [53, Remark 2], written with unscaled partial sums, has bag size three, and its feasible instances belong to $\mathfrak{D}$. Hence, unless $P = NP$, no polynomial-time algorithm returns an exact minimizer for every instance of $\mathfrak{D}$ with bag size three, and no choice of set-growth constants bounds κ_S by a polynomial in I on these instances (Proposition F.4 in Appendix F); a parameter such as κ_S is needed. Deciding whether a given minimizer of an instance of $\mathfrak{D}$ with bag size three is the only one is coNP-hard (last paragraph of Appendix F), so unless $P = NP$ Corollary 9.8 has no counterpart in $\mathfrak{D}$.

10 Structural limits

The algorithms of Sections 4–6 rest on four hypotheses: a product domain, upper bounds on coordinate curvature, weighted or point growth, and a correction that charges for rounding every coordinate. Sections 7–9 relax some of them: Section 7 replaces the corrections of eliminated coordinates by exact or certified value functions, Section 8 replaces the product domain by a feasible rounding law for totally unimodular constraints, and Sections 8 and 9 use set growth instead of growth at one minimizer. This section proves Theorem 1.2 and shows what fails when a hypothesis is dropped without such a replacement. Section 10.1 proves part (a), Section 10.2 part (d), and Section 10.3 parts (b) and (c). Section 10.4 shows that exact messages can be exponentially large; Section 10.5, that set growth does not bound the grids produced by filtering, and that under set growth a deterministic algorithm that queries values and derivatives needs at least of order $\varepsilon^{-n/2}$ queries; Section 10.6, that product domains are essential; and Section 10.7, that bag-local corrections combined with grid min-marginals are not valid lower bounds, and that matching separator moments of any finite order gives no local error bound in terms of negative curvature.

Del Pia and Khajavirad prove that continuous box QP is strongly NP-hard at treewidth two, even with bounded integer coefficients, and that quartic box problems on paths are strongly NP-hard [53]. Their instances need not have a unique minimizer, so their result does not by itself show that κ must enter the running time.

10.1 Conditioning cannot be dropped

An instance of Subset Sum consists of positive integers $a_1, \dots, a_m$ and a target a_0 ; it asks whether some subset of $\{a_1, \dots, a_m\}$ sums to a_0 . The following construction refines the simple weak NP-hardness reduction in [53, Remark 2]: a stronger concave penalty puts every minimizer at a vertex, a lexicographic linear term makes the minimizer unique, and the growth constant is computed explicitly.

Proposition 10.1 (Unique minimizers without conditioning). *Let $a_1, \dots, a_m$ be positive integers, $m \geq 2$, let $a = (a_1, \dots, a_m)$, and let a_0 be an integer with $0 \leq a_0 \leq A := \sum_i a_i$. Put $\lambda = 1/(m2^{m+1})$. On the box*

$$x \in [0, 1]^m, \quad y = (y_1, \dots, y_{m-1}) \in [-A, 2A]^{m-1}, \quad y_0 := 0, \quad y_m := a_0,$$

minimize

$$F(x, y) = \sum_{i=1}^m (y_i - y_{i-1} - a_i x_i)^2 + \|a\|^2 \sum_{i=1}^m x_i(1 - x_i) + \lambda \sum_{i=1}^m 2^{i-1} x_i.$$

Then:

1. The bags $\{y_{i-1}, y_i, x_i\}$, with the constants y_0, y_m omitted and arranged as a path in the order $i = 1, \dots, m$, form a tree decomposition with $p \leq 3$. All data have binary length polynomial in that of (a, a_0) .
2. F has a unique minimizer $z^* = (x^*, y^*)$, and $x^* \in \{0, 1\}^m$. If some $v \in \{0, 1\}^m$ satisfies $a^\top v = a_0$, then $a^\top x^* = a_0$ and $F^* < 1/(2m)$; otherwise $F^* \geq 1/m$.
3. $L = 4$ is a valid upper coordinate-curvature bound and

$$g = \frac{\lambda}{m(1 + 2(m-1)\|a\|^2)}$$

is a valid growth constant. Hence $\kappa \leq 2^{m+3}m^2(1 + 2m\|a\|^2)$ and $\log_2 \kappa = O(I)$.

Proof. Part 1 is immediate: y_i lies in the consecutive bags i and $i+1$, each x_i in one bag, and each square and unary term lies in a bag.

Elimination of y . For $x \in [0, 1]^m$ let $e(x) = a^\top x - a_0$, $\sigma_0(x) = 0$, and

$$\sigma_k(x) = \sum_{j \leq k} a_j x_j - \frac{k}{m} e(x) \quad (k = 1, \dots, m),$$

so that $\sigma_m(x) = a_0$. For a feasible y put $\delta_k = y_k - \sigma_k(x)$, with $\delta_0 = \delta_m = 0$. Then $y_i - y_{i-1} - a_i x_i = -e(x)/m + \delta_i - \delta_{i-1}$, and $\sum_i (\delta_i - \delta_{i-1}) = 0$. Expanding the squares gives the identity

$$F(x, y) = \Phi(x) + \sum_{i=1}^m (\delta_i - \delta_{i-1})^2, \quad \Phi(x) = \frac{e(x)^2}{m} + \|a\|^2 \sum_i x_i(1 - x_i) + \lambda \sum_i 2^{i-1} x_i. \quad (10.1)$$

Because $0 \leq \sum_{j \leq k} a_j x_j \leq A$ and $|e(x)| \leq A$, the vector $(\sigma_1(x), \dots, \sigma_{m-1}(x))$ lies in $[-A, 2A]^{m-1}$. Since $\delta_k = \sum_{i \leq k} (\delta_i - \delta_{i-1})$, the Cauchy-Schwarz inequality gives $\delta_k^2 \leq k \sum_i (\delta_i - \delta_{i-1})^2$. Summing over $k \leq m-1$ gives the path inequality

$$\sum_{i=1}^m (\delta_i - \delta_{i-1})^2 \geq \frac{2}{m(m-1)} \sum_{k=1}^{m-1} \delta_k^2 \quad (\delta_0 = \delta_m = 0), \quad (10.2)$$

which holds for all reals $\delta_0, \dots, \delta_m$ with $\delta_0 = \delta_m = 0$.

The reduced function. Its Hessian is $\nabla^2 \Phi = \frac{2}{m} a a^\top - 2\|a\|^2 I \preceq 0$, so Φ is concave. At a vertex v , $\Phi(v) = e(v)^2/m + \lambda \chi(v)$ with $\chi(v) = \sum_i 2^{i-1} v_i$, a bijection of $\{0, 1\}^m$ onto $\{0, \dots, 2^m - 1\}$. If $\Phi(v) = \Phi(v')$, then $(e(v)^2 - e(v')^2)/m = \lambda(\chi(v') - \chi(v))$; the left side lies in $m^{-1}\mathbb{Z}$ and the right side has absolute value less than $\lambda 2^m = 1/(2m)$, so both vanish and $v = v'$. Let x^* be the unique minimizing vertex and $\Phi^* = \Phi(x^*)$. We claim that $\Phi(v) - \Phi^* \geq \lambda$ for every vertex $v \neq x^*$. If $e(v)^2 = e(x^*)^2$, the difference is a positive integer multiple of λ . Otherwise $e(v)^2 - e(x^*)^2$ is a nonzero integer. It cannot be negative, since then $\Phi(v) - \Phi^* \leq -1/m + \lambda(2^m - 1) < 0$. Hence it is at least one and $\Phi(v) - \Phi^* \geq 1/m - \lambda(2^m - 1) > 1/(2m) \geq \lambda$.

For $x \in [0, 1]^m$ let Y have independent coordinates with $\Pr(Y_i = 1) = x_i$. By concavity and Jensen's inequality, $\Phi(x) = \Phi(\mathbb{E}Y) \geq \mathbb{E}\Phi(Y) \geq \Phi^* + \lambda \Pr(Y \neq x^*)$. Since x^* is binary, $\Pr(Y_i \neq x_i^*) = |x_i - x_i^*| \leq 1$, so $\Pr(Y \neq x^*) \geq \max_i |x_i - x_i^*| \geq \frac{1}{m} \|x - x^*\|^2$. Therefore

$$\Phi(x) - \Phi^* \geq \frac{\lambda}{m} \|x - x^*\|^2 \quad (x \in [0, 1]^m). \quad (10.3)$$

Uniqueness and growth. By (10.1) and (10.3), $z^* = (x^*, \sigma(x^*))$ is the unique minimizer and $F^* = \Phi^*$. Let $z = (x, y)$ be feasible and $h = x - x^*$. Then $\sigma_k(x) - \sigma_k(x^*) = \sum_j c_{kj} a_j h_j$ with $c_{kj} = [j \leq k] - k/m \in [-1, 1]$, so $\|\sigma(x) - \sigma(x^*)\|^2 \leq (m-1)\|a\|^2\|h\|^2$ and

$$\|z - z^*\|^2 \leq \|h\|^2 + 2\|\delta\|^2 + 2\|\sigma(x) - \sigma(x^*)\|^2 \leq (1 + 2(m-1)\|a\|^2)\|h\|^2 + 2\|\delta\|^2.$$

By (10.1)–(10.3), $F(z) - F^* \geq (\lambda/m)\|h\|^2 + \frac{2}{m(m-1)}\|\delta\|^2$. Since $g(1 + 2(m-1)\|a\|^2) = \lambda/m$ and $g \leq \lambda/m < 1/(m(m-1))$, this gives $F(z) - F^* \geq g\|z - z^*\|^2$. On coordinate lines, $\partial^2 F / \partial y_k^2 = 4$ and $\partial^2 F / \partial x_i^2 = 2a_i^2 - 2\|a\|^2 \leq 0$, so $L = 4$ is valid and $\kappa \leq 4/g \leq 2^{m+3}m^2(1 + 2m\|a\|^2)$. Both m and $\log\|a\|^2$ are $O(I)$.

Decision. If $a^\top v = a_0$ for a vertex v , then $F^* \leq \Phi(v) \leq \lambda(2^m - 1) < 1/(2m)$, and, as shown above, $e(x^*)^2 \leq e(v)^2 = 0$. If no vertex solves the instance, every vertex has $e(v)^2 \geq 1$, and $F^* = \Phi(x^*) \geq 1/m$. $\square$

Corollary 10.2 (The dependence on κ is not polylogarithmic). *Unless $P = NP$, no algorithm computes, for every rational continuous box QP with a supplied tree decomposition of maximum bag size at most three and a unique minimizer, the optimal value within 2^{-q} in time polynomial in $I + q + \log \kappa$. In particular, no algorithm achieves the guarantee of Theorem 5.7 with $f(3, \kappa)$ bounded by a polynomial in $\log \kappa$.*

Proof. Apply such an algorithm to the instances of Proposition 10.1 with $q = \lceil \log_2(4m) \rceil$. An approximation within $1/(4m)$ separates $F^* < 1/(2m)$ from $F^* \geq 1/m$, and $\log \kappa = O(I)$. This would decide Subset Sum, which is NP-complete [75], in polynomial time. $\square$

For fixed p , the factor $f(p, \kappa) = c_0(c_1 p \sqrt{\kappa})^p \kappa (1 + \log_2 \kappa)^2$ in Theorem 5.7 is bounded by a polynomial in κ , and the proof of Theorem 6.14 bounds $f_1(p, \kappa)$ by $f(p, \kappa)$ times a power of $1 + \log_2 \kappa$. Hence, for fixed p , instances with a unique minimizer and $\kappa \leq I^{O(1)}$ are solved exactly in polynomial time, while by Proposition 10.1 the problem is NP-hard on instances with $\kappa \leq 2^{O(I)}$. Adding decoupled terms t_j^2 with $t_j \in [0, 1]$, each in a bag of its own, keeps the minimizer unique, leaves p and κ unchanged, and increases I by an amount of our choice; so for every fixed $\delta > 0$ the problem is NP-hard already on instances with $\kappa \leq 2^{I^\delta}$.

The strongly NP-hard instances of [53, Theorem 3] have bags of size three and coefficients bounded by absolute constants. Remark G.1 in Appendix G.1 shows that, already for a family of two-item instances on which their minimizer is unique, κ and the ratio ν/g of negative curvature to growth are exponential in the number of variables. These instances are therefore consistent with Corollary 10.2, and they do not contradict an algorithm parameterized by κ or by ν/g .

10.2 The value-oracle model

CT (Algorithm 5.6) uses only exact values of F and the curvature bounds. In a model where F is available only through evaluations, the exponent $p/2$ of κ in its running time cannot be improved. The construction is the classical hidden-well argument [146], adapted to the present hypotheses.

Proposition 10.3 (Value-oracle lower bound). *Let $p \geq 1$, $L > 0$, $\kappa \geq 8p$, $g = L/\kappa$, and $X = [0, 1]^p$. For $c \in X$ put*

$$F_c(x) = \min\left\{gp, \frac{L}{2}\|x - c\|^2\right\}.$$

Each F_c has upper coordinate curvature L , the unique minimizer c with $F_c(c) = 0$, and growth $F_c(x) \geq g\|x - c\|^2$ on X ; a single bag of size p is a valid decomposition. Suppose a deterministic algorithm receives p , L , κ and value access to F_c for an unknown c , and always returns $y \in X$

with $F_c(y) < Lp/\kappa$. Then on some F_c it makes at least $(\kappa/(8p))^{p/2} - 1$ evaluations. A randomized algorithm that makes at most Q evaluations succeeds, for some c , with probability at most $(Q + 1)(8p/\kappa)^{p/2}$.

Proof. On a coordinate line, $\frac{L}{2}\|x - c\|^2 - \frac{L}{2}x_i^2$ is affine in x_i and $gp - \frac{L}{2}x_i^2$ is concave; their minimum is concave. Thus $t \mapsto F_c - Lt^2/2$ is concave on every coordinate line. For $x \in X$, $\|x - c\|^2 \leq p$, so $gp \geq g\|x - c\|^2$; and $L/2 \geq g$ because $\kappa \geq 2$. This proves growth, and c is the unique zero.

Moreover $F_c(x) = gp$ unless $\|x - c\| < r$, where $r^2 = 2gp/L = 2p/\kappa$. Since $\kappa \geq 8p$, $2r \leq 1$. Let $\mathcal{C}$ consist of the points of X whose coordinates lie in $\{0, 2r, 4r, \dots\}$. Then $|\mathcal{C}| = (\lfloor 1/(2r) \rfloor + 1)^p \geq (2r)^{-p} = (\kappa/(8p))^{p/2}$, and the open balls $B(c, r)$, $c \in \mathcal{C}$, are pairwise disjoint. The guarantee $F_c(y) < Lp/\kappa = gp$ means $y \in B(c, r)$.

Consider the run of the deterministic algorithm in which every answer equals gp , with evaluation points $x^1, x^2, \dots$ (finitely or infinitely many). For $c \in \mathcal{C}$ let t_c be the index of the first evaluation point in $B(c, r)$, and $t_c = \infty$ if there is none. Since $F_c = gp$ outside $B(c, r)$, the run on F_c agrees with the constant run up to and including evaluation t_c . If $t_c = \infty$, the two runs coincide, so the constant run stops and its output lies in $B(c, r)$; as the balls are disjoint, this happens for at most one c . Each point lies in at most one ball, so the finite t_c are distinct. At least $|\mathcal{C}| - 1$ of them are finite, so the largest is at least $|\mathcal{C}| - 1$, and the run on the corresponding F_c makes at least $|\mathcal{C}| - 1$ evaluations.

For a randomized algorithm, choose c uniformly in $\mathcal{C}$ and fix the random bits. Consider the first Q evaluation points of the run on the constant answers. If $B(c, r)$ contains none of them, the run on F_c agrees with the constant run on these evaluations; since it makes at most Q evaluations, it stops there with the output of the constant run, and it succeeds only if that output lies in $B(c, r)$. Hence at most $Q + 1$ choices of c succeed. Averaging over the random bits gives success probability at most $(Q + 1)/|\mathcal{C}|$ for a uniform c , hence for some c . $\square$

Remark 10.4 (Comparison with the algorithm). With one bag ($n = p$, $N = 1$), each stage of CT evaluates F at most at K^p grid points. By Lemma 5.5 and the trial schedule, $K = O(\sqrt{\kappa} \log(p + 2))$ nodes per coordinate in the successful trial, and the earlier trials cost no more. Thus the exponent $p/2$ of κ is attained, and Proposition 10.3 shows that it cannot be improved in this model; the remaining gap is a factor $(C\sqrt{p} \log(p + 2))^p$ times the number of stages. The lower bound is also exponential in p once $\kappa \geq 32p$.

10.3 Width and the exponent of κ

For explicit quadratics we use ETH and rETH, as stated before Theorem 1.2, and the strong exponential-time hypothesis (SETH): for every $\epsilon > 0$ there is a k such that no algorithm decides k -SAT on n' variables in time $(2 - \epsilon)^{n'}$ [44, Chapter 14]. The next construction is a variant of the box formulation of maximum independent set [1], with a binary tie-break that makes the minimizer unique.

Proposition 10.5 (Exponential dependence on the width at bounded conditioning). *For a graph with vertex set $[n]$, $n \geq 1$, and edge set E let*

$$\Psi(x) = 2^n \left(- \sum_i x_i + 2 \sum_{ij \in E} x_i x_j \right) + \sum_i 2^{i-1} x_i + \frac{1}{2n} \sum_i (x_i^2 - x_i), \quad x \in [0, 1]^n.$$

Then Ψ has a unique minimizer x^ , which is binary, $\lfloor \Psi^*/2^n \rfloor = -\alpha$, where $\Psi^* = \min \Psi$ and α is the independence number of the graph, and $\Psi(x) - \Psi^* \geq \frac{1}{2n} \|x - x^*\|^2$ on $[0, 1]^n$. With $L_i = L = 1/n$ this gives $\kappa \leq 2$ and $\bar{\kappa} \leq 2$. The same holds on $\{0, 1\}^n$. Every tree decomposition of the graph*

is a decomposition of Ψ . Consequently, computing F^* within $1/2$ for box quadratics with $\kappa \leq 2$ (continuous, or all-binary) is NP-hard, and, assuming ETH, no algorithm does it in time $2^{o(p)} I^{O(1)}$.

Proof. On binary points $\Psi = 2^n \Phi + \chi$ with $\Phi = -\sum x_i + 2 \sum_E x_i x_j$ integral and $\chi = \sum 2^{i-1} x_i \in \{0, \dots, 2^n - 1\}$ injective. A binary minimizer therefore minimizes Φ and is unique. If the support of a binary x contains both endpoints of an edge, deleting such an endpoint u from the support changes Φ by $1 - 2d(u) \leq -1$, where $d(u) \geq 1$ is the number of neighbors of u in the support; hence $\min \Phi = -\alpha$.

For $x \in [0, 1]^n$ write $\Psi = \Psi_0 + \frac{1}{2^n} \sum (x_i^2 - x_i)$, where $\Psi_0 = 2^n \Phi + \chi$ is multilinear. Let Y have independent coordinates with $\Pr[Y_i = 1] = x_i$. Then $\Psi_0(x) = \mathbb{E} \Psi_0(Y) = \mathbb{E} \Psi(Y)$, and since $\Psi(v) \geq \Psi(x^*) + 1$ for binary $v \neq x^*$, $\Psi_0(x) - \Psi(x^*) \geq \Pr[Y \neq x^*]$. With $\delta_i = |x_i - x_i^*| \in [0, 1]$, $\Pr[Y = x^*] = \prod_i (1 - \delta_i) \leq 1 - \max_i \delta_i$, so

$$\Psi_0(x) - \Psi(x^*) \geq \max_i \delta_i \geq \frac{1}{n} \sum_i \delta_i \geq \frac{1}{n} \|x - x^*\|^2. \quad (10.4)$$

Also $x_i(1 - x_i) \leq \delta_i$ because $x_i^* \in \{0, 1\}$. Hence $\Psi(x) - \Psi(x^*) \geq (\frac{1}{n} - \frac{1}{2n}) \sum_i \delta_i \geq \frac{1}{2n} \|x - x^*\|^2$, so x^* is the unique minimizer over $[0, 1]^n$ and $\Psi^* = \Psi(x^*)$. The only squared terms are $\frac{1}{2n} x_i^2$, so $L = 1/n$ and $L/g \leq 2$; in the weighted norm, $\Psi(x) - \Psi^* \geq \frac{1}{2} \|x - x^*\|_{\mathbf{L}}^2$, so $\bar{\kappa} \leq 2$.

Since $n \geq 1$, we have $\alpha \geq 1$, so $\Phi(x^*) = -\alpha < \Phi(0)$ and $x^* \neq 0$. Thus $\Psi^* = -2^n \alpha + \chi(x^*)$ with $\chi(x^*) \in [1, 2^n - 1]$, and every number $\tilde{\Psi}$ with $|\tilde{\Psi} - \Psi^*| \leq 1/2$ satisfies $\lfloor \tilde{\Psi}/2^n \rfloor = -\alpha$, which also gives $\lfloor \Psi^*/2^n \rfloor = -\alpha$. The instance has $O(n^2)$ coefficients of $O(n)$ bits, so $I = n^{O(1)}$, and the single bag $[n]$ gives $p = n$. Computing α is NP-hard [75], and an algorithm with running time $2^{o(p)} I^{O(1)}$ would compute it in time $2^{o(n)}$, contradicting ETH [101]. $\square$

Under SETH, the same reduction, applied with a supplied tree decomposition of the graph, excludes running times $(2 - \epsilon)^p I^{O(1)}$ for every $\epsilon > 0$, by the lower bound of [127] for independent set.

If $L = 0$, CT is exact at its first stage, whose grids consist of box endpoints, and it runs in time $2^p \text{poly}(I)$. The multilinear part Ψ_0 of the instance above has $L = 0$, the same unique binary minimizer, and growth constant $g = 1/n$ by (10.4), so $\kappa = 1$; the same reduction applies to it. For $L = 0$ the dependence on p is therefore optimal up to the base of the exponential under ETH, and optimal under SETH.

The next proposition concerns the exponent of κ . Its reduction uses random weights, so it needs rETH.

Proposition 10.6 (The exponent of κI cannot be $o(p)$). *Assume rETH. There are no computable f and computable nondecreasing unbounded $\psi \geq 1$ such that some algorithm, on every integer box quadratic with a supplied tree decomposition of maximum bag size p and a unique minimizer, outputs a feasible point within $1/2$ of F^* in time $f(p) (\kappa I)^{p/\psi(p)}$. In particular, no algorithm has running time $f(p) \kappa^{a(p)} I^C$ with a computable f , a constant C , and $a(p) \leq p/\psi(p)$ for such a ψ . The bound of Theorem 5.7 has $a(p) = p/2 + O(1)$ and $C = 5$.*

The proof is in Appendix G.2. It encodes multicolored clique with k color classes in the standard way, with one integer variable of domain size N_0 per color class and one pair constraint for each pair of classes. What is new is a quadratic encoding of the pair constraints with a unique minimizer, obtained by random weights and isolation, and with κ polynomial in kN_0 . Under rETH, therefore, the exponent $p/2 + O(1)$ of κ in Theorem 5.7 cannot be improved to $o(p)$ on integer instances, in the sense $a(p) \leq p/\psi(p)$, while the degree in I stays fixed. The fixed degree matters: on the hard instances both κ and I are bounded by a polynomial in kN_0 , so the reduction cannot separate the exponent of κ from the degree in I , and it does not exclude a bound such as $\kappa^3 I^p$. The constant

in the exponent, the necessity of the factor p^p , and whether the same holds for purely continuous instances or with a deterministic reduction, are open.

10.4 Exact messages can be exponentially large

For forests, exact value functions of box QP are piecewise quadratic with linearly many pieces [53]. Dynamic programming with exact piecewise-quadratic messages on trees is also used for total-variation problems [116, 120]; for nonconvex piecewise-quadratic costs the worst-case bound of [120] is exponential. The next proposition gives a lower bound for the message itself: at bag size three, exact messages can have exponentially many pieces even when κ is bounded and the minimizer is unique. This is why CT uses grids rather than exact piecewise-quadratic messages.

Proposition 10.7 (Exponential messages at bounded conditioning). *For $m \geq 2$ let $\xi_t \in [0, 2^t - 1]$ and $z_t \in [0, 1]$, $t = 1, \dots, m$, put $\xi_0 := 0$, $\rho_t = \xi_t - 2\xi_{t-1} - z_t$, and*

$$\Psi_m(\xi, z) = \xi_m^2 + \sum_{t=1}^m \rho_t^2 + \frac{1}{8} \sum_{t=1}^m z_t(1 - z_t).$$

Then:

1. *the bags $\{\xi_{t-1}, \xi_t, z_t\}$ (ξ_0 omitted) form a path decomposition with $p = 3$, and $I = O(m^2)$;*
2. *$\Psi_m(\xi, z) \geq \frac{1}{8}(\|\xi\|^2 + \|z\|^2)$ on the box, so zero is the unique minimizer and $g = 1/8$ is valid;*
3. *$L = 10$ is the exact maximum coordinate curvature, so $\kappa \leq 80$; moreover $\nabla^2 \Psi_m \succeq -\frac{1}{4}I$;*
4. *let $V(v) = \min\{\Psi_m(\xi, z) : \xi_m = v\}$ for $v \in [0, 2^m - 1]$, the value function of ξ_m . Up to the term v^2 , it is the message sent across the separator $\{\xi_m\}$ to a leaf bag $\{\xi_m\}$ carrying ξ_m^2 . If $[0, 2^m - 1]$ is covered by N_{pc} intervals on each of which V coincides with a quadratic polynomial, or if $V = \min_{l \leq N_{\text{pc}}} \varphi_l$ for quadratic polynomials φ_l , then $N_{\text{pc}} \geq 2^{m-1}$.*

By Theorem 5.7, CT nevertheless solves this family to accuracy 2^{-q} in time polynomial in $m + q$.

The proof is in Appendix G.3. For part 4, $V(v) - v^2$ vanishes exactly at the 2^m integers of $[0, 2^m - 1]$ and is positive elsewhere, so each quadratic piece agrees with V at no more than two of them.

10.5 Point growth cannot be replaced by set growth

Sections 8 and 9 use set growth (Definition 3.2). The next proposition shows that set growth alone does not bound the grids that filtering produces.

Proposition 10.8 (Set growth does not bound filtered grids). *For $n \geq 2$ let $F(x) = \sum_{i=1}^{n-1} (x_i - x_{i+1})^2$ on $X = [0, 1]^n$, with the path decomposition of bags $\{x_i, x_{i+1}\}$. The optimal set is the diagonal $\mathcal{S} = \{t\mathbf{1} : t \in [0, 1]\}$, $F^* = 0$, and $F(x) \geq \frac{2}{n(n-1)} \text{dist}(x, \mathcal{S})^2$; for $n = 2$, $F = 2 \text{dist}(x, \mathcal{S})^2$. Every valid upper coordinate curvature satisfies $L_i \geq 2$, and with the smallest valid ones, $\kappa_S \leq 2n(n-1)$. Let the corrections be $d_i(v) = L_i w_i(v)^2 / 8$ (Definition 4.1).*

- (a) *For every grid G of X , every j and every node $v \in G_j$, the min-marginal satisfies $m_j(v) \leq -\frac{L_j}{8} w_j(v)^2$. In particular $\beta \leq -\frac{L_j}{8} W_j^2$, where W_j is the largest length of a grid interval of G_j .*

- (b) The filtering step (Definition 4.4), applied to a grid of X with a threshold $U \geq F^*$, removes nothing. Hence every run of filtering with thresholds $U_j \geq F^*$ that starts from a grid of X , in particular every run of TRIAL (Algorithm 5.3) and hence of CT, uses grids of X at all stages, and if it certifies accuracy ε , its last grid has at least $1 + 1/(2\sqrt{\varepsilon})$ nodes in every coordinate.
- (c) In every valid path certificate (Definition 4.6) with $F(\hat{x}) - \beta \leq \varepsilon$, every grid interval of $G_i^{(0)}$ that is not contained in $X_i^{(1)}$ (every grid interval of $G_i^{(0)}$ if $k = 0$) has length at most $2\sqrt{\varepsilon}$.

The proof is in Appendix G.3. For (a), it sets the other coordinates, outward along the path from j , to the grid node nearest to the previous coordinate; because $L_i \geq 2$, the correction of each such coordinate is at least the square term of its edge to the previous one.

Part (b) covers the grids that CT returns. A general path certificate may remove intervals whose endpoint min-marginals lie in $[\beta, 0)$, and then its last grid can be small; for such certificates, part (c) bounds only the stage-0 grid. For $n = 2$ and $\varepsilon = 1/50$, the uniform grid of step $1/5$ on X , followed by the box $[0, 1/5]^2$ with the grid $\{0, 1/5\}^2$, $\beta = -1/50$ and $\hat{x} = 0$, is a valid path certificate of accuracy ε : all stage-0 min-marginals equal $-1/50$. Its last grid has two nodes per coordinate, while (b) requires at least five for runs of filtering, and its removed intervals have length $1/5 \leq 2\sqrt{\varepsilon}$. We do not know whether every valid path certificate for this family has $\varepsilon^{-\Omega(1)}$ nodes in total. For finitely many optimal values per coordinate, unions of uniform cells give grids with $O(r\sqrt{n\kappa_S})$ nodes per coordinate (Theorem 9.4). The proposition concerns a continuum of optimal coordinate values. It shows a limitation of filtered grids, not a hard problem: here F is a sum of squares.

Proposition 10.9 (Value-oracle barrier for unknown set growth). *Fix $n \geq 1$, $X = [0, 1]^n$ and the coordinate-curvature bound $L = 1$. Consider a deterministic algorithm that queries exact values, gradients and Hessians of an unknown C^2 function f at adaptively chosen points and outputs either a rational interval of width at most ε containing $\min_X f$, or a point $\hat{x}$ with $f(\hat{x}) - \min_X f \leq \varepsilon$. Suppose it is correct for every C^2 function f with $\partial_{ii}f \leq 1$ on X for all i that has set growth, $f(x) - \min_X f \geq g_S \text{dist}(x, \arg \min_X f)^2$ on X for some $g_S > 0$. Then on $f \equiv 0$, for which $\kappa_S = 1$, it makes at least $(1/(2\sqrt{6\varepsilon}) - 1)^n - 1$ queries.*

The proof, in Appendix G.3, hides a C^2 bump of depth greater than ε in a subcube that contains none of the points queried on $f \equiv 0$. This is the classical hidden-bump argument of information-based complexity [146], adapted to set growth. Under point growth, CT uses a number of nodes per coordinate that does not depend on the accuracy, so its number of evaluations grows with the accuracy only through the number of stages, which is linear in $\log(1/\varepsilon)$ (Algorithm 5.6 and Remark 10.4). Under growth towards an unknown set, Proposition 10.9 shows that every deterministic algorithm that queries values, gradients and Hessians needs $\Omega(\varepsilon^{-n/2})$ queries, already for $n = p = 1$ on an instance with $\kappa_S = 1$. The barrier concerns information, not the input class: the bumps are not quadratics. An algorithm for explicit quadratics must therefore exploit the coefficients, for example through a certificate such as Lemma 9.10, or the structure of Section 9.2.

10.6 Product domains are essential

The next construction is the running-sum encoding of Subset Sum of Bienstock and Muñoz [24, Appendix A]; see also [38, Example 1.1]. What we add is a unique minimizer and the growth constant, so that $\kappa = 1$. An extra item x_0 , whose size is the target, keeps the feasible set nonempty.

Proposition 10.10 (Local affine constraints). *Let $a_0, a_1, \dots, a_m$ be positive integers with $a_i \leq a_0$. Consider binary $x_0, \dots, x_m$, continuous $y_0, \dots, y_{m+1} \in [0, 1]$, and the constraints*

$$y_0 = 0, \quad y_{m+1} = 1, \quad y_{i+1} = y_i + \frac{a_i}{a_0} x_i \quad (i = 0, \dots, m),$$

with objective $F = 2^{m+1}x_0 + \sum_{i=1}^m 2^{i-1}x_i$. Each constraint lies in the bag $\{y_i, x_i, y_{i+1}\}$, and these bags form a path with $p = 3$. The feasible set is nonempty, the minimizer z^* is unique, and $F(z) - F^* \geq \|z - z^*\|^2 / (2m + 3)$ for every feasible z . Since F is affine, every $L > 0$ is valid; with $L = 1/(2m + 3)$, $\kappa = 1$. Moreover $x_0^* = 0$ if and only if some subset of $\{a_1, \dots, a_m\}$ sums to a_0 . Consequently, unless $P = NP$, no algorithm solves such problems in time $f(p, \kappa) \text{poly}(I)$ for any function f , even when the optimal value is required only within $1/2$.

The proof, in Appendix G.3, is a direct check: the constraints force $\sum_{i=0}^m a_i x_i = a_0$, and the weights 2^{i-1} and 2^{m+1} give distinct feasible points distinct integer values.

In the product setting an affine objective is solved exactly by endpoint dynamic programming. The difficulty here comes entirely from the constraints. Independent coordinate rounding, the basis of Proposition 4.2, does not preserve the running-sum equations.

After scaling y by a_0 , the instances of Proposition 10.10 satisfy Definition 8.1 with a path constraint matrix, mesh unit $\eta = 1$, box widths $s = a_0$ and two values per discrete coordinate; the objective is affine, so every $\bar{L} > 0$ is valid, and $\bar{L} \leq g_S$ gives $\kappa_c = 1$. By the discussion after Theorem 8.9, the pseudopolynomial level-0 grid of TU-GRID (Algorithm 8.6) therefore cannot be removed unless $P = NP$.

10.7 Local corrections and local moments

Proposition C.2 in Appendix C shows that a correction charged only to the coordinates of one bag, while the other coordinates are optimized on the grid, cannot be valid for any fixed allowance: the missing error grows with the number of outside coordinates and depends on the separator value. The valid local alternative is exact or certified conditional recourse (Theorem 7.4).

A different route to a local error bound keeps the exact positive curvature and discretizes only the negative part: for a continuous box quadratic with $\nabla^2 F \succeq -\nu I$ one would like an error bound in which L is replaced by ν . Such a bound holds for relaxations that condition all coordinates on one global cell (Lemma G.3), but a global cell index can take K^n values. Sparse relaxations keep exact information only on bags and match moments across separators, cell by cell (Definition G.4). Proposition G.5 shows that this does not suffice for any fixed order k of the matched moments: for every k there are instances with bag size three, a unique minimizer, $\nabla^2 F \succeq -2I$ and a growth constant that depends only on k , whose order- k relaxations have feasible points more than $1/8$ below F^* that occupy only cells of arbitrarily small length. A ν -based local error bound must therefore use information beyond finite-order separator moments, for example inequalities that couple both sides of a separator. Appendix G.4 gives the definitions, the construction and the proofs.

11 Implementation and computational illustration

The experiments in this section test whether the predicted grid sizes appear in practice and whether the certificates are cheap enough to check. They are not a benchmark of competitive performance: the implementation is a reference code in exact rational arithmetic, and all instance families with continuous coordinates are synthetic (planted, random or constructed). The only library instances we ran are binary (Section 11.8), so we do not test whether modeled instances with continuous nonconvex terms have small width and a moderate condition number. Appendix H contains the tables and details that this section summarizes.

11.1 Implementation

The reference implementation is written in Python with exact rational arithmetic (Python’s `fractions` module); floating-point input is rejected. It handles explicit box QPs $\frac{1}{2}x^\top Hx + b^\top x + c$ with continuous and integer coordinates, supplied or heuristically computed (minimum-fill) tree decompositions with arbitrary branching, and empty separators. It implements the corrected grids of Section 4 with corrections $\max\{H_{ii}, 0\}w^2/8$, the two-node grids $\{\ell_i, u_i\}$ for coordinates with $H_{ii} \leq 0$, two-pass dynamic programming with all min-marginals, filtering, and CT (Algorithm 5.6) in the common-mesh form of Lemma 5.9. It differs from the analysis in seven ways, none of which affects validity (Appendix H lists them); the most important is that all coordinates share the mesh $h_j = s2^{-j}$, so the analysis that applies is Lemma 5.9 with the Euclidean condition number κ , not Theorem 5.7. Exact coordinate descent from ℓ , u and the box midpoint provides the first incumbent before stage 0, and each trial starts with the center at the current incumbent.

Experiments E1–E3 below run single trials, that is, TRIAL (Algorithm 5.3) with the common mesh, a fixed grading θ ($\theta = 0$ for uniform grids), no cap on the nodes per coordinate and a fixed number of stages, in E1 also without filtering; the “theorem” grading is the largest $2^{-\mu}$ with $8\kappa\theta^2 \leq 1$ for the certified upper bound on κ , which the algorithm itself does not know. Experiment E4 runs EX (Algorithm 6.13); the default schedule of E2, experiments E5 and E6, the supplementary experiment S1, the chain and the recourse runs use CT. In E2, E5, E6 and the chain no trial was aborted by the cap: the only restarts occurred in E2, and each followed a trial that reached its stage limit.

Certificates store the grids, the directed messages and the filtering decisions of every stage, which is more than Theorem 4.7 requires. A separate checker recomputes every table and message, checks the Bellman equalities, the min-marginals, every exclusion, the incumbent values and the final status, and never calls the grid generator, the dynamic program or the filtering code of the solver; it shares model parsing, decomposition validation and objective evaluation with the solver. The checker is independently written, not formally verified. On the runs of E1–E6, the chain and the plain-grid recourse runs below with a solve time of at least 0.1 seconds, replay took between 0.81 and 2.54 times as long as the solve. Code, inputs, raw results and reproduction scripts (one command per experiment group), together with the solver files they import, are provided as supplementary material.

All runs below used one thread per process and at most four processes on an Intel Xeon w5-2565X under Windows Subsystem for Linux (WSL2) with Python 3.13, on a machine shared with unrelated long-running jobs (load average 11 to 19). Timings are single measurements and indicative only. Every certificate on which a result in Sections 11.2–11.7 rests was replayed by its checker, and every replay was valid; the CT runs of S1 that only measure when the gap falls below a threshold (Section 11.4) were not replayed.

11.2 Accuracy independence, conditioning and dimension

The planted family consists of nonconvex quadratics on $[-1, 1]^n$ with Hessian $H = 2I + C$, where C has path, random-tree or band sparsity. About a quarter of the coordinates are at a bound at the planted minimizer x^* , all with a strictly inward derivative of the same size; the other coordinates of x^* are of the form $k/21$ with $|k| \leq 10$, and on them the Hessian block $H_{J_0 J_0}$ has smallest eigenvalue $4/\kappa_{\text{target}}$, up to a rounding of the coupling scale to a multiple of 2^{-12} . Entries of C that touch an active coordinate have size 2 to 4, and H is indefinite in every instance (its smallest eigenvalue is at most -1.16). Parts (a) and (b) of Lemma 3.3 bracket κ to within 12% (Appendix H). The family is benign in one respect: once x^* is known, (3.5) certifies it directly; the grid solver does not use this.

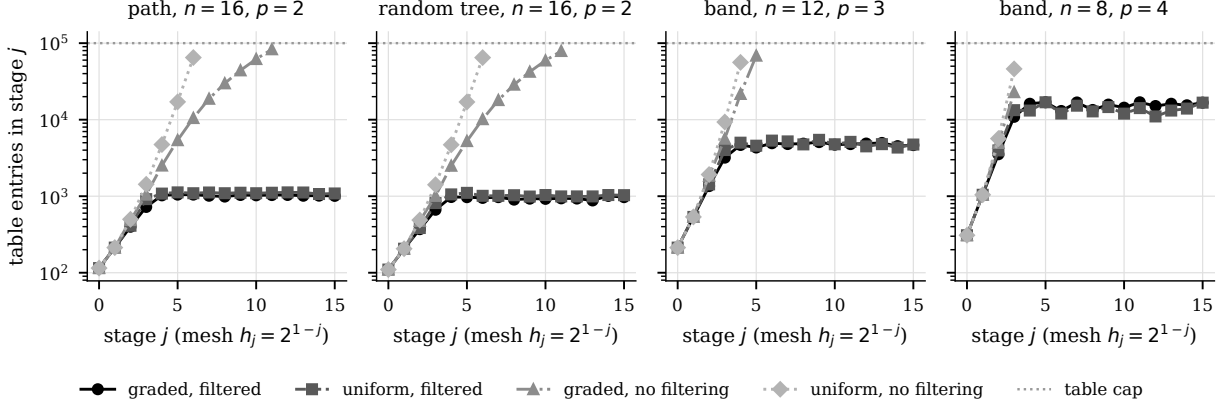


Figure 1: Experiment E1: table entries per stage against the stage index j (mesh $h_j = 2^{1-j}$), median of three seeds, for graded and uniform grids with and without filtering; $\theta = 1/8$, $\kappa \in [3.99, 4.45]$. A stage is plotted only if all three runs completed it. Runs without filtering stop at the cap of 10^5 table entries per stage (dotted line).

The minimizers are mostly off the grids: over the runs of E1–E3 with $\kappa_{\text{target}} \geq 4$, 347 of 3,177 free coordinates of x^* , counted once per run (11%), were nodes of the last grid.

E1: accuracy. Figure 1 shows the effect of filtering in single trials with $\theta = 1/8$. On the path with $n = 16$, filtered graded grids use about 1,000 table entries per stage from stage 4 to stage 15, while unfiltered graded grids reach 83,161 entries at stage 11 and unfiltered uniform grids 64,879 at stage 6, after which both hit the cap. Filtered uniform grids also plateau (about 1,100 entries): filtering, not the grading, removes the dependence on the accuracy. This is predicted by (G1)–(G4) of Lemma 5.9, which also hold for $\theta = 0$, so the retained radius is still at most $4.2\sqrt{n\kappa}h_j$ and the next uniform grid has at most $16.8\sqrt{n\kappa} + 3$ nodes per continuous coordinate. In all 876 stages of the filtered graded runs of E1 and the runs of E2 and E3 with the theorem’s grading (all with $8\kappa\theta^2 \leq 1$), the gap $D(y_j)$, the retained radius and the number of nodes per coordinate were at most 0.28, 0.18 and 0.05 times their bounds (G3)–(G5) in Lemma 5.9 (Appendix H).

E2 and E3: conditioning and dimension. Figure 2 varies κ on the path with $n = 16$, in single trials of 12 stages. With the theorem’s grading the plateau grows from 9 to 49 nodes per coordinate, and the retained radius grows more slowly than $\sqrt{\kappa}$ and stays far below its bound. With the fixed grading $\theta = 1/4$ the condition $8\kappa\theta^2 \leq 1$ fails for every instance with $\kappa_{\text{target}} \geq 4$; the trial still contracted for $\kappa_{\text{target}} \leq 32$, but for $\kappa_{\text{target}} \geq 64$ the last-stage gap divided by $Ln h_j^2$ was 4.7, 15 and 61. CT with $\varepsilon = 2^{-20}$ handled this automatically: it succeeded in its first trial ($\theta = 1/4$) for $\kappa_{\text{target}} \leq 32$ and restarted once or twice for larger κ_{target} , earlier than Lemma 5.9 guarantees. For $\kappa_{\text{target}} = 2$ the generator sets the couplings between free coordinates to zero ($H_{J_0 J_0} = 2I$); the free coordinates are then separable, and the initial coordinate descent already returns x^* . Experiment E3 varies n on paths with $n = 4, 8, \dots, 128$ and $\kappa_{\text{target}} = 4$ ($\kappa \in [3.99, 4.45]$), in single trials of 11 stages. The plateau, measured as in Figure 2 (left), grew from 7 to 26 nodes per free coordinate for filtered uniform grids, from 9 to 19 with the theorem’s grading $\theta = 1/8$, and from 9 to 17 with $\theta = 1/4$. This range of n is too small to separate $\log n$ from $\sqrt{n}$; Corollary 5.11 shows that uniform grids need order $\sqrt{n\kappa}$ nodes on some instances.

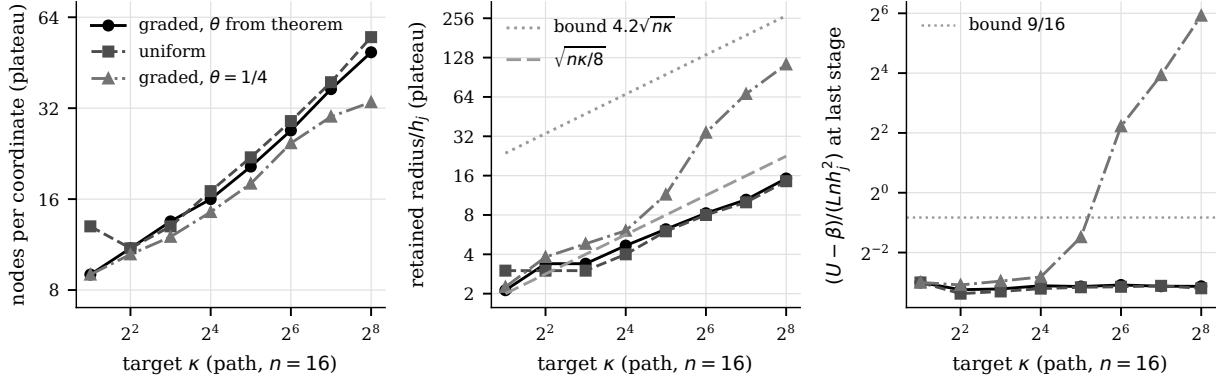


Figure 2: Experiment E2: path with $n = 16$ and $\kappa_{\text{target}} = 2, 4, \dots, 256$; the certified κ lies in $[0.97 \kappa_{\text{target}}, 1.12 \kappa_{\text{target}}]$. Left: nodes per coordinate, the maximum over free coordinates and the median over the last four stages and the three seeds. Middle: retained radius over h_j , same statistics, with the bound $4.2\sqrt{n\kappa}$ of Lemma 5.9 and the reference $\sqrt{n\kappa}/8$, both evaluated at the lower end of the certified κ interval. Right: gap $U - \beta$ at the last stage over Lnh_j^2 , with the bound $9/16$.

11.3 The expanding-box chain

Table 3 in Appendix H reports CT on the family Ψ_m of Proposition 10.7 (Example 5.13) for $m = 2, 4, 8, \dots, 64$ and target gap 10^{-3} . The exact Bellman message for ξ_m would need at least 2^{m-1} quadratic pieces, but the largest coordinate grid grows only from 5 to 11 nodes as n grows from 4 to 128, and the number of stages grows linearly in m , from 9 to 74, because the largest box width, $2^m - 1$, enters the stage count logarithmically. All runs succeeded in the first trial ($\theta = 1/4$), although $8\kappa\theta^2 \leq 1$ fails: $\Psi_m = 1$ at the point with $z_1 = 1$ and all other coordinates 0, so every growth constant is at most 1 and $\kappa \geq 10$. The minimizer 0 is the lower corner of the box, the solver's first descent start. Started at the upper corner instead, the runs used the same numbers of stages, the same largest grid (except for $m = 2$: 6 instead of 5 nodes) and between 2% fewer and 8% more table entries.

11.4 Exact output and localized acceptance

Experiment E4 ran EX on twenty small instances: twelve random unplanted mixed-integer QPs ($n = 3, \dots, 6$, one or two integer coordinates), three with two isolated minimizers, three planted instances ($n = 6$) and two with segments of minimizers, with limits of 3,000 stages and 30 seconds (5 seconds for the last two). An independent enumeration of integer assignments and continuous faces gives the reference optima. Exact output was certified for 18 instances, all with the correct value and an optimal point. Three of them ($n = 3$) have no continuous coordinate with positive curvature and needed at most one stage. Because our implementation of EX starts at $q = 4$, doubles q after each round and restarts the grid solver in every round, the number of stages of the other 15 is close to the precision q of the last round, a power of two near the required number of bits $\log_2(\Omega'W)$, where W is the denominator of the optimal value and $\Omega' \geq \Omega$ the implementation's height constant: 67 to 534 stages for 49 to 342 bits (Appendix H); with $\Omega' = \Omega$ from (6.1), 21 to 315 bits would suffice. The two instances with segments of minimizers stopped at the time limit with certified gaps of about 2^{-11} , against about 2^{-53} required. They have set growth (Remark 6.10) but no point growth, so Theorem 6.14 guarantees only termination; their objective contains $a(x_0 - x_1)^2$, a multiple of the instance of Proposition 10.8 with $n = 2$, on which every CT run that certifies

accuracy ε ends with at least $1 + 1/(2\sqrt{\varepsilon})$ nodes per coordinate.

S1: localized acceptance. The supplementary experiment S1 tests the localized acceptance rule of Proposition 6.17 on 30 further random instances generated as in E4, with $n = 4, 6, 8, 12, 16$. After every stage of one replayed CT run with $\varepsilon = 2^{-60}$, the test was applied to the face candidate of Definition 6.18, which Corollary 6.19 covers, and to the stationary point of the face of the stage incumbent. Both accepted the same 29 instances, the face candidate within nine stages and the incumbent rule within five; on 12 of them the incumbent rule passes even with X in place of the narrowed box, that is, without the filtering history. The accepted values agreed with those of EX on all 29 instances, and on the 12 instances with $n \leq 6$ also with the enumeration. The one failure has two optimal vertices that differ in a concave coordinate, a tie outside the scope of Corollary 6.19. EX, as implemented, used up to 542 stages on these instances. An optimal candidate passes the test of Proposition 6.6 once the certified gap is below $1/(\Omega'W)$; on the five instances on which EX used the most stages, one CT run reached this threshold after 40 to 72 stages with $\Omega' = \Omega$ and 139 to 157 with the implementation’s constant.

11.5 Comparison with a global solver

Experiment E5 compares CT with SCIP 10.0 [98, 179] on 21 planted instances with $\kappa_{\text{target}} = 4$ ($\kappa \in [3.99, 4.45]$), with $n = 8$ to 128 and bag sizes 2 to 4. SCIP ran single-threaded on the epigraph formulation $\min\{t : t \geq F(x)\}$, with the coefficients rounded to double precision, absolute gap 10^{-6} and a 20-second limit. CT used $\varepsilon = 10^{-6}$ and a 60-second limit; it reached a replayed gap below 10^{-6} on all 21 instances in its first trial, in at most 6.9 seconds. SCIP reported its gap as reached on the 12 instances with $n \leq 16$ and stopped at the time limit on all nine paths with $n \geq 32$ (Table 4 in Appendix H). Its dual bounds were valid, but its reported interval did not contain the optimum in any run: every reported primal bound lay below the optimum, by $1.8 \cdot 10^{-6}$ to $1.6 \cdot 10^{-5}$. SCIP’s incumbents lay slightly outside the box: every coordinate that is at a bound in x^* violated that bound by about 10^{-8} , which SCIP’s feasibility tolerance permits. Moving such a coordinate outward decreases F at the rate of its inward derivative, and these violations account for 51% to 94% of the shortfall; a violation of the epigraph constraint by $9 \cdot 10^{-7}$ accounts for the rest. Projected onto the box, the incumbents were optimal to within $1.3 \cdot 10^{-14}$, and on the instances with $n \leq 16$ their exact gap to the dual bound was $2.8 \cdot 10^{-6}$ to $6.8 \cdot 10^{-6}$, so the requested 10^{-6} was not reached. This is the usual behavior of floating-point solvers, and rigorous bounds need safe computations [149] or an exact check. A longer time limit, tighter tolerances or an objective offset did not change these outcomes (Appendix H). The comparison uses one formulation, default settings and a family that favors decompositions of small width; it is not a performance ranking.

11.6 Random instances without planted structure

Experiment E6 uses 40 unplanted random continuous box QPs on $[-1, 1]^n$ with path or band sparsity (bag sizes 2 and 3), $n = 8, 12, 16, 24$ and five seeds each. All Hessians are indefinite, and nothing about the minimizer or the conditioning is prescribed. CT was run at $\varepsilon = 2^{-10}, 2^{-20}, \dots, 2^{-50}$ and reached every target in its first trial ($\theta = 1/4$). The largest grid had 8 to 18 nodes per coordinate. From $\varepsilon = 2^{-10}$ to 2^{-50} it stayed the same on 27 of the 40 instances and changed by at most 5 nodes on the others, while the number of stages grew linearly in $\log(1/\varepsilon)$, from 7–9 to 27–29 (Table 5 in Appendix H). At the stationary point $\hat{x}$ of the face of the final incumbent, Lemma 3.3(a) certified point growth on 11 instances, with certified $\kappa \geq 4.5$; there $8\kappa\theta^2 > 1$ for $\theta = 1/4$, so the first trial succeeded although Lemma 5.9 does not guarantee it. On the other 29 instances growth and κ

remain unknown; on 28 of them the localized test of Proposition 6.17 proved a candidate optimal, and on the remaining one no candidate was proved optimal. For $n = 8$ all candidates agree with the enumeration.

11.7 Recourse

Six designed instances compare plain corrected grids (CT) with the recourse pipeline at target gap 2^{-10} and a 20-second limit (Table 6 in Appendix H). On two affine stars ($n = 5$ and 17), recognition of the globally affine responses and an exact solve of the reduced problem took at most 0.07 seconds, while plain grids stopped at the time limit with gaps 0.019 and 1.23. On the piecewise instances $\phi_M - z^2$ of Example C.11, the certified convex value factor removed the dependence on M , whereas plain grids needed 498 table entries for $M = 1$ and 35,600 for $M = 100$. On a dense instance with $n = 33$ whose residual is concave in each coordinate and submodular, the minimum-cut oracle of Section 7.4 certified the target gap in 0.08 seconds; plain grids, with a decomposition of bag size 33, stopped at the table limit before the first stage. All certificates were replayed; the checkers of the recourse pipeline reuse the solver’s grid, table and lifting code and are therefore not independent in the sense of Section 11.1. These are designed examples of the structures in Section 7, not hard instances.

11.8 Limits

The cost of the method is exponential in the bag size, and exact rational arithmetic dominates the running time; the band with bag size four already takes several seconds. On two binary QPLIB instances [71] the first grid is exact, so the method is plain dynamic programming over the decomposition. QPLIB_3852 (231 variables, minimum-fill bag size 20) was solved exactly in one stage, with $5.7 \cdot 10^6$ table entries, 156 seconds and 4.4 GB of memory; the certified bounds coincide at -234 , the library’s value after the sign change to minimization. We did not replay this certificate, because the reference checker is too slow at this size. For QPLIB_5881 (120 variables, bag size 94) the first stage would need about $8 \cdot 10^{28}$ table entries. Large width, not the accuracy, is the binding limit; Section 10 shows that this is unavoidable in the worst case.

12 Conclusions and open problems

Small treewidth alone does not make sparse mixed-integer problems tractable. Upper bounds on the coordinate curvature turn finite dynamic programming into a valid lower bound and its min-marginals into a safe filter, and quadratic growth keeps the filtered grids at a size independent of the accuracy. For mixed-integer box quadratic programs with a tree decomposition of maximum bag size p , CT (Algorithm 5.6) certifies accuracy 2^{-q} in $f(p, \bar{\kappa})(I + q + 1)^5$ bit operations under weighted growth, and EX (Algorithm 6.13) returns an exact minimizer in $f_1(p, \kappa)(I + 1)^{C_1}$ bit operations under point growth, without knowledge of the growth constant; the certificates are checked by recomputation in exact arithmetic and are valid without any growth assumption. By Theorem 1.2, neither parameter can be dropped unless $P = NP$, and under rETH the exponent of κ in a bound $f(p)\kappa^{a(p)} \text{poly}(I)$ cannot be $o(p)$.

A corrected-grid bound on a subbox is a valid node bound for spatial branch-and-bound, and the min-marginal filter is certified bound tightening; whether either pays off inside a solver is untested. A replayable certificate also turns a floating-point incumbent into a certified one: the exact value of its projection onto the box and a grid certificate give a rigorous gap, which matters because floating-point solvers satisfy bounds and constraints only within tolerances (Section 11). In the

experiments, grid sizes stay nearly constant as the accuracy grows, also on random instances on which growth could not be certified, and the width of the available decomposition, not the accuracy, is the binding limit.

Several questions remain open.

- (1) *Negative curvature.* Let g be a point-growth constant, $\nu = \max\{0, -\lambda_{\min}(\nabla^2 F)\}$ and $\kappa_\nu = \max\{1, \nu/g\}$. Can continuous box QP be solved to accuracy 2^{-q} in time $f'(p, \kappa_\nu) \text{poly}(I + q)$ for a computable function f' ? Theorem 7.9 and Corollary C.10 (Appendix C) give such a bound, with p the bag size after elimination, when certified responses of private convex blocks reduce every coordinate curvature to at most $C_0\nu$ for a constant C_0 . Proposition 7.11 shows that recourse alone cannot ensure this, and Proposition G.5 (Appendix G.4) that matching local moments of any finite order is not enough. These obstructions suggest measuring curvature only where near-optimal points can lie, or using inequalities that couple both sides of a separator.
- (2) *Constants in the exponent.* Under weighted growth, each bag table of CT has at most $(C\sqrt{\kappa} \log(n+2))^p$ entries. The factor p^p in $f(p, t) = c_0(c_1 p \sqrt{t})^p t(1 + \log_2 t)^2$ comes only from moving the logarithmic factor into the polynomial part (inequality (5.4)); the value-oracle lower bound of Proposition 10.3 has no such factor. Can the logarithmic factor, and with it p^p , be avoided, and is the exponent $p/2$ of κ optimal for explicit inputs, as it is for value oracles?
- (3) *Coupling constraints.* Is there a feasible rounding scheme for totally unimodular, or more general, sparse constraints that works with graded grids? Curvature-only corrections on misaligned grids and on a common graded pattern give invalid bounds (Proposition E.11 and Example E.12 in Appendix E), so a different correction is needed. A partial step, which we have not worked out, would keep the aligned uniform mesh only on coordinates that occur in a constraint and use graded grids on the others; this should replace the factor $n_c^{p/2}$ of Theorem 8.9 by $n^{p_c/2}$, where p_c is the largest number of constrained continuous coordinates in a bag.
- (4) *Optimal sets.* Suppose that the optimal set is finite and that r bounds the number of optimal values per coordinate. Can exact output then be obtained in $f'(p, \kappa_S, r) \text{poly}(I)$ bit operations for a computable function f' ? For optimal sets with continuous components, the grids produced by filtering can need a number of nodes that grows with the accuracy (Proposition 10.8); which compact descriptions beyond the classes of Section 9 can be computed with a decomposition?
- (5) *Exact polynomial output.* For continuous polynomial objectives the minimizer can be irrational. Certificates that keep correlations between coordinates, rather than signs over coordinate rectangles, seem necessary under weak complementarity (Proposition B.6 in Appendix B.2); no construction of parameterized size is known.

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A Proofs for Section 5

This appendix proves Lemma 5.2, the bound $K(\theta, n_P)$ in Lemma 5.5, the bit-length claims in the proof of Theorem 5.7, Lemma 5.9, Proposition 5.10, Corollary 5.11, and the claims of Examples 5.12 and 5.13.

Proof of Lemma 5.2. (a) Consider the side towards u'_i ; the other is symmetric. The grid interval between the nodes at distances $t_k < t_{k+1}$ has length at most $\sigma(t_k)$. For $i \in I_C$ this is $h + \theta t_k$. For $i \in I_Z$ and a length of at least two, $\sigma(t_k) \geq 2$, so $\sigma(t_k) = \lfloor h + \theta t_k \rfloor \leq h + \theta t_k$; intervals of length one are ignored in w_i . The node at distance t_k has adjacent intervals starting at distances t_{k-1} and t_k , both at most $t_k = |v - c_i|$; the intervals adjacent to the center have length at most h .

(b) Let $t_1 < \dots < t_m = R$ be the nodes on that side, and $t_0 = 0$. Every step except the last is unclipped, and an unclipped step from t has length $\sigma(t) \geq (\hat{h} + \theta t)/3$: for $i \in I_C$ trivially; for $i \in I_Z$ with $h \geq 1$ because $\lfloor y \rfloor \geq y/2$ for $y \geq 1$; for $i \in I_Z$ with $h < 1$, either $\theta t < 2$ and $\sigma \geq 1 > (1 + \theta t)/3$, or $\theta t \geq 2$ and $\lfloor h + \theta t \rfloor \geq \theta t/2 \geq (1 + \theta t)/3$. Hence $t_{k+1} + \hat{h}/\theta \geq (1 + \theta/3)(t_k + \hat{h}/\theta)$ for $k \leq m - 2$, and $(1 + \theta/3)^{m-1} \leq (t_{m-1} + \hat{h}/\theta)\theta/\hat{h} < 1 + \theta R/\hat{h}$. So $m - 1 < \ln(1 + \theta R/\hat{h})/\ln(1 + \theta/3)$, and $\ln(1 + x) \geq \frac{23}{24}x$ for $0 \leq x = \theta/3 \leq \frac{1}{12}$ gives $m \leq \lceil \frac{72}{23\theta} \ln(1 + \theta R/\hat{h}) \rceil$, which implies the bound.

(c) Each side has length $R \leq h$, and $\sigma(0) \geq R$ (for $i \in I_Z$, R is an integer and $\lfloor h \rfloor \geq R$), so the first step reaches the endpoint. $\square$

Proof of the bound $K(\theta, n_P)$ in Lemma 5.5. The proof of Lemma 5.5 bounds every grid after stage 0 by $1 + 2^{\lceil \frac{4}{\theta} \ln(\frac{5}{4} + 4\sqrt{2n_P}) \rceil}$ nodes. Since $\lceil x \rceil < x + 1$ and $3 \leq 3/(4\theta)$, this is at most $\theta^{-1}\varphi(n_P)$, where $\varphi(m) = \frac{3}{4} + 8 \ln(\frac{5}{4} + 4\sqrt{2m})$. Now $\varphi(1) < 16.3 \leq 10 \lceil \log_2 3 \rceil$, $\varphi(2) < 18.6 \leq 10 \log_2 4$, and for $m \geq 2$, $\varphi'(m) \leq 4/m$ while $\frac{d}{dm} 10 \log_2(m + 2) \geq 7.2/m$. Hence $\varphi(m) \leq 10 \log_2(m + 2)$ for $m \geq 2$, and $\varphi(m) \leq 10 \lceil \log_2(m + 2) \rceil$ for all $m \geq 1$, so the bound is at most $K(\theta, n_P)$. $\square$

Bit lengths in Theorem 5.7. Let Γ_X be the product of the denominators of the box endpoints and Γ_F that of the coefficients, after substituting fixed coordinates; both have $O(I)$ bits, and $|\varpi_i|, |E| = O(I)$. Fix a trial μ and let $\alpha = j_{\max} + \max_i |\varpi_i| + |E| + \mu K_\mu$. Every node of every grid of the trial has a denominator dividing $\Gamma_X 2^\alpha$. Indeed, the initial center and the original endpoints have denominators dividing Γ_X , and a new node is a center coordinate plus or minus

$$h_{ij} \frac{(1 + \theta)^k - 1}{\theta} = 2^{E-j-\varpi_i} \frac{(2^\mu + 1)^k - 2^{\mu k}}{2^{\mu(k-1)}}, \quad k \leq K_\mu,$$

or an integer step for $i \in I_Z$, or a clipped endpoint of the current box; centers and box endpoints are nodes of earlier stages of the same trial or original data. So the bound does not compound from stage to stage; it depends on the stage limit only through the term $j_{\max}$ in α . Nodes lie in $\bar{X}$, so their numerators have $O(I + \alpha)$ bits. Every factor value, correction $L_i w^2/8$, table entry and message has a denominator dividing $8\Gamma_F \Gamma_X^2 4^\alpha$ and is a sum of at most $I + n$ terms, each a coefficient times a product of at most two nodes or of two differences of nodes (a message is the value of such a sum at a minimizing assignment). Over this common denominator every number is an integer of $b = O(I + \alpha)$ bits, the dynamic program uses only integer additions and comparisons, and a factor evaluation costs $O(b^2)$ bit operations per monomial. Since $j_{\max} + 1 \leq c(I + q + 1)$ and $\lceil \log_2(n_P + 2) \rceil \leq I + 1$, we have $\mu K_\mu \leq 10\mu 2^\mu(I + 1)$ and $b \leq c(1 + \mu 2^\mu)(I + q + 1)$. With $2^{\mu^*} \leq 6\sqrt{\kappa}$

and $\mu^* \leq 3(1 + \log_2 \bar{\kappa})$, the table work of Theorem 5.7 times b^2 gives the stated $f(p, \bar{\kappa})(I + q + 1)^5$. The certificate stores the grids of one trial and is checked by the same dynamic programs. Without growth the schedule still halts, and every number it forms is bounded as above for its trial.

Proof of Lemma 5.9. We follow the proofs of Lemmas 5.4 and 5.5 with $a_j = Lnh_j^2$ and $\|\cdot\|_{\mathbf{L}}^2$ replaced by $L\|\cdot\|^2$. Write $h = h_j$ and $\rho = \sqrt{n\kappa}$, and note that $L \leq \kappa g$ and $L\theta^2/2 \leq L/(16\kappa) \leq g/16$. Point growth makes x^* the unique minimizer, so $x^* \in X^{(j)}$ and $\beta_j \leq F^*$ follow as in Lemma 5.4(i).

(G1) For $i \in P$, Lemma 5.2(a) with mesh h and $L_i \leq L$ give $d_i(z_i) \leq \frac{L_i}{8}(h + \theta|z_i - c_i|)^2 \leq \frac{L}{4}h^2 + \frac{L\theta^2}{4}(z_i - c_i)^2$, and $(z_i - c_i)^2 \leq 2(z_i - x_i^*)^2 + 2(c_i - x_i^*)^2$; for $i \notin P$, $d_i = 0$. Summing over $i \in P$ and using $n_P \leq n$ gives (G1).

(G2) and (G3). We first show by induction on j that $\|c - x^*\|^2 \leq 4\kappa nh^2$. For $j = 0$ and any $c^{(0)} \in X$, $\|c^{(0)} - x^*\|^2 \leq \sum_i s_i^2 \leq ns^2 = nh_0^2$; no other step uses the choice of $c^{(0)}$. Let $Y = \|y_j - x^*\|^2$. Point growth, $\beta_j \leq F^*$, $Q(y_j) = \beta_j$, (G1) and the bound on $\|c - x^*\|^2$ give

$$gY \leq F(y_j) - F^* \leq F(y_j) - \beta_j = D(y_j) \leq \frac{L}{4}nh^2 + \frac{g}{16}(Y + 4\kappa nh^2),$$

so, with $L \leq \kappa g$, $Y \leq \frac{16}{15}(\frac{1}{4} + \frac{1}{4})\kappa nh^2 = \frac{8}{15}\kappa nh^2$. The center of stage $j + 1$ is y_j and $h_{j+1} = h/2$, so $\|c^{(j+1)} - x^*\|^2 \leq \frac{32}{15}\kappa nh_{j+1}^2 \leq 4\kappa nh_{j+1}^2$, which completes the induction and proves the first two bounds of (G2). By (G1),

$$D(y_j) \leq \frac{L}{4}nh^2 + \frac{L}{16\kappa}\left(\frac{8}{15} + 4\right)\kappa nh^2 = \frac{8}{15}Lnh^2 \leq \frac{9}{16}Lnh^2.$$

Since $U \leq F(y_j)$ after step (ii) and $F^* \geq \beta_j$, this proves (G3). For a grid point z with $Q(z) \leq U$ let $Z = \|z - x^*\|^2$; then

$$gZ \leq F(z) - F^* = Q(z) - F^* + D(z) \leq U - \beta_j + D(z) \leq D(y_j) + D(z) \leq \frac{13}{16}Lnh^2 + \frac{g}{16}(Z + 4\kappa nh^2),$$

so $Z \leq \frac{16}{15}(\frac{13}{16} + \frac{4}{16})\kappa nh^2 = \frac{17}{15}\kappa nh^2$, the last bound of (G2).

(G4) Let $i \in P$, let $[a, a']$ be a grid interval of coordinate i retained at stage j , let $v \in \{a, a'\}$ satisfy $m_i(v) \leq U$, and let the grid point z attain $m_i(v)$, so $z_i = v$ and $Q(z) \leq U$. By (G2), $|v - x_i^*| \leq \sqrt{17/15}\rho h$, $|(y_j)_i - x_i^*| \leq \rho h$ and $|c_i - x_i^*| \leq 2\rho h$; hence $|v - (y_j)_i| \leq (1 + \sqrt{17/15})\rho h$ and $|v - c_i| \leq (2 + \sqrt{17/15})\rho h$. An integer interval of length one lies within $|v - (y_j)_i| + 1 < 4.2\rho h + 1$ of $(y_j)_i$. Otherwise the interval has length at most $w_i(v) \leq h + \theta|v - c_i|$ by Lemma 5.2(a), and $h \leq \rho h$ and $\theta\rho = \theta\sqrt{\kappa}\sqrt{n} \leq \rho/\sqrt{8}$ give that each of its points is within

$$(1 + \sqrt{17/15})\rho h + \rho h + \frac{2 + \sqrt{17/15}}{\sqrt{8}}\rho h = (2 + \sqrt{17/15})\left(1 + \frac{1}{\sqrt{8}}\right)\rho h < 4.15\rho h$$

of $(y_j)_i$. So every retained interval, and hence their hull $X_i^{(j+1)}$, lies within $4.2\rho h + [i \in I_Z]$ of $(y_j)_i$.

(G5) At stage 0, $h_0 = s \geq s_i$, so every grid has at most three nodes by Lemma 5.2(c), and $3 < 8\theta^{-1}$; coordinates outside P have two nodes. At stage $j + 1$ the grid of $i \in P$ is centered at $(y_j)_i \in X_i^{(j+1)}$ (Proposition 4.5) with mesh $h/2$, and by (G4) each side has length at most $R = 4.2\rho h + [i \in I_Z]$. With $\hat{h}$ as in Lemma 5.2(b) for mesh $h/2$, $\theta R/\hat{h} \leq 8.4\theta\rho + \theta \leq 8.4\sqrt{n}/\sqrt{8} + \frac{1}{4} < 3\sqrt{n} + \frac{1}{4}$, so the grid has at most $1 + 2\lceil \frac{4}{\theta} \ln(\frac{5}{4} + 3\sqrt{n}) \rceil \leq \theta^{-1}\psi(n)$ nodes, with $\psi(m) = \frac{3}{4} + 8\ln(\frac{5}{4} + 3\sqrt{m})$, using $3 \leq 3/(4\theta)$. Now $\psi(1) < 12.4 \leq 8\lceil \log_2 3 \rceil$, $\psi(2) < 14.4 \leq 8\log_2 4$, and for $m \geq 2$, $\psi'(m) \leq 4/m$ while $\frac{d}{dm} 8\log_2(m+2) \geq 5.7/m$. Hence $\psi(m) \leq 8\lceil \log_2(m+2) \rceil$ for all $m \geq 1$.

CT with a common mesh. If $P = \emptyset$ it is CT (Algorithm 5.6). Otherwise every recorded path is a valid certificate by Corollary 4.8, and a successful stage gives $F(\hat{x}) = U \leq \beta_j + \varepsilon$. For termination,

take $\mu \geq 2$ with $\theta = 2^{-\mu} \leq h_{j_{\max}}/s$, so that $\theta s_i \leq h_j$ for all $j \leq j_{\max}$ and i . Unclipped steps are at least h_j for $i \in I_C$, and at least $\max\{1, \lfloor h_j \rfloor\} \geq h_j/2$ for $i \in I_Z$ with $h_j \geq 1$; so a side of length at most $s_i \leq h_j/\theta$ takes at most $2/\theta$ steps. For $i \in I_Z$ with $h_j < 1$ the grid has at most $s_i + 1 < 1/\theta + 1$ nodes. So every grid has at most $1 + 4/\theta < 8\theta^{-1} \lceil \log_2(n+2) \rceil$ nodes, and the trial is not aborted. At stage $j_{\max}$, Lemma 5.2(a) gives $w_i(v) \leq h_{j_{\max}} + \theta s_i \leq 2h_{j_{\max}}$ for every node v , so $U - \beta_{j_{\max}} \leq D(y_{j_{\max}}) \leq \sum_{i \in P} L_i h_{j_{\max}}^2/2 \leq \frac{1}{2} L n h_{j_{\max}}^2 \leq \frac{8}{9} \varepsilon$, and the trial succeeds. Under point growth, the trial $\mu = \max\{2, \lceil \log_4(8\kappa) \rceil\}$ has $8\kappa\theta^2 \leq 1$; by (G5) it is not aborted, and by (G3) it succeeds at stage $j_{\max}$ at the latest, because $\frac{9}{16} L n h_{j_{\max}}^2 = \frac{9}{16} L n s^2 4^{-j_{\max}} \leq \varepsilon$. $\square$

Proof of Proposition 5.10. The Hessian is block diagonal with blocks $\begin{pmatrix} \Lambda & \Lambda-2g \\ \Lambda-2g & \Lambda \end{pmatrix}$, whose eigenvalues $2\Lambda - 2g$ and $2g$ are positive. Hence $F(x) \geq g\|x\|^2$ with equality along $u_k = -v_k$, so $x^* = 0$, $F^* = 0$ and g is the largest growth constant. Each diagonal entry is Λ . As g decreases from $\Lambda/2$ to 0, $\kappa = \Lambda/g$ increases continuously from 2 to ∞ .

The corrected objective separates over pairs, $Q = \sum_k Q_k(u_k, v_k)$ with $Q_k(u, v) = \frac{\Lambda}{2}(u^2 + v^2) + (\Lambda - 2g)uv - \frac{\Lambda}{8}(w(u)^2 + w(v)^2)$, where w denotes w_i in the relevant coordinate. Therefore the min-marginal of u_1 is $m_{u_1}(a) = \min_v Q_1(a, v) + \sum_{k \geq 2} \min Q_k$, the minima taken over the grids. By hypothesis $w(0) \geq h$ in every coordinate, so $\min Q_k \leq Q_k(0, 0) \leq -\Lambda h^2/4$ for each $k \geq 2$, and $\sum_{k \geq 2} \min Q_k \leq -(m-1)\Lambda h^2/4 = -(n-2)\Lambda h^2/8$.

Fix a node a with $|a| \leq R_0$ and put $t^* = -(\Lambda - 2g)a/\Lambda$; then $|t^*| \leq |a| \leq R_0$, so t^* lies in the box of v_1 and in some grid interval of G_{v_1} , of length λ say. Let v_a be the endpoint of that interval nearest to t^* . Then $|v_a - t^*| \leq \lambda/2$ and $w(v_a) \geq \lambda$. Completing the square, $\frac{\Lambda}{2}v^2 + (\Lambda - 2g)av = \frac{\Lambda}{2}(v - t^*)^2 - \frac{(\Lambda - 2g)^2 a^2}{2\Lambda}$, so

$$Q_1(a, v_a) \leq \frac{\Lambda^2 - (\Lambda - 2g)^2}{2\Lambda} a^2 + \frac{\Lambda}{2} \left(\frac{\lambda}{2}\right)^2 - \frac{\Lambda}{8} \lambda^2 = \frac{2g(\Lambda - g)}{\Lambda} a^2,$$

where we also dropped the nonpositive term $-\frac{\Lambda}{8}w(a)^2$. Thus

$$m_{u_1}(a) \leq \frac{2g(\Lambda - g)}{\Lambda} a^2 - \frac{(n-2)\Lambda h^2}{8},$$

which is at most 0 $\leq U$ when $a^2 \leq (n-2)\Lambda^2 h^2/(16g(\Lambda - g)) = (n-2)h^2\kappa^2/(16(\kappa - 1))$. Because $\kappa^2/(\kappa - 1) \geq \kappa$, the hypothesis $a^2 \leq (n-2)\kappa h^2/16$ suffices. The coordinates v_k and the other pairs are symmetric. An interval with an endpoint a satisfying $m_i(a) \leq U$ passes the filtering test (4.4). $\square$

Proof of Corollary 5.11. Uniform rule, center. Suppose a stage is centered at 0 with box $X' \ni 0$. Its grid in coordinate i consists of the multiples of h_j in X'_i and the endpoints of X'_i . All grid intervals have length at most h_j , and $w_i(0)$ is at least the length of every interval: if some interval is not adjacent to 0, the side of 0 that contains it is longer than h_j , so the interval between 0 and $\pm h_j$ on that side has length h_j . Hence $w_i(v) \leq w_i(0)$ for all nodes v , so $D(x) \leq D(0)$ and, for every grid point $x \neq 0$, $Q(x) - Q(0) \geq F(x) - F(0) \geq g\|x\|^2 > 0$. The corrected minimizer is 0, and so is the next center; by induction every stage is centered at 0.

Uniform rule, filtered box. Put $\rho_j = \min\{1, 2(k_0 + 1)h_j\}$; we show by induction that $X^{(j)} \supseteq [-\rho_j, \rho_j]^n$. This holds for $j = 0$, since $X^{(0)} = [-1, 1]^n$. At stage 0 ($h_0 = 2$) the grid is $\{-1, 0, 1\}$ in every coordinate, with $w_i(0) = 1$; by Proposition 5.10 with $h = 1$ and $R_0 = 1$ the node 0 passes, so both intervals are retained and $X^{(1)} = [-1, 1]^n \supseteq [-\rho_1, \rho_1]^n$. Let $j \geq 1$. Then $h_j \leq 1$, hence $h_j \leq \rho_j$, so $\pm h_j$ are nodes and $w_i(0) = h_j$. Proposition 5.10 with $h = h_j$ and $R_0 = \rho_j$ retains every interval with an endpoint kh_j , where $|k| \leq k_0$ and $|k|h_j \leq \rho_j$. If $(k_0 + 1)h_j \leq 1$,

then $(k_0 + 1)h_j \leq \rho_j$, and the retained intervals $[kh_j, (k + 1)h_j]$, $-k_0 - 1 \leq k \leq k_0$, give $X^{(j+1)} \supseteq [-(k_0 + 1)h_j, (k_0 + 1)h_j]^n = [-\rho_{j+1}, \rho_{j+1}]^n$. If $(k_0 + 1)h_j > 1$, then $\rho_j = 1$ and $X^{(j)} = [-1, 1]^n$; moreover $1/h_j$ is an integer with $1/h_j < k_0 + 1$, so the nodes $\pm 1 = \pm(1/h_j)h_j$ pass, the intervals $[1 - h_j, 1]$ and $[-1, -1 + h_j]$ are retained, and $X^{(j+1)} = [-1, 1]^n \supseteq [-\rho_{j+1}, \rho_{j+1}]^n$. This completes the induction. When $(k_0 + 1)h_j \leq 1$ and stage $j + 1$ is run, its grid is centered at 0 with spacing $h_j/2$ and contains every multiple of $h_j/2$ in $[-(k_0 + 1)h_j, (k_0 + 1)h_j]$, that is, $4(k_0 + 1) + 1$ nodes; and $k_0 + 1 \geq \sqrt{(n - 2)\kappa/16}$ gives $4k_0 + 5 \geq \sqrt{(n - 2)\kappa} + 1$.

Graded rule. Apply Proposition 5.10 with $h = h_j$ and $R_0 = R$. Every point $t \in [-R, R]$ lies in a grid interval $[a, b]$; if $t \geq 0$ then $0 \leq a \leq R$ (the node 0 lies in the grid), and if $t < 0$ then $-R \leq b \leq 0$, so the interval has an endpoint of absolute value at most R and is retained. Hence the filtered box contains $[-R, R]$ in every coordinate. The next grid starts from its center c' inside this box and uses unclipped steps $h' + \theta t$ with $h' = h_j/2$, so after k steps the distance from c' is at most $h'((1 + \theta)^k - 1)/\theta$. One side of c' has length at least R within the box and must be covered; hence at least $\ln(1 + \theta R/h')/\ln(1 + \theta)$ nodes lie on that side, besides c' . $\square$

Proof of the claims in Example 5.12. (a) Let $d = (1 - u, 1 - v) \in [0, 1]^2$. At $(1, 1)$ the gradient of $u^2 + v^2 - 4uv + \frac{u+v}{4}$ is $(-\frac{7}{4}, -\frac{7}{4})$ and its quadratic part is $d_1^2 + d_2^2 - 4d_1d_2 \geq -(d_1^2 + d_2^2)$, so, using $d_i \geq d_i^2$,

$$u^2 + v^2 - 4uv + \frac{u+v}{4} + \frac{3}{2} \geq \frac{7}{4}(d_1 + d_2) - (d_1^2 + d_2^2) \geq \frac{3}{4}(d_1^2 + d_2^2).$$

With $(\omega - \frac{1}{2})^2 \leq 2(\omega - \frac{u}{2})^2 + \frac{1}{2}d_1^2$ this gives $\phi - \phi^* \geq \frac{3}{4}(d_1^2 + d_2^2) + (\omega - \frac{u}{2})^2 \geq \frac{1}{2}(d_1^2 + d_2^2 + (\omega - \frac{1}{2})^2)$. The coupling is nonnegative and vanishes at x^* ; summing over blocks gives (a). (b) The diagonal Hessian entries are $\frac{5}{2} + \frac{2\deg(b)}{16\Delta_G} \leq \frac{21}{8}$ for u_b and 2 for v_b, ω_b ; with (a), $\kappa \leq \frac{21}{8}/\frac{1}{2}$. (c) At such a point $\bar{x}$ the coupling gradient in u_b has absolute value at most $\frac{2\deg(b)}{16\Delta_G} \leq \frac{1}{8}$. Hence $\partial_{u_b} F \geq \frac{1}{8}$ and $\partial_{v_b} F = \frac{1}{4}$ at $u_b = v_b = 0$, $\partial_{u_b} F \leq -\frac{7}{4} + \frac{1}{8}$ and $\partial_{v_b} F = -\frac{7}{4}$ at $u_b = v_b = 1$, and $\partial_{\omega_b} F = 0$. For feasible x let $d = x - \bar{x}$ and $\delta = \sum_b (|d_{u_b}| + |d_{v_b}|)$. The first-order term is at least $\delta/8$; the second-order term is at least $\sum_b (d_{\omega_b} - \frac{1}{2}d_{u_b})^2 - \delta^2$. Thus $F(x) - F(\bar{x}) \geq \delta/8 - \delta^2 + \sum_b (d_{\omega_b} - \frac{1}{2}d_{u_b})^2 > 0$ for $0 < \delta < \frac{1}{8}$, and $F(x) - F(\bar{x}) = \sum_b d_{\omega_b}^2 > 0$ for $\delta = 0 \neq d$. (d) Attach to a bag containing u_b a new bag $\{u_b, v_b, \omega_b\}$. $\square$

Proof of the claims in Example 5.13. With states $\xi_t \in [0, 2^t - 1]$, controls $z_t \in [0, 1]$ and $\xi_0 = 0$, the objective of Proposition 10.7 is $\Psi_m = \xi_m^2 + \sum_{t=1}^m (\xi_t - 2\xi_{t-1} - z_t)^2 + \frac{1}{8} \sum_{t=1}^m z_t(1 - z_t)$. Its second derivatives along the coordinates are $2 + 8 = 10$ for ξ_t with $t < m$, $2 + 2 = 4$ for ξ_m and $2 - \frac{1}{4} = \frac{7}{4}$ for z_t . Put $\varsigma_t = 2^{-t}\xi_t \in [0, 1 - 2^{-t}]$. The rescaling multiplies the curvature of the t th state by 4^t , so the rescaled curvatures are $10 \cdot 4^t$ for ς_t with $t < m$, 4^{m+1} for ς_m and $\frac{7}{4}$ for z_t , and $L = 4^{m+1}$. Proposition 10.7 shows that $\Psi_m \geq \frac{1}{8}(\|\xi\|^2 + \|z\|^2) \geq \frac{1}{8}(\|\varsigma\|^2 + \|z\|^2)$, since $\xi_t^2 = 4^t \varsigma_t^2$; so $\frac{1}{8}$ is a growth constant of the rescaled instance. The feasible point with $\varsigma_1 = \delta \in (0, \frac{1}{2}]$ and all other coordinates 0 has $\xi_1 = 2\delta$ and $\Psi_m = (2\delta)^2 + (4\delta)^2 = 20\delta^2$ (here $m \geq 2$), so every growth constant g is at most 20 and has $L/g \geq 4^{m+1}/20 = 4^m/5$; the constant $\frac{1}{8}$ gives $L/g = 32 \cdot 4^m$. The weighted condition number is invariant under the rescaling, so it stays at most 80. $\square$

B Exact output: localized acceptance and polynomial boundary optima

This appendix proves Corollary 6.19 and, for the explicit polynomials of Section 6.6, gives a procedure with exact implicit output when the minimizer may be irrational and lie on the boundary (Theorem B.4), together with two propositions showing that this procedure needs the margin term in its cost bound and strict complementarity (Propositions B.5 and B.6).

B.1 Proof of Corollary 6.19

We use the notation of Section 6.5 and Corollary 6.19, and the bounds of Lemma 5.9 for a trial of CT (Algorithm 5.6) with a common mesh with $8\kappa\theta^2 \leq 1$; the center of stage j is $c = c^{(j)}$, and $x^* \in X^{(j)}$.

Proof of Corollary 6.19. Facts about x^ .* By Lemma 3.3(b), $\nabla_{J_0} F(x^*) = 0$ and $H_{J_0 J_0} \succeq 2gI$; in particular $H_{ii} \geq 2g > 0$ for $i \in J_0$, so $J_0 \subseteq P$. Let $i \notin P$. If $\ell_i < x_i^* < u_i$, the two points obtained from x^* by setting coordinate i to ℓ_i and to u_i are feasible and differ from x^* , so their values exceed F^* ; but F is concave along coordinate i , so $F^* = F(x^*)$ is at least the smaller of the two values, a contradiction. So $x_i^* \in \{\ell_i, u_i\}$ for $i \notin P$. The coordinates therefore split into $I_Z \cap P$, the coordinates $i \notin P$ (all at a bound), J_0 and A .

Fix a stage j with $h_j \leq h^*$ that ends by filtering, and put $\omega_j = \sqrt{n\kappa} h_j \leq \omega^*$. Then $\omega_j \leq \frac{1}{2}$ if $I_Z \cap P \neq \emptyset$, $\omega_j \leq \delta_X/6$ if $I_C \cup (I_Z \setminus P) \neq \emptyset$, and $\omega_j \leq \lambda_A/(5\Gamma)$ if $A \neq \emptyset$. By (G2), $|(y_j)_i - x_i^*| \leq \omega_j$ and $|c_i - x_i^*| \leq 2\omega_j$ for every i , and every node v of G_i with $m_i(v) \leq U$ satisfies $|v - x_i^*|^2 \leq \frac{17}{15}\omega_j^2$, because the grid point that attains $m_i(v)$ has $Q \leq U$.

Coordinates $i \notin P$. Here $G_i = \{\ell_i, u_i\}$ and x_i^* is one of the two nodes. The other node v has $|v - x_i^*| = u_i - \ell_i \geq \delta_X \geq 6\omega_j > \sqrt{17/15}\omega_j$, so $m_i(v) > U$. The node x_i^* has $m_i(x_i^*) \leq F(x^*) = F^* \leq U$ by Proposition 4.2(c), because $x^* \in X^{(j)}$. Hence $\tilde{X}_i = \{x_i^*\}$.

Coordinates $i \in I_Z \cap P$. Every node $v \neq x_i^*$ has $|v - x_i^*| \geq 1 > \sqrt{17/15} \cdot \frac{1}{2} \geq \sqrt{17/15}\omega_j$, so $m_i(v) > U$. The center c_i is an integer with $|c_i - x_i^*| \leq 2\omega_j \leq 1$, and $h_j \leq \omega_j \leq \frac{1}{2}$. The grid steps $\max\{1, \lfloor h_j + \theta t \rfloor\}$ at distance $t \leq 5$ from c_i equal one, because $h_j + \theta t \leq \frac{1}{2} + \frac{5}{4} < 2$. So every integer of $X_i^{(j)}$ within distance 6 of c_i is a node. By (G4) and (G2), $X_i^{(j+1)}$ lies within $1 + 4.2\omega_j + \omega_j \leq 3.6$ of x_i^* , hence within 5 of c_i . So every integer of $X_i^{(j+1)}$ is a node, and none lies strictly inside a grid interval. Thus $Z_i \cap X_i^{(j+1)} \subseteq \{x_i^*\}$, and x_i^* belongs to it because $x^* \in X^{(j+1)}$ and $m_i(x_i^*) \leq F^* \leq U$. Hence $\tilde{X}_i = \{x_i^*\}$.

Continuous coordinates $i \in P$. Here $\tilde{X}_i = X_i^{(j+1)}$, which contains x_i^* and, by (G4) and (G2), lies within $4.2\omega_j + \omega_j < \delta_X$ of x_i^* . It has positive length, because X_i does and filtering replaces an interval of positive length by a hull of grid intervals of positive length. So $i \in J_+$, and $\tilde{X}_i$ contains a bound of X_i exactly when $i \in A$, namely the bound x_i^* . Hence $J_+ = J_0 \cup A$.

The face candidate. Since $y_j \in \tilde{X}$, the face candidate has $\tilde{x}_i = (y_j)_i = x_i^*$ for every integer coordinate and every $i \notin P$, and $\tilde{x}_i = x_i^*$ for $i \in A$, while the free set is $J_f = J_0$. The matrix $H_{J_0 J_0} \succeq 2gI$ is nonsingular, and x^* satisfies $\nabla_{J_0} F(x^*) = 0$ and agrees with $\tilde{x}$ outside J_0 , so $\tilde{x} = x^*$.

The test. We have $x^* \in \tilde{X}$ and $F(x^*) = F^* \leq U$. For $i \in J_0$, $\zeta_i = \partial_i F(x^*) = 0$, so case (i), (ii) or (iii) applies with $\mu_i = 0$. For $i \in A$, x_i^* is an endpoint of $\tilde{X}_i$, and since x^* minimizes F along coordinate i in X_i , $\zeta_i \geq 0$ at ℓ_i and $\zeta_i \leq 0$ at u_i ; so case (i) or (ii) applies. The interval $\tilde{X}_i$ lies on one side of x_i^* , so its length is at most $5.2\omega_j$ and $\mu_i \geq \lambda_A/(5.2\omega_j) \geq \frac{5}{5.2}\Gamma \geq \Gamma/2$.

Let $N = H_{J_+ J_+} + 2 \text{diag}(\mu_{J_+})$. If $A = \emptyset$, then $N = H_{J_0 J_0} \succeq 2gI$. If $J_0 = \emptyset$, then $N \succeq (2 \min_{i \in A} \mu_i - \|H_{AA}\|_2)I \succeq 0$. Otherwise $H_{J_0 J_0} \succ 0$, and $N \succeq 0$ if the Schur complement $H_{AA} + 2 \text{diag}(\mu_A) - H_{AJ_0} H_{J_0 J_0}^{-1} H_{J_0 A}$ is positive semidefinite. Its smallest eigenvalue is at least $2 \min_{i \in A} \mu_i - \|H_{AA}\|_2 - \|H_{J_0 A}\|_2^2/(2g) = 2 \min_{i \in A} \mu_i - \Gamma \geq 0$, using $\|H_{J_0 J_0}^{-1}\|_2 \leq 1/(2g)$. All hypotheses of Proposition 6.17 hold.

Stage count. The least j with $h_j = s2^{-j} \leq h^*$ is at most $\max\{0, \lceil \log_2(s/h^*) \rceil\}$. By Corollary 6.4(a), the isolated minimizer x^* and all endpoints lie in $\rho^{-1}\mathbb{Z}$ for an integer $\rho \leq R$ with $\Delta \mid \rho$. A nonzero difference $x_i^* - e$ therefore has absolute value at least $1/\rho$, so $\delta_X \geq 1/R$; and $\partial_i F(x^*) = \sum_k H_{ik} x_k^* + b_i \in (\Delta\rho)^{-1}\mathbb{Z}$, so $\lambda_A \geq 1/(\Delta R)$. If $A \neq \emptyset$, then $L > 0$, hence $L \geq 2/\Delta$ and $1/g \leq \kappa/L \leq \Delta\kappa/2$; with $\|H\|_2 \leq n \max_{ik} |H_{ik}|$ this gives $\log_2 \Gamma \leq \text{poly}(I) + \log_2 \kappa$. Since $\log_2 s$, $\log_2 R$ and $\log_2 \Delta$ are polynomial in I , the least such j is at most $\text{poly}(I) + O(\log \kappa)$. $\square$

B.2 Exact implicit output for polynomial boundary optima

This subsection makes precise what exact output means for the explicit polynomials of Section 6.6 when the minimizer can be irrational and lie on the boundary. Throughout, F is an explicit rational polynomial of fixed degree $d \geq 2$ with factor scopes in the bags, on a mixed box X , with a verified rational bound $L \geq \max\{0, \max_i \sup_{\bar{X}} \partial_{ii} F\}$, for example the largest bound of Lemma 6.20. Fix rationals $C_2, C_3 \geq 1$ with

$$\sum_{j \in I_C} \sup_{\bar{X}} |\partial_{ij} F| \leq C_2, \quad \sum_{j, k \in I_C} \sup_{\bar{X}} |\partial_{ijk} F| \leq C_3 \quad (i \in I_C).$$

Monomialwise interval bounds give such numbers with bit length $\text{poly}(I)$ at fixed d .

In this subsection every coordinate uses the single curvature bound L , that is, $L_i := L$ for all i ; these are valid upper coordinate curvatures. If $L > 0$, then P contains every coordinate, so every coordinate is gridded and filtered, and the localization bound (G4) of Lemma 5.9 applies to every coordinate. If $L = 0$, the first stage uses endpoint grids, is exact and has zero gap. We use TRIAL (Algorithm 5.3) with a common mesh $h_j = s2^{-j}$ as in Lemma 5.9. Under point growth (3.2) with $8\kappa\theta^2 \leq 1$, (G4) places every retained interval of stage j within $4.2\sqrt{n\kappa}h_j \leq 5\sqrt{n\kappa}h_j$ of $(y_j)_i$, plus one for integer coordinates; we use the radius $5\sqrt{n\kappa}h_j$ below.

Three sound tests. The trials are run with one extra filter.

Lemma B.1 (Integer-label filter). *(i) Suppose at some stage every grid interval of an integer coordinate i has length one. Then deleting the nodes v with $m_i(v) > U$ and taking the integer hull of the remaining nodes removes no point with objective at most U , and keeps the incumbent and the corrected minimizer. The stages remain a valid path certificate (Definition 4.6): the second alternative of condition (C1) holds for every grid interval of coordinate i that the step removes. (ii) Assume a unique minimizer x^* , point growth (3.2), and $8\kappa\theta^2 \leq 1$. At every stage with $h_j \leq 1/(10\sqrt{n\kappa})$, part (i) applies to every integer coordinate and fixes it to x_i^* .*

Proof. (i) Since every grid interval of coordinate i has length one, the integers of the current interval are exactly the nodes. For feasible x in the current box with $x_i = v$, Proposition 4.2(c) gives $F(x) \geq m_i(v)$. Hence a deleted node carries no point with value at most U , in particular no minimizer and not the incumbent. The corrected minimizer y survives because $m_i(y_i) \leq Q(y) = \beta \leq F^* \leq U$. A removed grid interval of coordinate i has length one, hence effective width zero, and each of its endpoints outside the new interval is a deleted node v with $m_i(v) > U \geq \beta$; this is the second alternative of (C1).

(ii) If $j = 0$, then $s = h_0 < 1$ and there is no integer coordinate. Otherwise, by (G4) at stage $j - 1$, the current integer interval lies within $1 + 5\sqrt{n\kappa}h_{j-1} = 1 + 10\sqrt{n\kappa}h_j \leq 2$ of the center $c_i = (y_{j-1})_i$. All steps $\max\{1, \lfloor h_j + \theta t \rfloor\}$ with $t \leq 2$ equal one, because $h_j + \theta t \leq \frac{1}{10} + \frac{1}{2} < 1$, so (i) applies. A retained node v has $m_i(v) \leq U$, so the grid point that attains $m_i(v)$ has $Q \leq U$, and (G2) gives $|v - x_i^*|^2 \leq \frac{17}{15}\kappa n h_j^2 < 1$; so $v = x_i^*$. The extra filter keeps x^* and the centers and only shrinks boxes; the proof of Lemma 5.9 uses nothing else about the boxes, so its bounds remain valid for the modified trial. $\square$

Lemma B.2 (Monotone face reduction). *Let X' be a product box with fixed integer coordinates, i a free continuous coordinate whose original bound ℓ_i is an endpoint of X'_i , and suppose $\partial_i F \geq 0$ on X' . Then $\min_{X'} F = \min_{X' \cap \{x_i = \ell_i\}} F$, and if $\partial_i F > 0$ on X' , every minimizer over X' has $x_i = \ell_i$. The same holds for u_i with $\partial_i F \leq 0$. With c the midpoint and r the largest half-width of the free continuous coordinates of X' , $\partial_i F(c) - C_2 r > 0$ certifies $\partial_i F > 0$ on X' , and $\partial_i F(c) + C_2 r < 0$ certifies $\partial_i F < 0$.*

Proof. Replacing x_i by ℓ_i keeps a point of X' in X' and, by the mean value theorem along the segment, does not increase F , and decreases it under a strict sign unless $x_i = \ell_i$. For the test, $|\partial_i F(x) - \partial_i F(c)| \leq \sum_j \sup |\partial_{ij} F| |x_j - c_j| \leq C_2 r$. $\square$

Any other sound sign enclosure may be used as well, for example Bernstein coefficient bounds [76] of each factor derivative on its own scope, which have at most d^p coefficients. Such monotonicity tests are standard in interval global optimization [5, 86, 87]; the termination proof below uses only the midpoint test.

Lemma B.3 (Uniform convexity patch). *Let X' be a box with fixed integer coordinates and some fixed continuous coordinates, J_f its free coordinates, c its midpoint and r the largest free half-width. If $H_{J_f J_f}(c) - (C_3 r + \sigma)I \succ 0$ for some $\sigma > 0$, then F restricted to X' is strongly convex in x_{J_f} and has a unique minimizer over X' .*

Proof. Each entry satisfies $|H_{jk}(x) - H_{jk}(c)| \leq r \sum_l \sup |\partial_{jkl} F|$, so every row sum of $|H_{J_f J_f}(x) - H_{J_f J_f}(c)|$ is at most $C_3 r$. The spectral norm of a symmetric matrix is at most its largest absolute row sum. Hence $H_{J_f J_f}(x) \succ \sigma I$ on X' . $\square$

The procedure. Trial $\mu \geq 2$ runs TRIAL with a common mesh, grading $\theta = 2^{-\mu}$, the cap $8\theta^{-1} \lceil \log_2(n+2) \rceil$ of Lemma 5.9(G5) and the filter of Lemma B.1, for stages $j = 0, 1, 2, \dots$ without a stage limit, aborting if the cap would be exceeded; step (iii) of TRIAL is replaced by the following test. After stage j the trial succeeds if $U = \beta_j$ (a zero certified gap; the incumbent is then an exact rational minimizer). Otherwise it filters, and if every integer coordinate of the retained box $X^{(j+1)}$ is a singleton, it copies $X^{(j+1)}$, applies sound reductions of Lemma B.2 to the copy until none applies, and succeeds if no free coordinate remains or if the test of Lemma B.3 holds with $\sigma_\mu = L/4^{\mu+1}$. The copy never influences later filtering. The procedure runs phases $k = 2, 3, \dots$; in phase k it restarts trial μ for each $2 \leq \mu \leq k$ and stops it after $2^{k-\mu}$ bit operations, stopping at the first success.

Theorem B.4 (Exact implicit boundary output with a complementarity margin). *(a) On success, the output is either the incumbent after a zero certified gap, which is then a rational minimizer, or a fixed integer assignment, a set of continuous coordinates fixed at original bounds, and a rational box for the remaining coordinates on which F is strongly convex; in the second case the unique minimizer $\hat{x}$ of F over this face box is a global minimizer of the original problem, and if no coordinate is free, $\hat{x}$ is an explicit rational minimizer. This holds with no assumption on the instance.*

(b) Suppose F has a unique minimizer x^ with point growth (3.2), and every coordinate in $A = J_\partial \cap I_C$, the continuous coordinates at which x^* is at a bound, is strictly complementary: its inward derivative, $\partial_i F(x^*)$ at a lower bound or $-\partial_i F(x^*)$ at an upper bound, is positive. Let λ_A be the smallest such inward derivative and $B_\lambda = \lceil \log_2 \max\{2, C_2/\lambda_A\} \rceil$ ($B_\lambda = 1$ if $A = \emptyset$). Then the procedure succeeds after $f'_d(p, \kappa) \text{poly}(I + B_\lambda)$ bit operations, for a computable function f'_d . Neither x^* , g , nor λ_A is supplied.*

Proof. (a) A zero certified gap makes the incumbent optimal (Theorem 4.7). Otherwise the retained box contains every minimizer (Proposition 4.5 and Lemma B.1(i)), and its integer coordinates are singletons, so its minimum is F^* . Each reduction preserves the minimum over the current box (Lemma B.2). On the final face box F has minimum F^* and, by Lemma B.3 or because it is a single point, a unique minimizer, which is therefore globally optimal.

(b) If $L = 0$, trial 2 succeeds at stage 0 with zero gap, and the phase accounting at the end of this proof applies with $\mu_* = 2$ and $j_* = 0$; assume $L > 0$. Let μ_* be the least $\mu \geq 2$ with $4^\mu \geq 8\kappa$,

so that $4^{\mu_*} \leq 32\kappa$ and trial μ_* satisfies $8\kappa\theta^2 \leq 1$; by Lemma 5.9(G5) it never reaches its cap. Let j_* be the least stage with

$$h_j \leq \frac{1}{10\sqrt{n\kappa}}, \quad 5\sqrt{n\kappa} h_j \leq \frac{\lambda_A}{4C_2}, \quad 5\sqrt{n\kappa} h_j \leq \frac{L}{8 \cdot 4^{\mu_*} C_3},$$

omitting the first condition without integer coordinates and the second if $A = \emptyset$. Then $j_* = \text{poly}(I) + O(\log \kappa + B_\lambda)$, since $\log s$, $\log C_2$, $\log C_3$ and $\log(1/L)$ are polynomial in I . We show that trial μ_* succeeds by stage j_* .

At stage j_* the integer coordinates are fixed to x^* (Lemma B.1(ii)) and every continuous retained interval has half-width $r \leq 5\sqrt{n\kappa} h_{j_*}$ (Lemma 5.9(G4)). A sound reduction never removes x^* : if a reduction that fixes coordinate i at a bound $\xi_i \in \{\ell_i, u_i\}$ is justified on a box $X' \ni x^*$, then setting x_i^* to ξ_i gives a point of X' with value at most F^* , which by uniqueness is x^* ; so $x_i^* = \xi_i$ and $i \in A$. Hence the current face box always contains x^* , agrees with it on fixed coordinates, and has $\|c - x^*\|_\infty \leq r$. For $i \in A$ at ℓ_i that is not yet fixed, ℓ_i is an endpoint of the current interval and $\partial_i F(c) - C_2 r \geq \lambda_A - 2C_2 r \geq \lambda_A/2 > 0$, so the midpoint test fixes it; the upper case is symmetric. When no reduction applies, the free set J_f consists exactly of the continuous coordinates that are interior at x^* . If $J_f = \emptyset$ the trial succeeds. Otherwise $x^* \pm tv \in X$ for $v \in \mathbb{R}^{J_f}$ and small t , so (3.2) and Taylor's formula give $\nabla_{J_f} F(x^*) = 0$ and $H_{J_f J_f}(x^*) \succeq 2gI$. By the row-sum bound, $H_{J_f J_f}(c) \succeq (2g - C_3 r)I$, and

$$H_{J_f J_f}(c) - (C_3 r + \sigma_{\mu_*})I \succeq (2g - 2C_3 r - \sigma_{\mu_*})I \succ 0,$$

because $2C_3 r \leq L/(4 \cdot 4^{\mu_*}) \leq g/32$ and $\sigma_{\mu_*} = L/(4 \cdot 4^{\mu_*}) \leq g/32$. Exact symmetric elimination detects this.

Through stage j_* , trial μ_* performs the table work of Lemma 4.3 with the cap $K = 8 \cdot 2^{\mu_*} \lceil \log_2(n+2) \rceil$ on numbers of bit length $\text{poly}(I + j_* + \mu_* K)$ at fixed d (Section 6.6), plus at most n^2 sign tests and one elimination per stage. Call this number of bit operations M_* ; it is at most a computable function of (p, κ) times $\text{poly}(I + B_\lambda)$. Phase k allots $\sum_{\mu=2}^k 2^{k-\mu} < 2^{k-1}$ operations, and trial μ_* completes in phase $k_* = \mu_* + \lceil \log_2 M_* \rceil$. The total is at most $2^{k_*} \leq 2 \cdot 2^{\mu_*} M_* = O(\sqrt{\kappa} M_*)$, plus clock overhead polylogarithmic in this bound, which proves (b). $\square$

The output is an exact implicit description, not a rational coordinate claim: for $(x^2 - \frac{1}{2})^2 + y + xy^2$ on $[0, 1]^2$ the face is $y = 0$ and the minimizer is $(1/\sqrt{2}, 0)$. Weak reductions preserve the minimum but not necessarily every minimizer, so without uniqueness the certificate proves global optimality of $\hat{x}$, not uniqueness in X . Trials must be dovetailed: a trial with $4^\mu < 8\kappa$ may run forever without exceeding its cap, and increasing μ once per unsuccessful stage would make the grading depend on λ_A .

The margin term and strict complementarity are both needed for certificates of this form, as the next two propositions show.

Proposition B.5 (The margin cost is intrinsic to rectangular sign certificates). *For $n \geq 1$ let $X = [0, \frac{1}{2}]^{n+2}$ with variables $x_0, \dots, x_n, y$, and*

$$\rho_0 = x_0 - \frac{1}{4}, \quad \rho_i = x_i - \frac{1}{4}x_{i-1}^2, \quad \chi = x_n - \frac{1}{8}x_{n-1}^2, \quad G_n = \sum_{i=0}^n \rho_i^2 + y^2 + 3y\chi.$$

Then G_n has degree four, a path decomposition with bags of size at most three, $L = \frac{35}{16}$, the unique minimizer $(a, 0)$ with $a_i = 2^{-(4 \cdot 2^i - 2)}$, $F^ = 0$, point growth with $g = \frac{9}{64}$ ($\kappa = \frac{140}{9}$), and strict complementarity at y with $\lambda_A = \frac{3}{2}a_n$. Every success of the procedure on G_n produces a rational*

number with reduced denominator at least $2^{4 \cdot 2^n - 2}$. Consequently a successful trial μ reaches a stage j with $j + \mu K \geq 4 \cdot 2^n - 2 - \log_2 \Gamma_X$, where K is its cap and $\Gamma_X = 2^{n+2}$ is the product of the denominators of the box endpoints.

Proof. The recursion $a_0 = \frac{1}{4}$, $a_i = \frac{1}{4}a_{i-1}^2$ gives the stated a_i , all in $(0, \frac{1}{2})$, and $\rho(a) = 0$. With $z = x - a$, $\rho_0 = z_0$ and $\rho_i = z_i - b_i z_{i-1}$ with $b_i = (x_{i-1} + a_{i-1})/4 \in [0, \frac{1}{4}]$, so $\|\rho\| \geq \frac{3}{4}\|z\|$ and $\Phi_n = \sum_i \rho_i^2 \geq \frac{9}{16}\|x - a\|^2$. Since $\chi = \frac{1}{2}(x_n + \rho_n)$,

$$G_n = \frac{1}{4}(\Phi_n + y^2) + \frac{3}{4}(\Phi_n - \rho_n^2) + \frac{3}{4}(\rho_n + y)^2 + \frac{3}{2}yx_n,$$

and every term is nonnegative on X . Thus $G_n \geq \frac{9}{64}(\|x - a\|^2 + y^2)$, with equality to zero only at $(a, 0)$. The diagonal second derivatives are $2 + \frac{3}{4}x_i^2 - x_{i+1} \leq \frac{35}{16}$ for $i < n - 1$, the same minus $\frac{3}{4}y$ for $i = n - 1$, and 2 for x_n and y . The bags are $\{x_{i-1}, x_i\}$ and $\{x_{n-1}, x_n, y\}$. Finally $\partial_y G_n(a, 0) = 3\chi(a) = 3(a_n - \frac{1}{2}a_n)$.

Consider a success. A zero-gap success outputs the unique minimizer, whose coordinate $a_n = 2^{-(4 \cdot 2^n - 2)}$ has the stated denominator. Otherwise the integer part is empty, and by the argument in the proof of Theorem B.4(b) no sound reduction fixes an x -coordinate, because all are interior at the unique minimizer. If y is not fixed, the patch test needs $H_{J_f J_f} \succ 0$ at $(a, 0)$, which lies in the face box; but the principal minor on $\{x_n, y\}$ is $\det(\frac{3}{2} \frac{3}{2}) = -5$. So y is fixed, and not at its upper bound, since that would need $\partial_y G_n \leq 0$ at $(a, 0)$. So it is fixed by a sound certificate of $\partial_y G_n \geq 0$ on a retained box $X' = \prod_i [\alpha_i, \alpha'_i] \times [0, \alpha'_y]$ that contains $(a, 0)$. At its point with $y = 0$, $x_n = \alpha_n$ and $x_{n-1} = \alpha'_{n-1}$, $\partial_y G_n = 3(\alpha_n - (\alpha'_{n-1})^2/8)$, so $\alpha_n \geq (\alpha'_{n-1})^2/8 \geq a_{n-1}^2/8 = a_n/2 > 0$; also $\alpha_n \leq a_n$. A rational in $(0, 2^{-(4 \cdot 2^n - 2)})$ has reduced denominator at least $2^{4 \cdot 2^n - 2}$. As in Appendix A, with $h_j = s2^{-j}$, box endpoints and grid nodes of stage j in trial μ have denominators dividing $\Gamma_X 2^{j+\mu K}$, which proves the last claim. $\square$

Here $I = O(n \log n)$ while every successful certificate of the above form needs exponentially many bits in n , at fixed d , p and κ . Thus the B_λ term of Theorem B.4 cannot be removed without a different kind of certificate, for instance one that keeps the correlation between x_{n-1} and x_n , such as the displayed identity.

Proposition B.6 (Weak complementarity defeats sign and patch tests). *On $X = [\frac{1}{2}, 1] \times [0, 1]^2$ with variables (x, w_1, w_2) let $u = x^2 - \frac{1}{2}$ and*

$$\Psi = u^2 + w_1^2 + w_2^2 + 3w_1w_2 + u(w_1 - w_2) = \frac{1}{2}(u^2 + w_1^2 + w_2^2) + \frac{1}{2}(u + w_1 - w_2)^2 + 4w_1w_2.$$

Then Ψ has degree four and one bag of size three; its unique minimizer is $x^ = (1/\sqrt{2}, 0, 0)$ with $F^* = 0$; point growth holds with $g = \frac{1}{2}$ and $L = 12$ ($\kappa = 24$); and $\nabla_w \Psi(x^*) = 0$, so both active coordinates are weakly complementary. For every rational box $X' \subseteq X$ containing x^* , none of $\partial_x \Psi$, $\partial_{w_1} \Psi$, $\partial_{w_2} \Psi$ has constant sign on X' , and $\nabla^2 \Psi(x^*)$ is not positive semidefinite. Hence no trial of the procedure ever succeeds on Ψ .*

Proof. The identity is direct expansion. On X it gives $\Psi \geq \frac{1}{2}(u^2 + \|w\|^2)$, and $u^2 = (x - x_1^*)^2(x + x_1^*)^2 \geq (x - x_1^*)^2$ because $x + x_1^* > 1$. So $\Psi \geq \frac{1}{2}\|(x, w) - x^*\|^2$, with zero only at x^* . Also $\partial_x^2 \Psi = 12x^2 - 2 + 2(w_1 - w_2) \leq 12$ and $\partial_{w_i}^2 \Psi = 2$. The x -interval of X' contains the irrational x_1^* in its interior. On the segment $w = 0$ inside X' , $\partial_x \Psi = 4xu$, $\partial_{w_1} \Psi = u$ and $\partial_{w_2} \Psi = -u$, and u changes sign at x_1^* . So no coordinate can be fixed by a sound test, and the patch test on all three coordinates would need $\nabla^2 \Psi(x^*) \succ 0$; but $v = (0, 1, -1)$ gives $v^T \nabla^2 \Psi(x^*) v = 2 + 2 - 6 = -2$. A zero certified gap is impossible because x^* is irrational. $\square$

The identity in Proposition B.6 is itself a short global certificate that $F^* = 0$ and that x^* is the unique minimizer. Certificates that keep such correlations, rather than certify signs over independent coordinate rectangles, are the natural route to exact output under point growth alone; no general construction of them with parameterized size is known.

C Value functions, local corrections and convex recourse: proofs and supplements

This appendix contains the proofs deferred from Sections 7.1–7.3 and the supplementary results summarized there. The notation is that of these sections. For a rational vector w we write $\langle w \rangle$ for its encoding length.

C.1 Proof of Theorem 7.2(b),(c)

(b) Let $\pi_L(I) \geq I$ be the polynomial bound on the encoding lengths of the L_i^V , and π_0 the polynomial such that a factor of F_0 or the oracle at a rational point w takes at most $\pi_0(I + \langle w \rangle)$ bit operations; its output then has encoding length at most $\pi_0(I + \langle w \rangle)$. As noted after Definition 3.2, point growth of V with constant g gives weighted growth of V with $\bar{\kappa} \leq \kappa_V$. As in the proof of Theorem 5.7(b), CT succeeds at a trial with $2^\mu \leq 6\sqrt{\kappa_V}$, and the number of table operations, factor evaluations and oracle calls over all trials is at most $c(j_{\max} + 1)pI^2(60p\sqrt{\kappa_V})^p$, now with $|\mathcal{A}| + r \leq 2I$ factors. The integer exponents that define the meshes h_{ij} in Section 5 are now $O(\pi_L(I))$, so $j_{\max} = O(\pi_L(I) + q)$, and the bit analysis of Appendix A, with $O(I)$ replaced by $O(\pi_L(I))$, shows that every grid node of a trial with grading $2^{-\mu}$ has encoding length at most $c(1 + \mu 2^\mu)(\pi_L(I) + q + 1)$; moreover $1 + \mu 2^\mu \leq c'\kappa_V$ for the trials that run. Only the cost of one operation must be bounded anew, because values of value factors have no common denominator. A table entry is a sum of at most $|\mathcal{A}| + r + n \leq 3I$ exact values and corrections at one grid point v , each of encoding length at most $\pi_0(I + \langle v \rangle)$, and $\langle v \rangle \leq cc'n\kappa_V(\pi_L(I) + q + 1)$. So each addition, comparison, factor evaluation and oracle call costs at most $(c''\kappa_V(I + q + 1))^{C'}$ bit operations, where C' and c'' depend only on π_0 and π_L . Multiplying by the number of operations, and using $I + q + 1 \geq 2$ to absorb constant factors into the exponent, gives the bound $f(p, \kappa_V)\kappa_V^C(I + q + 1)^C$.

(c) Each run of CT on V halts, because the proof of Theorem 5.7(a) uses only Definition 3.1. Its incumbent lifts to a feasible point $\hat{x} = (\hat{v}, y(\hat{v}))$ of (3.1) with $F(\hat{x}) - \beta \leq 2^{-q}$ and a certified $\beta \leq F^*$. REC and the test of Proposition 6.6 are applied to F at $\hat{x}$, so validity and termination follow as in the proof of Theorem 6.14. For the bound, let F have point growth with constant g . If $L^V = 0$, CT is exact at its first stage, $\hat{x}$ is optimal, and REC returns an optimal point, which passes the test (Lemma 6.8). Otherwise point growth of F is set growth with $\mathcal{S} = \{(v^*, y^*)\}$, and Theorem 6.9(a) applies once $2^{-q} \leq \varepsilon_{\mathcal{S}}$, that is, for some $q \leq \text{poly}(I) + O(\log \kappa_V)$, because $g \geq L^V/\kappa_V$ and $\log(1/L^V) \leq \text{poly}(I)$. With (b) and $O(\log q)$ runs, absorbing powers of $1 + \log \kappa_V$ into $\kappa_V^{C_1}$ proves the bound. $\square$

C.2 Supplements to Section 7.2

Proof of Theorem 7.4. (a) By (7.3) and Lemma 7.3 applied to x^* ,

$$\min_{\text{corners}} \underline{V} - e \leq \min_{\text{corners}} V_B - e_B(C) \leq F^* \leq U.$$

(b) Lemma 7.3 at x^* gives a corner v° of the minimizer's cell with $V_B(v^\circ) \leq F^* + e$. Hence $U \leq F(x^{v^\circ}) \leq \underline{V}(v^\circ) + \varepsilon_{\text{or}} \leq V_B(v^\circ) + \varepsilon_{\text{or}} \leq F^* + e + \varepsilon_{\text{or}}$. A retained cell has a corner v with $\underline{V}(v) \leq U + e$, and $V_B(v) \leq F(x^v) \leq \underline{V}(v) + \varepsilon_{\text{or}} \leq U + e + \varepsilon_{\text{or}} \leq F^* + 2e + 2\varepsilon_{\text{or}}$.

(c) For $x_B \in C$, Lemma 7.3 and $\underline{V} \leq V_B$ give $F(x) \geq \min_{\text{corners}} V_B - e \geq \min_{\text{corners}} \underline{V} - e \geq \lambda$. Every corner satisfies $\underline{V}(v) \geq F(x^v) - \varepsilon_{\text{or}} \geq U - \varepsilon_{\text{or}}$, so $\lambda \geq U - e - \varepsilon_{\text{or}}$. $\square$

If the cells of a multilevel scheme cover $\bar{X}_B$ at the top level and each retained cell is replaced by subcells covering it, then by Theorem 7.4(a) and induction some current cell contains x_B^* at every level, and by Lemma 7.3 every point whose bag projection lies in a cell discarded earlier has value above the incumbent of that level. So $\min\{\lambda, U\}$ is a certified lower bound on F^* at every level, as in Proposition 4.5.

Theorem 7.4 also certifies a gap; for that, valid lower bounds and feasible completions are needed. For filtering alone, exact values are not needed: only the oscillation of their error over the corners matters. A common additive error, such as a normalization constant of messages, is therefore harmless.

Proposition C.1 (Only the error oscillation matters). *Let $\mathcal{Q}$, e and ε_{or} be as in Theorem 7.4, and suppose some cell of $\mathcal{Q}$ contains x_B^* for a global minimizer x^* . Let $\mathcal{V}$ be the set of all corners and $\tilde{V} : \mathcal{V} \rightarrow \mathbb{R}$ a function such that, for some constants $\delta_- \leq \delta_+$ with $\delta_+ - \delta_- \leq \varepsilon_{\text{or}}$,*

$$\delta_- \leq V_B(v) - \tilde{V}(v) \leq \delta_+ \quad (v \in \mathcal{V}).$$

The constants need not be known. Retain C if and only if $\min_{\text{corners of } C} \tilde{V} \leq \min_{\mathcal{V}} \tilde{V} + e + \varepsilon_{\text{or}}$. Then every cell containing x_B^ for a global minimizer x^* is retained, and every retained cell has a corner with $V_B(v) \leq F^* + 2e + 2\varepsilon_{\text{or}}$.*

Proof. Let C^* contain x_B^* . By Lemma 7.3,

$$\min_{\text{corners of } C^*} \tilde{V} \leq \min_{\text{corners of } C^*} V_B - \delta_- \leq F^* + e - \delta_-,$$

while $V_B \geq F^*$ gives $\min_{\mathcal{V}} \tilde{V} \geq F^* - \delta_+$. The difference is at most $e + \delta_+ - \delta_- \leq e + \varepsilon_{\text{or}}$, so C^* is retained. If C is retained, it has a corner v with $V_B(v) \leq \tilde{V}(v) + \delta_+ \leq \min_{\mathcal{V}} \tilde{V} + e + \varepsilon_{\text{or}} + \delta_+$, and $\min_{\mathcal{V}} \tilde{V} \leq \min_{\text{corners of } C^*} \tilde{V} \leq F^* + e - \delta_-$. Thus $V_B(v) \leq F^* + 2e + \varepsilon_{\text{or}} + (\delta_+ - \delta_-) \leq F^* + 2e + 2\varepsilon_{\text{or}}$. $\square$

Grid min-marginals are not certified recourse. Grid min-marginals minimize over the outside coordinates on the grid, not over their original domain, and their error varies with the bag value. No allowance that depends only on the bag, the curvature and the mesh makes them valid.

Let $h = 2^{-j}$ and $G = h\mathbb{Z} \cap [0, 1]$. On $[0, 1]^{1+m}$ with coordinates $(x, y_1, \dots, y_m)$ and the uniform product grid G^{1+m} , put $U_G = \min_{G^{1+m}} F$, the grid value function $V^G(a) = \min\{F(z) : z \in G^{1+m}, z_x = a\}$ of the coordinate x , and, for the bag $B = \{x, y_1\}$, the grid min-marginal $m^F(v) = \min\{F(z) : z \in G^{1+m}, (z_x, z_{y_1}) = v\}$. A *constant-correction test* with constant c retains a bag cell C of the grid if and only if some corner v of C has $m^F(v) - c \leq U_G$. With a common curvature bound L , every node of a uniform grid has correction $Lh^2/8$ in every coordinate, so the corrected grid of Section 4 retains a grid interval $[a, a+h]$ of x if and only if $\min\{V^G(a), V^G(a+h)\} - c \leq U_G$ with $c = (1+m)Lh^2/8$; a bag-local budget would take $c = |B|Lh^2/8$. In both families below, the corner minimum of each relevant bag cell equals $\min\{V^G(a), V^G(a+h)\}$, where $[a, a+h]$ is the projection of the cell on x , so the thresholds below hold for both tests.

Proposition C.2 (Constant corrections must grow with dimension).

(A) *Let $j \geq 1$, $m \geq 1$ and $\epsilon > 0$ with $mh^2 > 4(1 + \epsilon)$, and*

$$F_A(x, y) = x^2 - hx \sum_{i=1}^m y_i + \sum_{i=1}^m y_i^2 + \left(\frac{mh^2}{4} - 1 - \epsilon\right)x.$$

The interaction graph is a star with bags $\{x, y_i\}$ ($p = 2$), $L = 2$ is valid, the unique minimizer is $x = 1$, $y_i = h/2$, with $F^* = -\epsilon$, and every growth constant is at most $\epsilon/(1 + mh^2/4)$. Moreover $U_G = 0$, the only bag cell containing $(1, h/2)$ is $C = [1 - h, 1] \times [0, h]$, and $\min_{\text{corners of } C} m^F = (1 - h)(mh^2/4 - h - \epsilon)$. So a constant-correction test discards C unless $c \geq (1 - h)(mh^2/4 - h - \epsilon)$; as $m \rightarrow \infty$, the ratio of $(1 + m)Lh^2/8$ to this bound tends to $(1 - h)^{-1}$.

(B) Let $j \geq 2$, let $d \geq 2/h$ be an integer, $m = d^2$, and

$$F_B(x, y) = \left(x - \frac{1}{2}\right)^2 + \sum_{i=1}^m \left(y_i - \frac{h}{2} - \frac{x - 1/2}{d}\right)^2.$$

It is a star with $p = 2$, $L = 4$ is valid, the unique minimizer $z^* = (1/2, h/2, \dots, h/2)$ is interior, and $F_B(z) - F^* \geq \frac{3 - \sqrt{5}}{2} \|z - z^*\|^2$, so $\kappa \leq 2(3 + \sqrt{5}) < 11$ for every d and h . Moreover $U_G = \frac{1}{2} - \frac{dh}{2} + \frac{d^2 h^2}{4}$, and both bag cells containing $(1/2, h/2)$ have $\min_{\text{corners}} m^F - U_G = dh(\frac{1}{2} - h) + 2h^2 - \frac{1}{2}$. So a constant-correction test discards every cell containing the minimizer unless $c \geq dh(\frac{1}{2} - h) + 2h^2 - \frac{1}{2}$, a quantity of order $\sqrt{m}h$ at fixed h .

Because the growth constant of family (A) degrades as m grows, family (A) refutes every budget that depends only on (p, L, h) ; family (B) refutes every budget that depends only on (p, κ, L, h) . Example C.3 after the proof works out the instance $h = \frac{1}{2}$, $m = 32$, $\epsilon = \frac{1}{16}$ of family (A).

Proof of Proposition C.2. (A) Write $\omega_A = mh^2/4 - 1 - \epsilon > 0$. All coordinate second derivatives equal 2 and F_A is quadratic, so $L = 2$ is valid. For fixed x , each leaf term $y^2 - hxy$ is strictly convex with minimizer $hx/2 \in [0, 1]$ and minimum $-h^2 x^2/4$. Hence $\min_y F_A(x, y) = (1 - mh^2/4)x^2 + \omega_A x =: V(x)$, which is strictly concave because $mh^2 > 4$, with $V(0) = 0$ and $V(1) = -\epsilon$. For $x \in (0, 1)$, strict concavity gives $V(x) > (1 - x)V(0) + xV(1) \geq -\epsilon$. So $x = 1$ is the unique minimizing center and $y_i = h/2$ the unique leaves; $F^* = -\epsilon$. The origin has $F_A = 0 = F^* + \epsilon$ and squared distance $1 + mh^2/4$ from the minimizer, which bounds g .

On the grid, $y(y - hx) \geq 0$ for $y \in G$ and $x \in [0, 1]$, since either $y = 0$ or $y \geq h \geq hx$. Also $x^2 + \omega_A x \geq 0$ for $x \geq 0$. Hence $F_A \geq 0$ on G^{1+m} with equality at the origin, so $U_G = 0$, and every leaf's grid minimum is 0. The point $(1, h/2)$ has x at the right endpoint and $h/2$ interior to $[0, h]$, so C is the only grid cell of the bag containing it. At its corners, $m^F(x, y_1) = x^2 + \omega_A x + (y_1^2 - hxy_1)$. For $x = 1 - h$ this equals $(1 - h)(1 - h + \omega_A) = (1 - h)(mh^2/4 - h - \epsilon)$ at $y_1 = 0$ and exceeds it by h^3 at $y_1 = h$. For $x = 1$ it equals $mh^2/4 - \epsilon$ at both $y_1 = 0, h$, which exceeds the previous value by $h(mh^2/4 - \epsilon) + h(1 - h) > 0$. This proves the corner minimum; it also equals $\min\{V^G(1 - h), V^G(1)\}$, because every leaf's grid minimum is attained at 0. Since $U_G = 0$ and $L = 2$, the threshold for c is as stated, and $(1 + m)h^2/4 / [(1 - h)(mh^2/4 - h - \epsilon)] \rightarrow (1 - h)^{-1}$.

(B) The Hessian H has $H_{xx} = 2 + 2m/d^2 = 4$, $H_{y_i y_i} = 2$, $H_{xy_i} = -2/d$, and zero elsewhere; thus $L = 4$ is valid. Vectors with zero x -component and leaf components summing to zero are eigenvectors with eigenvalue 2. On the span of e_x and $m^{-1/2} \mathbf{1}_y$, H acts as $\begin{pmatrix} 4 & -2 \\ -2 & 2 \end{pmatrix}$, with eigenvalues $3 \pm \sqrt{5}$. Since $F_B \geq 0$ vanishes only at $z^* \in (0, 1)^{1+m}$ and $F_B(z) = \frac{1}{2}(z - z^*)^\top H(z - z^*)$, growth holds with $g = (3 - \sqrt{5})/2$ and $\kappa = L/g = 2(3 + \sqrt{5})$.

Write $t = x - 1/2 \in h\mathbb{Z} \cap [-1/2, 1/2]$ for the center grid values (here $1/2 \in G$ because $j \geq 1$). A leaf's ideal value $\iota(t) = h/2 + t/d$ lies in $[0, h]$ because $|t|/d \leq 1/(2d) \leq h/4$. Its distance to G is therefore $h/2 - |t|/d$, and the grid value function of the center is

$$V^G(t) = t^2 + m \left(\frac{h}{2} - \frac{|t|}{d} \right)^2 = 2t^2 - dh|t| + \frac{d^2 h^2}{4}.$$

On the grid $|t| \leq 1/2 \leq dh/4$, and $\tau \mapsto 2\tau^2 - dh\tau$ decreases on $[0, dh/4]$, so $U_G = V^G(\pm 1/2) = \frac{1}{2} - \frac{dh}{2} + \frac{d^2h^2}{4}$. The minimizer's bag projection $(1/2, h/2)$ lies on the common edge of the cells $[1/2 - h, 1/2] \times [0, h]$ and $[1/2, 1/2 + h] \times [0, h]$ and in no other cell. At $t = 0$ both leaf nodes $0, h$ are at distance $h/2$ from ι , so $m^F = d^2h^2/4$. At $t = \pm h$ the nearer node gives $m^F = V^G(h) = 2h^2 - dh^2 + d^2h^2/4$ and the farther adds $2h^2/d$. Since $d \geq 2$, each cell has corner minimum $d^2h^2/4 - (d-2)h^2 = \min\{V^G(0), V^G(\pm h)\}$, and subtracting U_G gives the stated difference. For $h \leq 1/4$ this difference is at least $dh/4 + 2h^2 - 1/2$, which grows linearly in $d = \sqrt{m}$. Discarding both center intervals adjacent to $x = 1/2$ removes the minimizer. $\square$

Example C.3 (An instance with $h = \frac{1}{2}$). Take $h = \frac{1}{2}$, $m = 32$, $\epsilon = \frac{1}{16}$ in Proposition C.2(A): $F = x^2 - \frac{x}{2} \sum y_i + \sum y_i^2 + \frac{15}{16}x$ on $[0, 1]^{33}$. The exact leaf recourse is $y_i = x/4$, with $V(x) = -x^2 + \frac{15}{16}x$, so the unique minimizer is $x = 1$, $y_i = \frac{1}{4}$, value $-\frac{1}{16}$. On $G = \{0, \frac{1}{2}, 1\}$ the grid values V^G at $x = 0, \frac{1}{2}, 1$ are $0, \frac{23}{32}, \frac{31}{16}$, and $U_G = 0$. The four corner min-marginals of $C = [\frac{1}{2}, 1] \times [0, \frac{1}{2}]$ are $\frac{23}{32}, \frac{27}{32}, \frac{31}{16}, \frac{31}{16}$. The bag-local budget $\frac{1}{8}$ rejects C ; the global correction $\frac{33}{16}$ keeps it. The grid error $V^G(x) - V(x) = m \text{dist}(x/4, G)^2$ varies from 0 at $x = 0$ to $m/16$ at $x = 1$; no normalization of messages removes this variation.

C.3 Certificates: energy, existence, evaluation and height

In the setting of Lemma 7.7, put $\mathcal{W} = \{d : G_s d = 0 \ (s \in \mathcal{E})\}$. For every coordinate j ,

$$(B^\top CB)_{jj} = -\min_{d \in \mathcal{W}} (d^\top C d + 2d^\top \Gamma e_j). \quad (\text{C.1})$$

Indeed, the function $d \mapsto d^\top C d + 2d^\top \Gamma e_j$ is convex. By Lemma 7.7(b), $Be_j \in \mathcal{W}$, and by the w -linear part $CB + \Gamma + G^\top \Lambda = 0$ of (7.7), for $d \in \mathcal{W}$ half the gradient at Be_j satisfies $(CBe_j + \Gamma e_j)^\top d = -\sum_{s \in \mathcal{E}} (\Lambda_s e_j)(G_s d) = 0$. A convex function whose gradient at a point of a linear subspace is orthogonal to that subspace attains its minimum over the subspace there, so the minimum equals $(B^\top CB)_{jj} + 2(B^\top \Gamma)_{jj} = -(B^\top CB)_{jj}$. Formula (C.1) says that a piece cancels direct curvature exactly by the energy that the private response spends to follow a unit change of w_j , using only directions that keep every constraint with a nonzero multiplier active. If all private coordinates coupled to w_j are held at bounds with nonzero multipliers, the piece cancels nothing.

Proof of Proposition 7.8(b). Let $J = [t_-, t_+]$ with $t_- < t_+$; a point interval is trivial. The sets $J_{\Pi'} = \{t \in J : w + te_j \in \Pi'\}$ are closed intervals that cover J . The union of the nondegenerate ones is closed, and its complement in J is relatively open and contained in the finitely many degenerate ones, which are points; hence it is empty. The endpoints of the nondegenerate $J_{\Pi'}$ therefore split J into finitely many closed subintervals. Some nondegenerate $J_{\Pi'}$ contains the midpoint of a given subinterval, and then contains the whole subinterval, because no endpoint of $J_{\Pi'}$ lies strictly inside it. On that subinterval $\psi(w + te_j) = q_{\Pi'}(w + te_j)$ has second derivative $-(B_{\Pi'}^\top CB_{\Pi'})_{jj} \leq -\sigma_j$ by Lemma 7.7, so $\chi(t) = \psi(w + te_j) + \sigma_j t^2/2$ is concave there. The function χ is continuous, and at each breakpoint the jump $\chi'_+ - \chi'_-$ equals that of the concave function $t \mapsto \psi(w + te_j)$, which is nonpositive. So χ'_+ is nonincreasing, and χ is concave. $\square$

Proposition C.4 (Finite certificates exist). *Let $Y = [\ell^Y, u^Y]$ with $\ell^Y < u^Y$, and let $\Pi \subseteq \mathbb{R}^k$ be a rational box with nonempty interior. Then ψ has a certificate on Π in the sense of Definition 7.6.*

Proof. A pattern $J = (J_\ell, J_f, J_u)$ is a partition of the private coordinates. Let $\mathcal{M}_J$ be the set of $(w, y, \lambda^{\text{low}}, \lambda^{\text{up}})$ with $w \in \Pi$, $\ell^Y \leq y \leq u^Y$, $\lambda^{\text{low}}, \lambda^{\text{up}} \geq 0$, $Cy + \Gamma w + c = \lambda^{\text{low}} - \lambda^{\text{up}}$, and $y_i = \ell_i^Y$, $\lambda_i^{\text{up}} = 0$ for $i \in J_\ell$; $\lambda_i^{\text{low}} = \lambda_i^{\text{up}} = 0$ for $i \in J_f$; $y_i = u_i^Y$, $\lambda_i^{\text{low}} = 0$ for $i \in J_u$. Each $\mathcal{M}_J$ is a rational

polyhedron, and every point of it satisfies the KKT conditions, including complementarity. Its projection Π_J to the w coordinates is a rational polytope. Every $w \in \Pi$ has an optimal y and, since the constraints are linear, KKT multipliers; these fit some pattern. Thus the Π_J cover Π . The union of the full-dimensional Π_J is closed and contains Π minus finitely many lower-dimensional polytopes, a dense subset of Π ; hence it equals Π .

Triangulate each full-dimensional Π_J into k -simplices with rational vertices. At each vertex w' of a simplex Σ , choose a rational point $(w', y_{w'}, \lambda_{w'}) \in \mathcal{M}_J$. On Σ define $(\bar{y}, \lambda)$ by affine interpolation of these vertex values in barycentric coordinates. For $w \in \Sigma$ the point $(w, \bar{y}(w), \lambda(w))$ is a convex combination of points of the convex set $\mathcal{M}_J$, so it lies in $\mathcal{M}_J$. Hence $(\bar{y}, \lambda)$ is a certificate on Σ .

Let $\mathcal{H}$ be the finite set of hyperplanes spanned by facets of all these simplices. Starting from Π , split every current leaf whose interior meets a hyperplane of $\mathcal{H}$ by that hyperplane, until no leaf interior meets any of them. Both halves of every split have nonempty interior, and no hyperplane splits two nodes on one root path, so the process stops. Let Π' be a final leaf. Its interior is connected and misses every hyperplane of $\mathcal{H}$, so for each simplex Σ either $\text{int } \Pi' \subseteq \text{int } \Sigma$ or $\text{int } \Pi' \cap \Sigma = \emptyset$. The full-dimensional simplices cover Π , so the first case occurs for some Σ , and $\Pi' \subseteq \Sigma$. Restricting the certificate of Σ to Π' completes the proof. $\square$

The same argument applies to a polytope Y , with patterns given by sets of active rows.

Lemma C.5 (Exact tables with value factors). *For rational $w \in \mathbb{R}^{k_t}$, the value $\psi_t(w)$ and a minimizer $y^{(t)}(w)$ are rationals of encoding length $\text{poly}(I + \langle w \rangle)$, computable in that time, where I is the encoding length of the data. In CT applied to V , every message, bag-table entry and min-marginal is a sum of at most $m_0 = |\mathcal{A}| + r + |\mathcal{K}|$ factor values, value-factor values and corrections at one grid point v , hence has encoding length at most $m_0 \text{poly}(I + \langle v \rangle)$.*

Proof. A convex quadratic program with rational data is solvable exactly in polynomial time, and it has a rational optimal solution and value of polynomial size [119]. If a certificate is supplied, the value is instead $q_{\Pi'}(w)$ and the minimizer $\bar{y}(w)$ for a leaf Π' containing w , found by descending the partition tree. In the message recurrence and in the bag tables, every entry equals the value of a sum of factor values and unary corrections at one assignment, with each factor and each correction appearing at most once. If each of m_0 rationals has a numerator of absolute value at most $2^{b'}$ and a denominator at most $2^{b'}$, their sum has denominator at most $2^{m_0 b'}$ and numerator of absolute value at most $m_0 2^{m_0 b'}$. Comparisons and additions on such numbers are polynomial in their length. No common denominator across different active sets is needed. $\square$

Lemma C.6 (Height of a minimizer over a polytope). *Let $\mathcal{P} = \{x \in \mathbb{R}^{n'} : Ax \leq a\}$ be a nonempty bounded polyhedron with integral A and a , and let $F_{\mathcal{P}}(x) = (\frac{1}{2}x^T Hx + c^T x + c_0)/\Delta_{\mathcal{P}}$ with integral symmetric H , integral c and c_0 , and a positive integer $\Delta_{\mathcal{P}}$. Put $\beta_H = \max\{1, \max_{ij} |H_{ij}|, \max_{sj} |A_{sj}|\}$ and $\Lambda_0 = (2n'\beta_H^2)^{n'}$. Then some global minimizer x^* of $F_{\mathcal{P}}$ on $\mathcal{P}$ has $x^* = z/\rho$ with $z \in \mathbb{Z}^{n'}$ and an integer $1 \leq \rho \leq \Lambda_0$; if the minimizer is unique, it has this form. The minimum $\min_{\mathcal{P}} F_{\mathcal{P}}$ is a rational with denominator at most $2\Delta_{\mathcal{P}}\Lambda_0^2$.*

Proof. Global minimizers exist because $\mathcal{P}$ is compact. Among them choose x^* whose active set $\mathcal{E} = \{s : A_s x^* = a_s\}$ has maximal cardinality, and put $\mathcal{W} = \{d : A_{\mathcal{E}} d = 0\}$. For $d \in \mathcal{W}$ and small $\epsilon > 0$ both $x^* \pm \epsilon d$ lie in $\mathcal{P}$, so optimality gives $\nabla F_{\mathcal{P}}(x^*)^T d = 0$ and $d^T H d \geq 0$.

Suppose some $d \in \mathcal{W} \setminus \{0\}$ has $Hd \perp \mathcal{W}$. Then $d^T H d = 0$, so $F_{\mathcal{P}}(x^* + td) = F_{\mathcal{P}}(x^*)$ for all real t . As $\mathcal{P}$ is bounded, $\bar{t} = \max\{t \geq 0 : x^* + td \in \mathcal{P}\}$ is finite. The rows of $\mathcal{E}$ stay active along the line. For every $t > \bar{t}$ some row is violated at $x^* + td$; as there are finitely many rows, one row $s \notin \mathcal{E}$ is violated for t arbitrarily close to $\bar{t}$, and continuity and feasibility give $A_s(x^* + \bar{t}d) = a_s$. So $x^* + \bar{t}d$ is a global minimizer with a strictly larger active set, a contradiction. Hence no nonzero $d \in \mathcal{W}$

has $Hd \perp \mathcal{W}$. Since H is positive semidefinite on $\mathcal{W}$, it is then positive definite on $\mathcal{W}$: a nonzero $d \in \mathcal{W}$ with $d^\top Hd = 0$ would minimize the convex form $d \mapsto d^\top Hd$ over $\mathcal{W}$, so $Hd \perp \mathcal{W}$.

Choose a maximal linearly independent set $\mathcal{E}' \subseteq \mathcal{E}$ of rows, so that $\mathcal{W} = \{d : A_{\mathcal{E}'}d = 0\}$. Since $\nabla F_{\mathcal{P}}(x^*) \perp \mathcal{W}$, the vector $Hx^* + c$ lies in the row space of $A_{\mathcal{E}'}$: there is μ with $Hx^* + c + A_{\mathcal{E}'}^\top \mu = 0$, and $A_{\mathcal{E}'}x^* = a_{\mathcal{E}'}$. The saddle matrix $S = \begin{pmatrix} H & A_{\mathcal{E}'}^\top \\ A_{\mathcal{E}'} & 0 \end{pmatrix}$ is nonsingular: if $Hx + A_{\mathcal{E}'}^\top \mu' = 0$ and $A_{\mathcal{E}'}x = 0$, then $x \in \mathcal{W}$ and $x^\top Hx = -(A_{\mathcal{E}'}x)^\top \mu' = 0$, so $x = 0$, and then $\mu' = 0$ because the rows of $A_{\mathcal{E}'}$ are independent. The order of S is at most $2n'$, its entries are integers of absolute value at most β_H , and so each row S_i has norm at most $\sqrt{2n'}\beta_H$. Hadamard's inequality for rows, $|\det S| \leq \prod_i \|S_i\|$, gives $|\det S| \leq (\sqrt{2n'}\beta_H)^{2n'} = \Lambda_0$. Cramer's rule applied to $S(x^*, \mu) = (-c, a_{\mathcal{E}'})$, whose right side is integral, gives $x^* = z/\rho$ with $\rho = |\det S|$. Finally $F_{\mathcal{P}}(x^*) = (z^\top Hz + 2\rho c^\top z + 2\rho^2 c_0)/(2\rho^2 \Delta_{\mathcal{P}})$. If the minimizer is unique, it is x^* . $\square$

C.4 Proof of Theorem 7.9

Proof of Theorem 7.9. (i) A segment of $\bar{X}_{\mathcal{K}}$ parallel to e_i maps, for each t with $i \in E_t$, to a segment of Π_t parallel to $e_{j(t,i)}$, and leaves the other value factors constant. Along it, F_0 has second derivative $(H_0)_{ii}$, and $\psi_t + \sigma_{t,j(t,i)}v_i^2/2$ is concave by Proposition 7.8. A sum of concave functions is concave. The checking cost was discussed after Definition 7.6.

(ii) If $L^+ = 0$, V is concave along every coordinate segment of $\bar{X}_{\mathcal{K}}$. Replacing coordinates one at a time by a better endpoint does not increase V , so a vertex of $\bar{X}_{\mathcal{K}}$, which lies in $X_{\mathcal{K}}$, is optimal. Tree dynamic programming over the two endpoint nodes, with value-factor values from Lemma C.5, finds it; every bag table has at most 2^p entries. Private coordinates are recovered by one convex quadratic program per block.

(iii) *Approximation.* By (i), V has upper coordinate curvatures L_i^+ ; coordinates with $L_i \leq 0$ get the two endpoint nodes. Each ψ_t is assigned to a bag containing E_t . An exact convex quadratic program [119], or the leaf of a supplied certificate that contains the query point, gives ψ_t and a minimizer $y^{(t)}$ exactly in time polynomial in I and the encoding length of the point (Lemma C.5), with a fixed exponent. So Theorem 7.2(a),(b) apply with $L^V = L^+$; the encoding length of the L_i^+ , and hence $\log(1/L^+)$, is polynomial in I . There are at most rK^p value-factor evaluations per stage.

Exact output. Lemma C.6, applied to the integer slices of the retained box, bounds the denominator of F^* without uniqueness, and under growth the denominators of the continuous coordinates of v^* . The exact minimizer is obtained by rational reconstruction of the incumbent and accepted by Proposition 6.6 with this denominator bound. The proof below gives the procedure and its analysis.

(iv) $\sigma_t \geq 0$ by Proposition 7.8(b), so $L_i \leq (H_0)_{ii}$. Along e_i the quadratic F has second derivative $(H_0)_{ii}$, so $L' \geq \max\{(H_0)_{ii}, 0\} \geq L_i^+$ for every $i \in \mathcal{K}$, and $L' \geq L^+$. By Lemma 7.1(b), g is also a growth constant of V , as (iii) requires, and $\kappa(L^+, g) \leq \kappa(L', g)$. $\square$

Proof of the exact-output part of Theorem 7.9(iii). Fix the integer retained coordinates at any integer values. The remaining problem minimizes a quadratic over the bounded rational polytope formed by the continuous retained box and the sets Y_t . Scale its constraint rows to integers and multiply the objective by a common denominator $\Delta_{\mathcal{P}}$ of all coefficients of F . The fixed integer values change only the linear and constant terms, which stay integral, so the numbers β_H and $\Delta_{\mathcal{P}}$ of Lemma C.6 are the same for every integer slice. That lemma gives computable integers Λ_0 and $\Omega_0 = 2\Delta_{\mathcal{P}}\Lambda_0^2$ of polynomial encoding length: every slice optimum, in particular F^* , has denominator at most Ω_0 , and a unique minimizer has continuous coordinates with denominators at most Λ_0 .

Run CT on V with $\varepsilon = 2^{-q}$ for $q = 1, 2, 4, \dots$. After each run, with incumbent $\hat{v}$ and bound

β , replace each continuous coordinate of $\hat{v}$ by the unique rational with denominator at most Λ_0 within distance $1/(4\Lambda_0^2)$, if one exists; continued fractions find it in polynomial time. If every such coordinate is replaced and the result $\tilde{v}$ lies in $X_{\mathcal{K}}$, compute $y(\tilde{v})$ and $V(\tilde{v}) = a/W$ in lowest terms, and stop with $(\tilde{v}, y(\tilde{v}))$ if $V(\tilde{v}) - \beta < 1/(\Omega_0 W)$. By Proposition 6.6 with the denominator bound Ω_0 for F^* , a returned point is optimal: if $V(\tilde{v}) > F^*$, then $V(\tilde{v}) - \beta \geq V(\tilde{v}) - F^* \geq 1/(\Omega_0 W)$. This uses neither growth nor uniqueness.

If F has point growth with constant g , its minimizer (v^*, y^*) is unique, so the continuous coordinates of v^* have denominators at most Λ_0 . Once $\varepsilon \leq \min\{1/(2\Omega_0^2), g/(32\Lambda_0^4)\}$, Lemma 7.1(b) gives $\|\hat{v} - v^*\|^2 \leq (V(\hat{v}) - F^*)/g \leq 1/(32\Lambda_0^4)$. Every coordinate of $\hat{v}$ is then within $1/(4\Lambda_0^2) < 1$ of the corresponding coordinate of v^* , so integer coordinates agree, and two distinct rationals with denominators at most Λ_0 differ by at least $1/\Lambda_0^2$, so $\tilde{v} = v^*$. Then $V(\tilde{v}) = F^* = a/W$ with $W \leq \Omega_0$ and $V(\tilde{v}) - \beta \leq \varepsilon < 1/(\Omega_0 W)$, so the procedure stops. Since $1/g \leq \kappa_V/L^+$ and $\log(1/L^+)$, $\log \Lambda_0$, $\log \Omega_0$ are polynomial in I , this happens with $q \leq \text{poly}(I) + O(\log \kappa_V)$, and doubling overshoots by a factor at most two. Each run costs at most $f(p, \kappa_V) \kappa_V^C (I + q + 1)^C$ bit operations by the approximation part, the reconstruction and the evaluation of $V(\tilde{v})$ cost $\text{poly}(I + q)$, and there are $O(\log q)$ runs; absorbing the powers of $1 + \log \kappa_V$ into $\kappa_V^{C_1}$ gives the bound $f(p, \kappa_V) \kappa_V^{C_1} (I + 1)^{C_1}$. $\square$

C.5 Recognition of affine selectors

A globally affine response, that is, a one-leaf certificate (Remark 7.10), need not be supplied. Let $Y = [\ell^Y, u^Y]$ with $\ell^Y < u^Y$, let $\Pi = \{w : Aw \leq a\}$ be a rational polytope with nonempty interior and a rational interior point w_0 (for a scope box, its center), and let $\phi(w, y) = \frac{1}{2}y^\top C y + y^\top \Gamma w + c^\top y$ with $C \succeq 0$. An *affine optimal selector* is an affine map $\bar{y}$ with $\bar{y}(w) \in \arg \min_Y \phi(w, \cdot)$ for every $w \in \Pi$.

Theorem C.7 (Recognition of affine selectors). *One exact convex quadratic program and one linear program decide whether an affine optimal selector exists. If one exists, they return a rational selector of polynomial size together with affine multipliers forming a one-leaf certificate on Π . Singular C and nonunique responses are allowed.*

Proof. Common central gradient. Let $\hat{y}$ be a minimizer at w_0 . For any minimizer y' at w_0 , the function $t \mapsto \phi(w_0, \hat{y} + t(y' - \hat{y}))$ is a convex quadratic that is constant on $[0, 1]$, because the optimal set is convex. Its quadratic coefficient $(y' - \hat{y})^\top C (y' - \hat{y})$ vanishes, so $C(y' - \hat{y}) = 0$ since $C \succeq 0$. All central minimizers therefore have the same gradient $\zeta_0 = C\hat{y} + \Gamma w_0 + c$; this is a classical property of convex programs [132]. Let $I_\ell = \{i : \zeta_{0i} > 0\}$, $I_u = \{i : \zeta_{0i} < 0\}$ and $I_0 = \{i : \zeta_{0i} = 0\}$. By the box KKT conditions, every central minimizer has $y_i = \ell_i^Y$ for $i \in I_\ell$ and $y_i = u_i^Y$ for $i \in I_u$.

Necessity. We use one fact: if an affine function is nonnegative on Π and vanishes at an interior point, it vanishes identically. Let $\bar{y}$ be an affine optimal selector and write $\zeta(w) = C\bar{y}(w) + \Gamma w + c$, an affine map. Since $\bar{y}(w_0)$ is a central minimizer, $\zeta(w_0) = \zeta_0$. For $i \in I_\ell$, $\bar{y}_i - \ell_i^Y \geq 0$ on Π and vanishes at w_0 , so $\bar{y}_i \equiv \ell_i^Y$; optimality at every w then needs $\zeta_i \geq 0$ on Π . Symmetrically, $\bar{y}_i \equiv u_i^Y$ and $\zeta_i \leq 0$ for $i \in I_u$. Let $j \in I_0$, so $\zeta_j(w_0) = 0$. If $\bar{y}_j \equiv \ell_j^Y$, optimality needs $\zeta_j \geq 0$ on Π , so $\zeta_j \equiv 0$; the case $\bar{y}_j \equiv u_j^Y$ is symmetric. Otherwise $\bar{y}_j$ is not identically at a bound, so by the fact above it attains no bound at an interior point. Then optimality gives $\zeta_j = 0$ on the interior of Π , hence $\zeta_j \equiv 0$. Thus every affine optimal selector satisfies

$$\begin{aligned} \bar{y}_i &\equiv \ell_i^Y, \quad \zeta_i \geq 0 \text{ on } \Pi & (i \in I_\ell), \\ \bar{y}_i &\equiv u_i^Y, \quad \zeta_i \leq 0 \text{ on } \Pi & (i \in I_u), \\ \zeta_j &\equiv 0, \quad \ell_j^Y \leq \bar{y}_j \leq u_j^Y \text{ on } \Pi & (j \in I_0). \end{aligned} \tag{C.2}$$

Sufficiency. If an affine $\bar{y}$ satisfies (C.2), define lower multipliers ζ_i on I_ℓ and upper multipliers $-\zeta_i$ on I_u , and zero otherwise. These are affine and nonnegative on Π , and satisfy stationarity and complementarity identically. This is a certificate on Π ; by Lemma 7.7(a), $\bar{y}(w)$ is optimal for every $w \in \Pi$.

Linear program. Write $\bar{y}(w) = \bar{y}_0 + Bw$. Then $\zeta(w) = (C\bar{y}_0 + c) + (CB + \Gamma)w$ has coefficients linear in $(\bar{y}_0, B)$. Identities in (C.2) are linear equations. Each condition “an affine function is nonnegative on Π ” is replaced by its Farkas certificate $\pi \geq 0$, $A^\top \pi = -(\text{linear part})$, $a^\top \pi \leq (\text{constant part})$, which is linear in $(\bar{y}_0, B, \pi)$. For a box Π , auxiliary variables bounding $|B_{ij}|$ and $|(CB + \Gamma)_{ij}|$ give the exact ranges instead. The resulting system has polynomial size. It is feasible exactly when an affine optimal selector exists. Exact convex quadratic programming [119] and exact linear programming [82] run in polynomial time and return rational solutions of polynomial size. If $k = 0$, any central minimizer is a constant selector. $\square$

Fixing the active bounds of an arbitrary central minimizer is not enough when C is singular. For $(y_1 + y_2 - w)^2$ with $y \in [0, 1]^2$ and $w \in [0, 2]$, the central minimizer $(0, 1)$ would fix both coordinates, yet the common central gradient is zero and the theorem returns the selector $(w/2, w/2)$. For $(y - w)^2$ with $y \in [0, 1]$ and $w \in [-1, 2]$, the clipped response is not affine, and the linear program is infeasible.

A negative answer rejects the affine class for this block, not the optimization problem; the block may still enter Theorem 7.9 with a piecewise certificate or with $\sigma_t = 0$. Choosing a useful partition into private blocks is not addressed.

C.6 Affine condensation and a negative-curvature regime

Lemma C.8 (Growth transfer). *In the model of Definition 7.5, suppose $F(v, y) - F^* \geq g(\|v - v^*\|^2 + \|y - y^*\|^2)$ on the feasible set, with $g > 0$. Then for every $v \in X_K$ and every choice of optimal responses $y(v)$,*

$$V(v) - F^* \geq g(\|v - v^*\|^2 + \|y(v) - y^*\|^2).$$

If block t has a one-leaf certificate $\bar{y}_t$ on Π_t , then $y^{(t)} = \bar{y}_t(v_{E_t}^*)$.*

Proof. Evaluate the growth inequality at $(v, y(v))$ and use $F(v, y(v)) = V(v)$. Replacing $y^{*(t)}$ by the optimal response $\bar{y}_t(v_{E_t}^*)$ gives a point of value $V(v^*) = F^*$, which equals (v^*, y^*) because growth makes the minimizer unique. $\square$

Corollary C.9 (Affine condensation). *Suppose every block has a one-leaf certificate $\bar{y}_t$ on Π_t , the retained coordinates are continuous, and F has point growth with constant g . Let $\mathcal{J}$ be the matrix of the linear part of the affine map $v \mapsto (v, \bar{y}(v))$, where $\bar{y}(v)$ collects the responses $\bar{y}_t(v_{E_t})$, so that $\|\mathcal{J}d\|^2 = \|d\|^2 + \sum_t \|B_t d_{E_t}\|^2$. Let $H_{\text{red}} = \nabla^2 V$ and $\nu = \max\{0, -\lambda_{\min}(\nabla^2 F)\}$.*

(a) $V(v) - F^* \geq g\|\mathcal{J}(v - v^*)\|^2$ and $H_{\text{red}} \succeq -\nu \mathcal{J}^\top \mathcal{J}$.

(b) If $\delta \in \mathbb{R}^K$ is positive with $\mathcal{J}^\top \mathcal{J} \succeq \text{diag}(\delta)$, choose dyadic $\omega_i > 0$ with $1 \leq \omega_i^2 \delta_i < 4$ and substitute $v = \text{diag}(\omega)v'$. The new box is rational, the interaction graph is unchanged, the growth constant in v' is at least g , and the upper coordinate curvature in v' is at most $4 \max_i \max\{(H_{\text{red}})_{ii}, 0\}/\delta_i$.

Proof. (a) By Lemma C.8, $y(v) = \bar{y}(v)$ and $y^* = \bar{y}(v^*)$, so $\|v - v^*\|^2 + \|y(v) - y^*\|^2 = \|\mathcal{J}(v - v^*)\|^2$. Since $V(v) = F(v, \bar{y}(v))$ is a quadratic composed with an affine map, $H_{\text{red}} = \mathcal{J}^\top \nabla^2 F \mathcal{J} \succeq -\nu \mathcal{J}^\top \mathcal{J}$; by Lemma 7.7(b) it equals H_0 minus the matrices $B_t^\top C_t B_t$ placed in the rows and columns E_t . (b) Growth in v' is at least $g \sum_i \omega_i^2 \delta_i (v'_i - v'^*_i)^2 \geq g\|v' - v'^*\|^2$, and the diagonal entry in v' is $\omega_i^2 (H_{\text{red}})_{ii} < 4(H_{\text{red}})_{ii}/\delta_i$ when it is positive. $\square$

Rescaling an integer coordinate changes its lattice, so part (b) is stated for continuous coordinates. The inequality in (a) bounds negative curvature only; it gives no upper bound on $(H_{\text{red}})_{ii}$.

Corollary C.10 (A negative-curvature regime). *In Theorem 7.9(iii), if the full Hessian of F has $\nu = \max\{0, -\lambda_{\min}\} > 0$ and $L^+ \leq C_0\nu$, then $\kappa_V \leq \max\{1, C_0\} \max\{1, \nu/g\}$. Membership in this regime can be tested up to a factor two: a rational $\hat{\nu}$ with $\nu \leq \hat{\nu} < 2\nu$ is computable in polynomial time, and $L^+ \leq C_0\hat{\nu}/2$ implies $L^+ \leq C_0\nu$.*

Proof. The bound on κ_V is immediate, and $\hat{\nu}/2 < \nu$ gives the implication. For $\hat{\nu}$, write $H = \nabla^2 F$ and let Δ_H be a positive integer with $\Delta_H H$ integral. First test $H \succeq 0$ exactly; if it fails, $\nu > 0$. Start from $\hat{\nu}_0 = \max_i \sum_j |H_{ij}| \geq \|H\| \geq \nu$ and halve $\hat{\nu}$ while $H + (\hat{\nu}/2)I \succeq 0$. The final $\hat{\nu}$ satisfies $H + \hat{\nu}I \succeq 0$ and $H + (\hat{\nu}/2)I \not\succeq 0$, that is, $\nu \leq \hat{\nu} < 2\nu$. If $d = \text{rank } H$, the product of the nonzero eigenvalues is, up to sign, a coefficient of the characteristic polynomial with denominator dividing Δ_H^d , so its absolute value is at least Δ_H^{-d} . Each nonzero eigenvalue is at most $\hat{\nu}_0$ in absolute value, so $\nu \geq \Delta_H^{-d} \hat{\nu}_0^{1-d}$, and the number of halvings is $\text{poly}(I)$. Each test is an exact symmetric elimination. $\square$

The algorithm itself never uses ν or $\hat{\nu}$; it computes L^+ . The hypothesis $L^+ \leq C_0\nu$ is substantive: Proposition 7.11 shows that it can fail for every admissible choice of private blocks.

C.7 Examples

Example C.11 (Three pieces, cancellation $2M$). For $M \geq 1$ let $z \in [0, \frac{3}{4}]$ be retained, $y \in [0, 1]^2$ private, and $\phi_M(z, y) = M(y_1 + y_2 - z)^2 + (y_1 - 2z + \frac{1}{2})^2$. The retained-only part is $Mz^2 + (2z - \frac{1}{2})^2$, so $(H_0)_{zz} = 2M + 8$, and $C = \begin{pmatrix} 2M+2 & 2M \\ 2M & 2M \end{pmatrix} \succ 0$, $\Gamma = (-2M - 4, -2M)^\top$, $c = (1, 0)^\top$. The unique responses and values are

$$\begin{aligned} z \in [0, \frac{1}{4}] : & \quad y = (0, z), & \min_y \phi_M &= (\frac{1}{2} - 2z)^2, & B^\top CB &= 2M, \\ z \in [\frac{1}{4}, \frac{1}{2}] : & \quad y = (2z - \frac{1}{2}, \frac{1}{2} - z), & \min_y \phi_M &= 0, & B^\top CB &= 2M + 8, \\ z \in [\frac{1}{2}, \frac{3}{4}] : & \quad y = (\frac{(M+2)z-1/2}{M+1}, 0), & \min_y \phi_M &= \frac{M(z-1/2)^2}{M+1}, & B^\top CB &= \frac{2(M+2)^2}{M+1}. \end{aligned}$$

On the three intervals the private gradients are $(1 - 4z, 0)$, $(0, 0)$ and $(0, 2M(z - \frac{1}{2})/(M + 1))$, all with the signs required at the active lower bounds; in the third interval $y_1 \in [\frac{1}{2}, 1)$. These are certificates. By Proposition 7.8, $\sigma = 2M$, and the certified curvature is $2M + 8 - 2M = 8$, while the direct bound is $2M + 8$. The piece second derivatives $8, 0, 2M/(M + 1)$ agree with $(H_0)_{zz} - B^\top CB$. The response is unique and not affine, so no affine selector exists.

Proposition C.12 (A ladder with growing negative inertia). *For $m \geq 1$, $M \geq 1$ and $a = \frac{1}{5}$, let $z_i \in [0, \frac{3}{4}]$, $w_i \in [0, 1]$ be retained, $y_{1i}, y_{2i} \in [0, 1]$ private, and*

$$F = \sum_{i=1}^m \left[z_i^2 + \phi_M(z_i, y_{1i}, y_{2i}) - w_i^2 + 3w_i - (z_i - a)w_i \right] + \frac{1}{16} \sum_{i=1}^{m-1} \left[(z_{i+1} - z_i)^2 + (w_{i+1} - w_i)^2 \right].$$

Then F has the unique minimizer $z_i = a$, $w_i = 0$, $y_{1i} = 0$, $y_{2i} = a$, with $F^ = m/20$; growth holds with constant $\frac{1}{12}$, and no growth constant exceeds $\frac{33}{16}$. Its Hessian has exactly m negative eigenvalues and $2 \leq \nu \leq \sqrt{5}$. With the private blocks (y_{1i}, y_{2i}) , the retained bags $\{z_i, w_i, z_{i+1}, w_{i+1}\}$ have size at most four, and the certificates of Example C.11 give $L^+ \leq \frac{41}{4}$. The direct diagonal at z_i is at least $2M + 10$.*

Proof. For one index put $t = z - a$, $\alpha_1 = y_1$, $\alpha_2 = y_2 - a$, $\rho_1 = \alpha_1 + \alpha_2 - t$ and $\rho_2 = \alpha_1 - 2t$. Direct expansion gives

$$z^2 + \phi_M - \frac{1}{20} = t^2 + M\rho_1^2 + \rho_2^2 + \frac{\alpha_1}{5}.$$

Also $-w^2 + 3w - tw \geq w^2$, because $w(3 - 2w - t) \geq 0$ for $w \in [0, 1]$ and $t \leq \frac{11}{20}$. Since $\alpha_1 \geq 0$ and the path terms are nonnegative,

$$F - \frac{m}{20} \geq \sum_i (t_i^2 + M\rho_{1i}^2 + \rho_{2i}^2 + w_i^2).$$

From $\alpha_1 = \rho_2 + 2t$ and $\alpha_2 = \rho_1 - \rho_2 - t$, $\alpha_1^2 \leq 2\rho_2^2 + 8t^2$ and $\alpha_2^2 \leq 3(\rho_1^2 + \rho_2^2 + t^2)$, so $t^2 + \alpha_1^2 + \alpha_2^2 \leq 12(t^2 + \rho_1^2 + \rho_2^2) \leq 12(t^2 + M\rho_1^2 + \rho_2^2)$. Hence $F - F^* \geq \frac{1}{12}\|x - x^*\|^2$, and $F = F^*$ forces $t = \rho_1 = \rho_2 = w = 0$ and $\alpha_1 = \alpha_2 = 0$.

The terms $z_i^2 - w_i^2 - (z_i - a)w_i$ contribute the block $\begin{pmatrix} 2 & -1 \\ -1 & -2 \end{pmatrix}$ on each pair (z_i, w_i) , with eigenvalues $\pm\sqrt{5}$. All other quadratic terms, namely the squares in ϕ_M and the path terms, are positive semidefinite. So $\lambda_{\min} \geq -\sqrt{5}$. The direction with $w = m^{-1/2}(1, \dots, 1)$ and all other coordinates zero has Rayleigh quotient -2 , so $\nu \geq 2$. On the m -dimensional w -subspace the form is $-2I + \frac{1}{8}L_{\text{path}}$, where L_{path} is the Laplacian of the path, and $\frac{1}{8}\|L_{\text{path}}\| \leq \frac{1}{2}$, so it is negative definite; on the subspace $\{w = 0\}$, of codimension m , the form is a sum of squares. Hence the negative inertia is exactly m .

The retained diagonal at z_i is $(H_0)_{z_i z_i} = 2 + (2M + 8) + \frac{1}{8}\deg_i$ with $\deg_i \leq 2$, and at w_i it is $-2 + \frac{1}{8}\deg_i < 0$. Each block has $\sigma = 2M$ by Example C.11, so $L_{z_i} \leq 10 + \frac{1}{4} = \frac{41}{4}$ and $L_{w_i} < 0$. Raising w_1 from 0 to 1 at the minimizer increases F by at most $2 + \frac{1}{16}$, which shows that no growth constant exceeds $\frac{33}{16}$. $\square$

Thus the certified ratio is $L^+/g \leq 123$ for all m and M , whereas the direct coordinate ratio is at least $(2M + 10) \cdot 16/33$. Every instance also has the explicit nonnegative expansion above; the family separates parameters and is not offered as a hard benchmark.

Remark C.13 (The certified cancellation is unstable). For the block objective $M(y - w)^2 + \epsilon y$ with $y, w \in [0, 1]$, the response is $y = \max\{0, w - \epsilon/(2M)\}$. For $\epsilon = 0$ it is $y = w$, so $\sigma = 2M$. For every $\epsilon > 0$ the piece $[0, \epsilon/(2M)]$ has response $y = 0$, so $\sigma = 0$ by Proposition 7.8(c), and the value has second derivative $2M$ there. The full Hessian, and hence ν , does not depend on ϵ . The same happens near $z = 0$ in Example C.11 if ϵy_2 is added: on a short interval both private coordinates sit at their lower bounds, and $\sigma = 0$. The certified curvature is a maximum over pieces, however small the piece.

C.8 Proof of Proposition 7.11

Put $\xi_1 = x - 1 \in [-1, 1]$ and $\xi_2 = y - 1 \in [0, 2]$, so $F_M = M(\xi_1 - \xi_2)^2 - \frac{1}{8}(\xi_1 + \xi_2)^2 + 2\xi_2$. Using $M \geq 1$ and $2\xi_2 \geq \xi_2^2$ on $[0, 2]$,

$$F_M - \frac{1}{8}(\xi_1^2 + \xi_2^2) \geq \frac{3}{4}\xi_1^2 - \frac{9}{4}\xi_1\xi_2 + \frac{7}{4}\xi_2^2 = \frac{3}{4}(\xi_1 - \frac{3}{2}\xi_2)^2 + \frac{1}{16}\xi_2^2 \geq 0.$$

This proves growth $\frac{1}{8}$, uniqueness and $F^* = 0$. The Hessian $\begin{pmatrix} 2M-1/4 & -2M-1/4 \\ -2M-1/4 & 2M-1/4 \end{pmatrix}$ has eigenvalues $4M$ on $(1, -1)$ and $-\frac{1}{2}$ on $(1, 1)$; so $\nu = \frac{1}{2}$, and $\mathcal{K} = \emptyset$ is not admissible. The admissible sets are $\{x, y\}$, $\{y\}$ and $\{x\}$. Note $F_M(2, 2) = \frac{3}{2}$.

$\mathcal{K} = \{x, y\}$: $V = F_M$ has $L_{\mathcal{K}} = 2M - \frac{1}{4}$. Under $x = \omega_1 x'$, $y = \omega_2 y'$ the coordinate curvatures become $\omega_i^2(2M - \frac{1}{4})$, and the point $(2, 2)$ shows that every growth constant in the new coordinates is at most $\frac{3}{2}/(\omega_1^{-2} + \omega_2^{-2}) \leq \frac{3}{2} \min_i \omega_i^2$.

$\mathcal{K} = \{y\}$: F_M is strictly convex in x , and $\partial_x F_M(2, y) = 2M(2 - y) - y/4 \leq 0$ for $y \geq y_c = 4M/(2M + \frac{1}{4})$, where $1 < y_c < 2$. So on $[y_c, 3]$ the response is $x = 2$, and $V(y) = F_M(2, y)$ has second derivative $2M - \frac{1}{4}$. Hence $L_{\mathcal{K}} \geq 2M - \frac{1}{4}$, while $V(2) \leq \frac{3}{2}$ and $V(1) = 0$ give $g_{\mathcal{K}} \leq \frac{3}{2}$.

$\mathcal{K} = \{x\}$: F_M is strictly convex in y , and $\partial_y F_M(x, 1) = 2 - (2M + \frac{1}{4})(x - 1) \geq 0$ for $x \leq x_c = 1 + 2/(2M + \frac{1}{4})$, where $1 < x_c \leq 2$. So on $[0, x_c]$ the response is $y = 1$, and $V(x) = (M - \frac{1}{8})(x - 1)^2$ has second derivative $2M - \frac{1}{4}$. Again $V(2) \leq \frac{3}{2}$ gives $g_{\mathcal{K}} \leq \frac{3}{2}$.

In one dimension a rescaling multiplies curvature and growth by the same factor. In all cases $L_{\mathcal{K}}/g_{\mathcal{K}} \geq (2M - \frac{1}{4})/\frac{3}{2} = (4M - \frac{1}{2})/3 \geq M$. $\square$

The obstruction is a clipped piece: a private coordinate that cancels the stiffness hits a bound on a full-dimensional set of parameters. Such a piece can be arbitrarily short (Remark C.13). In the proof above, every upper bound on $g_{\mathcal{K}}$ comes from a point with objective gap at least $\frac{1}{8}$. A bound in terms of p and ν/g along this route therefore cannot use the global constants $L_{\mathcal{K}}$ and $g_{\mathcal{K}}$; one possibility is to measure curvature and growth only where near-optimal points can lie (Section 12, question 1).

D Cut-based recourse: exact output, concave–convex residuals and proofs

This appendix proves Theorems 7.15 and 7.18 and Proposition 7.19 of Section 7.4, the exact-output theorem for a cut residual (Theorem D.2), the claims of Remark 7.21 on concave–convex residuals, and the results deferred from Section 7.5. The first three subsections use the notation of Section 7.4: $F(x) = \frac{1}{2}x^T Hx + b^T x + c$ is a rational quadratic, $\mathcal{K}$ is a core of k continuous coordinates with domain $[0, 1]^k$, $\mathcal{R}$ is the residual, and α_i , δ_i and $y(z)$ are as defined before Proposition 7.14; the last uses that of Section 7.5.

D.1 Proofs for Section 7.4

The proofs below use the bag-cell error $e_B(C)$ of Section 7.2 with $B = \mathcal{K}$, and e_j , $\beta_V(C)$, $\mathcal{Q}_j$, $\mathcal{Q}'_j$, λ_j and U as in CORE (Algorithm 7.17).

Proof of Theorem 7.15. By Lemma 7.13 the conditional minimum is attained at some $y(z)$, and by Proposition 7.14 minimizing over z is a minimum cut problem. Every feasible flow has value at most the capacity of every cut (weak duality), so a flow of value $\text{cap}(S_z)$ proves that S_z is a minimum cut; checking it needs the capacity bounds, conservation at each residual node and one comparison. A maximum flow and a minimum cut of equal value exist and are found, for instance, by the Edmonds–Karp algorithm [64], whose number of arithmetic operations is polynomial in the number of nodes and arcs, here $|\mathcal{R}| + 2$ and at most $2|\mathcal{R}|$ plus twice the number of nonzero residual couplings. After multiplication by a common denominator of polynomial bit length, the capacities are integers, every augmentation is by an integer, and every flow value is an integer at most the total capacity, so all intermediate numbers have polynomial bit length. The coefficients of Proposition 7.14(b) are obtained by exact rational evaluation. $\square$

Proof of Theorem 7.18. Let C be a cell of side h_j and $x \in X$ with $x_{\mathcal{K}} \in C$. Lemma 7.3 with $B = \mathcal{K}$ gives $F(x) \geq \min\{V(v) : v \text{ a corner of } C\} - e_{\mathcal{K}}(C) \geq \beta_V(C)$, because the core coordinates are continuous and $e_{\mathcal{K}}(C) \leq e_j$.

(a) The unit cube contains $x_{\mathcal{K}}^*$. If $C \in \mathcal{Q}_j$ contains it, then $\beta_V(C) \leq F(x^*) = F^* \leq U$, so C is retained and every child of C that contains $x_{\mathcal{K}}^*$ belongs to $\mathcal{Q}_{j+1}$. (b), (c) These are Theorem 7.4 with $B = \mathcal{K}$, the cells $\mathcal{Q}_j$, $e = e_j$, $\underline{V} = V$ and $\varepsilon_{\text{or}} = 0$; its hypothesis that some cell of $\mathcal{Q}_j$ contains

$x_{\mathcal{K}}^*$ holds by (a), and $\lambda_j \leq F^*$ by the first paragraph applied to x^* . (d) Children of retained cells cover them, so the cells of $\mathcal{Q}_j$ together with the cells discarded at levels $i < j$ cover $[0, 1]^k$. If $x_{\mathcal{K}}$ lies in a cell of $\mathcal{Q}_j$, then $F(x) \geq \lambda_j$ by the first paragraph. If it lies in a cell C discarded at level i , then $F(x) \geq \beta_V(C) > U^{(i)} \geq U$, where $U^{(i)}$ is the incumbent value at level i . Each step uses only recorded values. $\square$

Proof of Proposition 7.19. Let N_j be the number of points $v \in h_j \mathbb{Z}^k \cap [0, 1]^k$ with $V(v) \leq F^* + 2e_j$. Level 0 makes 2^k queries. By Theorem 7.18(c), every retained level- j cell has a corner counted in N_j , and a point is a corner of at most 2^k cells, so $|\mathcal{Q}'_j| \leq 2^k N_j$. Each retained cell has 2^k children with 2^k corners each, so level $j + 1$ makes at most $8^k N_j$ queries. A point counted in N_j satisfies $g\|v - v^*\|^2 \leq 2e_j = kLh_j^2/4$, so each of its coordinates lies within $\frac{h_j}{2}\sqrt{k\kappa_{\mathcal{K}}}$ of that of v^* ; an interval of length $h_j\sqrt{k\kappa_{\mathcal{K}}}$ contains at most $1 + \sqrt{k\kappa_{\mathcal{K}}}$ points of $h_j \mathbb{Z}$. Hence $N_j \leq (1 + \sqrt{k\kappa_{\mathcal{K}}})^k$; sum over $j < J$. $\square$

D.2 Exact output with a cut residual

Let $\Delta_{\mathcal{K}} = \Lambda_c \Lambda_e^2$, where Λ_c is the least positive common denominator of the monomial coefficients of F and Λ_e that of the residual bounds after inward rounding; both have $O(I)$ bits. For a residual endpoint vector y put $F_y(v) = F(v, y)$ and $\Psi(y) = \min_{v \in [0, 1]^k} F_y(v)$.

Lemma D.1 (Uniform height over residual endpoint vectors). *Assume (R1). Let $\Omega_{\mathcal{K}}$ be the constant Ω of (6.1) computed for F_y on $[0, 1]^k$ with the common denominator $\Delta = \Delta_{\mathcal{K}}$. Then $\Omega_{\mathcal{K}}$ does not depend on y , $\log \Omega_{\mathcal{K}} = O(I^2)$, and for every residual endpoint vector y the value $\Psi(y)$ is a rational with reduced denominator at most $\Omega_{\mathcal{K}}$; so is F^* .*

Proof. The monomial coefficients of F_y are $\frac{1}{2}H_{ii}$ for $i \in \mathcal{K}$, H_{ij} for $i < j$ in $\mathcal{K}$, the linear coefficients $b_i + \sum_{j \in \mathcal{R}} H_{ij}y_j$ and the constant $c + \sum_{j \in \mathcal{R}} (b_j y_j + \frac{1}{2}H_{jj}y_j^2) + \sum_{j < l, j, l \in \mathcal{R}} H_{jl}y_j y_l$. Each is a sum of products of a coefficient of F with at most two residual bounds, so $\Delta_{\mathcal{K}}$ is a common denominator of all of them and of the endpoints 0, 1. The Hessian of F_y is $H_{\mathcal{K}\mathcal{K}}$ for every y , and the constant Ω of (6.1) depends only on Δ and the Hessian, so $\Omega_{\mathcal{K}}$ does not depend on y ; its bit length is $O(kI) = O(I^2)$, as in Section 6.1. Corollary 6.4(b), applied to the box QP F_y on $[0, 1]^k$, bounds the reduced denominator of $\Psi(y)$ by $\Omega_{\mathcal{K}}$. By Lemma 7.13, $F^* = \min_v \min_y F(v, y)$ with y ranging over residual endpoint vectors, so $F^* = \Psi(y^*)$ for some such y^* . $\square$

The square of Λ_e is needed: the term $\frac{1}{2}y_i y_j$ at residual endpoints $y_i = y_j = \frac{1}{2}$ equals $\frac{1}{8}$, while the data have common denominator 2.

Theorem D.2 (Exact output with a cut residual). *Assume (7.11) with continuous core $[0, 1]^k$ and rational data, and run CORE (Algorithm 7.17) with the oracle of Theorem 7.15, $L = \max_{i \in \mathcal{K}} \max\{H_{ii}, 0\}$, accuracy $\varepsilon = \Omega_{\mathcal{K}}^{-2}/2$ and level limit J the first level with $e_J \leq \varepsilon$. Let y_U be the residual part of the final incumbent and $\beta = \min\{\lambda_j, U\}$ the returned bound. Minimize F_{y_U} exactly over $[0, 1]^k$ by enumerating the 3^k faces of the cube, solving the stationarity system on each face whose free Hessian block is nonsingular, and keeping the solutions that lie in the face. The best candidate $\hat{v}$ makes $(\hat{v}, y_U)$ a global minimizer, and $(\hat{v}, y_U)$ passes the test of Proposition 6.6 with the denominator bound $\Omega_{\mathcal{K}}$ and this β . A checker needs only the trace of CORE, $\Omega_{\mathcal{K}}$, and the feasibility, exact value and reduced denominator of the output. If in addition V has core growth with constant g as in Proposition 7.19, the total bit complexity is $8^k(1 + \sqrt{k\kappa_{\mathcal{K}}})^k \text{poly}(I)$ with an absolute exponent, and g is not an input.*

Proof. By Theorem 7.18(b), CORE stops at level J at the latest, with $U - \beta \leq \varepsilon < \Omega_{\mathcal{K}}^{-2}$. Apply Lemma 6.3 to the box QP F_{y_U} on $[0, 1]^k$: let v° be a minimizer and v a vertex of its stationary polytope. By part (b), v is a minimizer, and by part (c) the Hessian block of F_{y_U} on the set $J_0(v)$ of coordinates of v in $(0, 1)$ is positive definite. So v is the unique solution of the stationarity system on the face of the cube that fixes the coordinates outside $J_0(v)$ at their values in v , and the enumeration contains it. All candidates lie in $[0, 1]^k$, so the best one attains $\Psi(y_U)$.

The oracle returns endpoint vectors, so y_U is one. With (v_U, y_U) the final incumbent, $F(\hat{v}, y_U) = \Psi(y_U) \leq F(v_U, y_U) = U$, and $\beta \leq F^*$ by Theorem 7.18(d). By Lemma D.1, F^* has denominator at most $\Omega_{\mathcal{K}}$ and $F(\hat{v}, y_U) = a/W$ in lowest terms with $W \leq \Omega_{\mathcal{K}}$. Hence $F(\hat{v}, y_U) - \beta \leq U - \beta < \Omega_{\mathcal{K}}^{-2} \leq 1/(\Omega_{\mathcal{K}}W)$, and Proposition 6.6 shows that $(\hat{v}, y_U)$ is a global minimizer. The checker verifies $\beta \leq F^*$ by Theorem 7.18(d) and then the inequality $F(\hat{v}, y_U) - \beta < 1/(\Omega_{\mathcal{K}}W)$, which proves optimality by Proposition 6.6.

For the complexity, if $L = 0$ then $J = 0$. Otherwise $\log(kL) = O(I)$ and $\log \Omega_{\mathcal{K}} = O(I^2)$, so $J = O(\log \Omega_{\mathcal{K}} + \log(kL) + 1) = O(I^2)$. Under core growth, Proposition 7.19 bounds the queries by $2^k + 8^k J(1 + \sqrt{k\kappa_{\mathcal{K}}})^k$. Each query is a cut problem whose data have bit length polynomial in $I + kJ$, hence $\text{poly}(I)$ by Theorem 7.15, with an absolute exponent. The face enumeration costs $3^k \text{poly}(I)$, and $2^k + 3^k \leq 2 \cdot 8^k$. $\square$

Without core growth the procedure still terminates, after at most $J = O(I^2)$ levels, but Example 7.20 shows that the number of queries can then be $2^{\Omega(kJ)}$.

D.3 Concave–convex residuals

This subsection proves the claims of Remark 7.21. Fix a rational core point v . Let $\mathcal{R} = \mathcal{R}_- \cup \mathcal{R}_+$ with $H_{ii} \leq 0$ for $i \in \mathcal{R}_-$, signs o with $o_i o_j H_{ij} \leq 0$ for all $i \neq j$ in $\mathcal{R}$, continuous coordinates in $\mathcal{R}_+$ and $H_{\mathcal{R}_+ \mathcal{R}_+} \succeq 0$. For $S \subseteq \mathcal{R}_-$ let $y_-(S) \in \mathbb{R}^{\mathcal{R}_-}$ have coordinates $\alpha_i + \delta_i[i \in S]$, and put

$$\Phi(S) = \min \left\{ F(v, y_-(S), y_+) : y_+ \in \prod_{i \in \mathcal{R}_+} [\ell_i, u_i] \right\}.$$

With $\mathcal{R}_+ = \emptyset$ this is the class (7.11); with $\mathcal{R}_- = \emptyset$ the residual problem is convex.

Proposition D.3 (Submodular conditional value). *(a) The minimum of $F(v, \cdot)$ over the continuous residual box and over the mixed residual domain equals $\min_{S \subseteq \mathcal{R}_-} \Phi(S)$. (b) Each $\Phi(S)$ is the value of a rational convex box QP. (c) For all $S, S' \subseteq \mathcal{R}_-$, $\Phi(S) + \Phi(S') \geq \Phi(S \cap S') + \Phi(S \cup S')$.*

Proof. (a) As in Lemma 7.13, move each coordinate of $\mathcal{R}_-$ to an endpoint without increasing F ; the resulting point is in the mixed domain because $\mathcal{R}_+$ is continuous. (b) After the coordinates in $\mathcal{R}_-$ are fixed, the objective in y_+ has Hessian $H_{\mathcal{R}_+ \mathcal{R}_+} \succeq 0$. (c) Substitute out the coordinates with $\delta_i = 0$ and put $y_i = \alpha_i + \delta_i t_i$ with $t \in [0, 1]^{\mathcal{R}_-}$. The transformed objective $\tilde{F}$ has off-diagonal Hessian entries $\delta_i \delta_j H_{ij} = o_i o_j (u_i - \ell_i)(u_j - \ell_j) H_{ij} \leq 0$. For $t, t' \in [0, 1]^{\mathcal{R}_-}$, $\tilde{F}(t) + \tilde{F}(t') \geq \tilde{F}(t \wedge t') + \tilde{F}(t \vee t')$: the terms in one coordinate cancel, because $\{t_i, t'_i\} = \{(t \wedge t')_i, (t \vee t')_i\}$; for a pair i, j the two sides agree if $t_i - t'_i$ and $t_j - t'_j$ do not have opposite signs, and otherwise $t_i t_j + t'_i t'_j - (t \wedge t')_i (t \wedge t')_j - (t \vee t')_i (t \vee t')_j = (t_i - t'_i)(t_j - t'_j) \leq 0$, which, multiplied by a nonpositive coefficient, is nonnegative. Let t^S and $t^{S'}$ attain $\Phi(S)$ and $\Phi(S')$ in the transformed coordinates of $\mathcal{R}_+$. Then $(\mathbf{1}_S, t^S) \wedge (\mathbf{1}_{S'}, t^{S'}) = (\mathbf{1}_{S \cap S'}, t^S \wedge t^{S'})$, similarly for $\vee$ with $S \cup S'$, and both completions are feasible. Hence $\Phi(S) + \Phi(S') \geq \tilde{F}(\mathbf{1}_{S \cap S'}, t^S \wedge t^{S'}) + \tilde{F}(\mathbf{1}_{S \cup S'}, t^S \vee t^{S'}) \geq \Phi(S \cap S') + \Phi(S \cup S')$. $\square$

Positive semidefiniteness is not used in (c); it makes each $\Phi(S)$ computable exactly in polynomial time [119]. Let $m = |\mathcal{R}_-|$ and $\hat{\Phi}(S) = \Phi(S) - \Phi(\emptyset)$. For an ordering π of $\mathcal{R}_-$ with prefixes

$S_0^\pi = \emptyset, S_1^\pi, \dots, S_m^\pi = \mathcal{R}_-$, the greedy vector $a^\pi \in \mathbb{R}^{\mathcal{R}_-}$ has $a_{\pi(l)}^\pi = \hat{\Phi}(S_l^\pi) - \hat{\Phi}(S_{l-1}^\pi)$. We write $a(S) = \sum_{i \in S} a_i$.

Proposition D.4 (Greedy-base certificate). (a) For every ordering π and every $S \subseteq \mathcal{R}_-$, $a^\pi(S) \leq \hat{\Phi}(S)$, and $a^\pi(\mathcal{R}_-) = \hat{\Phi}(\mathcal{R}_-)$. (b) If $\bar{a} = \sum_\pi \lambda_\pi a^\pi$ is a convex combination, then $\min_y F(v, y) \geq \Phi(\emptyset) + \sum_{i \in \mathcal{R}_-} \min\{0, \bar{a}_i\}$, the minimum taken over the residual domain. (c) Some convex combination of at most $m + 1$ greedy vectors attains equality in (b) [63]. Such a combination, a minimizing set S^* and exact minimizers of the convex box QPs involved can be computed in time polynomial in the input length. (d) Given the orderings, the weights, the $O(m^2)$ values $\Phi(S_l^\pi)$ with minimizers, and a set S^* with its completion, a checker verifies the bound in (b) and its attainment by S^* with polynomially many rational operations and no optimization. It checks a value $\Phi(S)$ by feasibility, exact evaluation and the sign conditions $\partial_i F = 0$ at interior coordinates, $\partial_i F \geq 0$ at lower bounds and $\partial_i F \leq 0$ at upper bounds for $i \in \mathcal{R}_+$, which suffice by convexity once $H_{\mathcal{R}_+ \mathcal{R}_+} \succeq 0$ and the sign conditions of the class have been verified exactly.

Proof. (a) Let $i = \pi(l) \in S$, and let $S_{<i}$ be the set of elements of S that precede i in π . Since $S_{<i} \subseteq S_{l-1}^\pi$ and $i \notin S_{l-1}^\pi$, submodularity (Proposition D.3(c)) applied to $S_{<i} \cup \{i\}$ and S_{l-1}^π gives $a_i^\pi = \hat{\Phi}(S_l^\pi) - \hat{\Phi}(S_{l-1}^\pi) \leq \hat{\Phi}(S_{<i} \cup \{i\}) - \hat{\Phi}(S_{<i})$. Summing over $i \in S$ telescopes to $a^\pi(S) \leq \hat{\Phi}(S)$, because $\hat{\Phi}(\emptyset) = 0$; the sum over all positions is $\hat{\Phi}(\mathcal{R}_-)$.

(b) By (a) and convexity, $\bar{a}(S) \leq \hat{\Phi}(S)$ for every S , so $\Phi(S) = \Phi(\emptyset) + \hat{\Phi}(S) \geq \Phi(\emptyset) + \bar{a}(S) \geq \Phi(\emptyset) + \sum_i \min\{0, \bar{a}_i\}$; conclude with Proposition D.3(a).

(c) Edmonds' theorem gives $\min_S \hat{\Phi}(S) = \max\{\sum_i \min\{0, \bar{a}_i\} : \bar{a} \in B(\hat{\Phi})\}$, where the base polytope $B(\hat{\Phi})$ has the greedy vectors as its vertices; by Carathéodory's theorem, the maximum of this concave piecewise-linear function is attained at a combination of at most $m + 1$ of them. For computation, consider the linear program $\max\{t : 0 \leq \xi \leq \mathbf{1}, t + (a^\pi)^\top \xi \leq 0 \text{ for all } \pi\}$, whose dual is $\min\{\sum_i r_i : \sum_\pi \lambda_\pi a^\pi + r \geq 0, \sum_\pi \lambda_\pi = 1, \lambda, r \geq 0\}$. For $\xi \in [0, 1]^m$, an ordering π with $\xi_{\pi(1)} \geq \dots \geq \xi_{\pi(m)}$ maximizes $(a^\pi)^\top \xi$ over all orderings [63], and with $\xi_{\pi(m+1)} := 0$

$$(a^\pi)^\top \xi = \sum_{l=1}^m (\xi_{\pi(l)} - \xi_{\pi(l+1)}) \hat{\Phi}(S_l^\pi) + (1 - \xi_{\pi(1)}) \hat{\Phi}(\emptyset)$$

is a convex combination of values of $\hat{\Phi}$, hence at least $\min \hat{\Phi}$; equality holds for $\xi = \mathbf{1}_S$ with S minimizing $\hat{\Phi}$. So the optimal value of the program is $-\min \hat{\Phi}$, a separation call costs $m + 1$ convex-QP evaluations, and for an optimal ξ every prefix with positive weight in this combination is a minimizing set. Each $\Phi(S)$ is the optimal value of a convex box QP with Hessian $H_{\mathcal{R}_+ \mathcal{R}_+}$ and has polynomial encoding length by Corollary 6.4(b); hence the polyhedron has polynomial facet complexity, and the ellipsoid method solves the program and finds an optimal basic dual solution in polynomial time [82, Theorem 6.4.9, Lemma 6.5.15]. A basic dual solution has at most $m + 1$ positive λ_π , and at optimality $r_i = \max\{0, -\bar{a}_i\}$ with $\bar{a} = \sum_\pi \lambda_\pi a^\pi$, so $\sum_i \min\{0, \bar{a}_i\} = \min \hat{\Phi}$.

(d) follows from (a), (b) and the optimality conditions for convex problems on a box. $\square$

D.4 Proofs for Section 7.5

Proof of Lemma 7.23. List G_i as $a_{i,0}, \dots, a_{i,k_i}$, in increasing order if $o_i = 1$ and in decreasing order if $o_i = -1$, so that $\delta_{i,l} = a_{i,l} - a_{i,l-1}$ has the sign of o_i . Encode $y_i = a_{i,l}$ by $z_i \in \{0, 1\}^{k_i}$ with $z_{i,l'} = 1$ exactly for $l' \leq l$. This is a bijection between G_i and the nonincreasing vectors $z_{i,1} \geq \dots \geq z_{i,k_i}$, and on these vectors

$$y_i = a_{i,0} + \sum_l \delta_{i,l} z_{i,l}, \quad \varphi(y_i) = \varphi(a_{i,0}) + \sum_l [\varphi(a_{i,l}) - \varphi(a_{i,l-1})] z_{i,l}$$

for every function φ on G_i . Apply the second formula to $\varphi_i(t) = \frac{1}{2}H_{ii}t^2 + b_it + \chi_i(t)$, and substitute the first into each cross term $H_{ij}y_iy_j$, $i < j$. On nonincreasing vectors this writes F_χ as a function $\Phi(z)$ of the binary vector $z = (z_{i,l})$ of the form of Proposition 7.14(a), after the pairs (i, l) are numbered, with pair coefficients $H_{ij}\delta_{i,l}\delta_{j,l'} \leq 0$ for $i \neq j$ and none within one coordinate. Take the network of Proposition 7.14(a) for Φ and add an arc $(i, l+1) \rightarrow (i, l)$ of infinite capacity for each i and $1 \leq l < k_i$. A cut S_z has finite capacity exactly when no such arc leaves it, that is, when z is nonincreasing in each coordinate; its capacity is then given by (7.12). The all-zero vector is nonincreasing, so a minimum cut is finite and decodes to a minimizer of F_χ , and a flow of equal value certifies it by weak duality. Fixing $y_i = v$ is the same construction with $G_i = \{v\}$, and the restricted function is still balanced. $\square$

Proof of the work bounds in Theorem 7.24(b). If $L = 0$, then $P = \emptyset$ and CT (Algorithm 5.6) makes one cut computation. Otherwise point growth with constant g gives weighted growth with $\gamma = g/L$, because $\|d\|_{\mathbf{L}}^2 \leq L\|d\|^2$, so $\bar{\kappa} \leq \kappa$. By the proof of Theorem 5.7(b), CT succeeds at the latest at the trial μ^* with $2^{\mu^*} \leq 6\sqrt{\kappa} \leq 6\sqrt{\kappa}$, so at most $\log_2(6\sqrt{\kappa})$ trials are run; the trial with $\theta = 2^{-\mu}$ has at most $K_\mu \leq K_{\mu^*} = O(\sqrt{\kappa} \log n)$ nodes per coordinate, and each trial has $O(I + q + 1)$ stages. A stage performs at most $1 + nK_\mu$ cut computations on networks with at most $nK_\mu + 2$ nodes and $3nK_\mu + n^2K_\mu^2$ arcs. As in Appendix A, grid nodes have denominators dividing $\Gamma_X 2^\alpha$ and the values of F and of the corrections at grid points have denominators dividing $8\Gamma_F \Gamma_X^2 4^\alpha$, with $\alpha = O(I + q + \mu K_\mu)$. Each capacity of Lemma 7.23 with $\chi_i = -d_i$ is a sum of $O(nK_\mu)$ terms, each a coefficient of F times at most two grid nodes or differences of grid nodes, a correction value, or half of such a term. So the capacities have a common denominator dividing $8\Gamma_F \Gamma_X^2 4^\alpha$ and numerators of $O(I + \alpha + \log(nK_\mu))$ bits. Give each arc of infinite capacity the capacity one plus the sum of all finite capacities; this changes neither the minimum cuts nor the maximum-flow value. After scaling by the common denominator, the Edmonds–Karp algorithm performs a number of augmentations polynomial in the network size, each on integers bounded by the total capacity. Multiplying these polynomial bounds gives the bound for CT. Under point growth, the proof of Theorem 6.14 shows that EX (Algorithm 6.13) stops at an accuracy $q = \text{poly}(I) + O(\log \kappa)$ after $O(\log q)$ calls of CT, and that REC then needs only Gaussian elimination; this gives the bound for EX. $\square$

E Coupling constraints: checks, proofs, exact output and examples

This appendix gives the checks of the model of Section 8.1, the proofs of Proposition 8.7 and Theorems 8.8 and 8.9, an example for the filter of Section 8.4, the algorithm TU-EXACT with the proof of Theorem 8.10, and the details for Section 8.7.

E.1 Checking the model

Remark E.1 (Directional versus coordinate curvature). With $\mathcal{W} = \mathbb{R}^{n_c}$, (8.2) implies, for each $z \in \mathcal{Z}$, the coordinate curvature condition of Definition 3.1 with $L_i = \bar{L}$ for the continuous coordinates, but not conversely (Example 8.5). The following checks of (8.2) are rational, take time polynomial in I , and change no factor scope. Two of them use that the largest eigenvalue of a symmetric matrix is at most its largest absolute row sum.

- (i) Explicit polynomial factors of fixed degree, any $\mathcal{W}$. For continuous coordinates i, i' write $\partial_{ii'} F = \sum_\alpha c_\alpha v^\alpha$, a sum over finitely many exponent vectors α , let $[\underline{\mu}_\alpha, \bar{\mu}_\alpha]$ be the range of the monomial v^α on the box $[\ell, u] \times \prod_k [\min Z_k, \max Z_k]$, computed as in the proof of Lemma 6.20,

and put

$$b_{ii'} = \sum_{\alpha} |c_{\alpha}| \max\{|\underline{\mu}_{\alpha}|, |\overline{\mu}_{\alpha}|\}.$$

Then $|\partial_{ii'} F| \leq b_{ii'}$ on that box, which contains $[\ell, u] \times \mathcal{Z}$. Hence every positive rational that is at least $\max_i \sum_{i'} b_{ii'}$ bounds the largest eigenvalue of $\nabla_{xx}^2 F(x, z)$ for all $x \in [\ell, u]$ and $z \in \mathcal{Z}$, and is a valid $\bar{L}$ for every subspace $\mathcal{W}$. There are n_c^2 bounds $b_{ii'}$, each a sum of at most as many terms as F has monomials, and for factors of degree at most d the range endpoints have bit length $O(dI)$. For a quadratic, $b_{ii'} = |(H_{xx})_{ii'}|$, and the bound is the largest absolute row sum of H_{xx} .

- (ii) Quadratic factors, $\mathcal{W} = \ker C$. Then $\bar{L}$ is valid if and only if $\Xi_C^T(\bar{L}I - H_{xx})\Xi_C \succeq 0$ for a basis matrix Ξ_C of $\ker C$; no orthonormality of Ξ_C is needed, since $w = \Xi_C c$ gives $w^T(\bar{L}I - H_{xx})w = c^T \Xi_C^T(\bar{L}I - H_{xx})\Xi_C c$. Alternatively, if C has full row rank and $\Pi_C = I - C^T(CC^T)^{-1}C$, every positive rational that is at least the largest absolute row sum of $\Pi_C H_{xx} \Pi_C$ is valid, because $\Pi_C w = w$ for $w \in \ker C$.

Proposition E.2 (Checking alignment without enumerating $\mathcal{Z}$). *Fix $z^0 \in \mathcal{Z}$ and let B_k be the k th column of B . Condition (8.1) holds if and only if ℓ/η and u/η are integral, $(b - Bz^0)/\eta \in \mathbb{Z}^m$, and $B_k(t - z_k^0)/\eta \in \mathbb{Z}^m$ for every k and every $t \in Z_k$. This takes $1 + \sum_k |Z_k|$ vector tests.*

Proof. For every $z \in \mathcal{Z}$, $(b - Bz)/\eta = (b - Bz^0)/\eta - \sum_k B_k(z_k - z_k^0)/\eta$, a sum of integral vectors under the stated tests. Conversely, if (8.1) holds, the tests are differences of two vectors $(b - Bz)/\eta$ for points of $\mathcal{Z}$ that differ in at most one coordinate. $\square$

E.2 Proofs for Sections 8.3–8.5

Proof of Proposition 8.7. (a) Apply Lemma 8.3 with $j = 0$ to any $(x, z) \in X$; its outcomes are feasible points of $G_0(\mathcal{D}^{(0)})$.

(b) Let $v = (x, z) \in X \cap \mathcal{D}^{(j)}$ and let Y be as in Lemma 8.3. Every outcome (Y, z) is a feasible grid point, so $\mu_j \leq \mathbb{E}F(Y, z) \leq F(x, z) + E_j$ by Lemma 8.4.

(c) Let $v = (x, z) \in (X \cap \mathcal{D}^{(j)}) \setminus \mathcal{D}^{(j+1)}$ and let Y be as in Lemma 8.3. If some value $z_k \notin \Lambda_k^{(j+1)}$, every outcome has the same value z_k , so $m_k(z_k) \leq F(Y, z) - E_j$ almost surely and $m_k(z_k) \leq \mathbb{E}F(Y, z) - E_j \leq F(v)$; removal means $m_k(z_k) > U_j$. Otherwise some $x_i \notin \mathcal{D}_i^{(j+1)}$. If $x_i \in h_j\mathbb{Z}$, then $Y_i = x_i$ almost surely and as before $m_i(x_i) \leq F(v)$. Since $x_i \in \mathcal{D}_i^{(j)}$, either $\{x_i\}$ is a single-point component, which was removed, so $m_i(x_i) > U_j$; or x_i is an endpoint of at least one cell, all such cells were removed, and each removed cell has both endpoint values above U_j , so again $m_i(x_i) > U_j$. If $x_i \notin h_j\mathbb{Z}$, then $Y_i \in \{t_i, t_i + h_j\}$ for the cell $[t_i, t_i + h_j] \subseteq \mathcal{D}_i^{(j)}$ containing x_i , which was removed, and $F(v) \geq \mathbb{E}F(Y, z) - E_j \geq \min\{m_i(t_i), m_i(t_i + h_j)\} > U_j$. In the hull variant the retained set is larger, so (c) holds a fortiori.

(d) The retained sets are finite unions of closed intervals with endpoints in $h_j\mathbb{Z} \subseteq h_{j+1}\mathbb{Z}$, as are their hulls. X is compact and F is continuous, so $\mathcal{S} \neq \emptyset$ when $X \neq \emptyset$. By induction $\mathcal{S} \subseteq \mathcal{D}^{(j)}$; minimizers have value $F^* \leq F(y^{(j)}) = U_j$, so by (c) they are not removed. The point $y^{(j)}$ has value U_j , so it is not removed either; its continuous coordinates lie in $h_j\mathbb{Z} \subseteq h_{j+1}\mathbb{Z}$, hence $y^{(j)} \in X \cap G_{j+1}(\mathcal{D}^{(j+1)})$ and $\mu_{j+1} \leq \mu_j$. In particular every $\mathcal{D}_i^{(j+1)}$ and $\Lambda_k^{(j+1)}$ is nonempty.

(e) $F^* \leq U_j$ because $y^{(j)} \in X$. By (d) a minimizer lies in $\mathcal{D}^{(j)}$, so (b) gives $\beta_j \leq F^*$; rounding that minimizer as in (b) gives $\mu_j \leq F^* + E_j$.

For the last claim, let $v \in X$. If $v \in \mathcal{D}^{(j)}$, then $F(v) \geq \beta_j$ by (b). Otherwise v is removed at a first level $j < J$, and $F(v) > U_j \geq U_J \geq \beta_J$ by (c); the inequality $U_j \geq U_J$ is checked directly from the recomputed values. Hence $F^* \geq \beta_J$. $\square$

Proof of Theorem 8.8. (a) A retained cell has an endpoint t with $m_i(t) \leq U_j$, a retained single point $\{t\}$ has $m_i(t) \leq U_j$, and a retained value t has $m_k(t) \leq U_j$. In each case $m_i(t) < \infty$, so there is a feasible grid point w with $w_i = t$ and, by Proposition 8.7(e),

$$F(w) = m_i(t) + E_j \leq U_j + E_j \leq F^* + 2E_j.$$

By (8.3), $\text{dist}(w, \mathcal{S})^2 \leq 2E_j/g_S = a_j^2$, so some $v \in \mathcal{S}$ has $|t - v_i| \leq \|w - v\| \leq a_j$. Points of a cell are within h_j of each of its endpoints.

(b) By (a), $\mathcal{D}_i^{(j+1)}$ lies in the union over $t \in \mathcal{S}_i$ of the intervals $[t - a_j - h_j, t + a_j + h_j]$. Each has length $2a_j + 2h_j = h_{j+1}(4a_j/h_j + 4)$, so it contains at most $\lfloor 4a_j/h_j \rfloor + 5$ points of $h_{j+1}\mathbb{Z}$, and $4a_j/h_j = 2\sqrt{n_c \bar{L}/g_S}$. Finally $\bar{L}/g_S \leq \kappa_c$.

(c) In the hull variant the argument of (a) applies to the retained cells, single points and values before the hull is taken. If $\mathcal{S} = \{v^*\}$, the retained cells and points lie in the interval $[v_i^* - a_j - h_j, v_i^* + a_j + h_j]$, and hence so does their hull. $\square$

Proof of Theorem 8.9. Correctness is Proposition 8.7, and the width is $U_J - \beta_J = E_J$. The bound on J follows from $E_j = n_c \bar{L} \eta^2 4^{-j}/8$. The table count is the tree computation of Section 8.3, summed over levels; the level-0 grid has $(u_i - \ell_i)/\eta + 1 \leq s/\eta + 1$ nodes in a continuous coordinate and at most K_Z values in a discrete one, and values are never added. The bound K is Theorem 8.8(b) together with $|\Lambda_k^{(j)}| \leq K_Z$. Grid points at level j lie in $h_j\mathbb{Z} \cap [\ell, u]$, so their coordinates are rationals with denominators dividing $\Delta_\eta 2^j$ and numerators bounded by $\Delta_\eta 2^j \max_i \max\{|\ell_i|, |u_i|\}$. Explicit fixed-degree factors evaluated at such points have polynomial bit length, and every table entry is a sum of at most $|\mathcal{A}|$ factor values or a minimum of such sums. $\square$

E.3 Union versus hull filtering

Example E.3 (Two minimizers per block: union versus hull). For one block take continuous $x_1, x_2, x_3 \in [0, 1]$ with the TU equality $x_1 + x_2 = x_3$ and

$$F_b(x) = (2x_1 - x_3)^2 + \frac{1}{4}x_3(1 - x_3).$$

Both terms are nonnegative on the feasible set X_b of the block, so $\mathcal{S}_b = \{(0, 0, 0), (\frac{1}{2}, \frac{1}{2}, 1)\}$ and $F^* = 0$. On X_b put $w = x_1 - x_3/2$, so $x_2 = x_3/2 - w$ and $F_b = 4w^2 + x_3(1 - x_3)/4$. If $x_3 \leq 1/2$, the squared distance to $(0, 0, 0)$ is $2w^2 + \frac{3}{2}x_3^2$ and $x_3(1 - x_3)/4 \geq x_3/8 \geq x_3^2/4$, so $F_b \geq 4w^2 + x_3^2/4 \geq \frac{1}{6}(2w^2 + \frac{3}{2}x_3^2)$. If $x_3 \geq 1/2$, the same computation with $1 - x_3$ in place of x_3 and the minimizer $(\frac{1}{2}, \frac{1}{2}, 1)$ gives the same inequality. Hence $g_S = 1/6$. The Hessian has rows $(8, 0, -4)$, $(0, 0, 0)$, $(-4, 0, 3/2)$, so $\bar{L} = 12$ is valid. For n_b independent blocks, $n_c = 3n_b$, and the bags are the blocks ($p = 3$). The optimal set is the product of the block sets, so squared distances to it add over blocks and $g_S = 1/6$ and $\bar{L} = 12$ are unchanged; every projection $\mathcal{S}_i$ has two points, and there are 2^{n_b} minimizers. With $\eta = 1$, the hull variant keeps $[0, 1]$ in the coordinate x_3 of every block at every level, because both 0 and 1 are optimal values, so its level- j grid has $2^j + 1$ nodes there. The union variant has at most $2(5 + \lfloor 2\sqrt{216n_b} \rfloor)$ nodes per coordinate at every level $j \geq 1$ by Theorem 8.8.

E.4 Stationary face polytopes and rational height

In this subsection and the next two, F is the quadratic of Section 8.6, with the notation of that subsection. For $z \in \mathcal{Z}$ put $\phi_z(x) = F(x, z)$, so that $\Delta \nabla \phi_z(x) = \hat{H}_{xx}x + c_z$; for a set $\mathcal{J}$ of rows, $\hat{A}_{\mathcal{J}}$ consists of the rows of $\hat{A}$ in $\mathcal{J}$. If z is integral and $x \in \rho^{-1}\mathbb{Z}^{n_c}$ for a positive integer ρ , then $\Delta \rho^2 F(x, z) \in \mathbb{Z}$, since Δ clears every monomial coefficient.

Definition E.4 (Stationary face polytope). For $\tilde{v} = (\tilde{x}, z) \in \mathcal{S}$ let $\mathcal{J}_0 = \{r : \hat{A}_r \tilde{x} = (\hat{b}_z)_r\}$ be its active rows and

$$\mathcal{P}(\tilde{v}) = \{x \in \mathcal{P}_z : \hat{A}_{\mathcal{J}_0} x = (\hat{b}_z)_{\mathcal{J}_0}, \nabla \phi_z(x) \perp \ker \hat{A}_{\mathcal{J}_0}\}.$$

Lemma E.5 (Stationary face polytopes). Let $\tilde{v} = (\tilde{x}, z) \in \mathcal{S}$ and $\mathcal{J}_0$ be as above.

(a) $\nabla \phi_z(\tilde{x}) \perp \ker \hat{A}_{\mathcal{J}_0}$, and $w^\top \hat{H}_{xx} w \geq 0$ for $w \in \ker \hat{A}_{\mathcal{J}_0}$.

(b) $\mathcal{P}(\tilde{v})$ is a nonempty polytope and $\mathcal{P}(\tilde{v}) \times \{z\} \subseteq \mathcal{S}$.

(c) Every vertex x° of $\mathcal{P}(\tilde{v})$ lies in $\rho^{-1}\mathbb{Z}^{n_c}$ for an integer ρ with $\Delta_\eta \mid \rho$ and $\rho \leq \Delta_\eta R_{\text{TU}}$.

Proof. (a) For $w \in \ker \hat{A}_{\mathcal{J}_0}$ and small $|t|$, $\tilde{x} + tw \in \mathcal{P}_z$: active rows stay active and inactive rows keep positive slack. The function $t \mapsto \phi_z(\tilde{x} + tw) = F^* + t \nabla \phi_z(\tilde{x})^\top w + \frac{t^2}{2} w^\top \hat{H}_{xx} w$ has a minimum at $t = 0$, which gives both claims, since $\hat{H}_{xx} = \Delta H_{xx}$ with $\Delta > 0$.

(b) By (a), $\tilde{x} \in \mathcal{P}(\tilde{v})$. The condition $\nabla \phi_z(x) \perp \ker \hat{A}_{\mathcal{J}_0}$ reads $\Xi_0^\top (\hat{H}_{xx} x + c_z) = 0$ for a basis matrix Ξ_0 of $\ker \hat{A}_{\mathcal{J}_0}$, which is linear, and $\mathcal{P}(\tilde{v}) \subseteq \mathcal{P}_z$ is bounded. For $x \in \mathcal{P}(\tilde{v})$ put $w = x - \tilde{x} \in \ker \hat{A}_{\mathcal{J}_0}$. Both $\nabla \phi_z(x)$ and $\nabla \phi_z(\tilde{x})$ are orthogonal to $\ker \hat{A}_{\mathcal{J}_0}$, hence so is their difference $H_{xx} w$, and

$$\phi_z(x) - \phi_z(\tilde{x}) = \nabla \phi_z(\tilde{x})^\top w + \frac{1}{2} w^\top H_{xx} w = 0.$$

Since $(x, z) \in X$, it is a minimizer.

(c) Let $\mathcal{J}_{x^\circ} \supseteq \mathcal{J}_0$ be the rows active at x° , let $\mathcal{E} \subseteq \mathcal{J}_{x^\circ}$ index linearly independent rows of $\hat{A}$ that span the rows of $\hat{A}_{\mathcal{J}_{x^\circ}}$. Then $\ker \hat{A}_{\mathcal{E}} = \ker \hat{A}_{\mathcal{J}_{x^\circ}} \subseteq \ker \hat{A}_{\mathcal{J}_0}$.

First, $\hat{H}_{xx}$ is positive definite on $\ker \hat{A}_{\mathcal{E}}$. Indeed, let $w \in \ker \hat{A}_{\mathcal{E}}$ with $w^\top \hat{H}_{xx} w = 0$. By (a) the quadratic form of $\hat{H}_{xx}$ is nonnegative on $\ker \hat{A}_{\mathcal{J}_0}$ and vanishes at w , so for $y \in \ker \hat{A}_{\mathcal{J}_0}$ and all real t , $0 \leq (w + ty)^\top \hat{H}_{xx} (w + ty) = 2t y^\top \hat{H}_{xx} w + t^2 y^\top \hat{H}_{xx} y$, whence $\hat{H}_{xx} w \perp \ker \hat{A}_{\mathcal{J}_0}$. Then $x^\circ \pm tw \in \mathcal{P}(\tilde{v})$ for small $t > 0$: rows in $\mathcal{J}_{x^\circ}$ stay active, the other rows keep positive slack, and the stationarity condition is preserved. As x° is a vertex, $w = 0$.

Next, $\nabla \phi_z(x^\circ) \perp \ker \hat{A}_{\mathcal{J}_0} \supseteq \ker \hat{A}_{\mathcal{E}}$, so $\hat{H}_{xx} x^\circ + c_z = \hat{A}_{\mathcal{E}}^\top \pi$ for some vector π . Thus $(x^\circ, -\pi)$ solves

$$\mathbf{K} \begin{pmatrix} x^\circ \\ -\pi \end{pmatrix} = \begin{pmatrix} -c_z \\ (\hat{b}_z)_{\mathcal{E}} \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} \hat{H}_{xx} & \hat{A}_{\mathcal{E}}^\top \\ \hat{A}_{\mathcal{E}} & 0 \end{pmatrix}.$$

The matrix $\mathbf{K}$ is nonsingular: if $\hat{H}_{xx} w + \hat{A}_{\mathcal{E}}^\top \pi' = 0$ and $\hat{A}_{\mathcal{E}} w = 0$, then $w^\top \hat{H}_{xx} w = -(\hat{A}_{\mathcal{E}} w)^\top \pi' = 0$, so $w = 0$, and then $\hat{A}_{\mathcal{E}}^\top \pi' = 0$ gives $\pi' = 0$ because $\hat{A}_{\mathcal{E}}$ has full row rank. The matrix $\mathbf{K}$ is integral and the right-hand side lies in $\Delta_\eta^{-1} \mathbb{Z}^{n_c + |\mathcal{E}|}$, so Cramer's rule gives $x^\circ \in \rho^{-1} \mathbb{Z}^{n_c}$ with $\rho = \Delta_\eta |\det \mathbf{K}|$, where $|\det \mathbf{K}|$ is a positive integer. By Hadamard's inequality for rows [99], $|\det \mathbf{K}|$ is at most the product of the row norms of $\mathbf{K}$. A row of $(\hat{H}_{xx} \hat{A}_{\mathcal{E}}^\top)$ has n_c entries of absolute value at most $C_{\hat{H}}$ and $|\mathcal{E}| \leq n_c$ entries in $\{0, \pm 1\}$, so its norm is at most $\sqrt{2n_c} C_{\hat{H}}$; a row of $(\hat{A}_{\mathcal{E}} \ 0)$ is a row of $\hat{A}$ and has norm at most $\sqrt{n_c}$. Hence $|\det \mathbf{K}| \leq (2n_c C_{\hat{H}}^2)^{n_c/2} n_c^{|\mathcal{E}|/2} \leq (\sqrt{2} n_c C_{\hat{H}})^{n_c} \leq R_{\text{TU}}$. $\square$

The argument follows Vavasis [177], with explicit constants that use only the continuous Hessian block and the $\{0, \pm 1\}$ structure of $\hat{A}$.

Corollary E.6 (Rational height with TU coupling). (a) Some global minimizer has continuous coordinates in $\rho^{-1}\mathbb{Z}$ for an integer $1 \leq \rho \leq \Delta_\eta R_{\text{TU}}$, and every isolated global minimizer has this property. (b) $F^* = a/W$ in lowest terms with $1 \leq W \leq \Omega_{\text{TU}}$.

Proof. (a) For $\tilde{v} \in \mathcal{S}$ the polytope $\mathcal{P}(\tilde{v})$ has a vertex, which is a minimizer by Lemma E.5(b) and has the stated height by Lemma E.5(c). If $\tilde{v}$ is isolated in $\mathcal{S}$, the convex set $\mathcal{P}(\tilde{v}) \times \{z\} \subseteq \mathcal{S}$ contains no other point near $\tilde{v}$, so $\mathcal{P}(\tilde{v}) = \{\tilde{x}\}$ and $\tilde{x}$ is its vertex. (b) Evaluating F at a minimizer of height $\rho \leq \Delta_\eta R_{\text{TU}}$ gives $\Delta \rho^2 F^* \in \mathbb{Z}$, so $W \leq \Delta (\Delta_\eta R_{\text{TU}})^2 = \Omega_{\text{TU}}$. $\square$

Corollary E.6 is an explicit-constant instance of the classical fact that quadratic programs over polyhedra have optimal solutions of polynomial size [177]; for mixed-integer quadratic programs see [55].

E.5 The algorithm TU-EXACT and snapping on rows

TU-EXACT, described in Section 8.6, is the following algorithm.

Algorithm E.7 (TU-EXACT: exact output with TU coupling). Run TU-GRID (Algorithm 8.6) without the stopping test of step (2). Between steps (2) and (3) of every level j , let $(y, z) = y^{(j)}$ and:

- (i) let $\mathcal{J} = \{r : (\hat{b}_z)_r - \hat{A}_r y \leq \tau_{\text{TU}}\}$, and compute a basis matrix $\Xi_{\mathcal{J}}$ of $\ker \hat{A}_{\mathcal{J}}$ by Gaussian elimination;
- (ii) find by exact linear programming a point $\hat{x}$ of

$$\mathcal{P}_z(\mathcal{J}) = \{x \in \mathcal{P}_z : \hat{A}_{\mathcal{J}}x = (\hat{b}_z)_{\mathcal{J}}, \Xi_{\mathcal{J}}^T(\hat{H}_{xx}x + c_z) = 0\},$$

or continue with step (3) if $\mathcal{P}_z(\mathcal{J})$ is empty;

- (iii) if $F(\hat{x}, z) - \beta_j < 1/(\Omega_{\text{TU}}W)$, where W is the reduced denominator of $F(\hat{x}, z)$, return $(\hat{x}, z)$, $F(\hat{x}, z)$ and the certificate of levels $0, \dots, j$; otherwise continue with step (3).

The large denominators of y enter only the comparisons that define $\mathcal{J}$. The rest of steps (i) and (ii) uses only the original data, the index set $\mathcal{J}$ and the integer vector z , so it takes time polynomial in I [82, Theorem 6.4.12].

Lemma E.8 (Snapping recovery on TU faces). *Let $(y, z) \in X$ with $\text{dist}((y, z), \mathcal{S}) \leq \tau_{\text{TU}}/(2\sqrt{n_c})$, and let $\mathcal{J}$ and $\mathcal{P}_z(\mathcal{J})$ be as in steps (i) and (ii) of TU-EXACT. Then $\mathcal{P}_z(\mathcal{J}) \neq \emptyset$ and $\mathcal{P}_z(\mathcal{J}) \times \{z\} \subseteq \mathcal{S}$.*

Proof. Write $\tau = \tau_{\text{TU}}$. Let $\tilde{v}$ be a point of $\mathcal{S}$ nearest to (y, z) ; it exists because $\mathcal{S}$ is compact. Since $\|(y, z) - \tilde{v}\| < 1$, the integer parts agree, so $\tilde{v} = (\tilde{x}, z)$ and $\|y - \tilde{x}\| \leq \tau/(2\sqrt{n_c})$. Let $\mathcal{J}_0$ be the rows active at $\tilde{x}$. Every row of $\hat{A}$ has norm at most $\sqrt{n_c}$, so for $r \in \mathcal{J}_0$, $0 \leq (\hat{b}_z)_r - \hat{A}_r y = \hat{A}_r(\tilde{x} - y) \leq \tau/2$; hence $\mathcal{J}_0 \subseteq \mathcal{J}$. For $r \in \mathcal{J} \setminus \mathcal{J}_0$ the slack at $\tilde{x}$ is at most $\tau + \tau/2$. On $\mathcal{P}(\tilde{v})$ the linear function

$$\psi(x) = \sum_{r \in \mathcal{J} \setminus \mathcal{J}_0} ((\hat{b}_z)_r - \hat{A}_r x)$$

is nonnegative, and $\psi(\tilde{x}) \leq m' \cdot \frac{3}{2}\tau = \frac{3}{8\Delta_{\eta}R_{\text{TU}}} < \frac{1}{\Delta_{\eta}R_{\text{TU}}}$. Its minimum over the polytope $\mathcal{P}(\tilde{v})$ is attained at a vertex x° . By Lemma E.5(c), $x^\circ \in \rho^{-1}\mathbb{Z}^{n_c}$ for an integer ρ with $\Delta_{\eta} \mid \rho$ and $\rho \leq \Delta_{\eta}R_{\text{TU}}$; as $\hat{A}$ is integral and $\hat{b}_z \in \Delta_{\eta}^{-1}\mathbb{Z}^{m'} \subseteq \rho^{-1}\mathbb{Z}^{m'}$, $\psi(x^\circ) \in \rho^{-1}\mathbb{Z}$. Since $0 \leq \psi(x^\circ) < 1/(\Delta_{\eta}R_{\text{TU}}) \leq 1/\rho$, $\psi(x^\circ) = 0$.

So x° satisfies every row of $\mathcal{J} \setminus \mathcal{J}_0$ with equality, and every row of $\mathcal{J}_0$ because $x^\circ \in \mathcal{P}(\tilde{v})$. Moreover $\ker \hat{A}_{\mathcal{J}} \subseteq \ker \hat{A}_{\mathcal{J}_0}$ and $\nabla \phi_z(x^\circ) \perp \ker \hat{A}_{\mathcal{J}_0}$, so $\Xi_{\mathcal{J}}^T(\hat{H}_{xx}x^\circ + c_z) = 0$. Hence $x^\circ \in \mathcal{P}_z(\mathcal{J})$. For any $x \in \mathcal{P}_z(\mathcal{J})$ let $w = x - x^\circ \in \ker \hat{A}_{\mathcal{J}}$. Both $\nabla \phi_z(x)$ and $\nabla \phi_z(x^\circ)$ are orthogonal to $\ker \hat{A}_{\mathcal{J}}$, hence so is $H_{xx}w$, and $\phi_z(x) = \phi_z(x^\circ) + \nabla \phi_z(x^\circ)^T w + \frac{1}{2}w^T H_{xx}w = \phi_z(x^\circ) = F^*$, the last equality by Lemma E.5(b). Since $x \in \mathcal{P}_z$, $(x, z) \in \mathcal{S}$. $\square$

No nonsingularity, strict complementarity or sign condition on multipliers is used: the condition $\Xi_{\mathcal{J}}^T(\hat{H}_{xx}x + c_z) = 0$ allows multipliers of either sign. A point returned from a distant y can be a nonglobal stationary point: for $F(x) = x_1 - x_1^2$ on $\{x \in [0, 1]^2 : x_1 = x_2\}$ and $y = (\frac{1}{2}, \frac{1}{2})$, steps (i) and (ii) return $(\frac{1}{2}, \frac{1}{2})$, with value $\frac{1}{4} > F^* = 0$. Acceptance therefore always goes through the test in step (iii).

E.6 Exact output

Proof of Theorem 8.10. (a) By Proposition 8.7(e), $\beta_j \leq F^*$, and $(\hat{x}, z) \in X$ because $\hat{x} \in \mathcal{P}_z$. By Corollary E.6(b), F^* is a rational with denominator at most Ω_{TU} , so Proposition 6.6 with $\Omega' = \Omega_{\text{TU}}$ shows that a point that passes the test in step (iii) is a global minimizer.

(b) By Proposition 8.7(e), $F(y^{(j)}) = U_j \rightarrow F^*$, since $E_j \rightarrow 0$. We claim $\text{dist}(y^{(j)}, \mathcal{S}) \rightarrow 0$. Otherwise a subsequence stays at distance at least some $c > 0$ from $\mathcal{S}$ and, X being compact, has a limit point $\bar{v} \in X$ with $F(\bar{v}) = F^*$ and $\text{dist}(\bar{v}, \mathcal{S}) \geq c$, which is impossible. So for all large j , $\text{dist}(y^{(j)}, \mathcal{S}) \leq \tau_{\text{TU}}/(2\sqrt{n_c})$ and $E_j < 1/\Omega_{\text{TU}}^2$. Then Lemma E.8 shows that step (ii) finds $\hat{x}$ with $(\hat{x}, z) \in \mathcal{S}$; its value is $F^* = a/W$ with $W \leq \Omega_{\text{TU}}$, and $F(\hat{x}, z) - \beta_j = F^* - \beta_j \leq U_j - \beta_j = E_j < 1/\Omega_{\text{TU}}^2 \leq 1/(\Omega_{\text{TU}}W)$, so the point is returned. This is the argument of Theorem 6.9(b).

(c) By (8.3) and Proposition 8.7(e), $g_S \text{dist}(y^{(j)}, \mathcal{S})^2 \leq U_j - F^* \leq E_j$, so the two conditions used in (b) hold as soon as $E_j \leq g_S \tau_{\text{TU}}^2/(4n_c)$ and $E_j \leq 1/(2\Omega_{\text{TU}}^2)$. Since $g_S \geq \bar{L}/\kappa_c$ and $E_j = n_c \bar{L} h_j^2/8$, it suffices that $h_j \leq \tau_{\text{TU}}/(n_c \sqrt{\kappa_c})$ and $h_j \leq 2/(\Omega_{\text{TU}} \sqrt{n_c \bar{L}})$, which hold for $j \geq J_{\text{ex}}$ because $h_j = \eta 2^{-j}$. The quantities $\log_2(1/\tau_{\text{TU}})$, $\log_2 \Omega_{\text{TU}}$, $\log_2 \eta$, $\log_2 n_c$ and $\log_2 \bar{L}$ are polynomial in I . The last statement follows from Theorems 8.8 and 8.9 and the paragraph after Algorithm E.7. $\square$

Remark E.9 (Unique minimizers without linear programming). Assume (8.3) with $\mathcal{S} = \{v^*\}$. Then the hull variant can return v^* by continued fractions alone. By Corollary E.6(a), $v_x^* \in \rho^{-1}\mathbb{Z}^{n_c}$ with $\rho \leq \Delta_\eta R_{\text{TU}}$. By Proposition 8.7(d) and Theorem 8.8(c), the retained hull in coordinate i contains v_i^* and has length at most $2a_j + 2h_j$, and every retained value of a discrete coordinate k lies within a_j of v_k^* . Two distinct rationals with denominators at most $\Delta_\eta R_{\text{TU}}$ differ by at least $(\Delta_\eta R_{\text{TU}})^{-2}$, so once $2a_j + 2h_j < (\Delta_\eta R_{\text{TU}})^{-2}$ and $a_j < 1$, the rational of least denominator in each continuous hull is v_i^* , and each value set is $\{v_k^*\}$, since values are integers. Once also $E_j < \Omega_{\text{TU}}^{-2}$, the candidate passes the test of step (iii) of TU-EXACT. A candidate is returned only if it is feasible and passes this test, so the test, not the reconstruction, carries the proof.

E.7 Details for Section 8.7

Lemma E.10 (TU-GRID keeps order $\sqrt{n_c}$ nodes per coordinate). *Let $n_c \geq 4$ be even, $1 \leq p \leq n_c$ and $\bar{L} > 0$ rational, and consider the model of Definition 8.1 with $n_d = 0$, $X = \{x \in [-1, 1]^{n_c} : x_1 + \dots + x_p \leq p\}$, $F(x) = \frac{\bar{L}}{2}\|x\|^2$, $\eta = 1$ and $\mathcal{W} = \mathbb{R}^{n_c}$. The row is redundant, so X is the box; F has the unique minimizer 0, (8.3) holds with $\mathcal{S} = \{0\}$ and $g_S = \bar{L}/2$, so $\kappa_c = 2$, and every admissible decomposition has a bag containing $\{1, \dots, p\}$. Let $k_0 = \lfloor \sqrt{(n_c - 2)/8} \rfloor$. At every level j with $(k_0 + 1)h_j \leq 1$ at which TU-GRID (Algorithm 8.6) filters, both variants retain $\mathcal{D}_i^{(j+1)} \supseteq [-(k_0 + 1)h_j, (k_0 + 1)h_j]$ for every i , so the grid of level $j + 1$ has at least $4k_0 + 5 \geq \sqrt{2(n_c - 2)} + 1$ nodes in every coordinate, and the bag containing $\{1, \dots, p\}$ has at least $(\sqrt{2(n_c - 2)} + 1)^p \geq n_c^{p/2}$ table entries. For every $\varepsilon \leq \bar{L}/64$, TU-GRID(ε) computes the tables of such a level $j + 1$.*

Proof. This is the instance of Proposition 5.10 with $n = n_c$, $\Lambda = \bar{L}$ and $g = \bar{L}/2$, for which the coupling terms vanish and $\kappa = 2$, together with a redundant row. For $\rho > 0$ write $C_\rho = [-\rho, \rho]^{n_c}$.

Comparison of min-marginals. Suppose that a level- j domain $\mathcal{D}^{(j)}$ contains C_ρ with $\rho \in h_j \mathbb{Z}$ and $\rho \geq h_j$. The points of $h_j \mathbb{Z}^{n_c} \cap C_\rho$ are feasible grid points, and 0 is among them, so $\mu_j = U_j = 0$. The uniform grid of mesh h_j on C_ρ is a grid for the subbox C_ρ in the sense of Section 4, with corrections $\bar{L}w_i^2/8 \leq \bar{L}h_j^2/8$: every coordinate grid contains 0 with both adjacent intervals of length h_j , and the corrections of a grid point sum to at most E_j . For a node a of coordinate i in C_ρ , the min-marginal $m_i(a)$ of TU-GRID is therefore at most the min-marginal of this grid at a : TU-GRID minimizes over a larger set of grid points and subtracts E_j , which is at least the correction subtracted at each point.

By Proposition 5.10 with $U = 0$, the latter is at most $0 = U_j$ whenever $|a| \leq \min\{\rho, h_j \sqrt{(n_c - 2)/8}\}$. Hence every cell of $\mathcal{D}_i^{(j)}$ with such an endpoint is retained, in both variants.

Induction, as in the proof of Corollary 5.11. Here $h_j = 2^{-j}$. Put $\rho_j = \min\{1, 2(k_0 + 1)h_j\}$, which lies in $h_j\mathbb{Z}$ and is at least h_j . We show $\mathcal{D}^{(j)} \supseteq C_{\rho_j}$. For $j = 0$, $\mathcal{D}^{(0)} = [-1, 1]^{n_c} = C_1$ and $\rho_0 = 1$. Suppose $\mathcal{D}^{(j)} \supseteq C_{\rho_j}$. If $(k_0 + 1)h_j \leq 1$, then $(k_0 + 1)h_j \leq \rho_j$, and the cells $[kh_j, (k + 1)h_j]$ with $-k_0 - 1 \leq k \leq k_0$ lie in C_{ρ_j} and have an endpoint of absolute value at most $k_0 h_j \leq \min\{\rho_j, h_j \sqrt{(n_c - 2)/8}\}$; they are retained, and $\mathcal{D}^{(j+1)} \supseteq C_{(k_0+1)h_j} = C_{\rho_{j+1}}$. If $(k_0 + 1)h_j > 1$, then $\rho_j = 1$, and every cell in $[-1, 1]$ has an endpoint of absolute value at most $1 - h_j < k_0 h_j$; all are retained, and $\mathcal{D}^{(j+1)} \supseteq C_1 \supseteq C_{\rho_{j+1}}$. This completes the induction.

Counting. When $(k_0 + 1)h_j \leq 1$, the interval $[-(k_0 + 1)h_j, (k_0 + 1)h_j]$ contains $4(k_0 + 1) + 1$ points of $h_{j+1}\mathbb{Z}$, and $k_0 + 1 \geq \sqrt{(n_c - 2)/8}$ gives $4k_0 + 5 \geq \sqrt{2(n_c - 2)} + 1$. The table of the bag containing $\{1, \dots, p\}$ is indexed by the product of the grids of its coordinates, and $(\sqrt{2(n_c - 2)} + 1)^2 = 2n_c - 3 + 2\sqrt{2(n_c - 2)} \geq n_c$. Finally, if $\varepsilon \leq \bar{L}/64$, the last level J satisfies $n_c \bar{L} 4^{-J}/8 = E_J \leq \varepsilon$, so $4^J \geq 8n_c$, $J \geq 1$ and $h_{J-1} = 2^{1-J} \leq (2n_c)^{-1/2}$. Since $k_0 + 1 \leq \sqrt{n_c/8} + 1 \leq \sqrt{2n_c}$ for $n_c \geq 1$, the level $J - 1$ satisfies $(k_0 + 1)h_{J-1} \leq 1$, and the table of level J is as stated. $\square$

Theorem 8.8 gives at most $5 + \lfloor 2\sqrt{2n_c} \rfloor$ nodes per coordinate on this instance, so the order $\sqrt{n_c}$ is attained.

Nonuniform alternatives. With a single order constraint, misaligned coordinate grids already give invalid bounds with curvature-only corrections.

Proposition E.11 (Misaligned grids give invalid bounds). *Let $G_1, G_2 \subseteq [0, 1]$ be finite sets containing 0 and 1, let $d_i : G_i \rightarrow [0, \infty)$ be arbitrary, let $L > 0$, and let $\bar{t} \in (0, 1)$ satisfy*

$$L(t - \bar{t})^2 > d_1(t) + d_2(t) \quad \text{for every } t \in G_1 \cap G_2. \quad (\text{E.1})$$

For $\lambda \geq 0$ let $X = \{x \in [0, 1]^2 : x_1 - x_2 \leq 0\}$ and $F_\lambda(x) = \frac{L}{2}((x_1 - \bar{t})^2 + (x_2 - \bar{t})^2) + \lambda(x_2 - x_1)$. Then $\nabla^2 F_\lambda = LI$, $F^ = 0$ is attained only at $(\bar{t}, \bar{t})$, and $F_\lambda(x) \geq \frac{L}{2}\|x - (\bar{t}, \bar{t})\|^2$ on X , for every λ . Let $\delta = \min\{y_2 - y_1 : y \in G_1 \times G_2, y_2 > y_1\}$. If $\lambda > (\max d_1 + \max d_2)/\delta$, then*

$$\min\{F_\lambda(y) - d_1(y_1) - d_2(y_2) : y \in (G_1 \times G_2) \cap X\} > F^*.$$

Proof. On X , $\lambda(x_2 - x_1) \geq 0$, which gives the growth inequality and $F^* = 0$ at $(\bar{t}, \bar{t})$ only. A feasible grid point either has $y_1 = y_2 = t \in G_1 \cap G_2$, with corrected value $L(t - \bar{t})^2 - d_1(t) - d_2(t) > 0$ by (E.1), or has $y_2 - y_1 \geq \delta$, with corrected value at least $\lambda\delta - \max d_1 - \max d_2 > 0$. $\square$

For instance, $G_1 = \{0, \frac{1}{2}, 1\}$, $G_2 = \{0, \frac{1}{3}, 1\}$, $\bar{t} = \frac{2}{5}$ and the corrections $d_i(t) = \frac{L}{8}w_i(t)^2$ of Section 4 satisfy (E.1), since $G_1 \cap G_2 = \{0, 1\}$, $\frac{4}{25} > \frac{1}{32} + \frac{1}{72}$ and $\frac{9}{25} > \frac{1}{32} + \frac{1}{18}$; here $\delta = \frac{1}{3}$, and every $\lambda > \frac{75}{288}L$ yields a corrected grid minimum above F^* .

Graded grids of Section 5 built around two different feasible centers behave in the same way. With $h = \frac{1}{16}$ and $\theta = \frac{1}{4}$, the graded grid of Definition 5.1 of $[0, 1]$ with center $\frac{3}{10}$ has the nodes

$$0, \frac{79}{1280}, \frac{51}{320}, \frac{19}{80}, \frac{3}{10}, \frac{29}{80}, \frac{141}{320}, \frac{689}{1280}, \frac{3381}{5120}, \frac{16649}{20480}, 1,$$

and the grid with center $\frac{7}{10}$ is its reflection $t \mapsto 1 - t$. The two grids have eleven nodes each and only 0 and 1 in common. With $\bar{t} = \frac{1}{2}$, the left side of (E.1) is $\frac{L}{4}$ at both common nodes, while every interval of either grid has length at most $\frac{3831}{20480} < \frac{1}{5}$, so $d_1 + d_2 < 2 \cdot \frac{L}{8} \cdot \frac{1}{25} = \frac{L}{100}$ there. The smallest positive difference $y_2 - y_1$ is $\delta = \frac{27}{1280}$.

The multiplier λ of the order constraint is invisible to the curvature L and to the growth constant. A correction that depends only on these and on the grids, and is small enough at the common nodes to satisfy (E.1), therefore cannot absorb it. Aligned uniform grids of mesh h escape because then $G_1 \cap G_2$ contains a node within $h/2$ of $\bar{t}$, and $Lh^2/4 \leq d_1 + d_2$ there.

Example E.12 (A graded grid fails on a sum equality). Let $X = \{x \in [0, 3]^3 : x_1 + x_2 - x_3 = 0\}$, which is TU with integral data, and $F(x) = \frac{L}{2}\|x - (1, 1, 2)\|^2$, so $F^* = 0$ at $(1, 1, 2) \in X$, $\nabla^2 F = LI$ and growth holds with $g = L/2$. Build the graded grid of Section 5 in each coordinate around the feasible center 0, with $h = 1$ and $\theta = \frac{1}{4}$: the steps $h + \theta t$ give the common grid $G = \{0, 1, \frac{9}{4}, 3\}$, clipped at 3, and the corrections $d(0) = \frac{L}{8}$, $d(1) = d(\frac{9}{4}) = \frac{25L}{128}$, $d(3) = \frac{9L}{128}$. A point of G^3 with $x_1 + x_2 = x_3$ has $x_3 \in \{0, 1, \frac{9}{4}, 3\}$; since $2, \frac{5}{4}, \frac{3}{4}$ and $\frac{13}{4}$ are not in G , the feasible grid points are $(0, 0, 0)$, $(1, 0, 1)$, $(0, 1, 1)$, $(\frac{9}{4}, 0, \frac{9}{4})$, $(0, \frac{9}{4}, \frac{9}{4})$, $(3, 0, 3)$ and $(0, 3, 3)$. Their values of $F - d(x_1) - d(x_2) - d(x_3)$ are $\frac{21}{8}L$, $\frac{31}{64}L$, $\frac{31}{64}L$, $\frac{51}{64}L$, $\frac{51}{64}L$, $\frac{175}{64}L$ and $\frac{175}{64}L$, so the corrected grid minimum is $\frac{31}{64}L > F^*$: with a sum row, a common graded pattern around a feasible center already gives an invalid bound, without any multiplier effect. The allowance $n_c \bar{L} w^2/8$ of Lemma 8.4, with $\bar{L} = L$ and the largest interval $w = \frac{5}{4}$ in place of h_j , equals $\frac{75}{128}L$ and also gives an invalid bound: the smallest value of F at these points is L , and $L - \frac{75}{128}L = \frac{53}{128}L > F^*$. The reason is visible in the cell $[0, 1] \times [0, 1] \times [1, \frac{9}{4}]$: its intersection with the plane $x_1 + x_2 = x_3$ is the triangle with vertices $(1, 0, 1)$, $(0, 1, 1)$ and $(1, 1, 2)$, and the last vertex is not a grid point, so the minimizer is not a convex combination of feasible grid points of its cell.

F Proofs and supplements for Section 9

This appendix contains the proofs for Section 9, including that of Theorem 9.4 on uniform cells. It also contains Example F.1 of tilted and disconnected optimal sets in $\mathfrak{D}$, the lemma on proximal stages used by DISC (Algorithm 9.13), Proposition F.4 on the hardness of discovery within $\mathfrak{D}$, and, in its last paragraph, the proof that deciding uniqueness of a given minimizer in $\mathfrak{D}$ is coNP-hard.

F.1 One center and two minimizers

Proof of Proposition 9.1. The instance. $F_M = x^2 + z(M - 2x)$. For $x < M/2$ the minimum over z is at $z = 0$ with value x^2 ; for $x > M/2$ it is at $z = M$ with value $(M - x)^2$; at $x = M/2$ the value is $M^2/4$. So $\mathcal{S}$ and F^* are as stated, and the Hessian $\begin{pmatrix} 2 & -2 \\ -2 & 0 \end{pmatrix}$ gives $L_x = 2$, $L_z = 0$. The map $(x, z) \mapsto (M - x, M - z)$ preserves F_M (a direct expansion) and swaps the minimizers, so for growth we may assume $x \leq M/2$. If $x \leq M/4$, then $M - 2x \geq M/2 \geq z/2$ and $F_M \geq x^2 + z^2/2 \geq \frac{1}{2}\|(x, z)\|^2$. If $M/4 < x \leq M/2$, then $F_M \geq x^2 \geq M^2/16$ while $\|(x, z)\|^2 \leq 5M^2/4$. In both cases $F_M \geq \frac{1}{20} \text{dist}((x, z), \mathcal{S})^2$, and $\kappa_S \leq L/g_S = 40$ with $g_S = 1/20$.

The bound (9.1). By symmetry assume $M - c_x \geq M/2$ (otherwise use the reflected argument with $z = 0$ and the endpoint 0). Let $c_x = v_0 < v_1 < \dots < v_k = M$ be the x -nodes on the right of the center, so $k \geq 1$, and put $\delta_m = v_m - c_x$. All steps except the last are unclipped: $\delta_{m+1} = (1 + \theta)\delta_m + h$ for $m < k - 1$, and $\delta_k \leq (1 + \theta)\delta_{k-1} + h$. The node M lies in G_z , $F_M(v, M) = (M - v)^2$, every correction is nonnegative, and $d_x(v) \geq w_x(v)^2/4$.

If $k = 1$, the node M has the adjacent interval $[c_x, M]$ of length at least $M/2$, so $\beta \leq Q(M, M) \leq -M^2/16$, and $M^2/16 \geq \max\{\theta^2 M^2/379, h^2/20\}$ since $\theta \leq 1/4$ and $h \leq M$.

If $k \geq 2$, put $\alpha = v_{k-1} - v_{k-2}$ and $\alpha' = M - v_{k-1}$. Then $Q(M, M) \leq -\alpha'^2/4$ and $Q(v_{k-1}, M) \leq \alpha'^2 - \alpha^2/4$. If $\alpha'^2 \geq \alpha^2/5$, the first is at most $-\alpha^2/20$; otherwise the second is below $-\alpha^2/20$; so $\beta \leq -\alpha^2/20$. Now $\alpha = h + \theta\delta_{k-2} \geq h$, and the recursion gives $\delta_{k-2} \geq (\delta_k - (2 + \theta)h)/(1 + \theta)^2$. If $h \leq M/16$, then with $\delta_k \geq M/2$ and $\theta \leq 1/4$, $\delta_{k-2} \geq (\frac{M}{2} - \frac{9M}{64}) \cdot \frac{16}{25} = \frac{23M}{100}$, so $\alpha \geq \frac{23}{100}\theta M$. If $h > M/16$,

then $\alpha \geq h > M/16 \geq \theta M/4$. Hence $\alpha \geq \max\{h, \frac{23}{100}\theta M\}$ and $\beta \leq -\max\{h^2, (0.23\theta M)^2\}/20$, which implies (9.1) because $0.23^2/20 > 1/379$.

Consequences. (a) Since $L_z = 0$, coordinate z is not in P : TRIAL uses $G_z = \{0, M\}$ and never filters z . Filtering with $U \geq F^*$ keeps both minimizers (Proposition 4.5), so the x -interval of every stage is $[0, M]$. At a stage $j \geq 1$ the mesh is $h_{xj} = h_{x0}2^{-j} < M$ and the center lies in X , so (9.1) applies. At stage 0 the center is $\ell = (0, 0)$ and $M \leq h_{x0} < 2M$ (Section 5), so $G_x = \{0, M\}$, $d_x(M) \geq M^2/4$ and $\beta \leq Q(M, M) \leq -M^2/4$; this is at most $-\max\{\theta^2 M^2/379, h_{x0}^2/20\}$ because $h_{x0}^2/20 < M^2/5$. With a common mesh, $h_j = M2^{-j} \leq M$ at every stage.

(b) By (a), $\beta \geq -\varepsilon$ gives $\theta M \leq \sqrt{379\varepsilon}$ and $h \leq \sqrt{20\varepsilon}$, so every step of the x -grid is at most $h + \theta M \leq (\sqrt{20} + \sqrt{379})\sqrt{\varepsilon} < 24\sqrt{\varepsilon}$. The side of $[0, M]$ from c_x to the farther endpoint has length at least $M/2$, so it takes at least $M/(48\sqrt{\varepsilon})$ steps and contains more than $M/(48\sqrt{\varepsilon})$ nodes. Since $|G_z| = 2$, the table of the bag $\{x, z\}$ has more than $M/(24\sqrt{\varepsilon})$ entries.

(c) Apply (b) with $\varepsilon = 1$. EX stops only when $F(\tilde{x}) - \beta < 1/(\Omega W) \leq 1$ for the bound β of the successful stage of CT, and $F(\tilde{x}) \geq F^* = 0$, so that stage has $\beta > -1$. The data of F_M are 1, -2 , M and the endpoints 0 and M , so $I = O(\log M)$. $\square$

F.2 Uniform cells

We use the notation of UC (Algorithm 9.2) and the properties of its stage cells stated after it in Section 9.1.

Proof of Theorem 9.4. Gap. Every stage cell of positive effective width has length at most h_j (continuous) or below $2h_j$ (integer), so $D(y) \leq \sum_i L_i(2h_j)^2/8 \leq \frac{1}{2}nLh_j^2$ for every $y \in G$. Hence $U - \beta_j \leq F(y^{(j)}) - \beta_j = D(y^{(j)}) \leq \frac{1}{2}nLh_j^2$, which is at most ε at $j = J$. Validity is Lemma 9.3.

Count. Fix a stage j and a cell $[a, a']$ retained by step (iii). One endpoint v has $m_i(v) \leq U$; let $z \in G$ attain $m_i(v)$, so $z_i = v$ and $Q(z) \leq U$. By Lemma 9.3, $\beta_j \leq F^*$, so $U \leq F(y^{(j)}) = \beta_j + D(y^{(j)}) \leq F^* + \frac{1}{2}nLh_j^2$ and

$$F(z) - F^* = Q(z) + D(z) - F^* \leq U + \frac{1}{2}nLh_j^2 - F^* \leq nLh_j^2.$$

Set growth gives $\text{dist}(z, \mathcal{S})^2 \leq nLh_j^2/g_S \leq n\kappa_S h_j^2$, so $|v - \bar{v}| \leq \sqrt{n\kappa_S} h_j$ for some $\bar{v} \in \mathcal{S}_i$. The endpoints of stage cells are points of $\mathcal{G}_{ij}$ or u_i , and consecutive points of $\mathcal{G}_{ij}$ are at least $h_j/2$ apart; so for each $\bar{v} \in \mathcal{S}_i$ at most $4\sqrt{n\kappa_S} + 2$ endpoints lie within $\sqrt{n\kappa_S} h_j$ of $\bar{v}$. Each endpoint belongs to at most two stage cells, so at most $2r(4\sqrt{n\kappa_S} + 2)$ cells are retained. Each of them contains at most one point of $\mathcal{G}_{i,j+1}$ in its interior, so the grid of stage $j + 1$ has at most $3 \cdot 2r(4\sqrt{n\kappa_S} + 2) = K_S$ nodes. Stage 0 has two nodes per coordinate.

Work and bit length. By Lemma 4.3 each stage takes $O(p(|\mathcal{A}| + n + N)K_S^p)$ table operations, and $J = O(\log(nLs^2/\varepsilon)) = O(I + q)$. Continuous nodes have the form $\ell_i + ks2^{-j}$ and integer nodes are integers, so all nodes have denominators dividing $\Delta 2^j$, with Δ as in Section 6.1, and bit length $O(I + j)$; factor values, corrections and messages have bit length polynomial in $I + j$. For exact output, Theorem 6.9(a) applies once $2^{-q} \leq \varepsilon_S$, and $\log_2(1/\varepsilon_S) \leq \text{poly}(I) + \log_2 \kappa_S$ as in the proof of Theorem 6.14 (if $L = 0$, UC is exact at stage 0). EX doubles q , so it makes $O(\log(I + \log \kappa_S))$ calls, each with $q \leq \text{poly}(I) + 2\log_2 \kappa_S$. $\square$

F.3 Coordinatewise concave quadratics

Proof of Lemma 9.6. Since $B_{t'}^\uparrow \subseteq B_t$ for a child t' , the bracket is a function of v_{B_t} , and $r_t \geq 0$ because M_t is its minimum over the labels of $B_t \setminus B_t^\uparrow$.

Telescoping. Let $v \in E([n])$. Every nonroot table M_t occurs in $\sum_t r_t(v_{B_t})$ once with a plus sign, in the residual of its parent, and once with a minus sign. Hence $\sum_t r_t(v_{B_t}) = \sum_t q_t(v_{B_t}) - M_{t_0} = F(v) - M_{t_0}$.

Expectation. Fix $x \in \bar{X}$ and let Y be the endpoint rounding of Section 9.2. Then $\mathbb{E}Y_i = x_i$, $\text{Var } Y_i = (x_i - \ell_i)(u_i - x_i)$, $\mathbb{E}Y_i Y_j = x_i x_j$ for $i \neq j$, and $\mathbb{E}Y_i^2 = x_i^2 + \text{Var } Y_i$. Therefore $\mathbb{E}F(Y) = F(x) + \frac{1}{2} \sum_i H_{ii}(x_i - \ell_i)(u_i - x_i)$, while $\mathbb{E}r_t(Y_{B_t}) = \sum_v r_t(v) \pi_t(v; x)$. Taking expectations in the telescoping identity gives (9.3) with M_{t_0} in place of F^* .

Value. Every term on the right of (9.3) is nonnegative on $\bar{X}$, so $F \geq M_{t_0}$ on $\bar{X} \supseteq X$. Trace back: choose labels of B_{t_0} attaining M_{t_0} ; for each child t' of a processed bag t , choose the labels of $B_{t'} \setminus B_{t'}^\uparrow$ attaining $M_{t'}(v_{B_{t'}^\uparrow})$. By running intersection, a variable belongs to $B_t \setminus B_t^\uparrow$ exactly for the bag t nearest the root among those containing it, and to $B_t^\uparrow$ for every other bag containing it. Hence every variable receives exactly one label, and the resulting endpoint assignment v^* has all residuals zero. By telescoping $F(v^*) = M_{t_0}$. Both endpoints of every X_i are feasible, so $v^* \in X$ and $F^* = M_{t_0}$.

Checking. The argument uses only the nonnegativity of the residuals and one endpoint assignment with zero residuals, so any tables with these two properties give (9.3).

Cost. A bag table has at most 2^p entries, and children are combined by addition. By induction on the tree, every message entry is the sum of the assigned factors over the subtree at one endpoint assignment. Its bit length is therefore bounded by the bit length of a common denominator of all endpoint monomial values plus $\log_2$ of the number of monomials times their maximum magnitude, both polynomial in I . $\square$

Proof of Corollary 9.8. (i) Condition (a) of Theorem 9.7 says $\chi_i(x) \neq *$ when $H_{ii} < 0$; and $\chi_i(x) = *$ is possible only if X_i has a point strictly between its endpoints. Condition (b) says that every v with $\pi_t(v; x) > 0$ has $r_t(v) = 0$, that is, $\text{cube}(\chi(x)_{B_t}) \subseteq Z_t$.

(ii) Replacing $*$ by an endpoint shrinks every cube. If χ is admissible and $x \in X \cap \text{face}(\chi)$, then $\chi(x)$ is obtained from χ by replacing some $*$ entries by endpoints, so x is optimal by (i). Conversely $x \in \mathcal{S}$ lies in $\text{face}(\chi(x))$. A pattern χ determines the set $\{x \in X : \chi(x) = \chi\}$, which is nonempty by the definition of Σ_i .

(iii) Each Adm_t is computed by testing at most 3^p patterns against at most 2^p residual entries. A constraint problem whose relations live on the bags of a tree decomposition is solved by nonserial dynamic programming over $(T, (B_t))$ in the min-sum, max-sum and sum-product semirings, with $O(p3^p)$ operations per bag [50]; each unary cost is charged once, at the bag nearest the root that contains its variable. The minimizer is unique if and only if exactly one pattern is admissible and it has no $*$, by the disjointness in (ii). The dimension of $\mathcal{S}$ is the maximum number of $*$ entries on continuous coordinates. When $\mathcal{S}$ is finite, $|\mathcal{S}| = \sum_\chi \prod_{i:\chi_i=*} |X_i \cap (\ell_i, u_i)|$. By (ii), $\text{dist}(y, \mathcal{S})^2$ is the minimum over admissible χ of the sum of the unary costs $(y_i - \ell_i)^2$, $(y_i - u_i)^2$ and $\text{dist}(y_i, X_i)^2$ for $\chi_i = \text{L}, \text{U}$ and $*$. A linear objective attains its minimum over each $X \cap \text{face}(\chi)$ at an endpoint pattern, so it is minimized by the same recursion over $\{\text{L}, \text{U}\}$ labels. All numbers are rationals of polynomial bit length. $\square$

F.4 The diagonal certificate and the proximal stage

As in Section 9.3, X is a continuous box and F a rational quadratic.

Proof of Lemma 9.10. (i) Both sides are quadratic polynomials in x that vanish at $\bar{x}$. Their Hessians are H and $H + \Lambda - \Lambda = H$. The i th partial derivative of the right side at $\bar{x}$ is $\frac{1}{2} \lambda_i(\bar{x})(u_i + \ell_i - 2\bar{x}_i)$. This equals $|\partial_i F(\bar{x})| = \partial_i F(\bar{x})$ when $\bar{x}_i = \ell_i$, equals $-|\partial_i F(\bar{x})| = \partial_i F(\bar{x})$ when $\bar{x}_i = u_i$, and is

$0 = \partial_i F(\bar{x})$ at an interior coordinate. Two quadratics with equal value, gradient and Hessian at one point coincide.

(ii) On X both terms on the right of (9.4) are nonnegative, so $\bar{x}$ is optimal, and $x \in X$ is optimal exactly when both vanish. A positive semidefinite quadratic form vanishes exactly on the kernel of its matrix.

(iii) For $x \in X$ the bracket in Ψ is at most $F(x)$, so $\Psi(\lambda) \leq F^*$. If (a) holds at $\bar{x}$, then by (i) the bracket with $\lambda = \lambda(\bar{x})$ equals $F(\bar{x}) + \frac{1}{2}(x - \bar{x})^\top (H + \Lambda(\bar{x}))(x - \bar{x})$, whose infimum over $\mathbb{R}^n$ is $F(\bar{x}) = F^*$; this gives (c). Suppose (c) holds for some λ , and let $\bar{x} \in \mathcal{S}$ be arbitrary. Write $\mathcal{L}(x)$ for the bracket. Then $F^* = \Psi(\lambda) \leq \mathcal{L}(\bar{x}) \leq F(\bar{x}) = F^*$. Hence $\lambda_i(\bar{x}_i - \ell_i)(u_i - \bar{x}_i) = 0$ for all i , and $\bar{x}$ minimizes the quadratic $\mathcal{L}$ over $\mathbb{R}^n$. So $\nabla^2 \mathcal{L} = H + \text{diag}(\lambda) \succeq 0$ and $0 = \partial_i \mathcal{L}(\bar{x}) = \partial_i F(\bar{x}) + \lambda_i(\bar{x}_i - \frac{\ell_i + u_i}{2})$. At an interior coordinate, complementarity gives $\lambda_i = 0 = \lambda_i(\bar{x})$. At $\bar{x}_i = \ell_i$ the stationarity equation gives $\lambda_i = 2\partial_i F(\bar{x})/(u_i - \ell_i) = \lambda_i(\bar{x})$, and similarly at $\bar{x}_i = u_i$. Thus $\lambda = \lambda(\bar{x})$ and $H + \Lambda(\bar{x}) \succeq 0$. Since $\bar{x} \in \mathcal{S}$ was arbitrary, this proves (b) and the uniqueness and invariance statements. Finally (b) implies (a) because $\mathcal{S} \neq \emptyset$. $\square$

Example F.1 (Tilted and disconnected optimal sets in $\mathfrak{D}$). Let $F = (x - y + t/2)^2 + \frac{1}{8}t(1 - t)$ on $[0, 1]^3$, and $a = (1, -1, \frac{1}{2})$. Both terms are nonnegative on the box and vanish together exactly when $t \in \{0, 1\}$ and $x - y + t/2 = 0$. So $F^* = 0$, and $\mathcal{S}$ consists of the two segments $\{(v, v, 0) : 0 \leq v \leq 1\}$ and $\{(v, v + \frac{1}{2}, 1) : 0 \leq v \leq \frac{1}{2}\}$. The Hessian $H = 2aa^\top - \text{diag}(0, 0, \frac{1}{4})$ has diagonal $(2, 2, \frac{1}{4})$ and is indefinite, since $w = (0, 1, 2)$ has $a^\top w = 0$ and $w^\top H w = -1$. At every minimizer $\partial_x F = \partial_y F = 0$ and $\partial_t F = \frac{1}{8}(1 - 2t)$, which is $\frac{1}{8}$ at $t = 0$ and $-\frac{1}{8}$ at $t = 1$. So every minimizer is a box KKT point with $\Lambda = \text{diag}(0, 0, \frac{1}{4})$, and $H + \Lambda = 2aa^\top \succeq 0$; hence $F \in \mathfrak{D}$.

Lemma F.2 (Proximal stages). Let $K_\theta = 8\theta^{-1} \lceil \log_2(n + 2) \rceil$. In $\text{PROX}(\hat{\kappa})$ (Algorithm 9.12), $\theta \leq 1/4$, $8\hat{\kappa}\theta^2 \leq 1$ and $\theta^{-1} \leq 6\sqrt{\hat{\kappa}}$, and:

- (i) every grid has at most K_θ nodes per coordinate;
- (ii) if F has set growth with a constant g_S such that $\kappa_S \leq \hat{\kappa}$, then a proximal stage whose center satisfies $\text{dist}(c, \mathcal{S})^2 \leq 4\hat{\kappa}nh^2$ returns y^+ with $F(y^+) - F^* \leq \frac{99}{256}Ln h^2$ and $\text{dist}(y^+, \mathcal{S})^2 \leq \frac{99}{256}\hat{\kappa}nh^2$;
- (iii) under the hypothesis of (ii) on g_S , $\text{dist}(y^{(j)}, \mathcal{S})^2 \leq \frac{99}{256}\hat{\kappa}nh_j^2$ for every $j \geq 0$;
- (iv) stage j costs $O(p(|\mathcal{A}| + n + N)K_\theta^p)$ table operations on rationals of bit length $O(I + j + \mu K_\theta)$.

Proof of Lemma F.2. Constants. Since $\mu \geq 2$, $\theta \leq 1/4$, and $4^{-\mu}\hat{\kappa} \leq 1/8$ is $8\hat{\kappa}\theta^2 \leq 1$. If $\mu = 2$, then $\theta^{-1} = 4 \leq 6\sqrt{\hat{\kappa}}$. Otherwise minimality of μ gives $4^{1-\mu}\hat{\kappa} > 1/8$, so $\theta^{-1} = 2^\mu < 2\sqrt{8\hat{\kappa}} < 6\sqrt{\hat{\kappa}}$.

(i) Each endpoint of X_i' is within ϱh of c_i , so by Lemma 5.2(b) each side of G_i has at most $\lceil (4/\theta) \ln(1 + \theta\varrho) \rceil$ nodes besides c_i . Now $\theta\varrho \leq 2\theta\sqrt{\hat{\kappa}n} + 2\theta \leq \sqrt{n/2} + \frac{1}{2}$, because $\theta\sqrt{\hat{\kappa}} \leq 1/\sqrt{8}$ and $\theta \leq \frac{1}{4}$, and $\frac{3}{2} + \sqrt{n/2} \leq n + 2$. Hence $|G_i| \leq 1 + 2\lceil (4/\theta) \ln(n + 2) \rceil \leq 3 + (8/\theta) \ln(n + 2)$. With $3 \leq 3/(4\theta)$, $8 \ln(n + 2) \leq 5.6 \log_2(n + 2)$ and $\lceil \log_2(n + 2) \rceil \geq 2$, this is at most $7\theta^{-1} \lceil \log_2(n + 2) \rceil \leq K_\theta$.

(ii) Let $x^\circ \in \mathcal{S}$ be nearest to c (it exists since $\mathcal{S}$ is compact), and put $\delta = \|x^\circ - c\|$, so $\delta^2 \leq 4\hat{\kappa}nh^2$. Each $|x_i^\circ - c_i| \leq \delta \leq 2\sqrt{\hat{\kappa}n}h \leq \varrho h$, so $x^\circ \in X'$. For each i choose a grid interval J_i of G_i that contains x_i° and whose endpoint a_i nearer to c_i satisfies $|a_i - c_i| \leq |x_i^\circ - c_i|$: if $x_i^\circ > c_i$, the interval $[a_i, a_i']$ with $a_i < x_i^\circ \leq a_i'$; if $x_i^\circ < c_i$, symmetrically; if $x_i^\circ = c_i$, any interval with endpoint c_i . By Lemma 5.2(a) at the node a_i , $w(J_i) \leq h + \theta|x_i^\circ - c_i|$. Let $C = \prod_i J_i$ and $\tilde{F}(y) = F(y) + \omega\|y - c\|^2$, which has the upper coordinate curvatures $L_i + 2\omega$. The proof of Proposition 4.2(a) uses only Definition 3.1, so it

applies to $\tilde{F}$ and gives a vertex v of C with $\tilde{F}(v) - \sum_i \frac{L_i + 2\omega}{8} w(J_i)^2 \leq \tilde{F}(x^\circ) = F^* + \omega\delta^2$. Since v_i is an endpoint of J_i , $d_i(v_i) \geq L_i w(J_i)^2/8$, and so

$$Q_\omega(y^+) \leq Q_\omega(v) \leq \tilde{F}(v) - \sum_i \frac{L_i}{8} w(J_i)^2 \leq F^* + \omega\delta^2 + \frac{\omega}{4} \sum_i w(J_i)^2.$$

With $w(J_i)^2 \leq 2h^2 + 2\theta^2(x_i^\circ - c_i)^2$ this gives

$$Q_\omega(y^+) \leq F^* + \omega \left[\left(1 + \frac{\theta^2}{2}\right) \delta^2 + \frac{nh^2}{2} \right].$$

By Lemma 5.2(a), $w_i(v) \leq h + \theta|v - c_i|$ for every node v , so $d_i(v) \leq \frac{L}{4}h^2 + \omega|v - c_i|^2$ and $D(y^+) \leq \frac{L}{4}nh^2 + \omega\|y^+ - c\|^2$. Therefore

$$\begin{aligned} F(y^+) - F^* &= Q_\omega(y^+) - F^* + D(y^+) - \omega\|y^+ - c\|^2 \\ &\leq \frac{Lnh^2}{4} \left[1 + 4\hat{\kappa}\theta^2 \left(1 + \frac{\theta^2}{2}\right) + \frac{\theta^2}{2} \right] \leq \frac{Lnh^2}{4} \cdot \frac{99}{64}, \end{aligned}$$

using $\delta^2 \leq 4\hat{\kappa}nh^2$, $\omega = L\theta^2/4$, $4\hat{\kappa}\theta^2 \leq \frac{1}{2}$ and $\theta^2 \leq \frac{1}{16}$. Set growth gives $\text{dist}(y^+, \mathcal{S})^2 \leq (F(y^+) - F^*)/g_S \leq \frac{99}{256}(L/g_S)nh^2 \leq \frac{99}{256}\hat{\kappa}nh^2$, since $L/g_S \leq \kappa_S \leq \hat{\kappa}$.

(iii) By induction on j , the center of stage j satisfies $\text{dist}(c^{(j)}, \mathcal{S})^2 \leq 4\hat{\kappa}nh_j^2$, so (ii) applies at stage j . For $j = 0$, $\text{dist}(\ell, \mathcal{S})^2 \leq ns^2 = nh_0^2 \leq 4\hat{\kappa}nh_0^2$. If (ii) applies at stage j , then $\text{dist}(c^{(j+1)}, \mathcal{S})^2 = \text{dist}(y^{(j)}, \mathcal{S})^2 \leq \frac{99}{256}\hat{\kappa}nh_j^2 < 2\hat{\kappa}nh_{j+1}^2$.

(iv) Let Δ be the common denominator of Section 6.1; it also clears s and L . By induction on j , every node of stage j lies in $\Gamma_j^{-1}\mathbb{Z}$ with $\Gamma_j = \Delta 2^{j+\mu K_\theta}$: the center is ℓ or a node of stage $j-1$; the box endpoints are ℓ_i , u_i or $c_i \pm \varrho h_j$ with $h_j = s2^{-j}$; an unclipped node is $c_i \pm t_k$ with $t_k = h_j[(2^\mu + 1)^k - 2^{\mu k}]/2^{\mu(k-1)}$ and $k < K_\theta$; and a clipped node is a box endpoint. Hence, at every grid point, $\Delta\Gamma_j^2 F$ is an integer (Section 6.1), and so are $8\Delta\Gamma_j^2 d_i$ and $4^{\mu+1}\Delta\Gamma_j^2 \omega\|y - c\|^2$. All table entries are therefore integers over the common denominator $8 \cdot 4^\mu \Delta\Gamma_j^2$, and their magnitudes are at most $2^{O(I)}$ times this denominator because all points lie in X . Messages are sums of at most $|\mathcal{A}| + 2n$ such values, so all numbers have bit length $O(I + j + \mu K_\theta)$. The count of table operations is that of Lemma 4.3 with at most K_θ nodes per coordinate. $\square$

Proof of Theorem 9.14. (a) DISC (Algorithm 9.13) stops only when the hypotheses of Lemma 9.10(ii) hold, and it checks them in exact rational arithmetic: the KKT conditions are sign tests, and positive semidefiniteness of $H + \Lambda(\bar{x})$ is decided by symmetric Gaussian elimination that rejects a negative pivot and a zero pivot with a nonzero entry in its row. By Lemma 9.10(ii), $\bar{x} \in \mathcal{S}$, $F^* = F(\bar{x})$ and (9.5) holds. So an accepted $\bar{x}$ is a point of $\mathcal{S}$ with $H + \Lambda(\bar{x}) \succeq 0$, and then $F \in \mathfrak{D}$ by Lemma 9.10(iii). Hence DISC does not stop if $F \notin \mathfrak{D}$.

(b) Remark 6.10 gives some $g_S > 0$. Let $\hat{\kappa} \geq \kappa_S$ be a guess. By Lemma F.2(iii) with $h_{J_{\hat{\kappa}}}^2 = s^2 4^{-J_{\hat{\kappa}}} \leq \tau^2/(4\hat{\kappa}n)$, $\text{dist}(y, \mathcal{S})^2 \leq \frac{99}{256} \cdot \frac{\tau^2}{4} < \frac{\tau^2}{4}$. By Lemma 6.8, REC(y) (Algorithm 6.7) does not fail, and every point it can return, in particular $\bar{x}$, lies in $\mathcal{S}$; so $\bar{x}$ is a box KKT point. Since $F \in \mathfrak{D}$, Lemma 9.10(iii) gives $H + \Lambda(\bar{x}) \succeq 0$ for every $\bar{x} \in \mathcal{S}$, and the test accepts, whichever optimal component contains $\bar{x}$. The guesses are powers of two, so the first guess $\hat{\kappa} \geq \kappa_S \geq 1$ satisfies $\hat{\kappa} \leq 2\kappa_S$.

(c) By (b), at most $2 + \log_2 \kappa_S$ guesses run, each with $\hat{\kappa} \leq 2\kappa_S$. For each, $J_{\hat{\kappa}} + 1 = O(\log(\hat{\kappa}ns^2/\tau^2)) + 1 = \text{poly}(I) + O(\log \hat{\kappa})$, because $\tau = 1/(4nR)$ and $\log R$ is polynomial in I . By Lemma F.2(i) and (iv), a stage costs $O(p(|\mathcal{A}| + n + N)K_\theta^\mu)$ operations on numbers of bit length $O(I + J_{\hat{\kappa}} + \mu K_\theta)$, where $\mu = O(\log(2\hat{\kappa}))$ and $K_\theta \leq 48\sqrt{\hat{\kappa}}\lceil \log_2(n+2) \rceil$. By (5.4) with n in place of

n_P , a bag table has at most $K_\theta^p \leq 2(48p\sqrt{\hat{\kappa}})^p(n+2) \leq 2(68p\sqrt{\kappa_S})^p(n+2)$ entries, since $48\sqrt{2} < 68$; and $\mu K_\theta \leq c\sqrt{\hat{\kappa}} \log(2\hat{\kappa}) I$ for an absolute constant c . Every factor that depends only on p and $\hat{\kappa}$ is nondecreasing in $\hat{\kappa}$; its value at $\hat{\kappa} = 2\kappa_S$ enters f' , and for fixed p these values are nondecreasing and polynomial in κ_S . The remaining factors are polynomial in I with an absolute exponent. REC uses the original data, so its linear system has size polynomial in I , and exact linear programming returns a vertex of polynomial bit length in polynomial time [82, Theorem 6.4.12]. The KKT and semidefiniteness tests are polynomial. $\square$

Remark F.3 (A false guess). If $\hat{\kappa} < \kappa_S$, PROX($\hat{\kappa}$) (Algorithm 9.12) may return the same nonoptimal point at every stage. For $F = x^2 + y^2 - 3xy + \frac{63}{128}(x+y)$ on $[0, 1]^2$, put $t = x+y$; then $F \geq t^2 - \frac{5}{4}t^2 + \frac{63}{128}t$, a concave function of $t \in [0, 2]$, so $(1, 1)$ is the unique minimizer with $F^* = -\frac{1}{64}$. The origin is a box KKT point with $F = 0$, so every set-growth constant satisfies $g_S \leq (F(0) - F^*)/\text{dist}(0, \mathcal{S})^2 = \frac{1}{128}$, and $\kappa_S \geq 256$ since $L = 2$. Exact computation shows that PROX($\hat{\kappa}$) returns the origin at every stage $j \leq 33$ for $\hat{\kappa} \in \{1, 2\}$. For $\hat{\kappa} = 4$, the points $(0, 0)$ and $(1, 1)$ both minimize Q_ω at stages 0 and 1, and PROX returns the origin at every stage $j \leq 33$ when ties are broken towards the current center. These stages cover the budgets of DISC: the least common denominator of Section 6.1 is $\Delta = 128$, so $R = 2^{23}$, $\tau = 2^{-26}$ and $J_{\hat{\kappa}} = 28, 28, 29$ for $\hat{\kappa} = 1, 2, 4$. Lemma 9.10 rejects the origin: $H + \Lambda = \begin{pmatrix} 191/64 & -3 \\ -3 & 191/64 \end{pmatrix}$ has eigenvalue $-\frac{1}{64}$. At $(1, 1)$ the matrix has diagonal $\frac{193}{64}$ and is positive definite, so $F \in \mathfrak{D}$, and DISC stops at a later guess by Theorem 9.14(b).

Proposition F.4 (Discovery within $\mathfrak{D}$ is hard without the growth parameter). *For integers $a_0, a_1, \dots, a_m$ with $A = \sum_{i=1}^m |a_i| \geq 1$ and $|a_0| \leq A$, let $\xi_0 = 0$ and*

$$F(x, \xi) = \sum_{i=1}^m (\xi_i - \xi_{i-1} - a_i x_i)^2 + (\xi_m - a_0)^2 + \sum_{i=1}^m x_i(1 - x_i)$$

on $x \in [0, 1]^m$, $\xi \in [-A, A]^m$. The bags $\{\xi_1, x_1\}$ and $\{\xi_{i-1}, \xi_i, x_i\}$ ($2 \leq i \leq m$) form a path decomposition with bag size at most 3. If some $\hat{x} \in \{0, 1\}^m$ has $\sum_{i=1}^m a_i \hat{x}_i = a_0$, then $F \in \mathfrak{D}$, $F^ = 0$, and $\mathcal{S}$ consists exactly of the points $(\hat{x}, \hat{\xi})$ with $\hat{x}$ such a solution and $\hat{\xi}$ its partial sums. Otherwise $F^* > 0$. Consequently, unless $P = NP$:*

- (i) *no polynomial-time algorithm returns an exact minimizer for every instance of $\mathfrak{D}$ with bag size three, even though $F^* = 0$ is known in advance for these instances;*
- (ii) *there is no polynomial bound $\kappa_S \leq \text{poly}(I)$ that holds, for a suitable set-growth constant, on every instance of this form in $\mathfrak{D}$.*

The objective is the Subset Sum reduction of [53, Remark 2] with unscaled partial sums; what the proposition adds is that its feasible instances belong to $\mathfrak{D}$.

Proof of Proposition F.4. $F \geq 0$ on the box, and $F = 0$ exactly when all squares and all products $x_i(1 - x_i)$ vanish, that is, for binary x with partial sums ξ and $\xi_m = a_0$; the partial sums lie in $[-A, A]$. If no solution exists, $F > 0$ on the compact box. In the feasible case let $\hat{z} = (\hat{x}, \hat{\xi})$ be optimal. The sum of squares is a quadratic Φ with $\Phi(\hat{z}) = 0$ and $\nabla\Phi(\hat{z}) = 0$, because each square is of an affine function vanishing at $\hat{z}$; so $\Phi(z) = \frac{1}{2}(z - \hat{z})^\top H_{\text{sq}}(z - \hat{z})$ with $H_{\text{sq}} = \nabla^2\Phi \succeq 0$. Also $\sum_i x_i(1 - x_i) = \frac{1}{2} \sum_i 2(x_i - 0)(1 - x_i)$. So $F(z) - F(\hat{z}) = \frac{1}{2}(z - \hat{z})^\top H_{\text{sq}}(z - \hat{z}) + \frac{1}{2} \sum_i 2x_i(1 - x_i)$. At $\hat{z}$ the gradient is $1 - 2\hat{x}_i = \pm 1$ in x_i and 0 in ξ , so $\hat{z}$ is a box KKT point with $\lambda_i(\hat{z}) = 2$ on x and 0 on ξ , and $H + \Lambda(\hat{z}) = H_{\text{sq}} \succeq 0$. Thus $F \in \mathfrak{D}$.

Subset Sum with binary-encoded integers is NP-complete [75], also when restricted to instances with $A \geq 1$ and $|a_0| \leq A$, because the other instances are decided trivially; and the instance F

has size polynomial in that of the Subset Sum instance. (i) An algorithm as in (i), stopped at its polynomial time bound and followed by a check that its output z is feasible with $F(z) = 0$, would decide Subset Sum. (ii) If such a polynomial bound existed, then by Theorem 9.14(c), whose bound is nondecreasing and polynomial in κ_S for fixed $p = 3$, DISC would stop within a polynomial time bound on every feasible instance; stopping DISC at that bound and checking whether it has output $F^* = 0$ would decide Subset Sum. $\square$

Uniqueness in $\mathfrak{D}$ is coNP-hard. Unless $P = NP$, Corollary 9.8 has no counterpart in $\mathfrak{D}$. Take $a_0 = 0$ in Proposition F.4. The origin is then a minimizer, so the instance belongs to $\mathfrak{D}$, and the origin is the only minimizer exactly when no nonempty subset of $a_1, \dots, a_m$ sums to zero. Deciding whether such a subset exists is NP-complete: Subset Sum with positive items $b_1, \dots, b_k$ and target $T > 0$ reduces to it by adding the item $-T$, because a nonempty subset with sum zero must contain $-T$. Hence deciding whether a given minimizer of an instance of $\mathfrak{D}$ with bag size three is the only one is coNP-hard, even when the description (9.5) is given.

G Proofs and supplements for Section 10

This appendix computes the conditioning of the bounded-coefficient reduction of Del Pia and Khajavirad, proves Proposition 10.6 and the propositions of Sections 10.4–10.6, and gives the local moment relaxations of Section 10.7.

G.1 Conditioning of the bounded-coefficient reduction

Remark G.1 (Conditioning of the bounded-coefficient reduction). Consider the construction of [53, Theorem 3] for the two-item instance $a_1 = c$, $a_2 = c - 1$ with target $a_0 = c$, where $c \geq 3$, with objective Ψ (their (20)–(24), all weights one) on the unit box. In their notation, $\ell = \lceil \log_2(2c) \rceil$, $D = 2^\ell \geq 2c$, and there are $5\ell + 2$ variables. Every term of Ψ is nonnegative. A zero of Ψ encodes a binary solution, and $(1, 0)$ is the only one; all copied and state variables are then determined by the recurrences. Hence Ψ has a unique minimizer v^* with value zero, and by Lemma 6.12 some growth constant $g > 0$ is valid.

Encode instead the choice $(0, 1)$ with every copy, digit and running-sum recurrence exact. Only the final square $(s_2 - w_\ell)^2 = ((c - 1 - c)/D)^2$ is nonzero, so $\Psi(y') = D^{-2}$. All ℓ copies of both choice bits change by one, so $\|y' - v^*\|^2 \geq 2\ell$ and every valid growth constant satisfies $g \leq 1/(2\ell D^2)$. The largest diagonal entry of $\nabla^2 \Psi$ is ten, attained by the digit-state variables. Therefore

$$\kappa \geq 20\ell D^2 \geq 80\ell c^2.$$

For $c = 2^k$, $k \geq 2$, there are $5k + 7$ variables and coefficients bounded by five, so $\kappa \geq 80(k + 1)4^k$ is exponential in the number of variables. The instance has $O(k)$ monomials and bags, listed by variable indices of $\Theta(\log k)$ bits, so $I = \Theta(k \log k)$ and $\kappa \geq 2^{\Omega(I/\log I)}$, consistent with Corollary 10.2.

Negative curvature is also large relative to growth. Let ν be the smallest number with $\nabla^2 \Psi \succeq -\nu I$. Encode the fractional choice $(1/c, 1)$; since $c \cdot c^{-1} + (c - 1) \cdot 1 = c$, every square residual can be kept at zero with all variables in $[0, 1]$, and $\Psi(y) = \varrho/c$ with $\varrho = 1 - 1/c$. The displacement $d = y - v^*$ annihilates every affine residual, so $d^T \nabla^2 \Psi d = -2(\varrho^2 + 1)$, coming from the two endpoint penalties. Every valid growth constant has $g \leq \varrho/(c\|d\|^2)$, while $\nu \geq 2(\varrho^2 + 1)/\|d\|^2$. Hence

$$\frac{\nu}{g} \geq 2c\left(\varrho + \frac{1}{\varrho}\right) \geq 4c.$$

This bound on ν/g also holds for arbitrary positive weights on the squares and a common positive weight on the two endpoint penalties: both estimates scale with the penalty weight, and the square weights do not enter. The reduction therefore does not contradict an algorithm parameterized by ν/g either.

G.2 Proof of Proposition 10.6

Lemma G.2 (Isolation [140]). *Let $\mathcal{F}$ be a nonempty family of subsets of a finite set $\mathcal{U}$, and let $w : \mathcal{U} \rightarrow \{1, \dots, \rho\}$ be uniformly random. Then the minimum of $w(Z) = \sum_{e \in Z} w(e)$ over $Z \in \mathcal{F}$ is attained by a unique set with probability at least $1 - |\mathcal{U}|/\rho$.*

Proof. For $e \in \mathcal{U}$ let $\alpha_e = \min\{w(Z) : Z \in \mathcal{F}, e \notin Z\} - \min\{w(Z \setminus \{e\}) : Z \in \mathcal{F}, e \in Z\}$ (with $\min \emptyset = +\infty$); α_e does not depend on $w(e)$. If two distinct minimizing sets exist, some e lies in exactly one of them, and then $w(e) = \alpha_e$. For fixed e this event has probability at most $1/\rho$, because $w(e)$ is uniform and independent of α_e . A union bound over $e \in \mathcal{U}$ gives probability at most $|\mathcal{U}|/\rho$ that the minimizing set is not unique. $\square$

Proof of Proposition 10.6. An instance of multicolored clique consists of $k \geq 2$ color classes, each identified with $\{1, \dots, N_0\}$, and, for each pair $c < d$ of classes, a list $E_{cd} = ((a_1^{cd}, b_1^{cd}), \dots, (a_{m_{cd}}^{cd}, b_{m_{cd}}^{cd}))$ of distinct pairs, the edges between the classes; it asks for $x_1, \dots, x_k$ with $(x_c, x_d) \in E_{cd}$ for all $c < d$. Let $E = \bigsqcup_{c < d} E_{cd}$, choose independent uniform weights $w(e) \in \{1, \dots, 2|E|\}$, and let $W_0 = 1 + \binom{k}{2} 2|E|$. Use integer variables $x_c \in [1, N_0]$ and, for each pair $c < d$ and $l = 1, \dots, m_{cd}$, binary ζ_l^{cd} , σ_l^{cd} and integer $y_l^{cd}, y'_l{}^{cd} \in [0, N_0]$, with $\sigma_0^{cd} = y_0^{cd} = y'_0{}^{cd} = 0$. Inside the brackets below the pair $c < d$ is fixed and the superscript cd is omitted, so that $m = m_{cd}$, $a_l = a_l^{cd}$, $\zeta_l = \zeta_l^{cd}$, and so on. Let

$$\Phi = \sum_{c < d} \left[\sum_l ((\sigma_l - \sigma_{l-1} - \zeta_l)^2 + (y_l - y_{l-1} - a_l \zeta_l)^2 + (y'_l - y'_{l-1} - b_l \zeta_l)^2) \right. \\ \left. + (\sigma_m - 1)^2 + (y_m - x_c)^2 + (y'_m - x_d)^2 \right], \quad \Psi_w = W_0 \Phi + \sum_{c < d} \left[\sum_l w(a_l, b_l) \zeta_l \right].$$

Decomposition. Take a central bag $\{x_1, \dots, x_k\}$; for each pair a bag $\{x_c, x_d, \sigma_m, y_m, y'_m\}$ adjacent to it, followed by the path of bags $\{\sigma_{l-1}, y_{l-1}, y'_{l-1}, \zeta_l, \sigma_l, y_l, y'_l\}$, $l = m, \dots, 1$, with the constants σ_0, y_0, y'_0 omitted. Every term lies in a bag, running intersection holds, and $p = \max\{k, 7\}$.

Encoding. $\Phi \geq 0$ is an integer. $\Phi = 0$ forces σ to increase from 0 to 1 in steps $\zeta_l \in \{0, 1\}$, so exactly one $\zeta_{l^*} = 1$, and then $(x_c, x_d) = (y_m, y'_m) = (a_{l^*}, b_{l^*}) \in E_{cd}$. Thus $\Phi = 0$ exactly at encodings of multicolored cliques, and each clique has exactly one encoding because the pairs in E_{cd} are distinct. At an encoding the linear term equals the weight of the clique's edge set, which is at most $W_0 - 1$. Every point with $\Phi \geq 1$ has $\Psi_w \geq W_0$. If a clique exists and the minimum-weight clique is unique, Ψ_w has a unique minimizer, with value at most $W_0 - 1$.

Conditioning. Suppose the minimum-weight clique is unique. The values of Ψ_w are integral and its minimizer z^* is unique, so $\Psi_w(z) - F^* \geq 1$ for $z \neq z^*$. All domains have length at most N_0 , and there are $n \leq k + 2k^2 N_0^2$ variables, so $\|z - z^*\|^2 \leq n N_0^2$ and $g = 1/(n N_0^2)$ is a valid growth constant. The diagonal Hessian entries are at most $W_0(2 + 4N_0^2 + 2k)$. Hence $\kappa \leq W_0(2 + 4N_0^2 + 2k)n N_0^2$, and both κ and I are at most $(k N_0)^{c_2}$ for an absolute constant c_2 .

Reduction. Run the assumed algorithm for $f(p)((k N_0)^{2c_2})^{p/\psi(p)}$ steps. If it outputs a feasible $\hat{z}$ with $\Psi_w(\hat{z}) < W_0$, then $\Phi(\hat{z}) = 0$ and $\hat{z}$ encodes a clique; answer yes. Otherwise answer no. There are no false positives. If a clique exists, Lemma G.2, with $\mathcal{U} = E$, $\rho = 2|E|$ and $\mathcal{F}$ the family of edge sets of multicolored cliques (which determine the cliques since $k \geq 2$), makes the

minimum-weight clique unique with probability at least $1/2$, and then the algorithm halts in time with $\Psi_w(\hat{z}) \leq W_0 - 1/2$. For $k \geq 7$ this is a one-sided-error randomized algorithm for multicolored clique with running time $f'(k)N_0^{2c_2k/\psi(k)+O(1)}$ for a computable f' .

Clique reduces to multicolored clique with the same k : take k copies of the vertex set as color classes, and join the copies of u and v in different classes when uv is an edge; the classes then have N_0 elements for a graph on N_0 vertices. Under ETH, Clique on graphs with N_0 vertices has no algorithm with running time $f(k)N_0^{o(k)}$ for a computable f [36], [44, Theorem 14.21]; the exponent $2c_2k/\psi(k) + O(1)$ is computable and $o(k)$. The proof of this lower bound composes the sparsification lemma with reductions from 3-SAT to Clique, all deterministic. Composing them and the reduction from Clique to multicolored clique with the algorithm above, run twice to reduce the error probability to $1/4$, gives a randomized algorithm for 3-SAT with error probability at most $1/3$ and running time $2^{o(n')}$ on n' variables, which contradicts rETH.

The reduction uses only instances with $p = k$, and k may be restricted to $k \geq k_0$ for any fixed k_0 , because finitely many values of k do not affect a bound $N_0^{o(k)}$. So the first claim holds even for algorithms whose time bound is required only for $p \geq k_0$. For the last claim, increasing C if necessary, we may assume that C is a positive integer. Suppose $a(p) \leq p/\psi(p)$ and put $\psi'(p) = \max\{1, \min\{\psi(p), p/C\}\}$. Then ψ' is computable, nondecreasing, unbounded and at least one, and for $p \geq C$ it equals $\min\{\psi(p), p/C\}$, so that $p/\psi'(p) \geq \max\{a(p), C\}$ and, because $\kappa, I \geq 1$, $f(p)\kappa^{a(p)}I^C \leq f(p)(\kappa I)^{p/\psi'(p)}$. The first claim with $k_0 = \max\{7, C\}$ excludes this running time. Finally, Theorem 5.7 with $q = 1$ runs in time $f(p, \kappa)(I + 2)^5 \leq 3^5 f(p, \kappa)I^5$, where $f(p, \kappa) = c_0(c_1 p \sqrt{\kappa})^p \kappa(1 + \log_2 \kappa)^2 \leq 4c_0(c_1 p)^p \kappa^{p/2+2}$ because $(1 + \log_2 \kappa)^2 \leq 4\kappa$. So $a(p) = p/2 + O(1)$ and $C = 5$. $\square$

G.3 Proofs for Sections 10.4–10.6

Proof of Proposition 10.7. Part 1. ξ_t lies in bags t and $t + 1$, and the endpoints $2^t - 1$ have t bits.

Part 2. On the box $\xi_t \geq 0$ and $z_t \geq 0$, so $\xi_{t-1} = (\xi_t - z_t - \rho_t)/2 \leq (\xi_t + |\rho_t|)/2$. By induction,

$$0 \leq \xi_t \leq 2^{t-m}\xi_m + \sum_{j=t+1}^m 2^{t-j}|\rho_j|.$$

The vector $(2^{t-m})_{t \leq m}$ has squared norm at most $4/3$. The map $y \mapsto (\sum_{j>t} 2^{t-j}y_j)_t$ is the sum over $l \geq 1$ of 2^{-l} times the shift by l places. Each shift has operator norm at most one, so the map has operator norm at most $\sum_{l \geq 1} 2^{-l} = 1$. Hence $\|\xi\| \leq (2/\sqrt{3})\xi_m + \|\rho\|$ and $\|\xi\|^2 \leq \frac{8}{3}\xi_m^2 + 2\|\rho\|^2$. Also $0 \leq z_t = \xi_t - 2\xi_{t-1} - \rho_t \leq \xi_t + |\rho_t|$, so $\|z\|^2 \leq 2\|\xi\|^2 + 2\|\rho\|^2$. Thus $\|\xi\|^2 + \|z\|^2 \leq 3\|\xi\|^2 + 2\|\rho\|^2 \leq 8\xi_m^2 + 8\|\rho\|^2 \leq 8\Psi_m$.

Part 3. The second derivatives are $2 + 8 = 10$ for ξ_t , $t < m$; $2 + 2 = 4$ for ξ_m ; and $2 - \frac{1}{4}$ for z_t . Writing Ψ_m as a sum of squares plus $-\frac{1}{8} \sum z_t^2$ plus linear terms gives $\nabla^2 \Psi_m \succeq -\frac{1}{4} \sum_t e_{z_t} e_{z_t}^\top \succeq -\frac{1}{4} I$, where e_{z_t} is the unit vector of the coordinate z_t .

Part 4. Let $V_0(v) = V(v) - v^2 = \min\{\sum_t \rho_t^2 + \frac{1}{8} \sum_t z_t(1 - z_t) : \xi_m = v\} \geq 0$. The minimum is attained by compactness. Hence $V_0(v) = 0$ exactly when some feasible point has $\xi_m = v$, all $\rho_t = 0$ and binary z . Then $\xi_t = 2\xi_{t-1} + z_t$ and $v = \sum_t 2^{m-t}z_t \in \{0, \dots, 2^m - 1\}$. Conversely, for an integer j in this range, its binary digits give $\xi_t = \lfloor j/2^{m-t} \rfloor \in [0, 2^t - 1]$. So V_0 vanishes exactly at the 2^m integers of $[0, 2^m - 1]$.

In the first representation, on an interval with nonempty interior, V_0 coincides with a polynomial of degree at most two. It is not identically zero, since $V_0 > 0$ off the integers. So it has at most two zeros, and the interval contains at most two integers. A degenerate interval contains at most one. Hence $2N_{\text{pc}} \geq 2^m$. In the second representation, each integer j has some l with $\varphi_l(j) = V(j)$. Then

$\varphi_l - v^2 \geq V_0 \geq 0$ on $[0, 2^m - 1]$ and vanishes at j . If $\varphi_l - v^2 \equiv 0$, then $V_0 \leq 0$, which is false; so $\varphi_l - v^2$ vanishes at most at two integers, and again $2N_{\text{pc}} \geq 2^m$. $\square$

Proof of Proposition 10.8. For $x \in X$, $\text{dist}(x, \mathcal{S})^2 \leq \sum_i (x_i - x_1)^2$, and $(x_i - x_1)^2 \leq (i - 1)F(x)$ by Cauchy–Schwarz, which gives the set-growth bound. For $n = 2$, $\text{dist}(x, \mathcal{S})^2 = (x_1 - x_2)^2/2$. The coordinate second derivatives are 2 for x_1, x_n and 4 otherwise, so every valid L_i is at least 2; with the smallest valid ones, $L = \max_i L_i \leq 4$, and $L = 2$ if $n = 2$, so $\kappa_S \leq 2n(n - 1)$.

(a) Fix j and $v \in G_j$, and put $v_j = v$. Choose the remaining coordinates outward along the path. If v_k has been chosen and $i = k \pm 1$ is the next index, let $[\alpha, \alpha']$ be a grid interval of G_i containing v_k , of length δ_i , and let v_i be its endpoint nearest to v_k . Such an interval exists because G_i contains 0 and 1. Then $(v_k - v_i)^2 \leq \delta_i^2/4$ and $w_i(v_i) \geq \delta_i$. Every edge of the path joins some $i \neq j$ to the index chosen just before it, so

$$Q(v) = F(v) - D(v) \leq \sum_{i \neq j} \frac{\delta_i^2}{4} - \frac{L_j}{8} w_j(v)^2 - \sum_{i \neq j} \frac{L_i}{8} \delta_i^2 \leq -\frac{L_j}{8} w_j(v)^2,$$

because $L_i \geq 2$. Hence $m_j(v) \leq Q(v) \leq -\frac{L_j}{8} w_j(v)^2$, and an endpoint v of a longest grid interval gives the bound on β .

(b) Let G be a grid of X , let $[a, a']$ be a grid interval of G_i , and let $t \in [a, a']$. The point $t\mathbf{1}$ is optimal and lies in X , so Proposition 4.2(c) at $x = t\mathbf{1}$ gives $\min\{m_i(a), m_i(a')\} \leq F(t\mathbf{1}) = 0 \leq U$, and the interval is retained. By induction, every stage of such a run has the box X . Suppose that the last stage certifies accuracy ε , that is, $U - \beta \leq \varepsilon$ with β the corrected minimum of the last grid and $U \geq F^* = 0$. By (a) and $L_j \geq 2$, $W_j^2/4 \leq -\beta \leq U - \beta \leq \varepsilon$, so $W_j \leq 2\sqrt{\varepsilon}$. Covering $[0, 1]$ by intervals of length at most $2\sqrt{\varepsilon}$ needs at least $1/(2\sqrt{\varepsilon})$ of them, so $|G_j| \geq 1 + 1/(2\sqrt{\varepsilon})$.

(c) Since $F(\hat{x}) \geq F^* = 0$, $\beta \geq -\varepsilon$. The grid $G^{(0)}$ is a grid of $X^{(0)} = X$. Let $J = [a, a']$ be a grid interval of $G_i^{(0)}$ of length $\delta > 0$ that is not contained in $X_i^{(1)}$. Both a and a' are endpoints of it, so $w_i(a), w_i(a') \geq \delta$, and (a) gives $\min\{m_i^{(0)}(a), m_i^{(0)}(a')\} \leq -\delta^2/4$. Condition (C1) of Definition 4.6 requires this minimum to be at least $\beta \geq -\varepsilon$ (its second alternative needs $w(J) = 0$, that is, $\delta = 0$), so $\delta \leq 2\sqrt{\varepsilon}$. If $k = 0$, (C2) and (a) give $-W_i^2/4 \geq \min_{G^{(0)}} Q^{(0)} \geq \beta \geq -\varepsilon$ for every i . $\square$

Proof of Proposition 10.9. For $c \in X$ and $0 < w \leq 1$ let $f_{c,w}(x) = -\frac{w^2}{6}(1 - \|x - c\|^2/w^2)^3$ when $\|x - c\| \leq w$ and $f_{c,w}(x) = 0$ otherwise. In the variable $z = (x - c)/w$ with $\rho = \|z\|^2$,

$$\partial_i f_{c,w} = w z_i (1 - \rho)^2, \quad \partial_{ij} f_{c,w} = \delta_{ij} (1 - \rho)^2 - 4z_i z_j (1 - \rho).$$

Value, gradient and Hessian vanish on $\rho = 1$, so $f_{c,w}$ is C^2 , and $\partial_{ii} f_{c,w} = (1 - \rho)(1 - \rho - 4z_i^2) \leq 1$. The minimizer c is unique with value $-w^2/6$. Inside the ball, $f_{c,w} + \frac{w^2}{6} = \frac{w^2}{6}(1 - (1 - \rho)^3) \geq \frac{w^2}{6}\rho = \frac{\|x - c\|^2}{6}$, because $1 - (1 - \rho)^3 - \rho = \rho(1 - \rho)(2 - \rho) \geq 0$. Outside it the gap is $w^2/6 \geq \frac{w^2}{6n}\|x - c\|^2$. Hence set growth holds with $g_S = w^2/(6n)$. The zero function has $\arg \min_X f = X$ and satisfies it with every g_S , in particular with $g_S = 1$.

Run the algorithm on $f \equiv 0$; let Q be its number of queries and add the output point, if any, to the query points. Suppose $Q < (1/(2\sqrt{6\varepsilon}) - 1)^n - 1$, and let $M = \lfloor (Q + 1)^{1/n} \rfloor + 1$. Then $M^n > Q + 1$ and $M < 1/(2\sqrt{6\varepsilon})$. Divide X into M^n closed subcubes of side $1/M$. The interior of one of them contains none of the at most $Q + 1$ points. Let c be its center and choose w with $\sqrt{6\varepsilon} < w < 1/(2M)$. The closed ball of radius w lies in the open subcube, so every oracle answer for $f_{c,w}$ at the recorded points equals the answer for 0. By determinism the algorithm produces the same transcript and output on $f_{c,w}$. An interval of width at most ε containing 0 does not contain $-w^2/6 < -\varepsilon$. An output point outside the ball has gap $w^2/6 > \varepsilon$ for $f_{c,w}$. Either way the algorithm is wrong on an admissible function. $\square$

Proof of Proposition 10.10. Summing the recurrences gives $\sum_{i=0}^m a_i x_i = a_0$. The choice $x_0 = 1$, $x_i = 0$ ($i \geq 1$) is feasible. Since all $a_i > 0$, it is the only feasible choice with $x_0 = 1$. A choice with $x_0 = 0$ is feasible exactly when its items sum to a_0 : its partial sums then lie in $[0, 1]$. For each binary x the constraints determine y . Distinct feasible points therefore have distinct binary vectors and distinct integer objective values. The dummy value 2^{m+1} exceeds every value $\leq 2^m - 1$ of a genuine subset. Hence the minimizer is unique, and $x_0^* = 0$ exactly for yes-instances. All $2m + 3$ coordinates lie in $[0, 1]$, so $\|z - z^*\|^2 \leq 2m + 3$, while $F(z) - F^* \geq 1$ for $z \neq z^*$. An approximation within $1/2$ separates $F^* \leq 2^m - 1$ from $F^* = 2^{m+1}$. The data have polynomial binary length, and Subset Sum is NP-complete [75]. $\square$

G.4 Local moment relaxations

This subsection gives the details behind Section 10.7. Throughout, $F(x) = \frac{1}{2}x^\top Hx + b^\top x + F(0)$ is a quadratic on a continuous box X with $H \succeq -\nu I$ for some $\nu \geq 0$. An error bound in which L is replaced by ν holds when all coordinates are conditioned on one global cell.

Lemma G.3 (Globally conditioned moments). *Let $\bar{x} \in C = \prod_i [\ell'_i, u'_i] \subseteq X$, and let Y be a symmetric matrix with $Y - \bar{x}\bar{x}^\top \succeq 0$ and $Y_{ii} \leq (\ell'_i + u'_i)\bar{x}_i - \ell'_i u'_i$ for all i . Then*

$$\frac{1}{2}\langle H, Y \rangle + b^\top \bar{x} + F(0) \geq F^* - \frac{\nu}{8} \sum_i (u'_i - \ell'_i)^2.$$

The same bound, averaged with the corresponding weights, holds for any convex combination of such pairs $(\bar{x}, Y)$ indexed by cells C .

Proof. Put $\Sigma = Y - \bar{x}\bar{x}^\top \succeq 0$. The left side equals $F(\bar{x}) + \frac{1}{2}\langle H, \Sigma \rangle \geq F(\bar{x}) - \frac{\nu}{2} \text{tr } \Sigma$. Also $\Sigma_{ii} = Y_{ii} - \bar{x}_i^2 \leq (u'_i - \bar{x}_i)(\bar{x}_i - \ell'_i) \leq (u'_i - \ell'_i)^2/4$, and $F(\bar{x}) \geq F^*$. Averaging preserves the inequality. $\square$

A global cell index can take K^n values when every coordinate is split into K intervals. Sparse relaxations keep exact information only on bags and match moments across separators.

Definition G.4 (Local moment relaxation of order k). Fix a tree decomposition and a finite partition of every X_i into intervals (cells). A feasible point μ consists of probability measures μ_t on $X_{B_t} = \prod_{i \in B_t} X_i$, one per bag, such that for every tree edge $\{t, u\}$, every product C of cells of the coordinates in S_{tu} , and every monomial $x_{S_{tu}}^\alpha$ with $|\alpha| \leq k$,

$$\int_{\{x_{S_{tu}} \in C\}} x_{S_{tu}}^\alpha d\mu_t = \int_{\{x_{S_{tu}} \in C\}} x_{S_{tu}}^\alpha d\mu_u.$$

Its value is $\sum_t \int q_t d\mu_t$, where q_t is the sum of the factors assigned to t . Point masses at points of X are feasible, so the optimal value of the relaxation is at most F^* . The *occupied width* $\omega_i(\mu)$ is the largest length of a cell of coordinate i that carries positive mass under some μ_t with $i \in B_t$.

Proposition G.5 (Every finite order fails). *For each integer $r \geq 1$ and rational $0 < h \leq 1/(4r)$ there is a continuous box $QP F_{r,h}$ in $2r - 1$ state variables with range $[-1, 1]$ and $2r - 1$ control variables with range $[0, 1]$, with a tree decomposition of maximum bag size three, a unique minimizer and $F^* = 1/4$, such that $\nabla^2 F_{r,h} \succeq -2I$ and $g_r = 1/(8r(1 + 8(2r - 1)^2))$ is a valid growth constant, both independent of h , with the following property. For every choice of cells in which $[0, rh]$ lies in one cell of length at most $2rh$ for each state variable, and 0 and 1 lie in cells of length at most h' for each control variable, the relaxation of order $2r - 1$ has a feasible point μ with value $\frac{1}{4} - \frac{2r+1}{16r}$ and*

$$\sum_i \omega_i(\mu)^2 \leq (4r - 2) \max\{2rh, h'\}^2.$$

Consequently there is no function ϕ such that every feasible point μ of the relaxation of order k has value at least $F^* - \phi(p, \bar{v}/g, k) \bar{v} \sum_i \omega_i(\mu)^2$, for all instances, all $\bar{v} \geq 0$ with $\nabla^2 F \succeq -\bar{v}I$, all growth constants g , all orders k and all cells.

Proof. Construction. Use state variables $s, y_1, \dots, y_{r-1}, z_1, \dots, z_{r-1}$ in $[-1, 1]$ and control variables $u_1, \dots, u_r, v_1, \dots, v_{r-1}$ in $[0, 1]$. Put $y_0 = z_0 = 0, y_r = s$, and

$$\rho_i = \frac{y_i - y_{i-1}}{h} - u_i \quad (1 \leq i \leq r), \quad \rho'_j = \frac{z_j - z_{j-1}}{h} - v_j \quad (1 \leq j < r), \quad \rho'_r = \frac{s - z_{r-1}}{h} - \frac{1}{2},$$

$$F_{r,h} = 2r \left(\sum_{i=1}^r \rho_i^2 + \sum_{j=1}^r \rho'_j{}^2 \right) + \sum_{i=1}^r u_i(1 - u_i) + \sum_{j=1}^{r-1} v_j(1 - v_j) + \lambda(\bar{u} + \bar{v}),$$

where $\bar{u} = \sum_i u_i, \bar{v} = \sum_j v_j$ and $\lambda = 1/(8r)$. The bags $B_i^L = \{y_{i-1}, y_i, u_i\}$ ($i \leq r$), $B_j^R = \{z_{j-1}, z_j, v_j\}$ ($j < r$) and $B_r^R = \{z_{r-1}, s\}$, with constants omitted, are arranged in the path $B_1^L, \dots, B_r^L, B_r^R, B_{r-1}^R, \dots, B_1^R$. The separator between B_r^L and B_r^R is $\{s\}$, and all bags have at most three elements.

Reduction. List the rescaled path nodes $\zeta_0 = 0, \zeta_k = y_k/h$ ($1 \leq k < r$), $\zeta_r = s/h, \zeta_{r+l} = z_{r-l}/h$ ($1 \leq l < r$), $\zeta_{2r} = 0$. Up to sign, the $2r$ residuals are $\zeta_k - \zeta_{k-1} - c_k$ with $c = (u_1, \dots, u_r, -\frac{1}{2}, -v_{r-1}, \dots, -v_1)$; note $\bar{c} := \sum_k c_k = \bar{u} - \bar{v} - \frac{1}{2}$. Let $\zeta^0(u, v)$ be the node values with every residual equal to $-\bar{c}/(2r)$, and $\delta = \zeta - \zeta^0$, with $\delta_0 = \delta_{2r} = 0$. Since the residuals sum to $-\bar{c}$, expanding the squares gives

$$F_{r,h} = \Psi(u, v) + 2r \sum_{k=1}^{2r} (\delta_k - \delta_{k-1})^2, \quad \Psi = \bar{c}^2 + \sum_i u_i(1 - u_i) + \sum_j v_j(1 - v_j) + \lambda(\bar{u} + \bar{v}). \quad (\text{G.1})$$

Since $|\bar{c}| < r$ and $|\sum_{l \leq k} c_l| \leq r$, every node of ζ^0 lies in $[-2r, 2r]$. Hence the corresponding state variables $h\zeta^0$ lie in $[-\frac{1}{2}, \frac{1}{2}]$ and are feasible.

The reduced function. With e_2 the second elementary symmetric polynomial,

$$\Psi - \frac{1}{4} = 2\Upsilon + \lambda(\bar{u} + \bar{v}), \quad \Upsilon = e_2(u) + e_2(v) - \bar{u}\bar{v} + \bar{v}. \quad (\text{G.2})$$

This follows from $\bar{c}^2 = (\bar{u} - \bar{v})^2 - (\bar{u} - \bar{v}) + \frac{1}{4}$, $\sum_i u_i(1 - u_i) = \bar{u} - \sum_i u_i^2$, and $\bar{u}^2 - \sum_i u_i^2 = 2e_2(u)$. The polynomial Υ is affine in each variable, so its minimum over the box is attained at a vertex. At a vertex with $\bar{u} = a$ and $\bar{v} = b$, $\Upsilon = \frac{1}{2}[(a - b)^2 - (a - b)] = \frac{1}{2}(a - b)(a - b - 1) \geq 0$, because $a - b$ is an integer. Therefore $\Psi \geq \frac{1}{4} + \lambda(\bar{u} + \bar{v})$, and (G.1) shows that the unique minimizer is $u = v = 0$, $\zeta = \zeta^0(0, 0)$, with $F^* = \frac{1}{4}$.

Growth and curvature. By (G.1), (G.2), $\bar{u} + \bar{v} \geq \|(u, v)\|^2$, and the path inequality (10.2) with $2r$ in place of m ,

$$F_{r,h} - \frac{1}{4} \geq \lambda\|(u, v)\|^2 + \frac{2}{2r-1}\|\delta\|^2.$$

Each node of ζ^0 is a partial sum of the c_l minus $k/(2r)$ times $\bar{c}$, so each of the $2r - 1$ free nodes moves by at most $2\|(u, v)\|_1 \leq 2\sqrt{2r-1}\|(u, v)\|$ when the controls move from 0 to (u, v) . Hence $\|\zeta^0(u, v) - \zeta^0(0, 0)\|^2 \leq 4(2r-1)^2\|(u, v)\|^2$. The state variables are h times the nodes and $h \leq 1$, so the squared distance to the minimizer is at most $2\|\delta\|^2 + (1 + 8(2r-1)^2)\|(u, v)\|^2$. This gives the growth constant $g_r = \lambda/(1 + 8(2r-1)^2) \leq 1/(2r-1)$. The Hessian is $4r$ times a sum of rank-one Gram terms minus 2 on the controls, so $\nabla^2 F_{r,h} \succeq -2I$.

The relaxation witness. On the left subtree take the law under which $\bar{u} = i$ with probability $\pi_i = \binom{2r}{2i} 2^{1-2r}$ ($0 \leq i \leq r$), $u = (1^i, 0^{r-i})$, $y_k = h \sum_{l \leq k} u_l$ and $s = hi$. On the right subtree

take $\bar{v} = j$ with probability $\pi'_j = \binom{2r}{2j+1} 2^{1-2r}$ ($0 \leq j < r$), $v = (1^j, 0^{r-1-j})$, $z_k = h \sum_{l \leq k} v_l$ and $s = h(j + \frac{1}{2})$. Both are probability laws, all residuals vanish on their supports, and all controls are binary. Let μ_t be the corresponding bag marginals. At every separator other than $\{s\}$ the two bag marginals come from one law, so they agree on every set and in all moments. At the separator $\{s\}$ the laws of $2s/h$ are those of a $\text{Bin}(2r, \frac{1}{2})$ variable conditioned on being even, respectively odd. Their moments of order α differ by a multiple of $\sum_{k=0}^{2r} (-1)^k \binom{2r}{k} k^\alpha$, which is zero for $\alpha < 2r$ as a finite difference of order $2r$ of a polynomial of degree α . Thus the moments of s on the two sides agree through order $2r - 1$.

All state atoms lie in $[0, rh]$, and all control atoms lie in $\{0, 1\}$. With cells as in the statement, s carries all its mass, on both sides, in the cell that contains $[0, rh]$, so the cellwise conditions at $\{s\}$ reduce to the moment equalities just proved. Hence μ is feasible for the relaxation of order $2r - 1$. By the symmetry $k \mapsto 2r - k$, $\mathbb{E}\bar{u} = r/2$ and $\mathbb{E}\bar{v} = (r - 1)/2$. Since $F_{r,h} = \lambda(\bar{u} + \bar{v})$ on the supports, the value of μ is $\lambda(r - \frac{1}{2}) = \frac{1}{4} - \frac{2r+1}{16r}$. Each state variable occupies only the cell that contains $[0, rh]$, and each control only the cells that contain 0 and 1; with $2r - 1$ variables of each kind this gives the bound on $\sum_i \omega_i(\mu)^2$.

Consequence. Given k , choose r with $2r - 1 \geq k$, so that μ is also feasible for the relaxation of order k , and fix $p = 3$, $\bar{v} = 2$ and $g = g_r$. The value of μ is $F^* - \frac{2r+1}{16r} < F^* - \frac{1}{8}$, whereas $\sum_i \omega_i(\mu)^2 \rightarrow 0$ as $h, h' \rightarrow 0$ with r fixed. $\square$

The two bag chains realize the even and the odd part of a binomial law on the separator, whose moments agree up to order $2r - 1$. The example does not contradict an error bound in terms of L , which is of order r/h^2 here. It also does not survive exact matching of separator laws: on a tree decomposition, bag laws that agree on every separator glue to a probability measure on X [180], whose value is at least F^* . The following remark shows that an inequality coupling both sides of a separator detects the gap.

Remark G.6 (The smallest case). The case $r = 2$ admits a four-variable form. On $[0, 1]^4$ with $0 < h \leq \frac{1}{2}$, let

$$F_h = 8\left(\frac{s}{h} - \frac{u_1 + u_2}{2}\right)^2 + 8\left(\frac{s}{h} - \frac{1}{4} - \frac{v}{2}\right)^2 + u_1(1 - u_1) + u_2(1 - u_2) + v(1 - v) + \frac{u_1 + u_2 + v}{16},$$

with bags $\{s, u_1, u_2\}$ and $\{s, v\}$. Direct expansion with $\delta = s/h - \frac{1}{8} - \frac{u_1 + u_2 + v}{4}$ gives $F_h - \frac{1}{4} = 16\delta^2 + 2[(1 - v)u_1u_2 + v(1 - u_1)(1 - u_2)] + (u_1 + u_2 + v)/16$. Hence the minimizer $(h/8, 0, 0, 0)$ is unique and $g = 1/22$ is valid, using $\|x - x^*\|^2 \leq 2\delta^2 + \frac{11}{8}\|(u_1, u_2, v)\|^2$ and $u_1 + u_2 + v \geq \|(u_1, u_2, v)\|^2$. Moreover $\nabla^2 F_h + 2I = 16\chi_1\chi_1^\top + 16\chi_2\chi_2^\top + 2e_s e_s^\top$ with $\chi_1 = (1/h, -\frac{1}{2}, -\frac{1}{2}, 0)$ and $\chi_2 = (1/h, 0, 0, -\frac{1}{2})$. Since $(0, 1, -1, 0)$ is orthogonal to χ_1, χ_2 and e_s , $\nu = 2$ exactly, while $L = 32/h^2$. The left measure with masses $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$ at $(s/h, u_1, u_2) = (0, 0, 0), (\frac{1}{2}, 1, 0), (\frac{1}{2}, 0, 1), (1, 1, 1)$ and the right measure with masses $\frac{1}{2}, \frac{1}{2}$ at $(s/h, v) = (\frac{1}{4}, 0), (\frac{3}{4}, 1)$ have equal moments of s/h through order three $(\frac{1}{2}, \frac{5}{16}, \frac{7}{32})$ but not four $(\frac{11}{64} \text{ versus } \frac{41}{256})$. Their value is $3/32$, a gap of $5/32$.

These bag moments admit a global positive semidefinite completion. With mean $(h/2, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and covariance $\frac{1}{16}aa^\top + \frac{3}{16}bb^\top$, $a = (h, 1, 1, 2)$, $b = (0, 1, -1, 0)$, the completion satisfies every McCormick inequality for $s \in [0, 2h]$ and the control boxes. It violates the valid inequality $u_1u_2 + v - u_1v - u_2v = (1 - v)u_1u_2 + v(1 - u_1)(1 - u_2) \geq 0$, a triangle inequality of the Boolean quadric polytope [150], with pseudo-expectation $-1/8$. This inequality couples variables on both sides of the separator.

H Additional computational results

This appendix gives the details and tables that Section 11 summarizes.

Implementation (Section 11.1). The implementation differs from the analysis in the following ways; none of them affects validity.

- (a) All coordinates share the mesh $h_j = s2^{-j}$, and the analysis that applies is Lemma 5.9 with the Euclidean condition number κ , not Theorem 5.7.
- (b) The cap of trial μ is $100 \cdot 2^\mu \lceil \log_2(n+2) \rceil$, 12.5 times the cap of Lemma 5.9, and the stage limit is the least $j_{\max}$ with $\frac{7}{8}Lns^24^{-j_{\max}} \leq \varepsilon$ instead of $\frac{9}{16}Lns^24^{-j_{\max}} \leq \varepsilon$, so a trial may run one stage longer.
- (c) Each trial starts with the center at the current incumbent instead of ℓ . Validity and the rate analysis allow any first center (the bound on $\|c - x^*\|$ in (G2) of Lemma 5.9 holds at $j = 0$ for every $c \in X$); the bit-length analysis of Appendix A does not cover it.
- (d) Before stage 0, two sweeps of exact coordinate descent from ℓ , u and the box midpoint provide the first incumbent; after each stage the same descent, within the current box, starts from the grid minimizer y_j . An optional presolve certifies convex instances by an exact $LDL^\top$ factorization and a rational KKT point.
- (e) The lower bound used in the success test and reported at the end is the largest bound of all stages and trials, including the initial interval bound, rather than β_j of the last stage.
- (f) The “uniform” grids of the experiments are the grids of Definition 5.1 with $\theta = 0$ around one center, not the unions of cells of UC (Algorithm 9.2).
- (g) The exact-output wrapper, which implements EX (Algorithm 6.13), uses the acceptance rule of Proposition 6.6 with the row-sum height constant of Remark 6.5, with a common denominator that also clears $H_{ij}/2$; this constant is never smaller than the one in (6.1). It starts at $q = 4$, doubles q after each round, restarts the grid solver from the full box in every round with the incumbent as its first center and, in place of ℓ , as a start of the descent, and tests three candidates after each round: the incumbent, its coordinatewise rational reconstruction (Remark 6.15) and the stationary point of REC (Algorithm 6.7).

Further modules, each with its own checker, implement exact linear and convex box-QP oracles with primal, dual or Farkas witnesses, affine recourse recognition and lifting, minimum-cut and convex value-factor recourse, the TU-constrained grids, the endpoint optimal-set descriptor and explicit polynomial factors. The convex value-factor and recourse checkers reuse the solver’s grid, table and lifting code, so only the main checker is independent in the sense of Section 11.1.

Planted family (Section 11.2). Lemma 3.3(a) gives the certified lower bound g_0 on the growth constant, by an exact $LDL^\top$ test of $H + 2M - 2g_0I$, and part (b) with a rational approximate eigenvector of $H_{J_0J_0}$ gives an upper bound; with $L = 2$ this brackets κ to within 12%.

The following observation shows when a free coordinate of x^* can be a grid node. Within a trial every node of coordinate i is either a dyadic rational or the sum of the first center $c_i^{(0)}$ and a dyadic rational, because the endpoints ± 1 , the meshes and the gradings are dyadic, filtering moves endpoints only to nodes, and every later center is a node. A coordinate $x_i^* = k/21$ with $k \neq 0$ is therefore a node only if $c_i^{(0)}$, which the coordinate descent provides, differs from x_i^* by a dyadic rational. In the 78 single-trial runs of E1 with filtering and of E3 with $\kappa_{\text{target}} = 4$, 153 free coordinates of x^* were nodes of the last grid: 146 had already been set to x_i^* by the initial descent, 3 had $x_i^* = 0$, and in the other 4 the first center differed from x_i^* by a nonzero dyadic rational.

Table 3: CT on the expanding-box chain Ψ_m (gap $\leq 10^{-3}$). “Nodes” is the largest number of nodes of a coordinate grid, “entries” the total number of table entries, and the last column the lower bound on the number of pieces of an exact message.

m	n	stages	nodes	entries	solve (s)	replay (s)	message pieces
2	4	9	5	808	0.01	0.02	≥ 2
4	8	12	6	3,143	0.06	0.10	≥ 8
8	16	16	8	9,403	0.16	0.29	≥ 128
16	32	25	9	36,123	0.76	1.28	$\geq 3.3 \cdot 10^4$
32	64	41	10	153,981	3.50	6.21	$\geq 2.1 \cdot 10^9$
64	128	74	11	585,760	14.8	28.7	$\geq 9.2 \cdot 10^{18}$

Experiments E1–E3 (Section 11.2). The graded runs of E1 with filtering and the runs of E2 and E3 with the theorem’s grading satisfy $8\kappa\theta^2 \leq 1$. At all 876 stages of these runs, the gap $D(y_j)$ was at most $0.16 Lnh_j^2$ (bound $\frac{9}{16} Lnh_j^2$), the retained radius at most 0.18 of $4.2\sqrt{n\kappa}h_j$, and the number of nodes per coordinate at most 5% of the cap $8\theta^{-1}\lceil\log_2(n+2)\rceil$ (and at most 0.4% of the implementation’s cap). The radius ratio uses the lower end of the certified interval for κ , which can only increase it. E3 was also run with $\kappa_{\text{target}} = 2$, for which the free coordinates are separable (Section 11.2); on these instances filtered uniform grids had exactly $4(\lfloor\sqrt{n/4}\rfloor + 1) + 1$ nodes per free coordinate in the last four stages. Section 11.2 reports only $\kappa_{\text{target}} = 4$ for E3, because the separable case does not test coupling.

Expanding-box chain (Section 11.3). Table 3 gives the results of CT on Ψ_m for each m .

Exact output (Section 11.4). In E4, the 15 certified instances that needed more than one stage needed 67–77, 132–144, 275 and 534 stages for 49–65, 80–120, 208 and 342 bits, with last-round precisions $q = 64, 128, 256$ and 512.

Comparison with SCIP (Section 11.5). Table 4 gives the results of E5 per instance class. The planted objectives have constants between 94 and $1.6 \cdot 10^3$ and optimal value 0, so the absolute target 10^{-6} is between $6 \cdot 10^{-10}$ and $1.1 \cdot 10^{-8}$ of the constant; the inward derivatives at the active coordinates are between 23 and 192.5. A rerun of the 21 SCIP runs reproduced every primal bound. For SCIP’s solution (x, t) , let $\bar{x}$ be the projection of x onto the box; then $F(\bar{x}) - t$ is the shortfall up to at most $1.3 \cdot 10^{-14}$. The rerun evaluated F exactly and split $F(\bar{x}) - t$ into the bound part $F(\bar{x}) - F(x)$ and the epigraph part $F(x) - t$. In every run the violated bounds were exactly those of the coordinates at a bound in x^* , each violated outward by about 10^{-8} , and SCIP’s own check accepted the solution. The bound part was 51% to 94% of the shortfall, and the epigraph part was between $9.00 \cdot 10^{-7}$ and $9.03 \cdot 10^{-7}$. In separate runs, giving SCIP 60 seconds (paths with $n \geq 32$), setting its primal and dual feasibility tolerances to 10^{-9} (paths with $n \geq 32$), or moving the objective constant into an offset (all 21 instances) did not change the outcome: SCIP stopped at the time limit in all 27 runs on paths with $n \geq 32$ (with 60 seconds, at dual bounds between $-7.5 \cdot 10^{-4}$ and $-5.1 \cdot 10^{-5}$); with the offset, the exact gap on the 12 instances with $n \leq 16$ was still $1.9 \cdot 10^{-6}$ to $5.8 \cdot 10^{-6}$; and its reported interval contained the optimum in none of the 39 runs.

Random instances (Section 11.6). The instances of E6 have diagonal entries $k/2$ with k uniform in $\{-4, \dots, 8\}$, interaction entries $\pm k/4$ with $k \in \{1, \dots, 6\}$, linear entries $k/4$ with $k \in \{-8, \dots, 8\}$, and band width 2 for the band instances. The candidate $\hat{x}$ is the stationary point of the face of

Table 4: Experiment E5: planted family, $\kappa \in [3.99, 4.45]$, three seeds per row. “Bag” is the maximum bag size. CT: maximum solve and replay times (gap $\leq 10^{-6}$ certified on all instances, 60-second limit). SCIP: status, maximum time (20-second limit), largest exact gap (exact value of SCIP’s incumbent, projected onto the box, minus its dual bound) and smallest dual bound. Because $F^* = 0$ and SCIP’s projected incumbents are optimal to within $1.3 \cdot 10^{-14}$, the exact gap equals minus the dual bound to the digits shown.

graph	n	bag	CT (s)		status	SCIP		
			solve	replay		time (s)	gap (exact)	dual bound
path	16	2	0.33	0.39	gap reported	5.4	$2.9 \cdot 10^{-6}$	$-2.9 \cdot 10^{-6}$
tree	16	2	0.30	0.43	gap reported	0.8	$5.5 \cdot 10^{-6}$	$-5.5 \cdot 10^{-6}$
band	12	3	1.33	2.12	gap reported	1.4	$6.8 \cdot 10^{-6}$	$-6.8 \cdot 10^{-6}$
band	8	4	6.66	11.1	gap reported	0.3	$5.5 \cdot 10^{-6}$	$-5.5 \cdot 10^{-6}$
path	32	2	0.99	1.20	time limit	20	$1.4 \cdot 10^{-4}$	$-1.4 \cdot 10^{-4}$
path	64	2	3.02	3.62	time limit	20	$3.0 \cdot 10^{-4}$	$-3.0 \cdot 10^{-4}$
path	128	2	6.90	9.39	time limit	20	$7.6 \cdot 10^{-4}$	$-7.6 \cdot 10^{-4}$

Table 5: Experiment E6: CT on unplanted random box QPs, five instances per row. “Bag” is the maximum bag size. “Growth” counts the instances on which Lemma 3.3(a) certifies point growth. “Nodes” is the largest number of nodes of a coordinate grid over all stages and the five instances, for each accuracy ε ; “stages” and the time (maximum, seconds) are for $\varepsilon = 2^{-50}$.

sparsity	n	bag	growth	nodes at $\varepsilon =$					stages	time
				2^{-10}	2^{-20}	2^{-30}	2^{-40}	2^{-50}		
path	8	2	3	16	16	16	16	16	27–28	0.20
path	12	2	3	15	15	16	16	16	28	0.29
path	16	2	0	12	13	15	15	15	28	0.42
path	24	2	0	16	17	17	17	17	29	0.67
band	8	3	2	10	10	10	10	10	27–28	0.57
band	12	3	2	13	14	14	14	14	28	0.81
band	16	3	1	15	16	18	18	18	28–29	5.22
band	24	3	0	15	15	15	15	15	29	1.43

the final incumbent at $\varepsilon = 2^{-50}$. When the sufficient condition of Lemma 3.3(a) holds at $\hat{x}$, it proves that $\hat{x}$ is the unique minimizer and gives a lower bound g_0 on the point-growth constant; an upper bound comes from part (b) and from the definition of growth at the points obtained by moving one active coordinate of $\hat{x}$ to its other endpoint. With $L = \max_i \max\{H_{ii}, 0\}$ this brackets κ . The condition held on 11 instances, with the certified intervals $[6.2, 6.6]$, $[11.6, 15.7]$, $[14.0, 22.3]$, $[6.1, 24.4]$, $[10.0, 28.6]$, $[8.1, 31.6]$, $[5.1, 55.8]$, $[9.5, 67.3]$, $[6.8, 141.9]$, $[4.5, 182.8]$ and $[8.0, 6662.2]$ for κ (rounded outward). On the other 29 instances the matrix $H + 2M$ was not positive definite. On 28 of them the localized test of Proposition 6.17, applied to the same certificate with the stationary point of the face of the stage incumbent as candidate, proved that candidate optimal; on one instance neither test succeeded. At $\varepsilon = 2^{-50}$ the 11 instances with certified growth had 9 to 11 nodes per coordinate, the other 29 had 8 to 18. The candidates are mostly off the grids: at $\varepsilon = 2^{-50}$, 42 of the 213 free coordinates of the candidates were nodes of the last grid. Table 5 gives the grid sizes per instance class.

Table 6: Plain corrected grids (CT) versus the recourse pipeline (gap target 2^{-10} , 20-second limit). Status: “exact”, exact minimizer certified; “certified”, gap at most 2^{-10} certified; “time limit” and “table limit”, run stopped by the time limit or because the next stage would need more than $2 \cdot 10^5$ table entries. “Gap” is the certified gap at the end of the plain run, “bag” the maximum bag size of the decomposition used by the plain grids, and “entries” the total number of table entries of the plain run; times in seconds.

instance	n	bag	plain grids					recourse		
			status	gap	entries	solve	replay	status	solve	replay
affine star	5	2	time limit	0.019	559,320	20.0	26.4	exact	0.005	0.001
affine star	17	2	time limit	1.23	823,569	20.0	50.9	exact	0.061	0.057
piecewise, $M = 1$	3	3	certified	$4 \cdot 10^{-4}$	498	0.017	0.017	certified	0.013	0.003
piecewise, $M = 100$	3	3	certified	$7 \cdot 10^{-4}$	35,600	0.63	1.12	certified	0.004	0.002
dense cut	7	7	certified	2^{-10}	2,038	0.044	0.125	exact	0.003	0.003
dense cut	33	33	table limit	2.96	0	0.021	0.009	certified	0.078	0.064

Recourse (Section 11.7). The affine stars have a core coordinate whose own term $-x_0^2$ is concave and 4 or 16 leaves coupled to it by stiff convex squares $64(y_i - x_0/2)^2$, with the variables in random order; recognition finds the globally affine responses, and the reduced problem is solved exactly. The piecewise instances are the functions $\phi_M - z^2$ with ϕ_M from Example C.11, for $M = 1$ and $M = 100$. The dense instances ($n = 7$ and 33) have one convex core coordinate and a residual that is concave in each coordinate and submodular, with a complete interaction graph, so the decomposition used by plain grids has a bag of size n . Table 6 gives the results.